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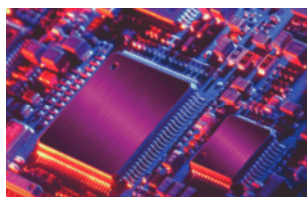
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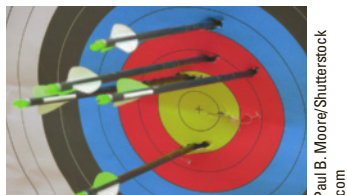
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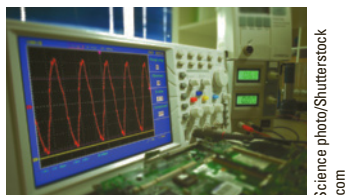


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TO THE STUDENT

Authors of books live with the hope that someone actually *reads* them. Contrary to what you might believe, almost everything in a typical college-level mathematics text is written for you and not the instructor. True, the topics covered in the text are chosen to appeal to instructors because they make the decision on whether to use it in their classes, but everything written in it is aimed directly at you, the student. So I want to encourage you—no, actually I want to *tell* you—to read this textbook! But do not read this text as you would a novel; you should not read it fast and you should not skip anything. Think of it as a *workbook*. By this I mean that mathematics should always be read with pencil and paper at the ready because, most likely, you will have to *work* your way through the examples and the discussion. Before attempting any problems in the section exercise sets, work through *all* the examples in that section. The examples are constructed to illustrate what I consider the most important aspects of the section, and therefore, reflect the procedures necessary to work most of the problems. When reading an example, copy it down on a piece of paper and do not look at the solution in the book. Try working it, then compare your results against the solution given, and, if necessary resolve any differences. I have tried to include most of the important steps in each example, but if something is not clear you should always try—and here is where the pencil and paper come in again—to fill in the details or missing steps. This may not be easy, but it is part of the learning process. The accumulation of facts followed by the slow assimilation of understanding simply cannot be achieved without a struggle.

Specifically for you, a *Student Resource Manual (SRM)* is available as an optional supplement. In addition to containing solutions of selected problems from the exercises sets, the *SRM* contains hints for solving problems, extra examples, and a review of those areas of algebra and calculus that I feel are particularly important to the successful study of differential equations. Bear in mind you do not have to purchase the *SRM*; you can review the appropriate mathematics from your old pre-calculus or calculus texts.

In conclusion, I wish you good luck and success. I hope you enjoy the text and the course you are about to embark on—as an undergraduate math major it was one of my favorites because I liked mathematics that connected with the physical world. If you have any comments, or if you find any errors as you read/work your way through the text, or if you come up with a good idea for improving either it or the *SRM*, please feel free to contact me through Cengage Learning:

spencer.arritt@cengage.com.

TO THE INSTRUCTOR

In case you are examining this text for the first time, *Differential Equations with Boundary-Value Problems*, Ninth Edition, can be used for either a one- or two-semester course that covers ordinary and partial differential equations. The shorter version of the text, *A First Course in Differential Equations with Modeling Applications*, Eleventh Edition, is intended for either a one-semester or one-quarter course in ordinary differential equations. This text ends with Chapter 9. For a one-semester course, it is assumed that the students have successfully completed at least two semesters of calculus. Since you are reading this, undoubtedly you have already examined

the table of contents for the topics that are covered. You will not find a “suggested syllabus” in this preface; I will not pretend to be so wise as to tell other teachers what to teach. I feel that there is plenty of material here to choose from and to form a course to your liking. The text strikes a reasonable balance between the analytical, qualitative, and quantitative approaches to the study of differential equations. As far as my “underlying philosophy” goes, it is this: An undergraduate text should be written with the students’ understanding kept firmly in mind, which means to me that the material should be presented in a straightforward, readable, and helpful manner, while keeping the level of theory consistent with the notion of a “first course.”

For those who are familiar with the previous editions, I would like to mention a few improvements made in this edition.

- Many exercise sets have been updated by the addition of new problems. Some of these problems involve new and, I think, interesting mathematical models.
- Additional examples, figures, and remarks have been added to many sections.
- Throughout the text I have given a greater emphasis to the concepts of piecewise-linear differential equations and solutions that involve nonelementary integrals.
- Appendix A, *Integral-Defined Functions*, is new to the text.
- The superposition principle has been added to the discussion in Section 12.4, *Wave Equation*.
- Section 12.6, *Nonhomogeneous Boundary-Value Problems*, has been rewritten.
- Modified Bessel functions are given a greater emphasis in Section 13.6, *Polar and Cylindrical Coordinates*.

Student Resources

- **Student Resource Manual (SRM)**, prepared by Warren S. Wright and Roberto Martinez (ISBN 978-1-305-96573-7, accompanies *A First Course in Differential Equations with Modeling Applications*, Eleventh Edition, and ISBN 978-1-305-96581-2 accompanies *Differential Equations with Boundary-Value Problems*, Ninth Edition) provides important review material from algebra and calculus, the solution of every third problem in each exercise set (with the exception of the Discussion Problems and Computer Lab Assignments), relevant command syntax for the computer algebra systems *Mathematica* and *Maple*, and lists of important concepts, as well as helpful hints on how to start certain problems.
- **MindTap for Differential Equations with Boundary-Value Problems**, Ninth Edition, is a digital representation of your course that provides you with the tools you need to better manage your limited time, stay organized, and be successful. You can complete assignments whenever and wherever you are ready to learn with course material specially customized for you by your instructor and streamlined in one proven, easy-to-use interface. With an array of study tools, you will get a true understanding of course concepts, achieve better grades, and set the groundwork for your future courses. Learn more at www.cengage.com/mindtap.

Instructor Resources

- **Instructor’s Solutions Manual (ISM)**, prepared by Warren S. Wright and Roberto Martinez, provides complete worked-out solutions for all problems in the text. It is available through the Instructor Companion website at cengage.com.
- **Cengage Learning Testing Powered by Cognero** is a flexible online system that allows you to author, edit, and manage test bank content, create multiple test versions in an instant, and deliver tests from your learning management system (LMS), your classroom, or wherever you want. This is available online at www.cengage.com/login.
- Turn the light on with **MindTap for Differential Equations with Boundary Value Problems**, Ninth Edition. Through personalized paths of dynamic

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ACKNOWLEDGEMENTS

Compiling a mathematics textbook such as this and making sure that its thousands of symbols and hundreds of equations are accurate is an enormous task, but since I am called “the author,” that is my job and responsibility. But many people besides myself have expended enormous amounts of time and energy in working toward its eventual publication. So I would like to take this opportunity to express my sincerest appreciation to everyone—most of them unknown to me—at Cengage Learning and at MPS North America who were involved in the publication of this edition. A special word of thanks goes to Spencer Arritt, Kathryn Schrupf, Jennifer Ridsen, Vernon Boes, and Jill Traut for their guidance in the labyrinth of the production process.

Finally, over the years, this text has been improved in a countless number of ways through the suggestions and criticisms of the reviewers. Thus it is fitting to conclude with an acknowledgement of my debt to the following generous people for sharing their expertise and experience.

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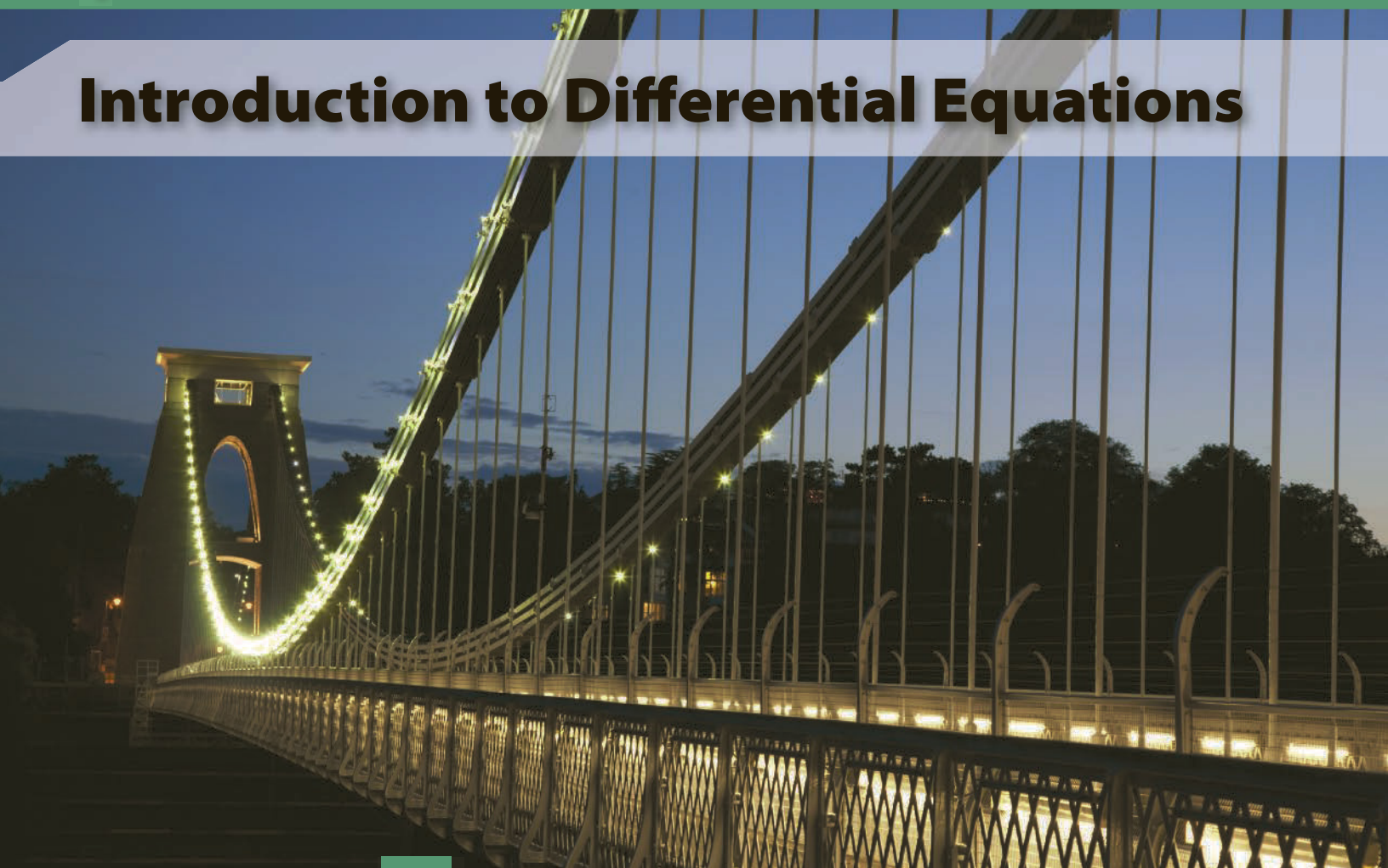
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Differential Equations with Boundary-Value Problems

Introduction to Differential Equations



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- 1.1 Definitions and Terminology
- 1.2 Initial-Value Problems
- 1.3 Differential Equations as Mathematical Models

CHAPTER 1 IN REVIEW

The words *differential* and *equations* suggest solving some kind of equation that contains derivatives y' , y'' , $\dots$. Analogous to a course in algebra, in which a good amount of time is spent solving equations such as $x^2 + 5x + 4 = 0$ for the unknown number x , in this course *one* of our tasks will be to solve differential equations such as $y'' + 2y' + y = 0$ for an unknown function $y = \phi(x)$. As the course unfolds, you will see there is more to the study of differential equations than just mastering methods that mathematicians over past centuries devised to solve them. But first things first. In order to read, study, and be conversant in a specialized subject you have to learn some of the terminology of that discipline. This is the thrust of the first two sections of this chapter. In the last section we briefly examine the link between differential equations and the real world.

1.1 Definitions and Terminology

INTRODUCTION The derivative dy/dx of a function $y = \phi(x)$ is itself another function $\phi'(x)$ found by an appropriate rule. The exponential function $y = e^{0.1x^2}$ is differentiable on the interval $(-\infty, \infty)$ and by the Chain Rule its first derivative is $dy/dx = 0.2xe^{0.1x^2}$. If we replace $e^{0.1x^2}$ on the right-hand side of the last equation by the symbol y , the derivative becomes

$$\frac{dy}{dx} = 0.2xy. \quad (1)$$

Now imagine that a friend of yours simply hands you equation (1)—you have no idea how it was constructed—and asks, *What is the function represented by the symbol y ?* You are now face to face with one of the basic problems in this course:

How do you solve an equation such as (1) for the function $y = \phi(x)$?

A DEFINITION The equation that we made up in (1) is called a **differential equation**. Before proceeding any further, let us consider a more precise definition of this concept.

DEFINITION 1.1.1 Differential Equation

An equation containing the derivatives of one or more unknown functions (or dependent variables), with respect to one or more independent variables, is said to be a **differential equation (DE)**.

To talk about them, we shall classify differential equations according to **type**, **order**, and **linearity**.

CLASSIFICATION BY TYPE If a differential equation contains only ordinary derivatives of one or more unknown functions with respect to a *single* independent variable, it is said to be an **ordinary differential equation (ODE)**. An equation involving partial derivatives of one or more unknown functions of two or more independent variables is called a **partial differential equation (PDE)**. Our first example illustrates several of each type of differential equation.

EXAMPLE 1 Types of Differential Equations

(a) The equations

$$\frac{dy}{dx} + 5y = e^x, \quad \frac{d^2y}{dx^2} - \frac{dy}{dx} + 6y = 0, \quad \text{and} \quad \frac{dx}{dt} + \frac{dy}{dt} = 2x + y \quad (2)$$

an ODE can contain more than one unknown function
↓ ↓

are examples of ordinary differential equations.

(b) The following equations are partial differential equations:*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - 2 \frac{\partial u}{\partial t}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \quad (3)$$

*Except for this introductory section, only ordinary differential equations are considered in *A First Course in Differential Equations with Modeling Applications*, Eleventh Edition. In that text the word *equation* and the abbreviation DE refer only to ODEs. Partial differential equations or PDEs are considered in the expanded volume *Differential Equations with Boundary-Value Problems*, Ninth Edition.

Notice in the third equation that there are two unknown functions and two independent variables in the PDE. This means u and v must be functions of *two or more* independent variables. ■

NOTATION Throughout this text ordinary derivatives will be written by using either the **Leibniz notation** dy/dx , d^2y/dx^2 , d^3y/dx^3 , $\dots$ or the **prime notation** y' , y'' , y''' , $\dots$. By using the latter notation, the first two differential equations in (2) can be written a little more compactly as $y' + 5y = e^x$ and $y'' - y' + 6y = 0$. Actually, the prime notation is used to denote only the first three derivatives; the fourth derivative is written $y^{(4)}$ instead of y'''' . In general, the n th derivative of y is written $d^n y/dx^n$ or $y^{(n)}$. Although less convenient to write and to typeset, the Leibniz notation has an advantage over the prime notation in that it clearly displays both the dependent and independent variables. For example, in the equation

$$\frac{d^2x}{dt^2} + 16x = 0$$

unknown function
↙ or dependent variable
↑ independent variable

it is immediately seen that the symbol x now represents a dependent variable, whereas the independent variable is t . You should also be aware that in physical sciences and engineering, Newton's **dot notation** (derogatorily referred to by some as the “fleyspeck” notation) is sometimes used to denote derivatives with respect to time t . Thus the differential equation $d^2s/dt^2 = -32$ becomes $\ddot{s} = -32$. Partial derivatives are often denoted by a **subscript notation** indicating the independent variables. For example, with the subscript notation the second equation in (3) becomes $u_{xx} = u_{tt} - 2u_t$.

CLASSIFICATION BY ORDER The **order of a differential equation** (either ODE or PDE) is the order of the highest derivative in the equation. For example,

$$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 4y = e^x$$

second order ↘ ↙ first order

is a second-order ordinary differential equation. In Example 1, the first and third equations in (2) are first-order ODEs, whereas in (3) the first two equations are second-order PDEs. A first-order ordinary differential equation is sometimes written in the **differential form**

$$M(x, y)dx + N(x, y)dy = 0.$$

EXAMPLE 2 Differential Form of a First-Order ODE

If we assume that y is the dependent variable in a first-order ODE, then recall from calculus that the differential dy is defined to be $dy = y'dx$.

(a) By dividing by the differential dx an alternative form of the equation $(y - x)dx + 4xdy = 0$ is given by

$$y - x + 4x \frac{dy}{dx} = 0 \quad \text{or equivalently} \quad 4x \frac{dy}{dx} + y = x.$$

(b) By multiplying the differential equation

$$6xy \frac{dy}{dx} + x^2 + y^2 = 0$$

by dx we see that the equation has the alternative differential form

$$(x^2 + y^2) dx + 6xy dy = 0. \quad \blacksquare$$

In symbols we can express an n th-order ordinary differential equation in one dependent variable by the general form

$$F(x, y, y', \dots, y^{(n)}) = 0, \quad (4)$$

where F is a real-valued function of $n + 2$ variables: $x, y, y', \dots, y^{(n)}$. For both practical and theoretical reasons we shall also make the assumption hereafter that it is possible to solve an ordinary differential equation in the form (4) uniquely for the highest derivative $y^{(n)}$ in terms of the remaining $n + 1$ variables. The differential equation

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}), \quad (5)$$

where f is a real-valued continuous function, is referred to as the **normal form** of (4). Thus when it suits our purposes, we shall use the normal forms

$$\frac{dy}{dx} = f(x, y) \quad \text{and} \quad \frac{d^2 y}{dx^2} = f(x, y, y')$$

to represent general first- and second-order ordinary differential equations.

EXAMPLE 3 Normal Form of an ODE

(a) By solving for the derivative dy/dx the normal form of the first-order differential equation

$$4x \frac{dy}{dx} + y = x \quad \text{is} \quad \frac{dy}{dx} = \frac{x - y}{4x}.$$

(b) By solving for the derivative y'' the normal form of the second-order differential equation

$$y'' - y' + 6 = 0 \quad \text{is} \quad y'' = y' - 6y. \quad \blacksquare$$

CLASSIFICATION BY LINEARITY An n th-order ordinary differential equation (4) is said to be **linear** if F is linear in $y, y', \dots, y^{(n)}$. This means that an n th-order ODE is linear when (4) is $a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x)$ or

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x). \quad (6)$$

Two important special cases of (6) are linear first-order ($n = 1$) and linear second-order ($n = 2$) DEs:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad \text{and} \quad a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x). \quad (7)$$

In the additive combination on the left-hand side of equation (6) we see that the characteristic two properties of a linear ODE are as follows:

- The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the first degree, that is, the power of each term involving y is 1.
- The coefficients $a_0, a_1, \dots, a_n$ of $y, y', \dots, y^{(n)}$ depend at most on the independent variable x .

A **nonlinear** ordinary differential equation is simply one that is not linear. Nonlinear functions of the dependent variable or its derivatives, such as $\sin y$ or $e^{y'}$, cannot appear in a linear equation.

EXAMPLE 4 Linear and Nonlinear ODEs

(a) The equations

$$(y - x) dx + 4x dy = 0, \quad y'' - 2y + y = 0, \quad x^3 \frac{d^3 y}{dx^3} + x \frac{dy}{dx} - 5y = e^x$$

are, in turn, *linear* first-, second-, and third-order ordinary differential equations. We have just demonstrated in part (a) of Example 2 that the first equation is linear in the variable y by writing it in the alternative form $4xy' + y = x$.

(b) The equations

$$\begin{array}{ccc} \text{nonlinear term:} & \text{nonlinear term:} & \text{nonlinear term:} \\ \text{coefficient depends on } y & \text{nonlinear function of } y & \text{power not 1} \\ \downarrow & \downarrow & \downarrow \\ (1 - y)y' + 2y = e^x, & \frac{d^2 y}{dx^2} + \sin y = 0, & \text{and } \frac{d^4 y}{dx^4} + y^2 = 0 \end{array}$$

are examples of *nonlinear* first-, second-, and fourth-order ordinary differential equations, respectively. ■

SOLUTIONS As was stated on page 2, one of the goals in this course is to solve, or find solutions of, differential equations. In the next definition we consider the concept of a solution of an ordinary differential equation.

DEFINITION 1.1.2 Solution of an ODE

Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a **solution** of the equation on the interval.

In other words, a solution of an n th-order ordinary differential equation (4) is a function ϕ that possesses at least n derivatives and for which

$$F(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)) = 0 \quad \text{for all } x \text{ in } I.$$

We say that ϕ *satisfies* the differential equation on I . For our purposes we shall also assume that a solution ϕ is a real-valued function. In our introductory discussion we saw that $y = e^{0.1x^2}$ is a solution of $dy/dx = 0.2xy$ on the interval $(-\infty, \infty)$.

Occasionally, it will be convenient to denote a solution by the alternative symbol $y(x)$.

INTERVAL OF DEFINITION You cannot think *solution* of an ordinary differential equation without simultaneously thinking *interval*. The interval I in Definition 1.1.2 is variously called the **interval of definition**, the **interval of existence**, the **interval of validity**, or the **domain of the solution** and can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

EXAMPLE 5 Verification of a Solution

Verify that the indicated function is a solution of the given differential equation on the interval $(-\infty, \infty)$.

(a) $\frac{dy}{dx} = xy^{1/2}; \quad y = \frac{1}{16}x^4$ (b) $y'' - 2y' + y = 0; \quad y = xe^x$

SOLUTION One way of verifying that the given function is a solution is to see, after substituting, whether each side of the equation is the same for every x in the interval.

(a) From

$$\text{left-hand side: } \frac{dy}{dx} = \frac{1}{16} (4 \cdot x^3) = \frac{1}{4} x^3,$$

$$\text{right-hand side: } xy^{1/2} = x \cdot \left(\frac{1}{16} x^4 \right)^{1/2} = x \cdot \left(\frac{1}{4} x^2 \right) = \frac{1}{4} x^3,$$

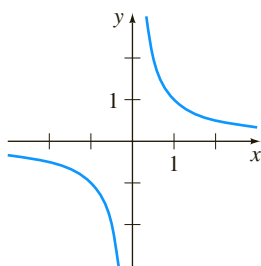
we see that each side of the equation is the same for every real number x . Note that $y^{1/2} = \frac{1}{4} x^2$ is, by definition, the nonnegative square root of $\frac{1}{16} x^4$.

(b) From the derivatives $y' = xe^x + e^x$ and $y'' = xe^x + 2e^x$ we have, for every real number x ,

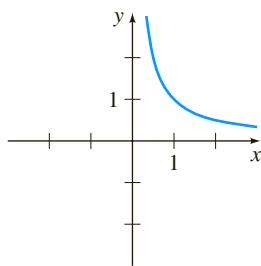
$$\text{left-hand side: } y'' - 2y' + y = (xe^x + 2e^x) - 2(xe^x + e^x) + xe^x = 0,$$

$$\text{right-hand side: } 0.$$

Note, too, that each differential equation in Example 5 possesses the constant solution $y = 0$, $-\infty < x < \infty$. A solution of a differential equation that is identically zero on an interval I is said to be a **trivial solution**.



(a) function $y = 1/x$, $x \neq 0$



(b) solution $y = 1/x$, $(0, \infty)$

FIGURE 1.1.1 In Example 6 the function $y = 1/x$ is not the same as the solution $y = 1/x$

SOLUTION CURVE The graph of a solution ϕ of an ODE is called a **solution curve**. Since ϕ is a differentiable function, it is continuous on its interval I of definition. Thus there may be a difference between the graph of the *function* ϕ and the graph of the *solution* ϕ . Put another way, the domain of the function ϕ need not be the same as the interval I of definition (or domain) of the solution ϕ . Example 6 illustrates the difference.

EXAMPLE 6 Function versus Solution

(a) The domain of $y = 1/x$, considered simply as a *function*, is the set of all real numbers x except 0. When we graph $y = 1/x$, we plot points in the xy -plane corresponding to a judicious sampling of numbers taken from its domain. The rational function $y = 1/x$ is discontinuous at 0, and its graph, in a neighborhood of the origin, is given in Figure 1.1.1(a). The function $y = 1/x$ is not differentiable at $x = 0$, since the y -axis (whose equation is $x = 0$) is a vertical asymptote of the graph.

(b) Now $y = 1/x$ is also a solution of the linear first-order differential equation $xy' + y = 0$. (Verify.) But when we say that $y = 1/x$ is a *solution* of this DE, we mean that it is a function defined on an interval I on which it is differentiable and satisfies the equation. In other words, $y = 1/x$ is a solution of the DE on *any* interval that does not contain 0, such as $(-3, -1)$, $(\frac{1}{2}, 10)$, $(-\infty, 0)$, or $(0, \infty)$. Because the solution curves defined by $y = 1/x$ for $-3 < x < -1$ and $\frac{1}{2} < x < 10$ are simply segments, or pieces, of the solution curves defined by $y = 1/x$ for $-\infty < x < 0$ and

$0 < x < \infty$, respectively, it makes sense to take the interval I to be as large as possible. Thus we take I to be either $(-\infty, 0)$ or $(0, \infty)$. The solution curve on $(0, \infty)$ is shown in Figure 1.1.1(b). ■

EXPLICIT AND IMPLICIT SOLUTIONS You should be familiar with the terms *explicit functions* and *implicit functions* from your study of calculus. A solution in which the dependent variable is expressed solely in terms of the independent variable and constants is said to be an **explicit solution**. For our purposes, let us think of an explicit solution as an explicit formula $y = \phi(x)$ that we can manipulate, evaluate, and differentiate using the standard rules. We have just seen in the last two examples that $y = \frac{1}{16}x^4$, $y = xe^x$, and $y = 1/x$ are, in turn, explicit solutions of $dy/dx = xy^{1/2}$, $y'' - 2y' + y = 0$, and $xy' + y = 0$. Moreover, the trivial solution $y = 0$ is an explicit solution of all three equations. When we get down to the business of actually solving some ordinary differential equations, you will see that methods of solution do not always lead directly to an explicit solution $y = \phi(x)$. This is particularly true when we attempt to solve nonlinear first-order differential equations. Often we have to be content with a relation or expression $G(x, y) = 0$ that defines a solution ϕ implicitly.

DEFINITION 1.1.3 Implicit Solution of an ODE

A relation $G(x, y) = 0$ is said to be an **implicit solution** of an ordinary differential equation (4) on an interval I , provided that there exists at least one function ϕ that satisfies the relation as well as the differential equation on I .

It is beyond the scope of this course to investigate the conditions under which a relation $G(x, y) = 0$ defines a differentiable function ϕ . So we shall assume that if the formal implementation of a method of solution leads to a relation $G(x, y) = 0$, then there exists at least one function ϕ that satisfies both the relation (that is, $G(x, \phi(x)) = 0$) and the differential equation on an interval I . If the implicit solution $G(x, y) = 0$ is fairly simple, we may be able to solve for y in terms of x and obtain one or more explicit solutions. See (iv) in the *Remarks*.

EXAMPLE 7 Verification of an Implicit Solution

The relation $x^2 + y^2 = 25$ is an implicit solution of the differential equation

$$\frac{dy}{dx} = -\frac{x}{y} \quad (8)$$

on the open interval $(-5, 5)$. By implicit differentiation we obtain

$$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = \frac{d}{dx}25 \quad \text{or} \quad 2x + 2y\frac{dy}{dx} = 0. \quad (9)$$

Solving the last equation in (9) for the symbol dy/dx gives (8). Moreover, solving $x^2 + y^2 = 25$ for y in terms of x yields $y = \pm\sqrt{25 - x^2}$. The two functions $y = \phi_1(x) = \sqrt{25 - x^2}$ and $y = \phi_2(x) = -\sqrt{25 - x^2}$ satisfy the relation (that is, $x^2 + \phi_1^2 = 25$ and $x^2 + \phi_2^2 = 25$) and are explicit solutions defined on the interval $(-5, 5)$. The solution curves given in Figures 1.1.2(b) and 1.1.2(c) are segments of the graph of the implicit solution in Figure 1.1.2(a).

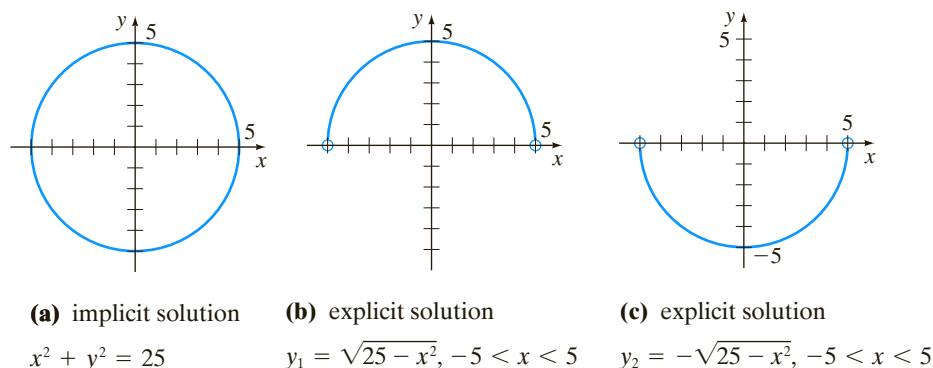


FIGURE 1.1.2 An implicit solution and two explicit solutions of (8) in Example 7

Because the distinction between an explicit solution and an implicit solution should be intuitively clear, we will not belabor the issue by always saying, “Here is an explicit (implicit) solution.”

FAMILIES OF SOLUTIONS The study of differential equations is similar to that of integral calculus. When evaluating an antiderivative or indefinite integral in calculus, we use a single constant c of integration. Analogously, we shall see in Chapter 2 that when solving a first-order differential equation $F(x, y, y') = 0$ we *usually* obtain a solution containing a single constant or parameter c . A solution of $F(x, y, y') = 0$ containing a constant c is a set of solutions $G(x, y, c) = 0$ called a **one-parameter family of solutions**. When solving an n th-order differential equation $F(x, y, y', \dots, y^{(n)}) = 0$ we seek an **n -parameter family of solutions** $G(x, y, c_1, c_2, \dots, c_n) = 0$. This means that a single differential equation can possess an infinite number of solutions corresponding to an unlimited number of choices for the parameter(s). A solution of a differential equation that is free of parameters is called a **particular solution**.

The parameters in a family of solutions such as $G(x, y, c_1, c_2, \dots, c_n) = 0$ are *arbitrary* up to a point. For example, proceeding as in (9) a relation $x^2 + y^2 = c$ formally satisfies (8) for any constant c . However, it is understood that the relation should always make sense in the real number system; thus, if $c = -25$ we cannot say that $x^2 + y^2 = -25$ is an implicit solution of the differential equation.

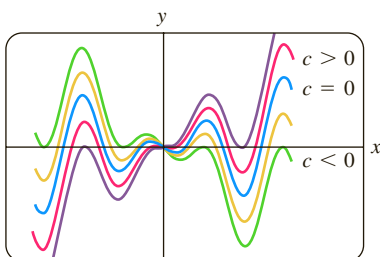


FIGURE 1.1.3 Some solutions of DE in part (a) of Example 8

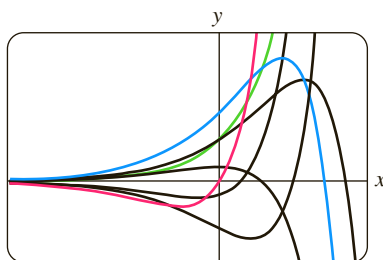


FIGURE 1.1.4 Some solutions of DE in part (b) of Example 8

EXAMPLE 8 Particular Solutions

(a) For all real values of c , the one-parameter family $y = cx - x \cos x$ is an explicit solution of the linear first-order equation

$$xy' - y = x^2 \sin x$$

on the interval $(-\infty, \infty)$. (Verify.) Figure 1.1.3 shows the graphs of some particular solutions in this family for various choices of c . The solution $y = -x \cos x$, the blue graph in the figure, is a particular solution corresponding to $c = 0$.

(b) The two-parameter family $y = c_1e^x + c_2xe^x$ is an explicit solution of the linear second-order equation

$$y'' - 2y' + y = 0$$

in part (b) of Example 5. (Verify.) In Figure 1.1.4 we have shown seven of the “double infinity” of solutions in the family. The solution curves in red, green, and blue are the graphs of the particular solutions $y = 5xe^x$ ($c_1 = 0$, $c_2 = 5$), $y = 3e^x$ ($c_1 = 3$, $c_2 = 0$), and $y = 5e^x - 2xe^x$ ($c_1 = 5$, $c_2 = 2$), respectively.

Sometimes a differential equation possesses a solution that is not a member of a family of solutions of the equation—that is, a solution that cannot be obtained by specializing *any* of the parameters in the family of solutions. Such an extra solution is called a **singular solution**. For example, we have seen that $y = \frac{1}{16}x^4$ and $y = 0$ are solutions of the differential equation $dy/dx = xy^{1/2}$ on $(-\infty, \infty)$. In Section 2.2 we shall demonstrate, by actually solving it, that the differential equation $dy/dx = xy^{1/2}$ possesses the one-parameter family of solutions $y = (\frac{1}{4}x^2 + c)^2$, $c \geq 0$. When $c = 0$, the resulting particular solution is $y = \frac{1}{16}x^4$. But notice that the trivial solution $y = 0$ is a singular solution since it is not a member of the family $y = (\frac{1}{4}x^2 + c)^2$; there is no way of assigning a value to the constant c to obtain $y = 0$.

In all the preceding examples we used x and y to denote the independent and dependent variables, respectively. But you should become accustomed to seeing and working with other symbols to denote these variables. For example, we could denote the independent variable by t and the dependent variable by x .

EXAMPLE 9 Using Different Symbols

The functions $x = c_1 \cos 4t$ and $x = c_2 \sin 4t$, where c_1 and c_2 are arbitrary constants or parameters, are both solutions of the linear differential equation

$$x'' + 16x = 0.$$

For $x = c_1 \cos 4t$ the first two derivatives with respect to t are $x' = -4c_1 \sin 4t$ and $x'' = -16c_1 \cos 4t$. Substituting x'' and x then gives

$$x'' + 16x = -16c_1 \cos 4t + 16(c_1 \cos 4t) = 0.$$

In like manner, for $x = c_2 \sin 4t$ we have $x'' = -16c_2 \sin 4t$, and so

$$x'' + 16x = -16c_2 \sin 4t + 16(c_2 \sin 4t) = 0.$$

Finally, it is straightforward to verify that the linear combination of solutions, or the two-parameter family $x = c_1 \cos 4t + c_2 \sin 4t$, is also a solution of the differential equation. ■

The next example shows that a solution of a differential equation can be a piecewise-defined function.

EXAMPLE 10 Piecewise-Defined Solution

The one-parameter family of quartic monomial functions $y = cx^4$ is an explicit solution of the linear first-order equation

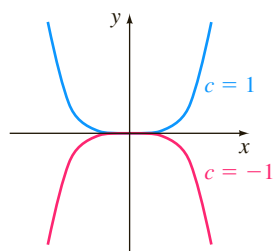
$$xy' - 4y = 0$$

on the interval $(-\infty, \infty)$. (Verify.) The blue and red solution curves shown in Figure 1.1.5(a) are the graphs of $y = x^4$ and $y = -x^4$ and correspond to the choices $c = 1$ and $c = -1$, respectively.

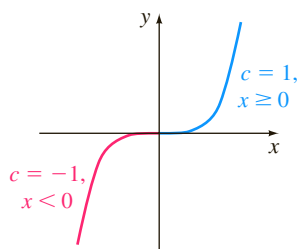
The piecewise-defined differentiable function

$$y = \begin{cases} -x^4, & x < 0 \\ x^4, & x \geq 0 \end{cases}$$

is also a solution of the differential equation but cannot be obtained from the family $y = cx^4$ by a single choice of c . As seen in Figure 1.1.5(b) the solution is constructed from the family by choosing $c = -1$ for $x < 0$ and $c = 1$ for $x \geq 0$. ■



(a) two explicit solutions



(b) piecewise-defined solution

FIGURE 1.1.5 Some solutions of DE in Example 10

SYSTEMS OF DIFFERENTIAL EQUATIONS Up to this point we have been discussing single differential equations containing one unknown function. But often

in theory, as well as in many applications, we must deal with systems of differential equations. A **system of ordinary differential equations** is two or more equations involving the derivatives of two or more unknown functions of a single independent variable. For example, if x and y denote dependent variables and t denotes the independent variable, then a system of two first-order differential equations is given by

$$\begin{aligned}\frac{dx}{dt} &= f(t, x, y) \\ \frac{dy}{dt} &= g(t, x, y).\end{aligned}\tag{10}$$

A **solution** of a system such as (10) is a pair of differentiable functions $x = \phi_1(t)$, $y = \phi_2(t)$, defined on a common interval I , that satisfy each equation of the system on this interval.

REMARKS

(i) It might not be apparent whether a first-order ODE written in differential form $M(x, y)dx + N(x, y)dy = 0$ is linear or nonlinear because there is nothing in this form that tells us which symbol denotes the dependent variable. See Problems 9 and 10 in Exercises 1.1.

(ii) We will see in the chapters that follow that a solution of a differential equation may involve an **integral-defined function**. One way of defining a function F of a single variable x by means of a definite integral is:

$$F(x) = \int_a^x g(t) dt.\tag{11}$$

If the integrand g in (11) is continuous on an interval $[a, b]$ and $a \leq x \leq b$, then the derivative form of the Fundamental Theorem of Calculus states that F is differentiable on (a, b) and

$$F'(x) = \frac{d}{dx} \int_a^x g(t) dt = g(x)\tag{12}$$

The integral in (11) is often **nonelementary**, that is, an integral of a function g that does not have an elementary-function antiderivative. Elementary functions include the familiar functions studied in a typical precalculus course:

constant, polynomial, rational, exponential, logarithmic, trigonometric, and inverse trigonometric functions,

as well as rational powers of these functions; finite combinations of these functions using addition, subtraction, multiplication, division; and function compositions. For example, even though e^{-t^2} , $\sqrt{1+t^3}$, and $\cos t^2$ are elementary functions, the integrals $\int e^{-t^2} dt$, $\int \sqrt{1+t^3} dt$, and $\int \cos t^2 dt$ are nonelementary. See Problems 25–28 in Exercises 1.1. Also see Appendix A.

(iii) Although the concept of a solution of a differential equation has been emphasized in this section, you should be aware that a DE does not necessarily have to possess a solution. See Problem 43 in Exercises 1.1. The question of whether a solution exists will be touched on in the next section.

(continued)

(iv) A few last words about implicit solutions of differential equations are in order. In Example 7 we were able to solve the relation $x^2 + y^2 = 25$ for y in terms of x to get two explicit solutions, $\phi_1(x) = \sqrt{25 - x^2}$ and $\phi_2(x) = -\sqrt{25 - x^2}$, of the differential equation (8). But don't read too much into this one example. Unless it is easy or important or you are instructed to, there is usually no need to try to solve an implicit solution $G(x, y) = 0$ for y explicitly in terms of x . Also do not misinterpret the second sentence following Definition 1.1.3. An implicit solution $G(x, y) = 0$ can define a perfectly good differentiable function ϕ that is a solution of a DE, yet we might not be able to solve $G(x, y) = 0$ using analytical methods such as algebra. The solution curve of ϕ may be a segment or piece of the graph of $G(x, y) = 0$. See Problems 49 and 50 in Exercises 1.1. Also, read the discussion following Example 4 in Section 2.2.

(v) It might not seem like a big deal to assume that $F(x, y, y', \dots, y^{(n)}) = 0$ can be solved for $y^{(n)}$, but one should be a little bit careful here. There are exceptions, and there certainly are some problems connected with this assumption. See Problems 56 and 57 in Exercises 1.1.

(vi) If every solution of an n th-order ODE $F(x, y, y', \dots, y^{(n)}) = 0$ on an interval I can be obtained from an n -parameter family $G(x, y, c_1, c_2, \dots, c_n) = 0$ by appropriate choices of the parameters c_i , $i = 1, 2, \dots, n$, we then say that the family is the **general solution** of the DE. In solving linear ODEs, we shall impose relatively simple restrictions on the coefficients of the equation; with these restrictions one can be assured that not only does a solution exist on an interval but also that a family of solutions yields all possible solutions. Nonlinear ODEs, with the exception of some first-order equations, are usually difficult or impossible to solve in terms of elementary functions. Furthermore, if we happen to obtain a family of solutions for a nonlinear equation, it is not obvious whether this family contains all solutions. On a practical level, then, the designation "general solution" is applied only to linear ODEs. Don't be concerned about this concept at this point, but store the words "general solution" in the back of your mind—we will come back to this notion in Section 2.3 and again in Chapter 4.

EXERCISES 1.1

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1–8 state the order of the given ordinary differential equation. Determine whether the equation is linear or nonlinear by matching it with (6).

1. $(1 - x)y'' - 4xy' + 5y = \cos x$

2. $x \frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^4 + y = 0$

3. $t^5 y^{(4)} - t^3 y'' + 6y = 0$

4. $\frac{d^2u}{dr^2} + \frac{du}{dr} + u = \cos(r + u)$

5. $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

6. $\frac{d^2R}{dt^2} = -\frac{k}{R^2}$

7. $(\sin \theta)y''' - (\cos \theta)y' = 2$

8. $\ddot{x} - \left(1 - \frac{x^2}{3}\right)\dot{x} + x = 0$

In Problems 9 and 10 determine whether the given first-order differential equation is linear in the indicated dependent variable by matching it with the first differential equation given in (7).

9. $(y^2 - 1)dx + xdy = 0$; in y ; in x

10. $u dv + (v + uv - ue^u) du = 0$; in v ; in u

In Problems 11–14 verify that the indicated function is an explicit solution of the given differential equation. Assume an appropriate interval I of definition for each solution.

11. $2y' + y = 0$; $y = e^{-x/2}$

12. $\frac{dy}{dt} + 20y = 24$; $y = \frac{6}{5} - \frac{6}{5}e^{-20t}$

13. $y'' - 6y' + 13y = 0$; $y = e^{3x} \cos 2x$

14. $y'' + y = \tan x$; $y = -(\cos x) \ln(\sec x + \tan x)$

In Problems 15–18 verify that the indicated function $y = \phi(x)$ is an explicit solution of the given first-order differential equation. Proceed as in Example 6, by considering ϕ simply as a function and give its domain. Then by considering ϕ as a solution of the differential equation, give at least one interval I of definition.

15. $(y - x)y' = y - x + 8$; $y = x + 4\sqrt{x + 2}$

16. $y' = 25 + y^2$; $y = 5 \tan 5x$

17. $y' = 2xy^2$; $y = 1/(4 - x^2)$

18. $2y' = y^3 \cos x$; $y = (1 - \sin x)^{-1/2}$

In Problems 19 and 20 verify that the indicated expression is an implicit solution of the given first-order differential equation. Find at least one explicit solution $y = \phi(x)$ in each case. Use a graphing utility to obtain the graph of an explicit solution. Give an interval I of definition of each solution ϕ .

19. $\frac{dX}{dt} = (X - 1)(1 - 2X)$; $\ln\left(\frac{2X - 1}{X - 1}\right) = t$

20. $2xy \, dx + (x^2 - y) \, dy = 0$; $-2x^2y + y^2 = 1$

In Problems 21–24 verify that the indicated family of functions is a solution of the given differential equation. Assume an appropriate interval I of definition for each solution.

21. $\frac{dP}{dt} = P(1 - P)$; $P = \frac{c_1 e^t}{1 + c_1 e^t}$

22. $\frac{dy}{dx} + 4xy = 8x^3$; $y = 2x^2 - 1 + c_1 e^{-2x^2}$

23. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$; $y = c_1 e^{2x} + c_2 x e^{2x}$

24. $x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 12x^2$;

$$y = c_1 x^{-1} + c_2 x + c_3 x \ln x + 4x^2$$

In Problems 25–28 use (12) to verify that the indicated function is a solution of the given differential equation. Assume an appropriate interval I of definition of each solution.

25. $x \frac{dy}{dx} - 3xy = 1$; $y = e^{3x} \int_1^x \frac{e^{-3t}}{t} dt$

26. $2x \frac{dy}{dx} - y = 2x \cos x$; $y = \sqrt{x} \int_4^x \frac{\cos t}{\sqrt{t}} dt$

27. $x^2 \frac{dy}{dx} + xy = 10 \sin x$; $y = \frac{5}{x} + \frac{10}{x} \int_1^x \frac{\sin t}{t} dt$

28. $\frac{dy}{dx} + 2xy = 1$; $y = e^{-x^2} + e^{-x^2} \int_0^x e^{t^2} dt$

29. Verify that the piecewise-defined function

$$y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

is a solution of the differential equation $xy' - 2y = 0$ on $(-\infty, \infty)$.

30. In Example 7 we saw that $y = \phi_1(x) = \sqrt{25 - x^2}$ and $y = \phi_2(x) = -\sqrt{25 - x^2}$ are solutions of $dy/dx = -x/y$ on the interval $(-5, 5)$. Explain why the piecewise-defined function

$$y = \begin{cases} \sqrt{25 - x^2}, & -5 < x < 0 \\ -\sqrt{25 - x^2}, & 0 \leq x < 5 \end{cases}$$

is not a solution of the differential equation on the interval $(-5, 5)$.

In Problems 31–34 find values of m so that the function $y = e^{mx}$ is a solution of the given differential equation.

31. $y' + 2y = 0$

32. $5y' = 2y$

33. $y'' - 5y' + 6y = 0$

34. $2y'' + 7y' - 4y = 0$

In Problems 35 and 36 find values of m so that the function $y = x^m$ is a solution of the given differential equation.

35. $xy'' + 2y' = 0$

36. $x^2y'' - 7xy' + 15y = 0$

In Problems 37–40 use the concept that $y = c$, $-\infty < x < \infty$, is a constant function if and only if $y' = 0$ to determine whether the given differential equation possesses constant solutions.

37. $3xy' + 5y = 10$

38. $y' = y^2 + 2y - 3$

39. $(y - 1)y' = 1$

40. $y'' + 4y' + 6y = 10$

In Problems 41 and 42 verify that the indicated pair of functions is a solution of the given system of differential equations on the interval $(-\infty, \infty)$.

41. $\frac{dx}{dt} = x + 3y$

42. $\frac{d^2x}{dt^2} = 4y + e^t$

$$\frac{dy}{dt} = 5x + 3y;$$

$$\frac{d^2y}{dt^2} = 4x - e^t;$$

$$x = e^{-2t} + 3e^{6t},$$

$$x = \cos 2t + \sin 2t + \frac{1}{5}e^t,$$

$$y = -e^{-2t} + 5e^{6t}$$

$$y = -\cos 2t - \sin 2t - \frac{1}{5}e^t$$

Discussion Problems

43. Make up a differential equation that does not possess any real solutions.

44. Make up a differential equation that you feel confident possesses only the trivial solution $y = 0$. Explain your reasoning.

45. What function do you know from calculus is such that its first derivative is itself? Its first derivative is a constant multiple k of itself? Write each answer in the form of a first-order differential equation with a solution.

46. What function (or functions) do you know from calculus is such that its second derivative is itself? Its second derivative is the negative of itself? Write each answer in the form of a second-order differential equation with a solution.

47. The function $y = \sin x$ is an explicit solution of the first-order differential equation $\frac{dy}{dx} = \sqrt{1 - y^2}$. Find an interval I of definition. [Hint: I is not the interval $(-\infty, \infty)$.]
48. Discuss why it makes intuitive sense to presume that the linear differential equation $y'' + 2y' + 4y = 5 \sin t$ has a solution of the form $y = A \sin t + B \cos t$, where A and B are constants. Then find specific constants A and B so that $y = A \sin t + B \cos t$ is a particular solution of the DE.

In Problems 49 and 50 the given figure represents the graph of an implicit solution $G(x, y) = 0$ of a differential equation $dy/dx = f(x, y)$. In each case the relation $G(x, y) = 0$ implicitly defines several solutions of the DE. Carefully reproduce each figure on a piece of paper. Use different colored pencils to mark off segments, or pieces, on each graph that correspond to graphs of solutions. Keep in mind that a solution ϕ must be a function and differentiable. Use the solution curve to estimate an interval I of definition of each solution ϕ .

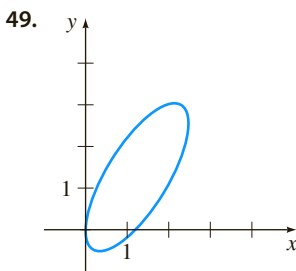


FIGURE 1.1.6 Graph for Problem 49

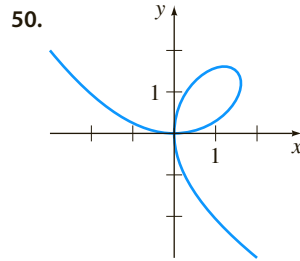


FIGURE 1.1.7 Graph for Problem 50

51. The graphs of members of the one-parameter family $x^3 + y^3 = 3cxy$ are called **folia of Descartes**. Verify that this family is an implicit solution of the first-order differential equation

$$\frac{dy}{dx} = \frac{y(y^3 - 2x^3)}{x(2y^3 - x^3)}.$$

52. The graph in Figure 1.1.7 is the member of the family of folia in Problem 51 corresponding to $c = 1$. Discuss: How can the DE in Problem 51 help in finding points on the graph of $x^3 + y^3 = 3xy$ where the tangent line is vertical? How does knowing where a tangent line is vertical help in determining an interval I of definition of a solution ϕ of the DE? Carry out your ideas and compare with your estimates of the intervals in Problem 50.
53. In Example 7 the largest interval I over which the explicit solutions $y = \phi_1(x)$ and $y = \phi_2(x)$ are defined is the open interval $(-5, 5)$. Why can't the interval I of definition be the closed interval $[-5, 5]$?
54. In Problem 21 a one-parameter family of solutions of the DE $P' = P(1 - P)$ is given. Does any solution curve pass through the point $(0, 3)$? Through the point $(0, 1)$?
55. Discuss, and illustrate with examples, how to solve differential equations of the forms $dy/dx = f(x)$ and $d^2y/dx^2 = f(x)$.
56. The differential equation $x(y')^2 - 4y' - 12x^3 = 0$ has the form given in (4). Determine whether the equation can be put into the normal form $dy/dx = f(x, y)$.

57. The normal form (5) of an n th-order differential equation is equivalent to (4) whenever both forms have exactly the same solutions. Make up a first-order differential equation for which $F(x, y, y') = 0$ is not equivalent to the normal form $dy/dx = f(x, y)$.
58. Find a linear second-order differential equation $F(x, y, y', y'') = 0$ for which $y = c_1x + c_2x^2$ is a two-parameter family of solutions. Make sure that your equation is free of the arbitrary parameters c_1 and c_2 .

Qualitative information about a solution $y = \phi(x)$ of a differential equation can often be obtained from the equation itself. Before working Problems 59–62, recall the geometric significance of the derivatives dy/dx and d^2y/dx^2 .

59. Consider the differential equation $dy/dx = e^{-x^2}$.
- Explain why a solution of the DE must be an increasing function on any interval of the x -axis.
 - What are $\lim_{x \rightarrow -\infty} dy/dx$ and $\lim_{x \rightarrow \infty} dy/dx$? What does this suggest about a solution curve as $x \rightarrow \pm\infty$?
 - Determine an interval over which a solution curve is concave down and an interval over which the curve is concave up.
 - Sketch the graph of a solution $y = \phi(x)$ of the differential equation whose shape is suggested by parts (a)–(c).
60. Consider the differential equation $dy/dx = 5 - y$.
- Either by inspection or by the method suggested in Problems 37–40, find a constant solution of the DE.
 - Using only the differential equation, find intervals on the y -axis on which a nonconstant solution $y = \phi(x)$ is increasing. Find intervals on the y -axis on which $y = \phi(x)$ is decreasing.
61. Consider the differential equation $dy/dx = y(a - by)$, where a and b are positive constants.
- Either by inspection or by the method suggested in Problems 37–40, find two constant solutions of the DE.
 - Using only the differential equation, find intervals on the y -axis on which a nonconstant solution $y = \phi(x)$ is increasing. Find intervals on which $y = \phi(x)$ is decreasing.
 - Using only the differential equation, explain why $y = a/2b$ is the y -coordinate of a point of inflection of the graph of a nonconstant solution $y = \phi(x)$.
 - On the same coordinate axes, sketch the graphs of the two constant solutions found in part (a). These constant solutions partition the xy -plane into three regions. In each region, sketch the graph of a nonconstant solution $y = \phi(x)$ whose shape is suggested by the results in parts (b) and (c).
62. Consider the differential equation $y' = y^2 + 4$.
- Explain why there exist no constant solutions of the DE.
 - Describe the graph of a solution $y = \phi(x)$. For example, can a solution curve have any relative extrema?

- (c) Explain why $y = 0$ is the y -coordinate of a point of inflection of a solution curve.
- (d) Sketch the graph of a solution $y = \phi(x)$ of the differential equation whose shape is suggested by parts (a)–(c).

Computer Lab Assignments

In Problems 63 and 64 use a CAS to compute all derivatives and to carry out the simplifications needed to verify that

the indicated function is a particular solution of the given differential equation.

63. $y^{(4)} - 20y''' + 158y'' - 580y' + 841y = 0$;

$$y = xe^{5x} \cos 2x$$

64. $x^3y''' + 2x^2y'' + 20xy' - 78y = 0$;

$$y = 20 \frac{\cos(5 \ln x)}{x} - 3 \frac{\sin(5 \ln x)}{x}$$

1.2 Initial-Value Problems

INTRODUCTION We are often interested in problems in which we seek a solution $y(x)$ of a differential equation so that $y(x)$ also satisfies certain prescribed side conditions, that is, conditions that are imposed on the unknown function $y(x)$ and its derivatives at a number x_0 . On some interval I containing x_0 the problem of solving an n th-order differential equation subject to n side conditions specified at x_0 :

$$\text{Solve: } \frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}) \quad (1)$$

$$\text{Subject to: } y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1},$$

where $y_0, y_1, \dots, y_{n-1}$ are arbitrary constants, is called an **n th-order initial-value problem (IVP)**. The values of $y(x)$ and its first $n-1$ derivatives at x_0 , $y(x_0) = y_0$, $y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$ are called **initial conditions (IC)**.

Solving an n th-order initial-value problem such as (1) frequently entails first finding an n -parameter family of solutions of the differential equation and then using the initial conditions at x_0 to determine the n constants in this family. The resulting particular solution is defined on some interval I containing the number x_0 .

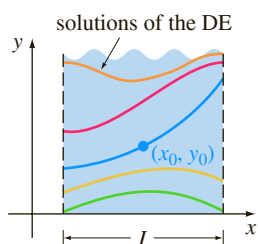


FIGURE 1.2.1 Solution curve of first-order IVP

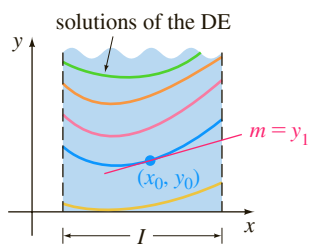


FIGURE 1.2.2 Solution curve of second-order IVP

GEOMETRIC INTERPRETATION The cases $n = 1$ and $n = 2$ in (1),

$$\text{Solve: } \frac{dy}{dx} = f(x, y) \quad (2)$$

$$\text{Subject to: } y(x_0) = y_0$$

and

$$\text{Solve: } \frac{d^2 y}{dx^2} = f(x, y, y') \quad (3)$$

$$\text{Subject to: } y(x_0) = y_0, y'(x_0) = y_1$$

are examples of **first-** and **second-order** initial-value problems, respectively. These two problems are easy to interpret in geometric terms. For (2) we are seeking a solution $y(x)$ of the differential equation $y' = f(x, y)$ on an interval I containing x_0 so that its graph passes through the specified point (x_0, y_0) . A solution curve is shown in blue in Figure 1.2.1. For (3) we want to find a solution $y(x)$ of the differential equation $y'' = f(x, y, y')$ on an interval I containing x_0 so that its graph not only passes through (x_0, y_0) but the slope of the curve at this point is the number y_1 . A solution curve is shown in blue in Figure 1.2.2. The words *initial conditions* derive from physical systems where the independent variable is time t and where $y(t_0) = y_0$

and $y'(t_0) = y_1$ represent the position and velocity, respectively, of an object at some beginning, or initial, time t_0 .

EXAMPLE 1 Two First-Order IVPs

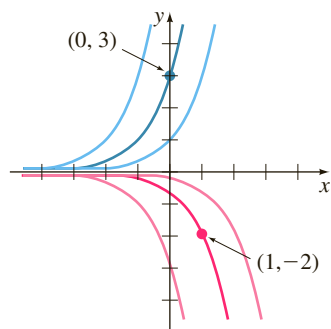


FIGURE 1.2.3 Solution curves of two IVPs in Example 1

(a) In Problem 45 in Exercises 1.1 you were asked to deduce that $y = ce^x$ is a one-parameter family of solutions of the simple first-order equation $y' = y$. All the solutions in this family are defined on the interval $(-\infty, \infty)$. If we impose an initial condition, say, $y(0) = 3$, then substituting $x = 0$, $y = 3$ in the family determines the constant $3 = ce^0 = c$. Thus $y = 3e^x$ is a solution of the IVP

$$y' = y, \quad y(0) = 3.$$

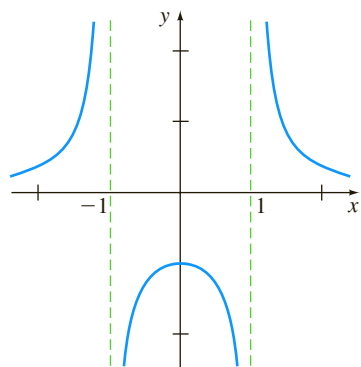
(b) Now if we demand that a solution curve pass through the point $(1, -2)$ rather than $(0, 3)$, then $y(1) = -2$ will yield $-2 = ce$ or $c = -2e^{-1}$. In this case $y = -2e^{x-1}$ is a solution of the IVP

$$y' = y, \quad y(1) = -2.$$

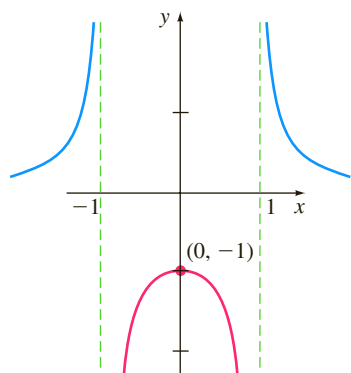
The two solution curves are shown in dark blue and dark red in Figure 1.2.3. ■

The next example illustrates another first-order initial-value problem. In this example notice how the interval I of definition of the solution $y(x)$ depends on the initial condition $y(x_0) = y_0$.

EXAMPLE 2 Interval I of Definition of a Solution



(a) function defined for all x except $x = \pm 1$



(b) solution defined on interval containing $x = 0$

In Problem 6 of Exercises 2.2 you will be asked to show that a one-parameter family of solutions of the first-order differential equation $y' + 2xy^2 = 0$ is $y = 1/(x^2 + c)$. If we impose the initial condition $y(0) = -1$, then substituting $x = 0$ and $y = -1$ into the family of solutions gives $-1 = 1/c$ or $c = -1$. Thus $y = 1/(x^2 - 1)$. We now emphasize the following three distinctions:

- Considered as a *function*, the domain of $y = 1/(x^2 - 1)$ is the set of real numbers x for which $y(x)$ is defined; this is the set of all real numbers except $x = -1$ and $x = 1$. See Figure 1.2.4(a).
- Considered as a *solution of the differential equation* $y' + 2xy^2 = 0$, the interval I of definition of $y = 1/(x^2 - 1)$ could be taken to be any interval over which $y(x)$ is defined and differentiable. As can be seen in Figure 1.2.4(a), the largest intervals on which $y = 1/(x^2 - 1)$ is a solution are $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$.
- Considered as a *solution of the initial-value problem* $y' + 2xy^2 = 0$, $y(0) = -1$, the interval I of definition of $y = 1/(x^2 - 1)$ could be taken to be any interval over which $y(x)$ is defined, differentiable, and contains the initial point $x = 0$; the largest interval for which this is true is $(-1, 1)$. See the red curve in Figure 1.2.4(b). ■

See Problems 3–6 in Exercises 1.2 for a continuation of Example 2.

EXAMPLE 3 Second-Order IVP

In Example 9 of Section 1.1 we saw that $x = c_1 \cos 4t + c_2 \sin 4t$ is a two-parameter family of solutions of $x'' + 16x = 0$. Find a solution of the initial-value problem

$$x'' + 16x = 0, \quad x\left(\frac{\pi}{2}\right) = -2, \quad x'\left(\frac{\pi}{2}\right) = 1. \quad (4)$$

FIGURE 1.2.4 Graphs of function and solution of IVP in Example 2

SOLUTION We first apply $x(\pi/2) = -2$ to the given family of solutions: $c_1 \cos 2\pi + c_2 \sin 2\pi = -2$. Since $\cos 2\pi = 1$ and $\sin 2\pi = 0$, we find that $c_1 = -2$. We next apply $x'(\pi/2) = 1$ to the one-parameter family $x(t) = -2 \cos 4t + c_2 \sin 4t$. Differentiating and then setting $t = \pi/2$ and $x' = 1$ gives $8 \sin 2\pi + 4c_2 \cos 2\pi = 1$, from which we see that $c_2 = \frac{1}{4}$. Hence $x = -2 \cos 4t + \frac{1}{4} \sin 4t$ is a solution of (4). ■

EXISTENCE AND UNIQUENESS Two fundamental questions arise in considering an initial-value problem:

Does a solution of the problem exist? If a solution exists, is it unique?

For the first-order initial-value problem (2) we ask:

- Existence** $\left\{ \begin{array}{l} \text{Does the differential equation } dy/dx = f(x, y) \text{ possess solutions?} \\ \text{Do any of the solution curves pass through the point } (x_0, y_0)? \end{array} \right.$
- Uniqueness** $\left\{ \begin{array}{l} \text{When can we be certain that there is precisely one solution curve} \\ \text{passing through the point } (x_0, y_0)? \end{array} \right.$

Note that in Examples 1 and 3 the phrase “a solution” is used rather than “the solution” of the problem. The indefinite article “a” is used deliberately to suggest the possibility that other solutions may exist. At this point it has not been demonstrated that there is a single solution of each problem. The next example illustrates an initial-value problem with two solutions.

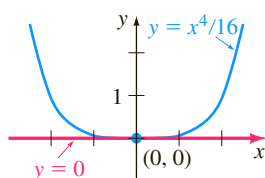


FIGURE 1.2.5 Two solution curves of the same IVP in Example 4

EXAMPLE 4 An IVP Can Have Several Solutions

Each of the functions $y = 0$ and $y = \frac{1}{16}x^4$ satisfies the differential equation $dy/dx = xy^{1/2}$ and the initial condition $y(0) = 0$, so the initial-value problem

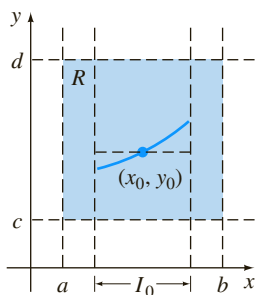
$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0$$

has at least two solutions. As illustrated in Figure 1.2.5, the graphs of both functions, shown in red and blue pass through the same point $(0, 0)$. ■

Within the safe confines of a formal course in differential equations one can be fairly confident that *most* differential equations will have solutions and that solutions of initial-value problems will *probably* be unique. Real life, however, is not so idyllic. Therefore it is desirable to know in advance of trying to solve an initial-value problem whether a solution exists and, when it does, whether it is the only solution of the problem. Since we are going to consider first-order differential equations in the next two chapters, we state here without proof a straightforward theorem that gives conditions that are sufficient to guarantee the existence and uniqueness of a solution of a first-order initial-value problem of the form given in (2). We shall wait until Chapter 4 to address the question of existence and uniqueness of a second-order initial-value problem.

THEOREM 1.2.1 Existence of a Unique Solution

Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$, $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f / \partial y$ are continuous on R , then there exists some interval $I_0: (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$, defined on I_0 , that is a solution of the initial-value problem (2).

FIGURE 1.2.6 Rectangular region R

The foregoing result is one of the most popular existence and uniqueness theorems for first-order differential equations because the criteria of continuity of $f(x, y)$ and $\partial f/\partial y$ are relatively easy to check. The geometry of Theorem 1.2.1 is illustrated in Figure 1.2.6.

EXAMPLE 5 Example 4 Revisited

We saw in Example 4 that the differential equation $dy/dx = xy^{1/2}$ possesses at least two solutions whose graphs pass through $(0, 0)$. Inspection of the functions

$$f(x, y) = xy^{1/2} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{x}{2y^{1/2}}$$

shows that they are continuous in the upper half-plane defined by $y > 0$. Hence Theorem 1.2.1 enables us to conclude that through any point (x_0, y_0) , $y_0 > 0$ in the upper half-plane there is some interval centered at x_0 on which the given differential equation has a unique solution. Thus, for example, even without solving it, we know that there exists some interval centered at 2 on which the initial-value problem $dy/dx = xy^{1/2}$, $y(2) = 1$ has a unique solution. ■

In Example 1, Theorem 1.2.1 guarantees that there are no other solutions of the initial-value problems $y' = y$, $y(0) = 3$ and $y' = y$, $y(1) = -2$ other than $y = 3e^x$ and $y = -2e^{x-1}$, respectively. This follows from the fact that $f(x, y) = y$ and $\partial f/\partial y = 1$ are continuous throughout the entire xy -plane. It can be further shown that the interval I on which each solution is defined is $(-\infty, \infty)$.

INTERVAL OF EXISTENCE/UNIQUENESS Suppose $y(x)$ represents a solution of the initial-value problem (2). The following three sets on the real x -axis may not be the same: the domain of the function $y(x)$, the interval I over which the solution $y(x)$ is defined or exists, and the interval I_0 of existence *and* uniqueness. Example 6 of Section 1.1 illustrated the difference between the domain of a function and the interval I of definition. Now suppose (x_0, y_0) is a point in the interior of the rectangular region R in Theorem 1.2.1. It turns out that the continuity of the function $f(x, y)$ on R by itself is sufficient to guarantee the existence of at least one solution of $dy/dx = f(x, y)$, $y(x_0) = y_0$, defined on some interval I . The interval I of definition for this initial-value problem is usually taken to be the largest interval containing x_0 over which the solution $y(x)$ is defined and differentiable. The interval I depends on both $f(x, y)$ and the **initial** condition $y(x_0) = y_0$. See Problems 31–34 in Exercises 1.2. The extra condition of continuity of the first partial derivative $\partial f/\partial y$ on R enables us to say that not only does a solution exist on some interval I_0 containing x_0 , but it is the *only* solution satisfying $y(x_0) = y_0$. However, Theorem 1.2.1 does not give any indication of the sizes of intervals I and I_0 ; *the interval I of definition need not be as wide as the region R , and the interval I_0 of existence and uniqueness may not be as large as I* . The number $h > 0$ that defines the interval I_0 : $(x_0 - h, x_0 + h)$ could be very small, so it is best to think that the solution $y(x)$ is *unique in a local sense*—that is, a solution defined near the point (x_0, y_0) . See Problem 51 in Exercises 1.2.

REMARKS

- (i) The conditions in Theorem 1.2.1 are sufficient but not necessary. This means that when $f(x, y)$ and $\partial f/\partial y$ are continuous on a rectangular region R , it must always follow that a solution of (2) exists and is unique whenever (x_0, y_0) is a point interior to R . However, if the conditions stated in the hypothesis of

Theorem 1.2.1 do not hold, then anything could happen: Problem (2) *may* still have a solution and this solution *may* be unique, or (2) may have several solutions, or it may have no solution at all. A rereading of Example 5 reveals that the hypotheses of Theorem 1.2.1 do not hold on the line $y = 0$ for the differential equation $dy/dx = xy^{1/2}$, so it is not surprising, as we saw in Example 4 of this section, that there are two solutions defined on a common interval $(-h, h)$ satisfying $y(0) = 0$. On the other hand, the hypotheses of Theorem 1.2.1 do not hold on the line $y = 1$ for the differential equation $dy/dx = |y - 1|$. Nevertheless it can be proved that the solution of the initial-value problem $dy/dx = |y - 1|$, $y(0) = 1$, is unique. Can you guess this solution?

(ii) You are encouraged to read, think about, work, and then keep in mind Problem 50 in Exercises 1.2.

(iii) Initial conditions are prescribed at a *single* point x_0 . But we are also interested in solving differential equations that are subject to conditions specified on $y(x)$ or its derivative at *two* different points x_0 and x_1 . Conditions such as

$$y(1) = 0, \quad y(5) = 0 \quad \text{or} \quad y(\pi/2) = 0, \quad y'(\pi) = 1$$

are called **boundary conditions (BC)**. A differential equation together with boundary conditions is called a **boundary-value problem (BVP)**. For example,

$$y'' + y = 0, \quad y'(0) = 0, \quad y'(\pi) = 0$$

is a boundary-value problem. See Problems 39–44 in Exercises 1.2.

When we start to solve differential equations in Chapter 2 we will solve only first-order equations and first-order initial-value problems. The mathematical description of many problems in science and engineering involve second-order IVPs or two-point BVPs. We will examine some of these problems in Chapters 4 and 5.

EXERCISES 1.2

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1 and 2, $y = 1/(1 + c_1 e^{-x})$ is a one-parameter family of solutions of the first-order DE $y' = y - y^2$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition.

1. $y(0) = -\frac{1}{3}$

2. $y(-1) = 2$

In Problems 3–6, $y = 1/(x^2 + c)$ is a one-parameter family of solutions of the first-order DE $y' + 2xy^2 = 0$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition. Give the largest interval I over which the solution is defined.

3. $y(2) = \frac{1}{3}$

4. $y(-2) = \frac{1}{2}$

5. $y(0) = 1$

6. $y(\frac{1}{2}) = -4$

In Problems 7–10, $x = c_1 \cos t + c_2 \sin t$ is a two-parameter family of solutions of the second-order DE $x'' + x = 0$. Find a solution of

the second-order IVP consisting of this differential equation and the given initial conditions.

7. $x(0) = -1, \quad x'(0) = 8$

8. $x(\pi/2) = 0, \quad x'(\pi/2) = 1$

9. $x(\pi/6) = \frac{1}{2}, \quad x'(\pi/6) = 0$

10. $x(\pi/4) = \sqrt{2}, \quad x'(\pi/4) = 2\sqrt{2}$

In Problems 11–14, $y = c_1 e^x + c_2 e^{-x}$ is a two-parameter family of solutions of the second-order DE $y'' - y = 0$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

11. $y(0) = 1, \quad y'(0) = 2$

12. $y(1) = 0, \quad y'(1) = e$

13. $y(-1) = 5, \quad y'(-1) = -5$

14. $y(0) = 0, \quad y'(0) = 0$

In Problems 15 and 16 determine by inspection at least two solutions of the given first-order IVP.

15. $y' = 3y^{2/3}$, $y(0) = 0$

16. $xy' = 2y$, $y(0) = 0$

In Problems 17–24 determine a region of the xy -plane for which the given differential equation would have a unique solution whose graph passes through a point (x_0, y_0) in the region.

17. $\frac{dy}{dx} = y^{2/3}$

18. $\frac{dy}{dx} = \sqrt{xy}$

19. $x \frac{dy}{dx} = y$

20. $\frac{dy}{dx} - y = x$

21. $(4 - y^2)y' = x^2$

22. $(1 + y^3)y' = x^2$

23. $(x^2 + y^2)y' = y^2$

24. $(y - x)y' = y + x$

In Problems 25–28 determine whether Theorem 1.2.1 guarantees that the differential equation $y' = \sqrt{y^2 - 9}$ possesses a unique solution through the given point.

25. $(1, 4)$

26. $(5, 3)$

27. $(2, -3)$

28. $(-1, 1)$

29. (a) By inspection find a one-parameter family of solutions of the differential equation $xy' = y$. Verify that each member of the family is a solution of the initial-value problem $xy' = y$, $y(0) = 0$.

(b) Explain part (a) by determining a region R in the xy -plane for which the differential equation $xy' = y$ would have a unique solution through a point (x_0, y_0) in R .

(c) Verify that the piecewise-defined function

$$y = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

satisfies the condition $y(0) = 0$. Determine whether this function is also a solution of the initial-value problem in part (a).

30. (a) Verify that $y = \tan(x + c)$ is a one-parameter family of solutions of the differential equation $y' = 1 + y^2$.

(b) Since $f(x, y) = 1 + y^2$ and $\partial f / \partial y = 2y$ are continuous everywhere, the region R in Theorem 1.2.1 can be taken to be the entire xy -plane. Use the family of solutions in part (a) to find an explicit solution of the first-order initial-value problem $y' = 1 + y^2$, $y(0) = 0$. Even though $x_0 = 0$ is in the interval $(-2, 2)$, explain why the solution is not defined on this interval.

(c) Determine the largest interval I of definition for the solution of the initial-value problem in part (b).

31. (a) Verify that $y = -1/(x + c)$ is a one-parameter family of solutions of the differential equation $y' = y^2$.

(b) Since $f(x, y) = y^2$ and $\partial f / \partial y = 2y$ are continuous everywhere, the region R in Theorem 1.2.1 can be taken to be the entire xy -plane. Find a solution from the family in part (a) that satisfies $y(0) = 1$. Then find a solution from the family in part (a) that satisfies $y(0) = -1$. Determine

the largest interval I of definition for the solution of each initial-value problem.

(c) Determine the largest interval I of definition for the solution of the first-order initial-value problem $y' = y^2$, $y(0) = 0$. [Hint: The solution is not a member of the family of solutions in part (a).]

32. (a) Show that a solution from the family in part (a) of Problem 31 that satisfies $y' = y^2$, $y(1) = 1$, is $y = 1/(2 - x)$.

(b) Then show that a solution from the family in part (a) of Problem 31 that satisfies $y' = y^2$, $y(3) = -1$, is $y = 1/(2 - x)$.

(c) Are the solutions in parts (a) and (b) the same?

33. (a) Verify that $3x^2 - y^2 = c$ is a one-parameter family of solutions of the differential equation $y \, dy/dx = 3x$.

(b) By hand, sketch the graph of the implicit solution $3x^2 - y^2 = 3$. Find all explicit solutions $y = \phi(x)$ of the DE in part (a) defined by this relation. Give the interval I of definition of each explicit solution.

(c) The point $(-2, 3)$ is on the graph of $3x^2 - y^2 = 3$, but which of the explicit solutions in part (b) satisfies $y(-2) = 3$?

34. (a) Use the family of solutions in part (a) of Problem 33 to find an implicit solution of the initial-value problem $y \, dy/dx = 3x$, $y(2) = -4$. Then, by hand, sketch the graph of the explicit solution of this problem and give its interval I of definition.

(b) Are there any explicit solutions of $y \, dy/dx = 3x$ that pass through the origin?

In Problems 35–38 the graph of a member of a family of solutions of a second-order differential equation $d^2y/dx^2 = f(x, y, y')$ is given. Match the solution curve with at least one pair of the following initial conditions.

(a) $y(1) = 1$, $y'(1) = -2$

(b) $y(-1) = 0$, $y'(-1) = -4$

(c) $y(1) = 1$, $y'(1) = 2$

(d) $y(0) = -1$, $y'(0) = 2$

(e) $y(0) = -1$, $y'(0) = 0$

(f) $y(0) = -4$, $y'(0) = -2$

35.

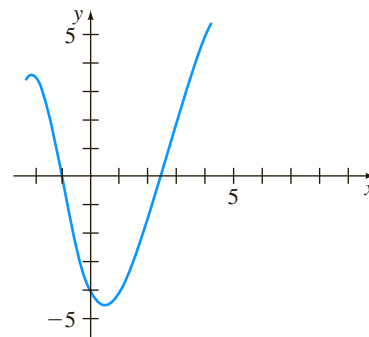


FIGURE 1.2.7 Graph for Problem 35

36.

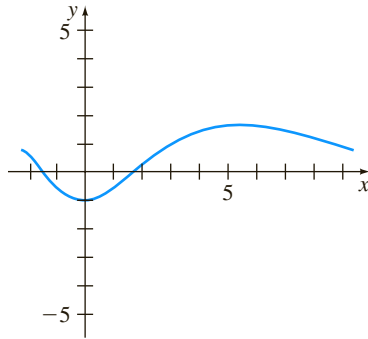


FIGURE 1.2.8 Graph for Problem 36

37.

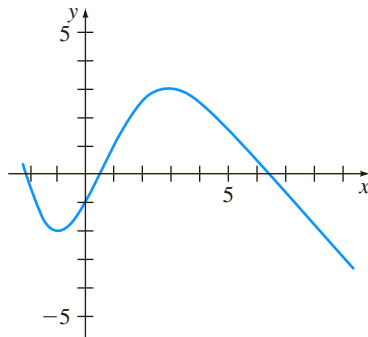


FIGURE 1.2.9 Graph for Problem 37

38.

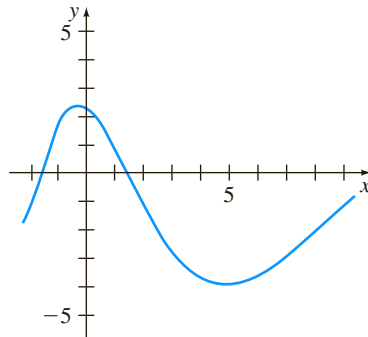


FIGURE 1.2.10 Graph for Problem 38

In Problems 39–44, $y = c_1 \cos 2x + c_2 \sin 2x$ is a two-parameter family of solutions of the second-order DE $y'' + 4y = 0$. If possible, find a solution of the differential equation that satisfies the given side conditions. The conditions specified at two different points are called boundary conditions.

39. $y(0) = 0, y(\pi/4) = 3$

40. $y(0) = 0, y(\pi) = 0$

41. $y'(0) = 0, y'(\pi/6) = 0$

42. $y(0) = 1, y'(\pi) = 5$

43. $y(0) = 0, y(\pi) = 2$

44. $y'(\pi/2) = 1, y'(\pi) = 0$

Discussion Problems

In Problems 45 and 46 use Problem 55 in Exercises 1.1 and (2) and (3) of this section.

45. Find a function whose graph at each point (x, y) has the slope given by $8e^{2x} + 6x$ and has the y -intercept $(0, 9)$.

46. Find a function whose second derivative is $y'' = 12x - 2$ at each point (x, y) on its graph and $y = -x + 5$ is tangent to the graph at the point corresponding to $x = 1$.

47. Consider the initial-value problem $y' = x - 2y, y(0) = \frac{1}{2}$. Determine which of the two curves shown in Figure 1.2.11 is the only plausible solution curve. Explain your reasoning.

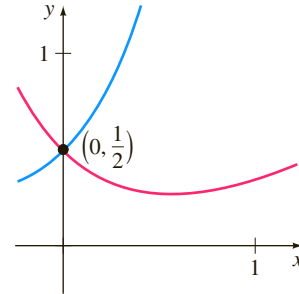


FIGURE 1.2.11 Graphs for Problem 47

48. Show that

$$x = \int_0^y \frac{1}{\sqrt{t^3 + 1}} dt$$

is an implicit solution of the initial-value problem

$$2 \frac{d^2y}{dx^2} - 3y^2 = 0, \quad y(0) = 0, \quad y'(0) = 1.$$

Assume that $y \geq 0$. [Hint: The integral is nonelementary. See (ii) in the Remarks at the end of Section 1.1.]

49. Determine a plausible value of x_0 for which the graph of the solution of the initial-value problem $y' + 2y = 3x - 6, y(x_0) = 0$ is tangent to the x -axis at $(x_0, 0)$. Explain your reasoning.

50. Suppose that the first-order differential equation $dy/dx = f(x, y)$ possesses a one-parameter family of solutions and that $f(x, y)$ satisfies the hypotheses of Theorem 1.2.1 in some rectangular region R of the xy -plane. Explain why two different solution curves cannot intersect or be tangent to each other at a point (x_0, y_0) in R .

51. The functions $y(x) = \frac{1}{16}x^4, -\infty < x < \infty$ and

$$y(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{16}x^4, & x \geq 0 \end{cases}$$

have the same domain but are clearly different. See Figures 1.2.12(a) and 1.2.12(b), respectively. Show that both functions are solutions of the initial-value problem $dy/dx = xy^{1/2}, y(2) = 1$ on the interval $(-\infty, \infty)$. Resolve the apparent contradiction between this fact and the last sentence in Example 5.

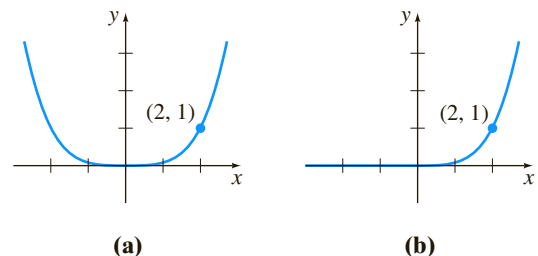


FIGURE 1.2.12 Two solutions of the IVP in Problem 51

1.3 Differential Equations as Mathematical Models

INTRODUCTION In this section we introduce the notion of a differential equation as a mathematical model and discuss some specific models in biology, chemistry, and physics. Once we have studied some methods for solving differential equations in Chapters 2 and 4, we return to and solve some of these models in Chapters 3 and 5.

MATHEMATICAL MODELS It is often desirable to describe the behavior of some real-life system or a phenomenon—whether physical, sociological, or even economic—in mathematical terms. The mathematical description of a system or a phenomenon is called a **mathematical model** and is constructed with certain goals in mind. For example, we may wish to understand the mechanisms of a certain ecosystem by studying the growth of animal populations in that system, or we may wish to date fossils by means of analyzing the decay of a radioactive substance, either in the fossil or in the stratum in which it was discovered.

Construction of a mathematical model of a system starts with

- (i) identification of the variables that are responsible for changing the system. We may choose not to incorporate all these variables into the model at first. In this step we are specifying the **level of resolution** of the model.

Next

- (ii) we make a set of reasonable assumptions, or hypotheses, about the system we are trying to describe. These assumptions will also include any empirical laws that may be applicable to the system.

For some purposes it may be perfectly within reason to be content with low-resolution models. For example, you may already be aware that in beginning physics courses, the retarding force of air friction is sometimes ignored in modeling the motion of a body falling near the surface of the Earth, but if you are a scientist whose job it is to accurately predict the flight path of a long-range projectile, you have to take into account air resistance and other factors such as the curvature of the Earth.

Since the assumptions made about a system frequently involve *a rate of change* of one or more of the variables, the mathematical depiction of all these assumptions may be one or more equations involving *derivatives*. In other words, the mathematical model may be a differential equation or a system of differential equations.

Once we have formulated a mathematical model that is either a differential equation or a system of differential equations, we are faced with the not insignificant problem of trying to solve it. *If* we can solve it, then we deem the model to be reasonable if its solution is consistent with either experimental data or known facts about the behavior of the system. But if the predictions produced by the solution are poor, we can either increase the level of resolution of the model or make alternative assumptions about the mechanisms for change in the system. The steps of the modeling process are then repeated, as shown in the diagram in Figure 1.3.1.

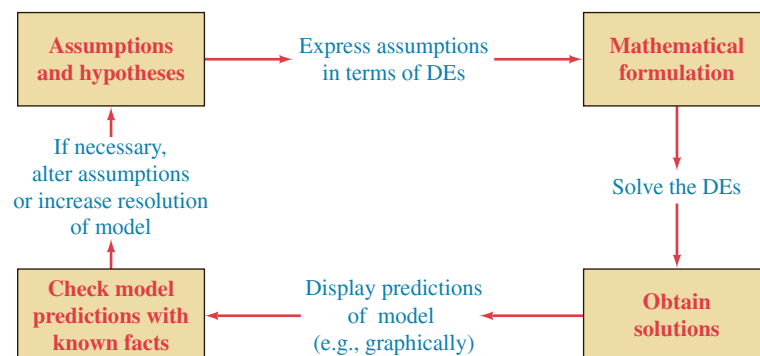


FIGURE 1.3.1 Steps in the modeling process with differential equations

Of course, by increasing the resolution, we add to the complexity of the mathematical model and increase the likelihood that we cannot obtain an explicit solution of the differential equation.

A mathematical model of a physical system will often involve the variable time t . A solution of the model then gives the **state of the system**; in other words, the values of the dependent variable (or variables) for appropriate values of t describe the system in the past, present, and future.

POPULATION DYNAMICS One of the earliest attempts to model human **population growth** by means of mathematics was by the English clergyman and economist **Thomas Malthus** (1766–1834) in 1798. Basically, the idea behind the Malthusian model is the assumption that the rate at which the population of a country grows at a certain time is proportional* to the total population of the country at that time. In other words, the more people there are at time t , the more there are going to be in the future. In mathematical terms, if $P(t)$ denotes the total population at time t , then this assumption can be expressed as

$$\frac{dP}{dt} \propto P \quad \text{or} \quad \frac{dP}{dt} = kP, \quad (1)$$

where k is a constant of proportionality. This simple model, which fails to take into account many factors that can influence human populations to either grow or decline (immigration and emigration, for example), nevertheless turned out to be fairly accurate in predicting the population of the United States during the years 1790–1860. Populations that grow at a rate described by (1) are rare; nevertheless, (1) is still used to model *growth of small populations over short intervals of time* (bacteria growing in a petri dish, for example).

RADIOACTIVE DECAY The nucleus of an atom consists of combinations of protons and neutrons. Many of these combinations of protons and neutrons are unstable—that is, the atoms decay or transmute into atoms of another substance. Such nuclei are said to be radioactive. For example, over time the highly radioactive radium, Ra-226, transmutes into the radioactive gas radon, Rn-222. To model the phenomenon of **radioactive decay**, it is assumed that the rate dA/dt at which the nuclei of a substance decay is proportional to the amount (more precisely, the number of nuclei) $A(t)$ of the substance remaining at time t :

$$\frac{dA}{dt} \propto A \quad \text{or} \quad \frac{dA}{dt} = kA. \quad (2)$$

Of course, equations (1) and (2) are exactly the same; the difference is only in the interpretation of the symbols and the constants of proportionality. For growth, as we expect in (1), $k > 0$, and for decay, as in (2), $k < 0$.

The model (1) for growth can also be seen as the equation $dS/dt = rS$, which describes the growth of capital S when an annual rate of interest r is compounded continuously. The model (2) for decay also occurs in biological applications such as determining the half-life of a drug—the time that it takes for 50% of a drug to be eliminated from a body by excretion or metabolism. In chemistry the decay model (2) appears in the mathematical description of a first-order chemical reaction. The point is this:

A single differential equation can serve as a mathematical model for many different phenomena.

*If two quantities u and v are proportional, we write $u \propto v$. This means that one quantity is a constant multiple of the other: $u = kv$.

Mathematical models are often accompanied by certain side conditions. For example, in (1) and (2) we would expect to know, in turn, the initial population P_0 and the initial amount of radioactive substance A_0 on hand. If the initial point in time is taken to be $t = 0$, then we know that $P(0) = P_0$ and $A(0) = A_0$. In other words, a mathematical model can consist of either an initial-value problem or, as we shall see later on in Section 5.2, a boundary-value problem.

NEWTON'S LAW OF COOLING/WARMING According to Newton's empirical law of cooling/warming, the rate at which the temperature of a body changes is proportional to the difference between the temperature of the body and the temperature of the surrounding medium, the so-called ambient temperature. If $T(t)$ represents the temperature of a body at time t , T_m the temperature of the surrounding medium, and dT/dt the rate at which the temperature of the body changes, then Newton's law of cooling/warming translates into the mathematical statement

$$\frac{dT}{dt} \propto T - T_m \quad \text{or} \quad \frac{dT}{dt} = k(T - T_m), \quad (3)$$

where k is a constant of proportionality. In either case, cooling or warming, if T_m is a constant, it stands to reason that $k < 0$.

SPREAD OF A DISEASE A contagious disease—for example, a flu virus—is spread throughout a community by people coming into contact with other people. Let $x(t)$ denote the number of people who have contracted the disease and $y(t)$ denote the number of people who have not yet been exposed. It seems reasonable to assume that the rate dx/dt at which the disease spreads is proportional to the number of encounters, or *interactions*, between these two groups of people. If we assume that the number of interactions is jointly proportional to $x(t)$ and $y(t)$ —that is, proportional to the product xy —then

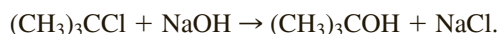
$$\frac{dx}{dt} = kxy, \quad (4)$$

where k is the usual constant of proportionality. Suppose a small community has a fixed population of n people. If one infected person is introduced into this community, then it could be argued that $x(t)$ and $y(t)$ are related by $x + y = n + 1$. Using this last equation to eliminate y in (4) gives us the model

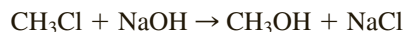
$$\frac{dx}{dt} = kx(n + 1 - x). \quad (5)$$

An obvious initial condition accompanying equation (5) is $x(0) = 1$.

CHEMICAL REACTIONS The disintegration of a radioactive substance, governed by the differential equation (1), is said to be a **first-order reaction**. In chemistry a few reactions follow this same empirical law: If the molecules of substance A decompose into smaller molecules, it is a natural assumption that the rate at which this decomposition takes place is proportional to the amount of the first substance that has not undergone conversion; that is, if $X(t)$ is the amount of substance A remaining at any time, then $dX/dt = kX$, where k is a negative constant since X is decreasing. An example of a first-order chemical reaction is the conversion of *t*-butyl chloride, $(\text{CH}_3)_3\text{CCl}$, into *t*-butyl alcohol, $(\text{CH}_3)_3\text{COH}$:



Only the concentration of the *t*-butyl chloride controls the rate of reaction. But in the reaction



one molecule of sodium hydroxide, NaOH , is consumed for every molecule of methyl chloride, CH_3Cl , thus forming one molecule of methyl alcohol, CH_3OH , and

one molecule of sodium chloride, NaCl . In this case the rate at which the reaction proceeds is proportional to the product of the remaining concentrations of CH_3Cl and NaOH . To describe this second reaction in general, let us suppose *one* molecule of a substance A combines with *one* molecule of a substance B to form *one* molecule of a substance C . If X denotes the amount of chemical C formed at time t and if α and β are, in turn, the amounts of the two chemicals A and B at $t = 0$ (the initial amounts), then the instantaneous amounts of A and B not converted to chemical C are $\alpha - X$ and $\beta - X$, respectively. Hence the rate of formation of C is given by

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X), \quad (6)$$

where k is a constant of proportionality. A reaction whose model is equation (6) is said to be a **second-order reaction**.

MIXTURES The mixing of two salt solutions of differing concentrations gives rise to a first-order differential equation for the amount of salt contained in the mixture. Let us suppose that a large mixing tank initially holds 300 gallons of brine (that is, water in which a certain number of pounds of salt has been dissolved). Another brine solution is pumped into the large tank at a rate of 3 gallons per minute; the concentration of the salt in this inflow is 2 pounds per gallon. When the solution in the tank is well stirred, it is pumped out at the same rate as the entering solution. See Figure 1.3.2. If $A(t)$ denotes the amount of salt (measured in pounds) in the tank at time t , then the rate at which $A(t)$ changes is a net rate:

$$\frac{dA}{dt} = \left(\begin{array}{c} \text{input rate} \\ \text{of salt} \end{array} \right) - \left(\begin{array}{c} \text{output rate} \\ \text{of salt} \end{array} \right) = R_{in} - R_{out}. \quad (7)$$

The input rate R_{in} at which salt enters the tank is the product of the inflow concentration of salt and the inflow rate of fluid. Note that R_{in} is measured in pounds per minute:

$$R_{in} = \begin{array}{c} \text{concentration} \\ \text{of salt} \\ \text{in inflow} \end{array} \cdot \begin{array}{c} \text{input rate} \\ \text{of brine} \end{array} = \begin{array}{c} \text{input rate} \\ \text{of salt} \end{array} = (2 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = (6 \text{ lb/min}).$$

Now, since the solution is being pumped out of the tank at the same rate that it is pumped in, the number of gallons of brine in the tank at time t is a constant 300 gallons. Hence the concentration of the salt in the tank as well as in the outflow is $c(t) = A(t)/300$ lb/gal, so the output rate R_{out} of salt is

$$R_{out} = \begin{array}{c} \text{concentration} \\ \text{of salt} \\ \text{in outflow} \end{array} \cdot \begin{array}{c} \text{output rate} \\ \text{of brine} \end{array} = \begin{array}{c} \text{output rate} \\ \text{of salt} \end{array} = \left(\frac{A(t)}{300} \text{ lb/gal} \right) \cdot (3 \text{ gal/min}) = \frac{A(t)}{100} \text{ lb/min}.$$

The net rate (7) then becomes

$$\frac{dA}{dt} = 6 - \frac{A}{100} \quad \text{or} \quad \frac{dA}{dt} + \frac{1}{100}A = 6. \quad (8)$$

If r_{in} and r_{out} denote general input and output rates of the brine solutions,* then there are three possibilities: $r_{in} = r_{out}$, $r_{in} > r_{out}$, and $r_{in} < r_{out}$. In the analysis leading to (8) we have assumed that $r_{in} = r_{out}$. In the latter two cases the number of gallons of brine in the tank is either increasing ($r_{in} > r_{out}$) or decreasing ($r_{in} < r_{out}$) at the net rate $r_{in} - r_{out}$. See Problems 10–12 in Exercises 1.3.

*Don't confuse these symbols with R_{in} and R_{out} , which are input and output rates of *salt*.

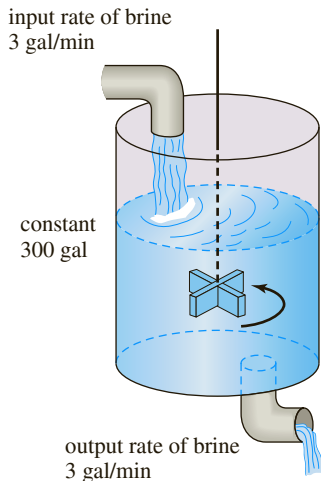


FIGURE 1.3.2 Mixing tank

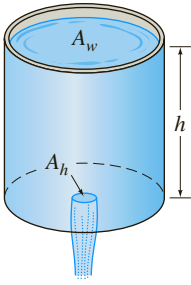


FIGURE 1.3.3 Draining tank

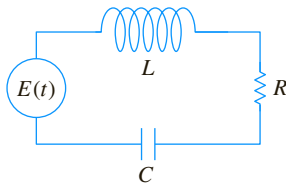
DRAINING A TANK In hydrodynamics, **Torricelli's law** states that the speed v of efflux of water through a sharp-edged hole at the bottom of a tank filled to a depth h is the same as the speed that a body (in this case a drop of water) would acquire in falling freely from a height h —that is, $v = \sqrt{2gh}$, where g is the acceleration due to gravity. This last expression comes from equating the kinetic energy $\frac{1}{2}mv^2$ with the potential energy mgh and solving for v . Suppose a tank filled with water is allowed to drain through a hole under the influence of gravity. We would like to find the depth h of water remaining in the tank at time t . Consider the tank shown in Figure 1.3.3. If the area of the hole is A_h (in ft^2) and the speed of the water leaving the tank is $v = \sqrt{2gh}$ (in ft/s), then the volume of water leaving the tank per second is $A_h\sqrt{2gh}$ (in ft^3/s). Thus if $V(t)$ denotes the volume of water in the tank at time t , then

$$\frac{dV}{dt} = -A_h\sqrt{2gh}, \quad (9)$$

where the minus sign indicates that V is decreasing. Note here that we are ignoring the possibility of friction at the hole that might cause a reduction of the rate of flow there. Now if the tank is such that the volume of water in it at time t can be written $V(t) = A_w h$, where A_w (in ft^2) is the *constant* area of the upper surface of the water (see Figure 1.3.3), then $dV/dt = A_w dh/dt$. Substituting this last expression into (9) gives us the desired differential equation for the height of the water at time t :

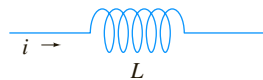
$$\frac{dh}{dt} = -\frac{A_h}{A_w}\sqrt{2gh}. \quad (10)$$

It is interesting to note that (10) remains valid even when A_w is not constant. In this case we must express the upper surface area of the water as a function of h —that is, $A_w = A(h)$. See Problem 14 in Exercises 1.3.

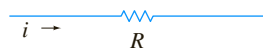


(a) LRC-series circuit

Inductor
inductance L : henries (h)
voltage drop across: $L \frac{di}{dt}$



Resistor
resistance R : ohms (Ω)
voltage drop across: iR



Capacitor
capacitance C : farads (f)
voltage drop across: $\frac{1}{C}q$



(b)

FIGURE 1.3.4 Symbols, units, and voltages. Current $i(t)$ and charge $q(t)$ are measured in amperes (A) and coulombs (C), respectively

SERIES CIRCUITS Consider the single-loop *LRC*-series circuit shown in Figure 1.3.4(a), containing an inductor, resistor, and capacitor. The current in a circuit after a switch is closed is denoted by $i(t)$; the charge on a capacitor at time t is denoted by $q(t)$. The letters L , R , and C are known as inductance, resistance, and capacitance, respectively, and are generally constants. Now according to **Kirchhoff's second law**, the impressed voltage $E(t)$ on a closed loop must equal the sum of the voltage drops in the loop. Figure 1.3.4(b) shows the symbols and the formulas for the respective voltage drops across an inductor, a capacitor, and a resistor. Since current $i(t)$ is related to charge $q(t)$ on the capacitor by $i = dq/dt$, adding the three voltages

$$\begin{array}{ccc} \text{inductor} & \text{resistor} & \text{capacitor} \\ L \frac{di}{dt} = L \frac{d^2q}{dt^2}, & iR = R \frac{dq}{dt}, & \text{and } \frac{1}{C}q \end{array}$$

and equating the sum to the impressed voltage yields a second-order differential equation

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = E(t). \quad (11)$$

We will examine a differential equation analogous to (11) in great detail in Section 5.1.

FALLING BODIES To construct a mathematical model of the motion of a body moving in a force field, one often starts with the laws of motion formulated by the English mathematician **Isaac Newton** (1643–1727). Recall from elementary physics that **Newton's first law of motion** states that a body either will remain at rest or will continue to move with a constant velocity unless acted on by an external force. In each case this is equivalent to saying that when the sum of the forces

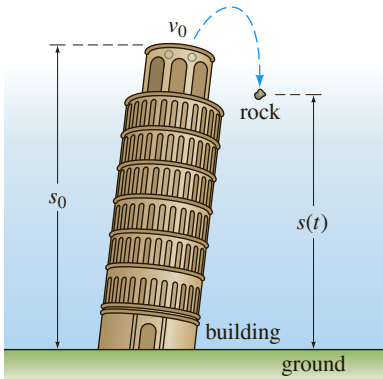


FIGURE 1.3.5 Position of rock measured from ground level

$F = \sum F_k$ —that is, the *net* or resultant force—acting on the body is zero, then the acceleration a of the body is zero. **Newton's second law of motion** indicates that when the net force acting on a body is not zero, then the net force is proportional to its acceleration a or, more precisely, $F = ma$, where m is the mass of the body.

Now suppose a rock is tossed upward from the roof of a building as illustrated in Figure 1.3.5. What is the position $s(t)$ of the rock relative to the ground at time t ? The acceleration of the rock is the second derivative d^2s/dt^2 . If we assume that the upward direction is positive and that no force acts on the rock other than the force of gravity, then Newton's second law gives

$$m \frac{d^2s}{dt^2} = -mg \quad \text{or} \quad \frac{d^2s}{dt^2} = -g. \quad (12)$$

In other words, the net force is simply the weight $F = F_1 = -W$ of the rock near the surface of the Earth. Recall that the magnitude of the weight is $W = mg$, where m is the mass of the body and g is the acceleration due to gravity. The minus sign in (12) is used because the weight of the rock is a force directed downward, which is opposite to the positive direction. If the height of the building is s_0 and the initial velocity of the rock is v_0 , then s is determined from the second-order initial-value problem

$$\frac{d^2s}{dt^2} = -g, \quad s(0) = s_0, \quad s'(0) = v_0. \quad (13)$$

Although we have not been stressing solutions of the equations we have constructed, note that (13) can be solved by integrating the constant $-g$ twice with respect to t . The initial conditions determine the two constants of integration. From elementary physics you might recognize the solution of (13) as the formula $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$.

FALLING BODIES AND AIR RESISTANCE Before the famous experiment by the Italian mathematician and physicist **Galileo Galilei** (1564–1642) from the leaning tower of Pisa, it was generally believed that heavier objects in free fall, such as a cannonball, fell with a greater acceleration than lighter objects, such as a feather. Obviously, a cannonball and a feather when dropped simultaneously from the same height *do* fall at different rates, but it is not because a cannonball is heavier. The difference in rates is due to air resistance. The resistive force of air was ignored in the model given in (13). Under some circumstances a falling body of mass m , such as a feather with low density and irregular shape, encounters air resistance proportional to its instantaneous velocity v . If we take, in this circumstance, the positive direction to be oriented downward, then the net force acting on the mass is given by $F = F_1 + F_2 = mg - kv$, where the weight $F_1 = mg$ of the body is force acting in the positive direction and air resistance $F_2 = -kv$ is a force, called **viscous damping**, acting in the opposite or upward direction. See Figure 1.3.6. Now since v is related to acceleration a by $a = dv/dt$, Newton's second law becomes $F = ma = m dv/dt$. By equating the net force to this form of Newton's second law, we obtain a first-order differential equation for the velocity $v(t)$ of the body at time t ,

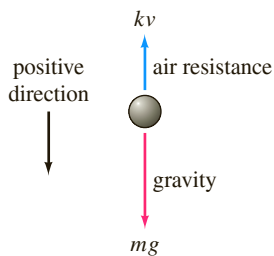
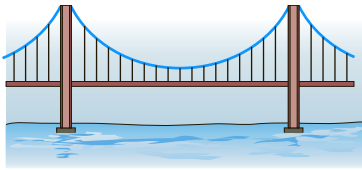


FIGURE 1.3.6 Falling body of mass m

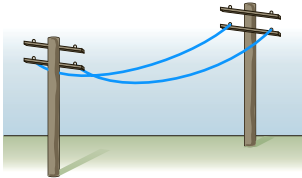
$$m \frac{dv}{dt} = mg - kv. \quad (14)$$

Here k is a positive constant of proportionality. If $s(t)$ is the distance the body falls in time t from its initial point of release, then $v = ds/dt$ and $a = dv/dt = d^2s/dt^2$. In terms of s , (14) is a second-order differential equation

$$m \frac{d^2s}{dt^2} = mg - k \frac{ds}{dt} \quad \text{or} \quad m \frac{d^2s}{dt^2} + k \frac{ds}{dt} = mg. \quad (15)$$



(a) suspension bridge cable



(b) telephone wires

FIGURE 1.3.7 Cables suspended between vertical supports

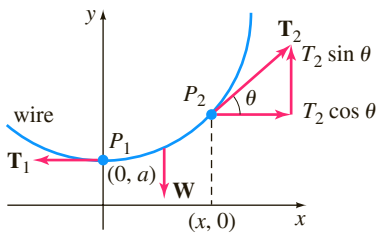


FIGURE 1.3.8 Element of cable

SUSPENDED CABLES Suppose a flexible cable, wire, or heavy rope is suspended between two vertical supports. Physical examples of this could be one of the two cables supporting the roadbed of a suspension bridge as shown in Figure 1.3.7(a) or a long telephone wire strung between two posts as shown in Figure 1.3.7(b). Our goal is to construct a mathematical model that describes the shape that such a cable assumes.

To begin, let's agree to examine only a portion or element of the cable between its lowest point P_1 and any arbitrary point P_2 . As drawn in blue in Figure 1.3.8, this element of the cable is the curve in a rectangular coordinate system with y -axis chosen to pass through the lowest point P_1 on the curve and the x -axis chosen a units below P_1 . Three forces are acting on the cable: the tensions $\mathbf{T}_1$ and $\mathbf{T}_2$ in the cable that are tangent to the cable at P_1 and P_2 , respectively, and the portion $\mathbf{W}$ of the total vertical load between the points P_1 and P_2 . Let $T_1 = |\mathbf{T}_1|$, $T_2 = |\mathbf{T}_2|$, and $W = |\mathbf{W}|$ denote the magnitudes of these vectors. Now the tension $\mathbf{T}_2$ resolves into horizontal and vertical components (scalar quantities) $T_2 \cos \theta$ and $T_2 \sin \theta$. Because of static equilibrium we can write

$$T_1 = T_2 \cos \theta \quad \text{and} \quad W = T_2 \sin \theta.$$

By dividing the last equation by the first, we eliminate T_2 and get $\tan \theta = W/T_1$. But because $dy/dx = \tan \theta$, we arrive at

$$\frac{dy}{dx} = \frac{W}{T_1}. \quad (16)$$

This simple first-order differential equation serves as a model for both the shape of a flexible wire such as a telephone wire hanging under its own weight and the shape of the cables that support the roadbed of a suspension bridge. We will come back to equation (16) in Exercises 2.2 and Section 5.3.

WHAT LIES AHEAD Throughout this text you will see three different types of approaches to, or analyses of, differential equations. Over the centuries differential equations would often spring from the efforts of a scientist or engineer to describe some physical phenomenon or to translate an empirical or experimental law into mathematical terms. As a consequence, a scientist, engineer, or mathematician would often spend many years of his or her life trying to find the solutions of a DE. With a solution in hand, the study of its properties then followed. This quest for solutions is called by some the *analytical approach* to differential equations. Once they realized that explicit solutions are at best difficult to obtain and at worst impossible to obtain, mathematicians learned that a differential equation itself could be a font of valuable information. It is possible, in some instances, to glean directly from the differential equation answers to questions such as

Does the DE actually have solutions? If a solution of the DE exists and satisfies an initial condition, is it the only such solution? What are some of the properties of the unknown solutions? What can we say about the geometry of the solution curves?

Such an approach is *qualitative analysis*. Finally, if a differential equation cannot be solved by analytical methods, yet we can prove that a solution exists, the next logical query is

Can we somehow approximate the values of an unknown solution?

Here we enter the realm of *numerical analysis*. An affirmative answer to the last question stems from the fact that a differential equation can be used as a cornerstone for constructing very accurate approximation algorithms. In Chapter 2 we start with qualitative considerations of first-order ODEs, then examine analytical stratagems for solving some special first-order equations, and conclude with an introduction to an elementary numerical method. See Figure 1.3.9.

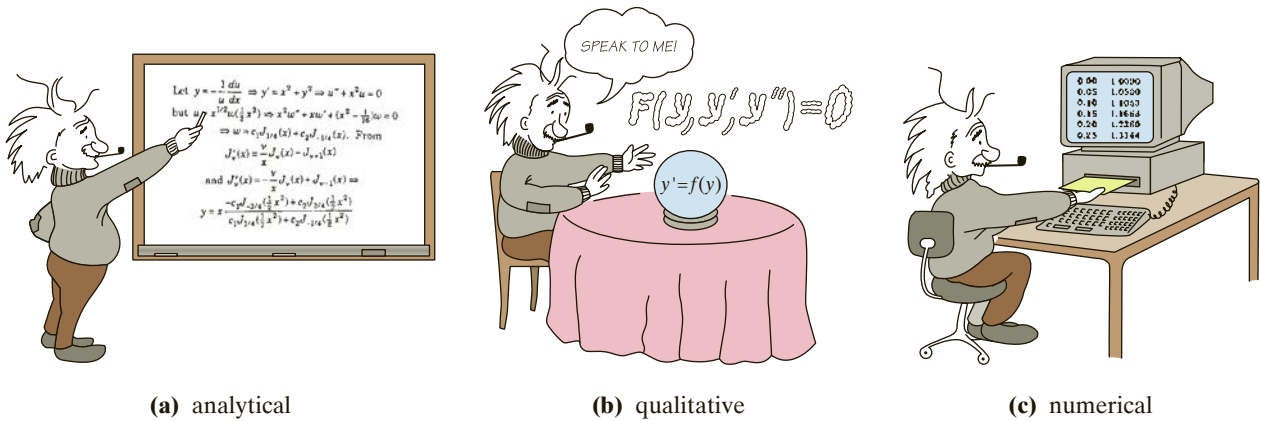


FIGURE 1.3.9 Different approaches to the study of differential equations

REMARKS

Each example in this section has described a dynamical system—a system that changes or evolves with the flow of time t . Since the study of dynamical systems is a branch of mathematics currently in vogue, we shall occasionally relate the terminology of that field to the discussion at hand.

In more precise terms, a **dynamical system** consists of a set of time-dependent variables, called **state variables**, together with a rule that enables us to determine (without ambiguity) the state of the system (this may be a past, present, or future state) in terms of a state prescribed at some time t_0 . Dynamical systems are classified as either discrete-time systems or continuous-time systems. In this course we shall be concerned only with continuous-time systems—systems in which *all* variables are defined over a continuous range of time. The rule, or mathematical model, in a continuous-time dynamical system is a differential equation or a system of differential equations. The **state of the system** at a time t is the value of the state variables at that time; the specified state of the system at a time t_0 is simply the initial conditions that accompany the mathematical model. The solution of the initial-value problem is referred to as the **response of the system**. For example, in the case of radioactive decay, the rule is $dA/dt = kA$. Now if the quantity of a radioactive substance at some time t_0 is known, say $A(t_0) = A_0$, then by solving the rule we find that the response of the system for $t \geq t_0$ is $A(t) = A_0 e^{(t-t_0)}$ (see Section 3.1). The response $A(t)$ is the single state variable for this system. In the case of the rock tossed from the roof of a building, the response of the system—the solution of the differential equation $d^2s/dt^2 = -g$, subject to the initial state $s(0) = s_0$, $s'(0) = v_0$, is the function $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$, $0 \leq t \leq T$, where T represents the time when the rock hits the ground. The state variables are $s(t)$ and $s'(t)$, which are the vertical position of the rock above ground and its velocity at time t , respectively. The acceleration $s''(t)$ is *not* a state variable, since we have to know only any initial position and initial velocity at a time t_0 to uniquely determine the rock's position $s(t)$ and velocity $s'(t) = v(t)$ for any time in the interval $t_0 \leq t \leq T$. The acceleration $s''(t) = a(t)$ is, of course, given by the differential equation $s''(t) = -g$, $0 < t < T$.

One last point: Not every system studied in this text is a dynamical system. We shall also examine some static systems in which the model is a differential equation.

EXERCISES 1.3

Answers to selected odd-numbered problems begin on page ANS-1.

Population Dynamics

- Under the same assumptions that underlie the model in (1), determine a differential equation for the population $P(t)$ of a country when individuals are allowed to immigrate into the country at a constant rate $r > 0$. What is the differential equation for the population $P(t)$ of the country when individuals are allowed to emigrate from the country at a constant rate $r > 0$?
- The population model given in (1) fails to take death into consideration; the growth rate equals the birth rate. In another model of a changing population of a community it is assumed that the rate at which the population changes is a *net* rate—that is, the difference between the rate of births and the rate of deaths in the community. Determine a model for the population $P(t)$ if both the birth rate and the death rate are proportional to the population present at time $t > 0$.
- Using the concept of net rate introduced in Problem 2, determine a model for a population $P(t)$ if the birth rate is proportional to the population present at time t but the death rate is proportional to the square of the population present at time t .
- Modify the model in Problem 3 for net rate at which the population $P(t)$ of a certain kind of fish changes by also assuming that the fish are harvested at a constant rate $h > 0$.

Newton's Law of Cooling/Warming

- A cup of coffee cools according to Newton's law of cooling (3). Use data from the graph of the temperature $T(t)$ in Figure 1.3.10 to estimate the constants T_m , T_0 , and k in a model of the form of a first-order initial-value problem: $dT/dt = k(T - T_m)$, $T(0) = T_0$.

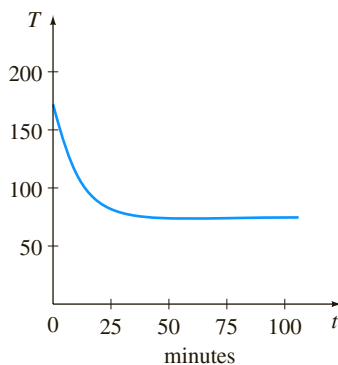


FIGURE 1.3.10 Cooling curve in Problem 5

- The ambient temperature T_m in (3) could be a function of time t . Suppose that in an artificially controlled environment, $T_m(t)$ is periodic with a 24-hour period, as illustrated in Figure 1.3.11. Devise a mathematical model for the temperature $T(t)$ of a body within this environment.

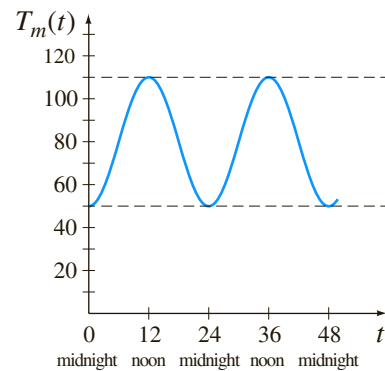


FIGURE 1.3.11 Ambient temperature in Problem 6

Spread of a Disease/Technology

- Suppose a student carrying a flu virus returns to an isolated college campus of 1000 students. Determine a differential equation for the number of people $x(t)$ who have contracted the flu if the rate at which the disease spreads is proportional to the number of interactions between the number of students who have the flu and the number of students who have not yet been exposed to it.
- At a time denoted as $t = 0$ a technological innovation is introduced into a community that has a fixed population of n people. Determine a differential equation for the number of people $x(t)$ who have adopted the innovation at time t if it is assumed that the rate at which the innovations spread through the community is jointly proportional to the number of people who have adopted it and the number of people who have not adopted it.

Mixtures

- Suppose that a large mixing tank initially holds 300 gallons of water in which 50 pounds of salt have been dissolved. Pure water is pumped into the tank at a rate of 3 gal/min, and when the solution is well stirred, it is then pumped out at the same rate. Determine a differential equation for the amount of salt $A(t)$ in the tank at time $t > 0$. What is $A(0)$?
- Suppose that a large mixing tank initially holds 300 gallons of water in which 50 pounds of salt have been dissolved. Another brine solution is pumped into the tank at a rate of 3 gal/min, and when the solution is well stirred, it is then pumped out at a *slower* rate of 2 gal/min. If the concentration of the solution entering is 2 lb/gal, determine a differential equation for the amount of salt $A(t)$ in the tank at time $t > 0$.
- What is the differential equation in Problem 10, if the well-stirred solution is pumped out at a *faster* rate of 3.5 gal/min?
- Generalize the model given in equation (8) of this section by assuming that the large tank initially contains N_0 number of gallons of brine, r_{in} and r_{out} are the input and output rates of the brine, respectively (measured in gallons per minute), c_{in} is the concentration of the salt in the inflow, $c(t)$ the concentration of

the salt in the tank as well as in the outflow at time t (measured in pounds of salt per gallon), and $A(t)$ is the amount of salt in the tank at time $t > 0$.

Draining a Tank

13. Suppose water is leaking from a tank through a circular hole of area A_h at its bottom. When water leaks through a hole, friction and contraction of the stream near the hole reduce the volume of water leaving the tank per second to $cA_h\sqrt{2gh}$, where c ($0 < c < 1$) is an empirical constant. Determine a differential equation for the height h of water at time t for the cubical tank shown in Figure 1.3.12. The radius of the hole is 2 in., and $g = 32 \text{ ft/s}^2$.

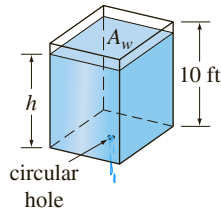


FIGURE 1.3.12 Cubical tank in Problem 13

14. The right-circular conical tank shown in Figure 1.3.13 loses water out of a circular hole at its bottom. Determine a differential equation for the height of the water h at time $t > 0$. The radius of the hole is 2 in., $g = 32 \text{ ft/s}^2$, and the friction/contraction factor introduced in Problem 13 is $c = 0.6$.

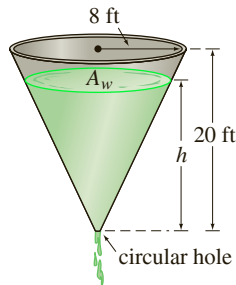


FIGURE 1.3.13 Conical tank in Problem 14

Series Circuits

15. A series circuit contains a resistor and an inductor as shown in Figure 1.3.14. Determine a differential equation for the current $i(t)$ if the resistance is R , the inductance is L , and the impressed voltage is $E(t)$.

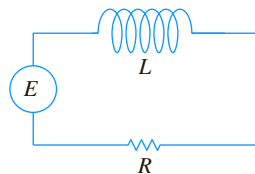


FIGURE 1.3.14 LR-series circuit in Problem 15

16. A series circuit contains a resistor and a capacitor as shown in Figure 1.3.15. Determine a differential equation for the charge $q(t)$ on the capacitor if the resistance is R , the capacitance is C , and the impressed voltage is $E(t)$.

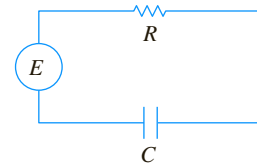


FIGURE 1.3.15 RC-series circuit in Problem 16

Falling Bodies and Air Resistance

17. For high-speed motion through the air—such as the skydiver shown in Figure 1.3.16, falling before the parachute is opened—air resistance is closer to a power of the instantaneous velocity $v(t)$. Determine a differential equation for the velocity $v(t)$ of a falling body of mass m if air resistance is proportional to the square of the instantaneous velocity. Assume the downward direction is positive.

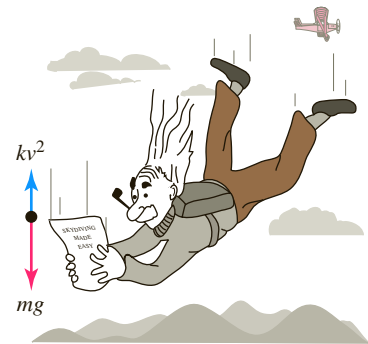


FIGURE 1.3.16 Air resistance proportional to square of velocity in Problem 17

Newton's Second Law and Archimedes' Principle

18. A cylindrical barrel s feet in diameter of weight w lb is floating in water as shown in Figure 1.3.17(a). After an initial depression the barrel exhibits an up-and-down bobbing motion along a vertical line. Using Figure 1.3.17(b), determine a differential equation for the vertical displacement $y(t)$ if the origin is taken to be on the vertical axis at the surface of the water when the barrel is at rest. Use **Archimedes' principle**: Buoyancy, or upward force of the water on the barrel, is equal to the weight of the water displaced. Assume that the downward direction is positive, that the weight density of water is 62.4 lb/ft^3 , and that there is no resistance between the barrel and the water.

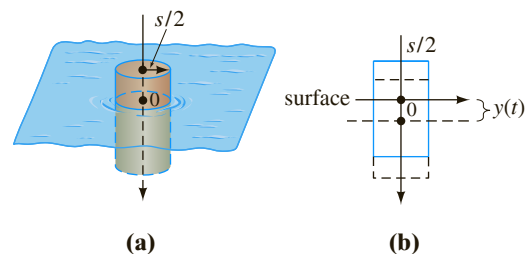


FIGURE 1.3.17 Bobbing motion of floating barrel in Problem 18

Newton's Second Law and Hooke's Law

19. After a mass m is attached to a spring, it stretches it s units and then hangs at rest in the equilibrium position as shown in Figure 1.3.18(b). After the spring/mass system has been set in motion, let $x(t)$ denote the directed distance of the mass beyond the equilibrium position. As indicated in Figure 1.3.18(c), assume that the downward direction is positive, that the motion takes place in a vertical straight line through the center of gravity of the mass, and that the only forces acting on the system are the weight of the mass and the restoring force of the stretched spring. Use **Hooke's law**: The restoring force of a spring is proportional to its total elongation. Determine a differential equation for the displacement $x(t)$ at time $t > 0$.

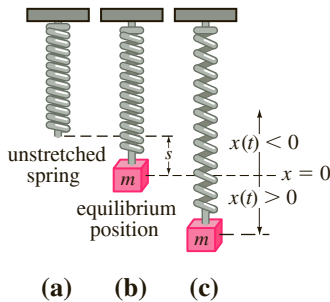


FIGURE 1.3.18 Spring/mass system in Problem 19

20. In Problem 19, what is a differential equation for the displacement $x(t)$ if the motion takes place in a medium that imparts a damping force on the spring/mass system that is proportional to the instantaneous velocity of the mass and acts in a direction opposite to that of motion?

Newton's Second Law and Variable Mass

When the mass m of a body is changing with time, Newton's second law of motion becomes

$$F = \frac{d}{dt}(mv), \quad (17)$$

where F is the net force acting on the body and mv is its momentum. Use (17) in Problems 21 and 22.

21. A small single-stage rocket is launched vertically as shown in Figure 1.3.19. Once launched, the rocket consumes its



FIGURE 1.3.19 Single-stage rocket in Problem 21

fuel, and so its total mass $m(t)$ varies with time $t > 0$. If it is assumed that the positive direction is upward, air resistance is proportional to the instantaneous velocity v of the rocket, and R is the upward thrust or force generated by the propulsion system, then construct a mathematical model for the velocity $v(t)$ of the rocket. [Hint: See (14) in Section 1.3.]

22. In Problem 21, the mass $m(t)$ is the sum of three different masses: $m(t) = m_p + m_v + m_f(t)$, where m_p is the constant mass of the payload, m_v is the constant mass of the vehicle, and $m_f(t)$ is the variable amount of fuel.
- Show that the rate at which the total mass $m(t)$ of the rocket changes is the same as the rate at which the mass $m_f(t)$ of the fuel changes.
 - If the rocket consumes its fuel at a constant rate λ , find $m(t)$. Then rewrite the differential equation in Problem 21 in terms of λ and the initial total mass $m(0) = m_0$.
 - Under the assumption in part (b), show that the burnout time $t_b > 0$ of the rocket, or the time at which all the fuel is consumed, is $t_b = m_f(0)/\lambda$, where $m_f(0)$ is the initial mass of the fuel.

Newton's Second Law and the Law of Universal Gravitation

23. By **Newton's universal law of gravitation** the free-fall acceleration a of a body, such as the satellite shown in Figure 1.3.20, falling a great distance to the surface is *not* the constant g . Rather, the acceleration a is inversely proportional to the square of the distance from the center of the Earth, $a = k/r^2$, where k is the constant of proportionality. Use the fact that at the surface of the Earth $r = R$ and $a = g$ to determine k . If the positive direction is upward, use Newton's second law and his universal law of gravitation to find a differential equation for the distance r .

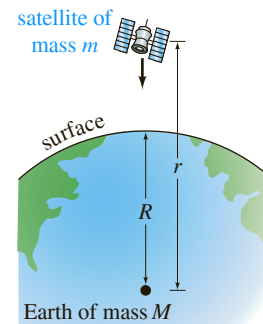


FIGURE 1.3.20 Satellite in Problem 23

24. Suppose a hole is drilled through the center of the Earth and a bowling ball of mass m is dropped into the hole, as shown in Figure 1.3.21. Construct a mathematical model that describes the

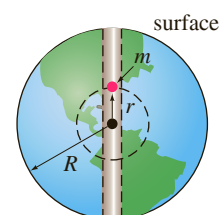


FIGURE 1.3.21 Hole through Earth in Problem 24

motion of the ball. At time t let r denote the distance from the center of the Earth to the mass m , M denote the mass of the Earth, M_r denote the mass of that portion of the Earth within a sphere of radius r , and δ denote the constant density of the Earth.

Additional Mathematical Models

- 25. Learning Theory** In the theory of learning, the rate at which a subject is memorized is assumed to be proportional to the amount that is left to be memorized. Suppose M denotes the total amount of a subject to be memorized and $A(t)$ is the amount memorized in time $t > 0$. Determine a differential equation for the amount $A(t)$.
- 26. Forgetfulness** In Problem 25 assume that the rate at which material is *forgotten* is proportional to the amount memorized in time $t > 0$. Determine a differential equation for the amount $A(t)$ when forgetfulness is taken into account.
- 27. Infusion of a Drug** A drug is infused into a patient's bloodstream at a constant rate of r grams per second. Simultaneously, the drug is removed at a rate proportional to the amount $x(t)$ of the drug present at time t . Determine a differential equation for the amount $x(t)$.
- 28. Tractrix** A motorboat starts at the origin and moves in the direction of the positive x -axis, pulling a waterskier along a curve C called a **tractrix**. See Figure 1.3.22. The waterskier, initially located on the y -axis at the point $(0, a)$, is pulled by a rope of constant length a that is kept taut throughout the motion. At time $t > 0$ the waterskier is at point $P(x, y)$. Assume that the rope is always tangent to C . Use the concept of slope to determine a differential equation for the path C of motion.

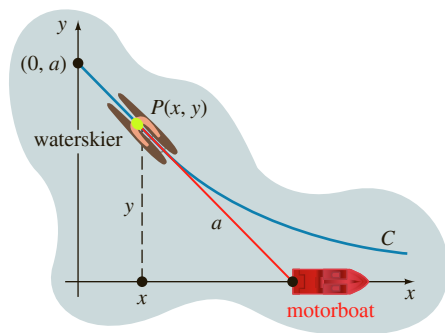


FIGURE 1.3.22 Waterskier in Problem 28

- 29. Reflecting Surface** Assume that when the plane curve C shown in Figure 1.3.23 is revolved about the x -axis, it generates

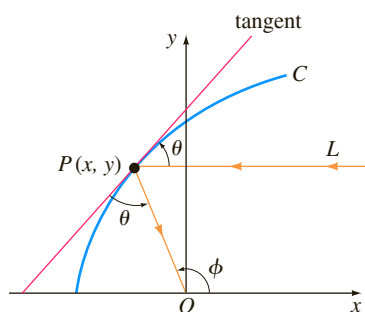


FIGURE 1.3.23 Reflecting surface in Problem 29

a surface of revolution with the property that all light rays L parallel to the x -axis striking the surface are reflected to a single point O (the origin). Use the fact that the angle of incidence is equal to the angle of reflection to determine a differential equation that describes the shape of the curve C . Such a curve C is important in applications ranging from construction of telescopes to satellite antennas, automobile headlights, and solar collectors. [Hint: Inspection of the figure shows that we can write $\phi = 2\theta$. Why? Now use an appropriate trigonometric identity.]

Discussion Problems

- 30.** Reread Problem 45 in Exercises 1.1 and then give an explicit solution $P(t)$ for equation (1). Find a one-parameter family of solutions of (1).
- 31.** Reread the sentence following equation (3) and assume that T_m is a positive constant. Discuss why we would expect $k < 0$ in (3) in both cases of cooling and warming. You might start by interpreting, say, $T(t) > T_m$ in a graphical manner.
- 32.** Reread the discussion leading up to equation (8). If we assume that initially the tank holds, say, 50 lb of salt, it stands to reason that because salt is being added to the tank continuously for $t > 0$, $A(t)$ should be an increasing function. Discuss how you might determine from the DE, without actually solving it, the number of pounds of salt in the tank after a long period of time.
- 33. Population Model** The differential equation $\frac{dP}{dt} = (k \cos t)P$, where k is a positive constant, is a model of human population $P(t)$ of a certain community. Discuss an interpretation for the solution of this equation. In other words, what kind of population do you think the differential equation describes?
- 34. Rotating Fluid** As shown in Figure 1.3.24(a), a right-circular cylinder partially filled with fluid is rotated with a constant angular velocity ω about a vertical y -axis through its center. The rotating fluid forms a surface of revolution S . To identify S , we first establish a coordinate system consisting of a vertical plane determined by the y -axis and an x -axis drawn perpendicular to the y -axis such that the point of intersection of the axes (the origin) is located at the lowest point on the surface S . We then seek a function $y = f(x)$ that represents the curve C of intersection of the surface S and the vertical coordinate plane. Let the point $P(x, y)$ denote the position of a particle of the rotating fluid of mass m in the coordinate plane. See Figure 1.3.24(b).

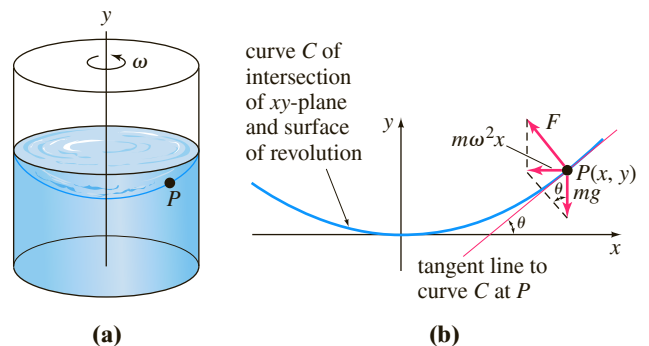


FIGURE 1.3.24 Rotating fluid in Problem 34

- (a) At P there is a reaction force of magnitude F due to the other particles of the fluid which is normal to the surface S . By Newton's second law the magnitude of the net force acting on the particle is $m\omega^2 x$. What is this force? Use Figure 1.3.24(b) to discuss the nature and origin of the equations

$$F \cos \theta = mg, \quad F \sin \theta = m\omega^2 x.$$

- (b) Use part (a) to find a first-order differential equation that defines the function $y = f(x)$.

- 35. Falling Body** In Problem 23, suppose $r = R + s$, where s is the distance from the surface of the Earth to the falling body. What does the differential equation obtained in Problem 23 become when s is very small in comparison to R ? [Hint: Think binomial series for

$$(R + s)^{-2} = R^{-2} (1 + s/R)^{-2}.]$$

- 36. Raindrops Keep Falling** In meteorology the term *virga* refers to falling raindrops or ice particles that evaporate before they reach the ground. Assume that a typical raindrop is spherical. Starting at some time, which we can designate as $t = 0$, the raindrop of radius r_0 falls from rest from a cloud and begins to evaporate.

- (a) If it is assumed that a raindrop evaporates in such a manner that its shape remains spherical, then it also makes sense to assume that the rate at which the raindrop evaporates—that is, the rate at which it loses mass—is proportional to its surface area. Show that this latter assumption implies that the rate at which the radius r of the raindrop decreases is a constant. Find $r(t)$. [Hint: See Problem 55 in Exercises 1.1.]

- (b) If the positive direction is downward, construct a mathematical model for the velocity v of the falling raindrop at time $t > 0$. Ignore air resistance. [Hint: Use the form of Newton's second law of motion given in (17).]

- 37. Let It Snow** The “snowplow problem” is a classic and appears in many differential equations texts, but it was probably made famous by Ralph Palmer Agnew:

One day it started snowing at a heavy and steady rate. A snowplow started out at noon, going 2 miles the first hour and 1 mile the second hour. What time did it start snowing?

Find the textbook *Differential Equations*, Ralph Palmer Agnew, McGraw-Hill Book Co., and then discuss the construction and solution of the mathematical model.

- 38. Population Dynamics** Suppose that $dP/dt = 0.15P(t)$ represents a mathematical model for the growth of a certain cell culture, where $P(t)$ is the size of the culture (measured in millions of cells) at time $t > 0$ (measured in hours). How fast is the culture growing at the time when the size of the culture reaches 2 million cells?
- 39. Radioactive Decay** Suppose that $dA/dt = -0.0004332 A(t)$ represents a mathematical model for the decay of radium-226, where $A(t)$ is the amount of radium (measured in grams) remaining at time $t > 0$ (measured in years). How much of the radium sample remains at the time when the sample is decaying at a rate of 0.002 grams per year?
- 40.** Reread this section and classify each mathematical model as linear or nonlinear.

Chapter 1 In Review

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1 and 2 fill in the blank and then write this result as a linear first-order differential equation that is free of the symbol c_1 and has the form $dy/dx = f(x, y)$. The symbol c_1 represents a constant.

1. $\frac{d}{dx} c_1 e^{10x} = \underline{\hspace{2cm}}$
2. $\frac{d}{dx} (5 + c_1 e^{-2x}) = \underline{\hspace{2cm}}$

In Problems 3 and 4 fill in the blank and then write this result as a linear second-order differential equation that is free of the symbols c_1 and c_2 and has the form $F(y, y'') = 0$. The symbols c_1 , c_2 , and k represent constants.

3. $\frac{d^2}{dx^2} (c_1 \cos kx + c_2 \sin kx) = \underline{\hspace{2cm}}$
4. $\frac{d^2}{dx^2} (c_1 \cosh kx + c_2 \sinh kx) = \underline{\hspace{2cm}}$

In Problems 5 and 6 compute y' and y'' and then combine these derivatives with y as a linear second-order differential equation that is free of the symbols c_1 and c_2 and has the form $F(y, y', y'') = 0$. The symbols c_1 and c_2 represent constants.

5. $y = c_1 e^x + c_2 x e^x$
6. $y = c_1 e^x \cos x + c_2 e^x \sin x$

In Problems 7–12 match each of the given differential equations with one or more of these solutions:

- | | | | |
|---------------------|---------------------|----------------|------------------|
| (a) $y = 0$, | (b) $y = 2$, | (c) $y = 2x$, | (d) $y = 2x^2$. |
| 7. $xy' = 2y$ | 8. $y' = 2$ | | |
| 9. $y' = 2y - 4$ | 10. $xy' = y$ | | |
| 11. $y'' + 9y = 18$ | 12. $xy'' - y' = 0$ | | |

In Problems 13 and 14 determine by inspection at least one solution of the given differential equation.

13. $y'' = y'$
14. $y' = y(y - 3)$

In Problems 15 and 16 interpret each statement as a differential equation.

15. On the graph of $y = \phi(x)$ the slope of the tangent line at a point $P(x, y)$ is the square of the distance from $P(x, y)$ to the origin.
16. On the graph of $y = \phi(x)$ the rate at which the slope changes with respect to x at a point $P(x, y)$ is the negative of the slope of the tangent line at $P(x, y)$.

17. (a) Give the domain of the function $y = x^{2/3}$.
 (b) Give the largest interval I of definition over which $y = x^{2/3}$ is a solution of the differential equation $3xy' - 2y = 0$.
18. (a) Verify that the one-parameter family $y^2 - 2y = x^2 - x + c$ is an implicit solution of the differential equation
 $(2y - 2)y' = 2x - 1$.
 (b) Find a member of the one-parameter family in part (a) that satisfies the initial condition $y(0) = 1$.
 (c) Use your result in part (b) to find an explicit function $y = \phi(x)$ that satisfies $y(0) = 1$. Give the domain of the function ϕ . Is $y = \phi(x)$ a solution of the initial-value problem? If so, give its interval I of definition; if not, explain.
19. The function $y = x - 2/x$ is a solution of the DE $xy' + y = 2x$. Find x_0 and the largest interval I for which $y(x)$ is a solution of the first-order IVP $xy' + y = 2x, y(x_0) = 1$.
20. Suppose that $y(x)$ denotes a solution of the first-order IVP $y' = x^2 + y^2, y(1) = -1$ and that $y(x)$ possesses at least a second derivative at $x = 1$. In some neighborhood of $x = 1$ use the DE to determine whether $y(x)$ is increasing or decreasing and whether the graph $y(x)$ is concave up or concave down.
21. A differential equation may possess more than one family of solutions.
 (a) Plot different members of the families $y = \phi_1(x) = x^2 + c_1$ and $y = \phi_2(x) = -x^2 + c_2$.
 (b) Verify that $y = \phi_1(x)$ and $y = \phi_2(x)$ are two solutions of the nonlinear first-order differential equation $(y')^2 = 4x^2$.
 (c) Construct a piecewise-defined function that is a solution of the nonlinear DE in part (b) but is not a member of either family of solutions in part (a).
22. What is the slope of the tangent line to the graph of a solution of $y' = 6\sqrt{y} + 5x^3$ that passes through $(-1, 4)$?

In Problems 23–26 verify that the indicated function is an explicit solution of the given differential equation. Give an interval of definition I for each solution.

23. $y'' + y = 2 \cos x - 2 \sin x$; $y = x \sin x + x \cos x$
 24. $y'' + y = \sec x$; $y = x \sin x + (\cos x) \ln(\cos x)$
 25. $x^2 y'' + xy' + y = 0$; $y = \sin(\ln x)$
 26. $x^2 y'' + xy' + y = \sec(\ln x)$;
 $y = \cos(\ln x) \ln(\cos(\ln x)) + (\ln x) \sin(\ln x)$

In Problems 27–30 use (12) of Section 1.1 to verify that the indicated function is a solution of the given differential equation. Assume an appropriate interval I of definition of each solution.

27. $\frac{dy}{dx} + (\sin x)y = x$; $y = e^{\cos x} \int_0^x t e^{-\cos t} dt$
 28. $\frac{dy}{dx} - 2xy = e^x$; $y = e^{x^2} \int_0^x e^{-t^2} dt$

29. $x^2 y'' + (x^2 - x)y' + (1 - x)y = 0$; $y = x \int_1^x \frac{e^{-t}}{t} dt$
 30. $y'' + y = e^{x^2}$; $y = \sin x \int_0^x e^{t^2} \cos t dt - \cos x \int_0^x e^{t^2} \sin t dt$

In Problems 31–34 verify that the indicated expression is an implicit solution of the given differential equation.

31. $x \frac{dy}{dx} + y = \frac{1}{y^2}$; $x^3 y^3 = x^3 + 1$
 32. $\left(\frac{dy}{dx}\right)^2 + 1 = \frac{1}{y^2}$; $(x - 5)^2 + y^2 = 1$
 33. $y'' = 2y(y')^3$; $y^3 + 3y = 1 - 3x$
 34. $(1 - xy)y' = y^2$; $y = e^{xy}$

In Problems 35–38, $y = c_1 e^{3x} + c_2 e^{-x} - 2x$ is a two-parameter family of the second-order DE $y'' - 2y' - 3y = 6x + 4$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

35. $y(0) = 0, y'(0) = 0$ 36. $y(0) = 1, y'(0) = -3$
 37. $y(1) = 4, y'(1) = -2$ 38. $y(-1) = 0, y'(-1) = 1$

39. The graph of a solution of a second-order initial-value problem $d^2y/dx^2 = f(x, y, y'), y(2) = y_0, y'(2) = y_1$, is given in Figure 1.R.1. Use the graph to estimate the values of y_0 and y_1 .

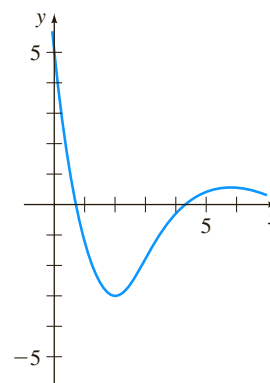


FIGURE 1.R.1 Graph for Problem 39

40. A tank in the form of a right-circular cylinder of radius 2 feet and height 10 feet is standing on end. If the tank is initially full of water and water leaks from a circular hole of radius $\frac{1}{2}$ inch at its bottom, determine a differential equation for the height h of the water at time $t > 0$. Ignore friction and contraction of water at the hole.

First-Order Differential Equations



Joggie Botma/Shutterstock.com

- 2.1 Solution Curves Without a Solution
- 2.2 Separable Equations
- 2.3 Linear Equations
- 2.4 Exact Equations
- 2.5 Solutions by Substitutions
- 2.6 A Numerical Method

CHAPTER 2 IN REVIEW

The history of mathematics is rife with stories of persons who devoted much of their lives to solving equations—algebraic equations at first and then eventually differential equations. In Sections 2.2–2.5 we will study some of the important analytical methods for solving first-order DEs. But before we start solving anything, you should be aware of two facts: A differential equation may have no solutions, and a differential equation may possess solutions yet there might not exist any analytical method for solving it. In Sections 2.1 and 2.6 we do not solve any DEs but show how to glean information about its solutions directly from the equation itself.

2.1

Solution Curves Without a Solution

INTRODUCTION Let us imagine for the moment that we have in front of us a first-order differential equation $dy/dx = f(x, y)$, and that we can neither find a solution nor invent a method for solving it analytically. This is not as bad a predicament as one might think, since the differential equation itself can be a fount of information about how its solutions “behave.”

We begin our study of first-order differential equations with two ways of analyzing a DE qualitatively. Both of these ways enable us to determine, in an approximate sense, what a solution curve must look like without actually solving the equation.

2.1.1 DIRECTION FIELDS

SOME FUNDAMENTAL QUESTIONS We saw in Section 1.2 that whenever $f(x, y)$ and $\partial f/\partial y$ satisfy certain continuity conditions, qualitative questions about existence and uniqueness of solutions can be answered. In this section we shall see that other qualitative questions about properties of solutions—How does a solution behave near a certain point? How does a solution behave as $x \rightarrow \infty$?—can often be answered when the function f depends solely on the variable y . We begin, however, with a simple concept from calculus:

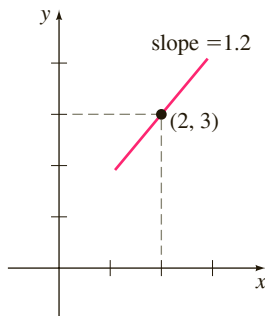
A derivative dy/dx of a differentiable function $y = y(x)$ gives slopes of tangent lines at points on its graph.

SLOPE Because a solution $y = y(x)$ of a first-order differential equation

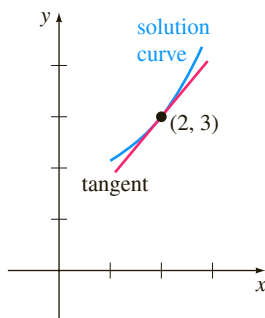
$$\frac{dy}{dx} = f(x, y) \quad (1)$$

is necessarily a differentiable function on its interval I of definition, it must also be continuous on I . Thus the corresponding solution curve on I must have no breaks and must possess a tangent line at each point $(x, y(x))$. The function f in the normal form (1) is called the **slope function** or **rate function**. The slope of the tangent line at $(x, y(x))$ on a solution curve is the value of the first derivative dy/dx at this point, and we know from (1) that this is the value of the slope function $f(x, y(x))$. Now suppose that (x, y) represents any point in a region of the xy -plane over which the function f is defined. The value $f(x, y)$ that the function f assigns to the point represents the slope of a line or, as we shall envision it, a line segment called a **lineal element**. For example, consider the equation $dy/dx = 0.2xy$, where $f(x, y) = 0.2xy$. At, say, the point $(2, 3)$ the slope of a lineal element is $f(2, 3) = 0.2(2)(3) = 1.2$. Figure 2.1.1(a) shows a line segment with slope 1.2 passing through $(2, 3)$. As shown in Figure 2.1.1(b), if a solution curve also passes through the point $(2, 3)$, it does so tangent to this line segment; in other words, the lineal element is a miniature tangent line at that point.

DIRECTION FIELD If we systematically evaluate f over a rectangular grid of points in the xy -plane and draw a line element at each point (x, y) of the grid with slope $f(x, y)$, then the collection of all these line elements is called a **direction field** or a **slope field** of the differential equation $dy/dx = f(x, y)$. Visually, the direction field suggests the appearance or shape of a family of solution curves of the differential equation, and consequently, it may be possible to see at a glance certain qualitative aspects of the solutions—regions in the plane, for example, in which a solution exhibits an unusual behavior. A single solution curve that passes through



(a) lineal element at a point



(b) lineal element is tangent to solution curve that passes through the point

FIGURE 2.1.1 A solution curve is tangent to lineal element at $(2, 3)$

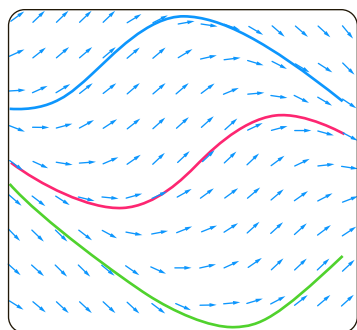
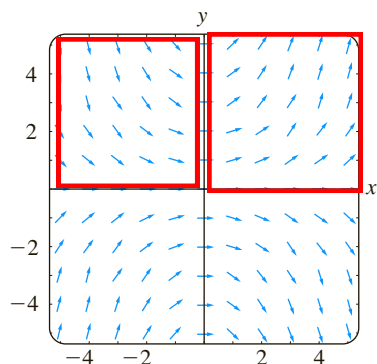
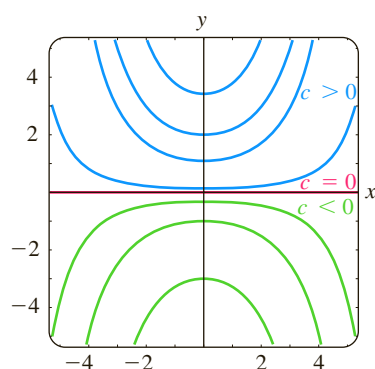


FIGURE 2.1.2 Solution curves following flow of a direction field



(a) direction field for $dy/dx = 0.2xy$



(b) some solution curves in the family $y = ce^{0.1x^2}$

FIGURE 2.1.3 Direction field and solution curves in Example 1

a direction field must follow the flow pattern of the field; it is tangent to a lineal element when it intersects a point in the grid. Figure 2.1.2 shows a computer-generated direction field of the differential equation $dy/dx = \sin(x + y)$ over a region of the xy -plane. Note how the three solution curves shown in color follow the flow of the field.

EXAMPLE 1 Direction Field

The direction field for the differential equation $dy/dx = 0.2xy$ shown in Figure 2.1.3(a) was obtained by using computer software in which a 5×5 grid of points (mh, nh) , m and n integers, was defined by letting $-5 \leq m \leq 5$, $-5 \leq n \leq 5$, and $h = 1$. Notice in Figure 2.1.3(a) that at any point along the x -axis ($y = 0$) and the y -axis ($x = 0$), the slopes are $f(x, 0) = 0$ and $f(0, y) = 0$, respectively, so the lineal elements are horizontal. Moreover, observe in the first quadrant that for a fixed value of x the values of $f(x, y) = 0.2xy$ increase as y increases; similarly, for a fixed y the values of $f(x, y) = 0.2xy$ increase as x increases. This means that as both x and y increase, the lineal elements almost become vertical and have positive slope ($f(x, y) = 0.2xy > 0$ for $x > 0, y > 0$). In the second quadrant, $|f(x, y)|$ increases as $|x|$ and y increase, so the lineal elements again become almost vertical but this time have negative slope ($f(x, y) = 0.2xy < 0$ for $x < 0, y > 0$). Reading from left to right, imagine a solution curve that starts at a point in the second quadrant, moves steeply downward, becomes flat as it passes through the y -axis, and then, as it enters the first quadrant, moves steeply upward—in other words, its shape would be concave upward and similar to a horseshoe. From this it could be surmised that $y \rightarrow \infty$ as $x \rightarrow \pm\infty$. Now in the third and fourth quadrants, since $f(x, y) = 0.2xy > 0$ and $f(x, y) = 0.2xy < 0$, respectively, the situation is reversed: A solution curve increases and then decreases as we move from left to right.

We saw in (1) of Section 1.1 that $y = e^{0.1x^2}$ is an explicit solution of the differential equation $dy/dx = 0.2xy$; you should verify that a one-parameter family of solutions of the same equation is given by $y = ce^{0.1x^2}$. For purposes of comparison with Figure 2.1.3(a) some representative graphs of members of this family are shown in Figure 2.1.3(b).

EXAMPLE 2 Direction Field

Use a direction field to sketch an approximate solution curve for the initial-value problem $dy/dx = \sin y$, $y(0) = -\frac{3}{2}$.

SOLUTION Before proceeding, recall that from the continuity of $f(x, y) = \sin y$ and $\partial f/\partial y = \cos y$, Theorem 1.2.1 guarantees the existence of a unique solution curve passing through any specified point (x_0, y_0) in the plane. Now we set our computer software again for a 5×5 rectangular region and specify (because of the initial condition) points in that region with vertical and horizontal separation of $\frac{1}{2}$ unit—that is, at points (mh, nh) , $h = \frac{1}{2}$, m and n integers such that $-10 \leq m \leq 10$, $-10 \leq n \leq 10$. The result is shown in Figure 2.1.4. Because the right-hand side of $dy/dx = \sin y$ is 0 at $y = 0$, and at $y = -\pi$, the lineal elements are horizontal at all points whose second coordinates are $y = 0$ or $y = -\pi$. It makes sense then that a solution curve passing through the initial point $(0, -\frac{3}{2})$ has the shape shown in the figure.

INCREASING/DECREASING Interpretation of the derivative dy/dx as a function that gives slope plays the key role in the construction of a direction field. Another telling property of the first derivative will be used next, namely, if $dy/dx > 0$ (or $dy/dx < 0$) for all x in an interval I , then a differentiable function $y = y(x)$ is increasing (or decreasing) on I .

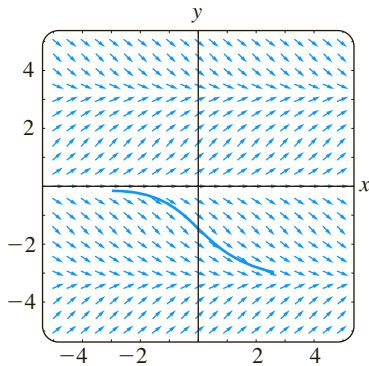


FIGURE 2.1.4 Direction field in
Example 2 on page 38

REMARKS

Sketching a direction field by hand is straightforward but time consuming; it is probably one of those tasks about which an argument can be made for doing it once or twice in a lifetime, but it is overall most efficiently carried out by means of computer software. Before calculators, PCs, and software the **method of isoclines** was used to facilitate sketching a direction field by hand. For the DE $dy/dx = f(x, y)$, any member of the family of curves $f(x, y) = c$, c a constant, is called an **isocline**. Lineal elements drawn through points on a specific isocline, say, $f(x, y) = c_1$ all have the same slope c_1 . In Problem 15 in Exercises 2.1 you have your two opportunities to sketch a direction field by hand.

2.1.2 AUTONOMOUS FIRST-ORDER DEs

AUTONOMOUS FIRST-ORDER DEs In Section 1.1 we divided the class of ordinary differential equations into two types: linear and nonlinear. We now consider briefly another kind of classification of ordinary differential equations, a classification that is of particular importance in the qualitative investigation of differential equations. An ordinary differential equation in which the independent variable does not appear explicitly is said to be **autonomous**. If the symbol x denotes the independent variable, then an autonomous first-order differential equation can be written as $f(y, y') = 0$ or in normal form as

$$\frac{dy}{dx} = f(y). \quad (2)$$

We shall assume throughout that the function f in (2) and its derivative f' are continuous functions of y on some interval I . The first-order equations

$$\begin{array}{ccc} \begin{array}{c} f(y) \\ \downarrow \end{array} & & \begin{array}{c} f(x, y) \\ \downarrow \end{array} \\ \frac{dy}{dx} = 1 + y^2 & \text{and} & \frac{dy}{dx} = 0.2xy \end{array}$$

are autonomous and nonautonomous, respectively.

Many differential equations encountered in applications or equations that are models of physical laws that do not change over time are autonomous. As we have already seen in Section 1.3, in an applied context, symbols other than y and x are routinely used to represent the dependent and independent variables. For example, if t represents time then inspection of

$$\frac{dA}{dt} = kA, \quad \frac{dx}{dt} = kx(n + 1 - x), \quad \frac{dT}{dt} = k(T - T_m), \quad \frac{dA}{dt} = 6 - \frac{1}{100}A,$$

where k , n , and T_m are constants, shows that each equation is time independent. Indeed, *all* of the first-order differential equations introduced in Section 1.3 are time independent and so are autonomous.

CRITICAL POINTS The zeros of the function f in (2) are of special importance. We say that a real number c is a **critical point** of the autonomous differential equation (2) if it is a zero of f —that is, $f(c) = 0$. A critical point is also called an **equilibrium point** or **stationary point**. Now observe that if we substitute the constant function $y(x) = c$ into (2), then both sides of the equation are zero. This means:

If c is a critical point of (2), then $y(x) = c$ is a constant solution of the autonomous differential equation.

A constant solution $y(x) = c$ of (2) is called an **equilibrium solution**; equilibria are the *only* constant solutions of (2).

As was already mentioned, we can tell when a nonconstant solution $y = y(x)$ of (2) is increasing or decreasing by determining the algebraic sign of the derivative dy/dx ; in the case of (2) we do this by identifying intervals on the y -axis over which the function $f(y)$ is positive or negative.

EXAMPLE 3 An Autonomous DE

The differential equation

$$\frac{dP}{dt} = P(a - bP),$$

where a and b are positive constants, has the normal form $dP/dt = f(P)$, which is (2) with t and P playing the parts of x and y , respectively, and hence is autonomous. From $f(P) = P(a - bP) = 0$ we see that 0 and a/b are critical points of the equation, so the equilibrium solutions are $P(t) = 0$ and $P(t) = a/b$. By putting the critical points on a vertical line, we divide the line into three intervals defined by $-\infty < P < 0$, $0 < P < a/b$, $a/b < P < \infty$. The arrows on the line shown in Figure 2.1.5 indicate the algebraic sign of $f(P) = P(a - bP)$ on these intervals and whether a nonconstant solution $P(t)$ is increasing or decreasing on an interval. The following table explains the figure.

Interval	Sign of $f(P)$	$P(t)$	Arrow
$(-\infty, 0)$	minus	decreasing	points down
$(0, a/b)$	plus	increasing	points up
$(a/b, \infty)$	minus	decreasing	points down

FIGURE 2.1.5 Phase portrait of the DE in Example 3

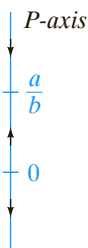
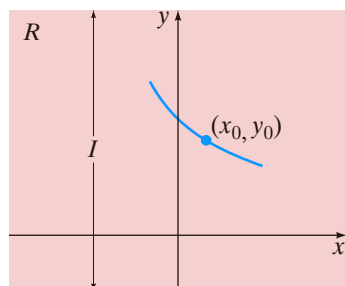


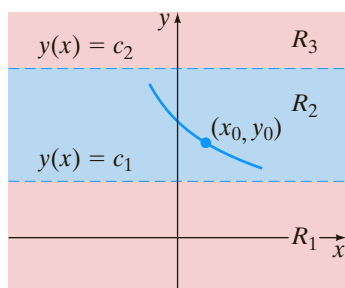
Figure 2.1.5 is called a **one-dimensional phase portrait**, or simply **phase portrait**, of the differential equation $dP/dt = P(a - bP)$. The vertical line is called a **phase line**.

SOLUTION CURVES Without solving an autonomous differential equation, we can usually say a great deal about its solution curves. Since the function f in (2) is independent of the variable x , we may consider f defined for $-\infty < x < \infty$ or for $0 \leq x < \infty$. Also, since f and its derivative f' are continuous functions of y on some interval I of the y -axis, the fundamental results of Theorem 1.2.1 hold in some horizontal strip or region R in the xy -plane corresponding to I , and so through any point (x_0, y_0) in R there passes only one solution curve of (2). See Figure 2.1.6(a). For the sake of discussion, let us suppose that (2) possesses exactly two critical points c_1 and c_2 and that $c_1 < c_2$. The graphs of the equilibrium solutions $y(x) = c_1$ and $y(x) = c_2$ are horizontal lines, and these lines partition the region R into three subregions R_1 , R_2 , and R_3 , as illustrated in Figure 2.1.6(b). Without proof here are some conclusions that we can draw about a nonconstant solution $y(x)$ of (2):

- If (x_0, y_0) is in a subregion R_i , $i = 1, 2, 3$, and $y(x)$ is a solution whose graph passes through this point, then $y(x)$ remains in the subregion R_i for all x . As illustrated in Figure 2.1.6(b), the solution $y(x)$ in R_2 is bounded below by c_1 and above by c_2 , that is, $c_1 < y(x) < c_2$ for all x . The solution curve stays within R_2 for all x because the graph of a nonconstant solution of (2) cannot cross the graph of either equilibrium solution $y(x) = c_1$ or $y(x) = c_2$. See Problem 33 in Exercises 2.1.
- By continuity of f we must then have either $f(y) > 0$ or $f(y) < 0$ for all x in a subregion R_i , $i = 1, 2, 3$. In other words, $f(y)$ cannot change signs in a subregion. See Problem 33 in Exercises 2.1.



(a) region R



(b) subregions R_1 , R_2 , and R_3 of R

FIGURE 2.1.6 Lines $y(x) = c_1$ and $y(x) = c_2$ partition R into three horizontal subregions

- Since $dy/dx = f(y(x))$ is either positive or negative in a subregion R_i , $i = 1, 2, 3$, a solution $y(x)$ is strictly monotonic—that is, $y(x)$ is either increasing or decreasing in the subregion R_i . Therefore $y(x)$ cannot be oscillatory, nor can it have a relative extremum (maximum or minimum). See Problem 33 in Exercises 2.1.
- If $y(x)$ is *bounded above* by a critical point c_1 (as in subregion R_1 where $y(x) < c_1$ for all x), then the graph of $y(x)$ must approach the graph of the equilibrium solution $y(x) = c_1$ either as $x \rightarrow \infty$ or as $x \rightarrow -\infty$. If $y(x)$ is *bounded*—that is, bounded above and below by two consecutive critical points (as in subregion R_2 where $c_1 < y(x) < c_2$ for all x)—then the graph of $y(x)$ must approach the graphs of the equilibrium solutions $y(x) = c_1$ and $y(x) = c_2$, one as $x \rightarrow \infty$ and the other as $x \rightarrow -\infty$. If $y(x)$ is *bounded below* by a critical point (as in subregion R_3 where $c_2 < y(x)$ for all x), then the graph of $y(x)$ must approach the graph of the equilibrium solution $y(x) = c_2$ either as $x \rightarrow \infty$ or as $x \rightarrow -\infty$. See Problem 34 in Exercises 2.1.

With the foregoing facts in mind, let us reexamine the differential equation in Example 3.

EXAMPLE 4 Example 3 Revisited

The three intervals determined on the P -axis or phase line by the critical points 0 and a/b now correspond in the tP -plane to three subregions defined by:

$$R_1: -\infty < P < 0, \quad R_2: 0 < P < a/b, \quad \text{and} \quad R_3: a/b < P < \infty,$$

where $-\infty < t < \infty$. The phase portrait in Figure 2.1.7 tells us that $P(t)$ is decreasing in R_1 , increasing in R_2 , and decreasing in R_3 . If $P(0) = P_0$ is an initial value, then in R_1 , R_2 , and R_3 we have, respectively, the following:

- For $P_0 < 0$, $P(t)$ is bounded above. Since $P(t)$ is decreasing, $P(t)$ decreases without bound for increasing t , and so $P(t) \rightarrow 0$ as $t \rightarrow -\infty$. This means that the negative t -axis, the graph of the equilibrium solution $P(t) = 0$, is a horizontal asymptote for a solution curve.
- For $0 < P_0 < a/b$, $P(t)$ is bounded. Since $P(t)$ is increasing, $P(t) \rightarrow a/b$ as $t \rightarrow \infty$ and $P(t) \rightarrow 0$ as $t \rightarrow -\infty$. The graphs of the two equilibrium solutions, $P(t) = 0$ and $P(t) = a/b$, are horizontal lines that are horizontal asymptotes for any solution curve starting in this subregion.
- For $P_0 > a/b$, $P(t)$ is bounded below. Since $P(t)$ is decreasing, $P(t) \rightarrow a/b$ as $t \rightarrow \infty$. The graph of the equilibrium solution $P(t) = a/b$ is a horizontal asymptote for a solution curve.

In Figure 2.1.7 the phase line is the P -axis in the tP -plane. For clarity the original phase line from Figure 2.1.5 is reproduced to the left of the plane in which the subregions R_1 , R_2 , and R_3 are shaded. The graphs of the equilibrium solutions $P(t) = a/b$ and $P(t) = 0$ (the t -axis) are shown in the figure as blue dashed lines; the solid graphs represent typical graphs of $P(t)$ illustrating the three cases just discussed. ■

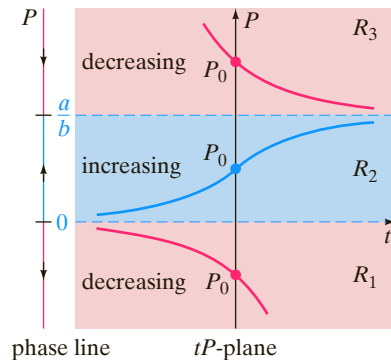


FIGURE 2.1.7 Phase portrait and solution curves in Example 4

In a subregion such as R_1 in Example 4, where $P(t)$ is decreasing and unbounded below, we must necessarily have $P(t) \rightarrow -\infty$. Do *not* interpret this last statement to mean $P(t) \rightarrow -\infty$ as $t \rightarrow \infty$; we could have $P(t) \rightarrow -\infty$ as $t \rightarrow T$, where $T > 0$ is a finite number that depends on the initial condition $P(t_0) = P_0$. Thinking in dynamic terms, $P(t)$ could “blow up” in finite time; thinking graphically, $P(t)$ could have a vertical asymptote at $t = T > 0$. A similar remark holds for the subregion R_3 .

The differential equation $dy/dx = \sin y$ in Example 2 is autonomous and has an infinite number of critical points, since $\sin y = 0$ at $y = n\pi$, n an integer. Moreover, we now know that because the solution $y(x)$ that passes through $(0, -\frac{\pi}{2})$ is bounded above and below by two consecutive critical points ($-\pi < y(x) < 0$) and is

decreasing ($\sin y < 0$ for $-\pi < y < 0$), the graph of $y(x)$ must approach the graphs of the equilibrium solutions as horizontal asymptotes: $y(x) \rightarrow -\pi$ as $x \rightarrow \infty$ and $y(x) \rightarrow 0$ as $x \rightarrow -\infty$.

EXAMPLE 5 Solution Curves of an Autonomous DE

The autonomous equation $dy/dx = (y - 1)^2$ possesses the single critical point 1. From the phase portrait in Figure 2.1.8(a) we conclude that a solution $y(x)$ is an increasing function in the subregions defined by $-\infty < y < 1$ and $1 < y < \infty$, where $-\infty < x < \infty$. For an initial condition $y(0) = y_0 < 1$, a solution $y(x)$ is increasing and bounded above by 1, and so $y(x) \rightarrow 1$ as $x \rightarrow \infty$; for $y(0) = y_0 > 1$ a solution $y(x)$ is increasing and unbounded.

Now $y(x) = 1 - 1/(x + c)$ is a one-parameter family of solutions of the differential equation. (See Problem 4 in Exercises 2.2.) A given initial condition determines a value for c . For the initial conditions, say, $y(0) = -1 < 1$ and $y(0) = 2 > 1$, we find, in turn, that $y(x) = 1 - 1/(x + \frac{1}{2})$, and $y(x) = 1 - 1/(x - 1)$. As shown in Figures 2.1.8(b) and 2.1.8(c), the graph of each of these rational functions possesses

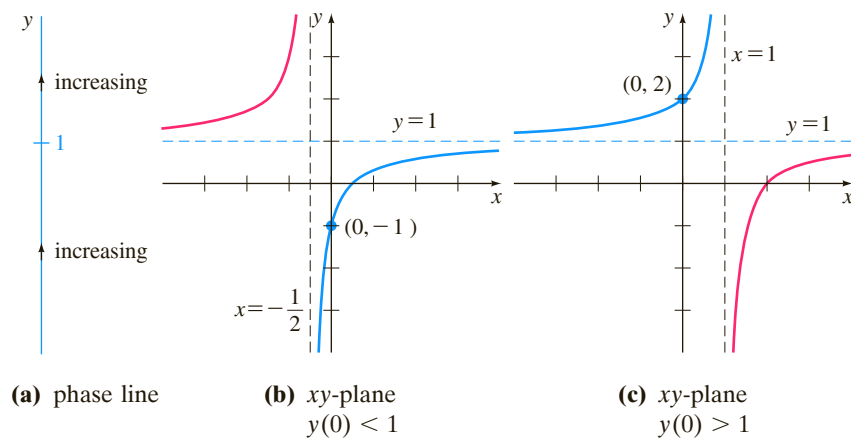


FIGURE 2.1.8 Behavior of solutions near $y = 1$ in Example 5

a vertical asymptote. But bear in mind that the solutions of the IVPs

$$\frac{dy}{dx} = (y - 1)^2, \quad y(0) = -1 \quad \text{and} \quad \frac{dy}{dx} = (y - 1)^2, \quad y(0) = 2$$

are defined on special intervals. The two solutions are, respectively,

$$y(x) = 1 - \frac{1}{x + \frac{1}{2}}, \quad -\frac{1}{2} < x < \infty \quad \text{and} \quad y(x) = 1 - \frac{1}{x - 1}, \quad -\infty < x < 1.$$

The solution curves are the portions of the graphs in Figures 2.1.8(b) and 2.1.8(c) shown in blue. As predicted by the phase portrait, for the solution curve in Figure 2.1.8(b), $y(x) \rightarrow 1$ as $x \rightarrow \infty$; for the solution curve in Figure 2.1.8(c), $y(x) \rightarrow \infty$ as $x \rightarrow 1$ from the left.

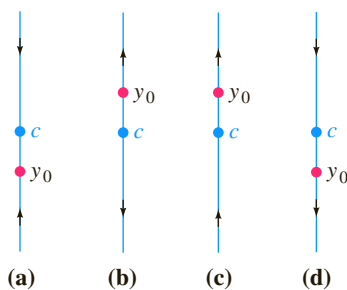


FIGURE 2.1.9 Critical point c is an attractor in (a), a repeller in (b), and semi-stable in (c) and (d).

ATTRACTORS AND REPELLERS Suppose that $y(x)$ is a nonconstant solution of the autonomous differential equation given in (1) and that c is a critical point of the DE. There are basically three types of behavior that $y(x)$ can exhibit near c . In Figure 2.1.9 we have placed c on four vertical phase lines. When both arrowheads on either side of the dot labeled c point toward c , as in Figure 2.1.9(a), all

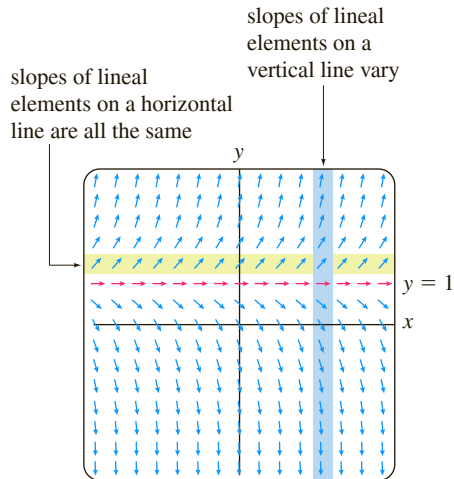


FIGURE 2.1.10 Direction field for an autonomous DE

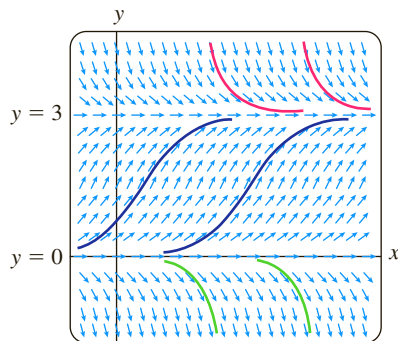


FIGURE 2.1.11 Translated solution curves of an autonomous DE

solutions $y(x)$ of (1) that start from an initial point (x_0, y_0) sufficiently near c exhibit the asymptotic behavior $\lim_{x \rightarrow \infty} y(x) = c$. For this reason the critical point c is said to be **asymptotically stable**. Using a physical analogy, a solution that starts near c is like a charged particle that, over time, is drawn to a particle of opposite charge, and so c is also referred to as an **attractor**. When both arrowheads on either side of the dot labeled c point *away* from c , as in Figure 2.1.9(b), all solutions $y(x)$ of (1) that start from an initial point (x_0, y_0) move away from c as x increases. In this case the critical point c is said to be **unstable**. An unstable critical point is also called a **repeller**, for obvious reasons. The critical point c illustrated in Figures 2.1.9(c) and 2.1.9(d) is neither an attractor nor a repeller. But since c exhibits characteristics of both an attractor and a repeller—that is, a solution starting from an initial point (x_0, y_0) sufficiently near c is attracted to c from one side and repelled from the other side—we say that the critical point c is **semi-stable**. In Example 3 the critical point a/b is asymptotically stable (an attractor) and the critical point 0 is unstable (a repeller). The critical point 1 in Example 5 is semi-stable.

AUTONOMOUS DES AND DIRECTION FIELDS If a first-order differential equation is autonomous, then we see from the right-hand side of its normal form $dy/dx = f(y)$ that slopes of lineal elements through points in the rectangular grid used to construct a direction field for the DE depend solely on the y -coordinate of the points. Put another way, lineal elements passing through points on any *horizontal* line must all have the same slope and therefore are parallel; slopes of lineal elements along any *vertical* line will, of course, vary. These facts are apparent from inspection of the horizontal yellow strip and vertical blue strip in Figure 2.1.10. The figure exhibits a direction field for the autonomous equation $dy/dx = 2(y - 1)$. The red lineal elements in Figure 2.1.10 have zero slope because they lie along the graph of the equilibrium solution $y = 1$.

TRANSLATION PROPERTY You may recall from precalculus mathematics that the graph of a function $y = f(x - k)$, where k is a constant, is the graph of $y = f(x)$ rigidly translated or shifted horizontally along the x -axis by an amount $|k|$; the translation is to the right if $k > 0$ and to the left if $k < 0$. It turns out that under the conditions stipulated for (2), solution curves of an autonomous first-order DE are related by the concept of translation. To see this, let's consider the differential equation $dy/dx = y(3 - y)$, which is a special case of the autonomous equation considered in Examples 3 and 4. Because $y = 0$ and $y = 3$ are equilibrium solutions of the DE, their graphs divide the xy -plane into three subregions R_1 , R_2 , and R_3 :

$$R_1: -\infty < y < 0, \quad R_2: 0 < y < 3, \quad \text{and} \quad R_3: 3 < y < \infty.$$

In Figure 2.1.11 we have superimposed on a direction field of the DE six solution curves. The figure illustrates that all solution curves of the same color, that is, solution curves lying within a particular subregion R_i , look alike. This is no coincidence but is a natural consequence of the fact that lineal elements passing through points on any horizontal line are parallel. That said, the following **translation property** of an autonomous DE should make sense:

If $y(x)$ is a solution of an autonomous differential equation $dy/dx = f(y)$, then $y_1(x) = y(x - k)$, k a constant, is also a solution.

Thus, if $y(x)$ is a solution of the initial-value problem $dy/dx = f(y)$, $y(0) = y_0$, then $y_1(x) = y(x - x_0)$ is a solution of the IVP $dy/dx = f(y)$, $y(x_0) = y_0$. For example, it is easy to verify that $y(x) = e^x$, $-\infty < x < \infty$, is a solution of the IVP $dy/dx = y$, $y(0) = 1$ and so a solution $y_1(x)$ of, say, $dy/dx = y$, $y(5) = 1$ is $y(x) = e^x$ translated 5 units to the right:

$$y_1(x) = y(x - 5) = e^{x-5}, \quad -\infty < x < \infty.$$

EXERCISES 2.1

Answers to selected odd-numbered problems begin on page ANS-1.

2.1.1 Direction Fields

In Problems 1–4 reproduce the given computer-generated direction field. Then sketch, by hand, an approximate solution curve that passes through each of the indicated points. Use different colored pencils for each solution curve.

1. $\frac{dy}{dx} = x^2 - y^2$

(a) $y(-2) = 1$

(b) $y(3) = 0$

(c) $y(0) = 2$

(d) $y(0) = 0$

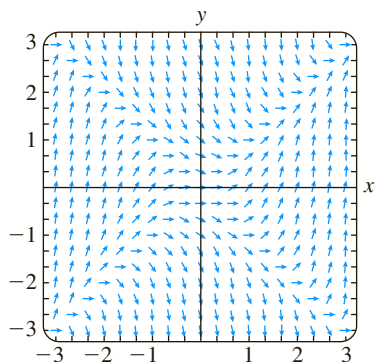


FIGURE 2.1.12 Direction field for Problem 1

2. $\frac{dy}{dx} = e^{-0.01xy^2}$

(a) $y(-6) = 0$

(b) $y(0) = 1$

(c) $y(0) = -4$

(d) $y(8) = -4$

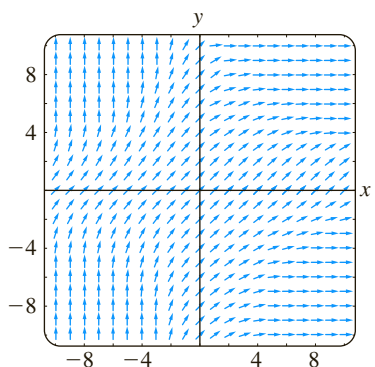


FIGURE 2.1.13 Direction field for Problem 2

3. $\frac{dy}{dx} = 1 - xy$

(a) $y(0) = 0$

(b) $y(-1) = 0$

(c) $y(2) = 2$

(d) $y(0) = -4$

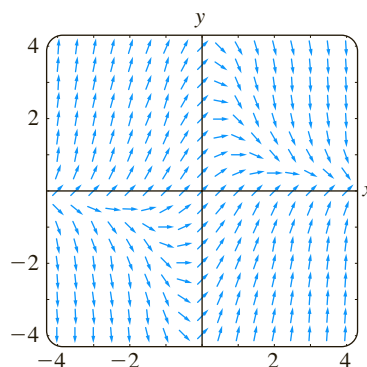


FIGURE 2.1.14 Direction field for Problem 3

4. $\frac{dy}{dx} = (\sin x) \cos y$

(a) $y(0) = 1$

(b) $y(1) = 0$

(c) $y(3) = 3$

(d) $y(0) = -\frac{5}{2}$

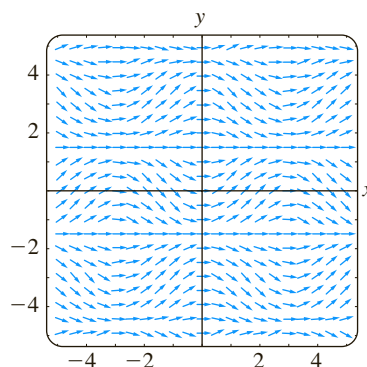


FIGURE 2.1.15 Direction field for Problem 4

In Problems 5–12 use computer software to obtain a direction field for the given differential equation. By hand, sketch an approximate solution curve passing through each of the given points.

5. $y' = x$

(a) $y(0) = 0$

(b) $y(0) = -3$

6. $y' = x + y$

(a) $y(-2) = 2$

(b) $y(1) = -3$

7. $y \frac{dy}{dx} = -x$

(a) $y(1) = 1$

(b) $y(0) = 4$

8. $\frac{dy}{dx} = \frac{1}{y}$

(a) $y(0) = 1$

(b) $y(-2) = -1$

9. $\frac{dy}{dx} = 0.2x^2 + y$

(a) $y(0) = \frac{1}{2}$

(b) $y(2) = -1$

10. $\frac{dy}{dx} = xe^y$

(a) $y(0) = -2$

(b) $y(1) = 2.5$

11. $y' = y - \cos \frac{\pi}{2}x$ 12. $\frac{dy}{dx} = 1 - \frac{y}{x}$
- (a) $y(2) = 2$ (a) $y(-\frac{1}{2}) = 2$
- (b) $y(-1) = 0$ (b) $y(\frac{3}{2}) = 0$

In Problems 13 and 14 the given figure represents the graph of $f(y)$ and $f(x)$, respectively. By hand, sketch a direction field over an appropriate grid for $dy/dx = f(y)$ (Problem 13) and then for $dy/dx = f(x)$ (Problem 14).

13.

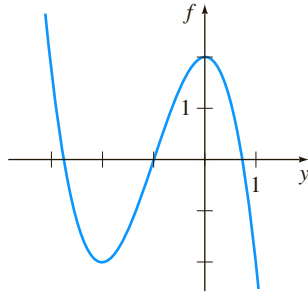


FIGURE 2.1.16 Graph for Problem 13

14.

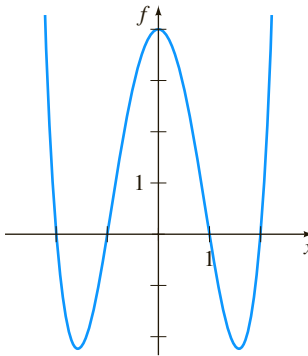


FIGURE 2.1.17 Graph for Problem 14

15. In parts (a) and (b) sketch **isoclines** $f(x, y) = c$ (see the *Remarks* on page 39) for the given differential equation using the indicated values of c . Construct a direction field by carefully drawing lineal elements with the appropriate slope at chosen points on each isocline. In each case, use this rough direction field to sketch an approximate solution curve for the IVP consisting of the DE and the initial condition $y(0) = 1$.
- (a) $dy/dx = x + y$; c an integer satisfying $-5 \leq c \leq 5$
- (b) $dy/dx = x^2 + y^2$; $c = \frac{1}{4}, c = 1, c = \frac{9}{4}, c = 4$

Discussion Problems

16. (a) Consider the direction field of the differential equation $dy/dx = x(y - 4)^2 - 2$, but do not use technology to obtain it. Describe the slopes of the lineal elements on the lines $x = 0$, $y = 3$, $y = 4$, and $y = 5$.
- (b) Consider the IVP $dy/dx = x(y - 4)^2 - 2$, $y(0) = y_0$, where $y_0 < 4$. Can a solution $y(x) \rightarrow \infty$ as $x \rightarrow \infty$? Based on the information in part (a), discuss.

17. For a first-order DE $dy/dx = f(x, y)$ a curve in the plane defined by $f(x, y) = 0$ is called a **nullcline** of the equation, since a lineal element at a point on the curve has zero slope. Use computer software to obtain a direction field over a rectangular grid of points for $dy/dx = x^2 - 2y$, and then superimpose the graph of the nullcline $y = \frac{1}{2}x^2$ over the direction field. Discuss the behavior of solution curves in regions of the plane defined by $y < \frac{1}{2}x^2$ and by $y > \frac{1}{2}x^2$. Sketch some approximate solution curves. Try to generalize your observations.

18. (a) Identify the nullclines (see Problem 17) in Problems 1, 3, and 4. With a colored pencil, circle any lineal elements in Figures 2.1.12, 2.1.14, and 2.1.15 that you think may be a lineal element at a point on a nullcline.
- (b) What are the nullclines of an autonomous first-order DE?

2.1.2 Autonomous First-Order DEs

19. Consider the autonomous first-order differential equation $dy/dx = y - y^3$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when y_0 has the given values.
- (a) $y_0 > 1$ (b) $0 < y_0 < 1$
- (c) $-1 < y_0 < 0$ (d) $y_0 < -1$
20. Consider the autonomous first-order differential equation $dy/dx = y^2 - y^4$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when y_0 has the given values.
- (a) $y_0 > 1$ (b) $0 < y_0 < 1$
- (c) $-1 < y_0 < 0$ (d) $y_0 < -1$

In Problems 21–28 find the critical points and phase portrait of the given autonomous first-order differential equation. Classify each critical point as asymptotically stable, unstable, or semi-stable. By hand, sketch typical solution curves in the regions in the xy -plane determined by the graphs of the equilibrium solutions.

21. $\frac{dy}{dx} = y^2 - 3y$ 22. $\frac{dy}{dx} = y^2 - y^3$
23. $\frac{dy}{dx} = (y - 2)^4$ 24. $\frac{dy}{dx} = 10 + 3y - y^2$
25. $\frac{dy}{dx} = y^2(4 - y^2)$ 26. $\frac{dy}{dx} = y(2 - y)(4 - y)$
27. $\frac{dy}{dx} = y \ln(y + 2)$ 28. $\frac{dy}{dx} = \frac{ye^y - 9y}{e^y}$

In Problems 29 and 30 consider the autonomous differential equation $dy/dx = f(y)$, where the graph of f is given. Use the graph to locate the critical points of each differential equation. Sketch a phase portrait of each differential equation. By hand, sketch typical solution curves in the subregions in the xy -plane determined by the graphs of the equilibrium solutions.

29.

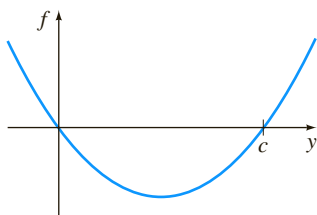


FIGURE 2.1.18 Graph for Problem 29

30.

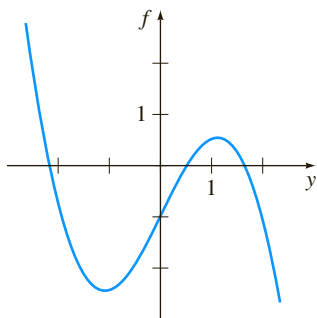


FIGURE 2.1.19 Graph for Problem 30

Discussion Problems

31. Consider the autonomous DE $dy/dx = (2/\pi)y - \sin y$. Determine the critical points of the equation. Discuss a way of obtaining a phase portrait of the equation. Classify the critical points as asymptotically stable, unstable, or semi-stable.
32. A critical point c of an autonomous first-order DE is said to be **isolated** if there exists some open interval that contains c but no other critical point. Can there exist an autonomous DE of the form given in (2) for which *every* critical point is nonisolated? Discuss; do not think profound thoughts.
33. Suppose that $y(x)$ is a nonconstant solution of the autonomous equation $dy/dx = f(y)$ and that c is a critical point of the DE. Discuss: Why can't the graph of $y(x)$ cross the graph of the equilibrium solution $y = c$? Why can't $f(y)$ change signs in one of the subregions discussed on page 40? Why can't $y(x)$ be oscillatory or have a relative extremum (maximum or minimum)?
34. Suppose that $y(x)$ is a solution of the autonomous equation $dy/dx = f(y)$ and is bounded above and below by two consecutive critical points $c_1 < c_2$, as in subregion R_2 of Figure 2.1.6(b). If $f(y) > 0$ in the region, then $\lim_{x \rightarrow \infty} y(x) = c_2$. Discuss why there cannot exist a number $L < c_2$ such that $\lim_{x \rightarrow \infty} y(x) = L$. As part of your discussion, consider what happens to $y'(x)$ as $x \rightarrow \infty$.
35. Using the autonomous equation (2), discuss how it is possible to obtain information about the location of points of inflection of a solution curve.
36. Consider the autonomous DE $dy/dx = y^2 - y - 6$. Use your ideas from Problem 35 to find intervals on the y -axis for which solution curves are concave up and intervals for which solution

curves are concave down. Discuss why *each* solution curve of an initial-value problem of the form $dy/dx = y^2 - y - 6$, $y(0) = y_0$, where $-2 < y_0 < 3$, has a point of inflection with the same y -coordinate. What is that y -coordinate? Carefully sketch the solution curve for which $y(0) = -1$. Repeat for $y(0) = 2$.

37. Suppose the autonomous DE in (2) has no critical points. Discuss the behavior of the solutions.

Mathematical Models

38. **Population Model** The differential equation in Example 3 is a well-known population model. Suppose the DE is changed to

$$\frac{dP}{dt} = P(aP - b),$$

where a and b are positive constants. Discuss what happens to the population P as time t increases.

39. **Population Model** Another population model is given by

$$\frac{dP}{dt} = kP - h,$$

where h and k are positive constants. For what initial values $P(0) = P_0$ does this model predict that the population will go extinct?

40. **Terminal Velocity** In Section 1.3 we saw that the autonomous differential equation

$$m \frac{dv}{dt} = mg - kv,$$

where k is a positive constant and g is the acceleration due to gravity, is a model for the velocity v of a body of mass m that is falling under the influence of gravity. Because the term $-kv$ represents air resistance, the velocity of a body falling from a great height does not increase without bound as time t increases. Use a phase portrait of the differential equation to find the limiting, or terminal, velocity of the body. Explain your reasoning.

41. Suppose the model in Problem 40 is modified so that air resistance is proportional to v^2 , that is,

$$m \frac{dv}{dt} = mg - kv^2.$$

See Problem 17 in Exercises 1.3. Use a phase portrait to find the terminal velocity of the body. Explain your reasoning.

42. **Chemical Reactions** When certain kinds of chemicals are combined, the rate at which the new compound is formed is modeled by the autonomous differential equation

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X),$$

where $k > 0$ is a constant of proportionality and $\beta > \alpha > 0$. Here $X(t)$ denotes the number of grams of the new compound formed in time t .

- (a) Use a phase portrait of the differential equation to predict the behavior of $X(t)$ as $t \rightarrow \infty$.
- (b) Consider the case when $\alpha = \beta$. Use a phase portrait of the differential equation to predict the behavior of $X(t)$ as $t \rightarrow \infty$ when $X(0) < \alpha$. When $X(0) > \alpha$.

- (c) Verify that an explicit solution of the DE in the case when $k = 1$ and $\alpha = \beta$ is $X(t) = \alpha - 1/(t + c)$. Find a solution that satisfies $X(0) = \alpha/2$. Then find a solution that satisfies $X(0) = 2\alpha$. Graph these two solutions. Does the behavior of the solutions as $t \rightarrow \infty$ agree with your answers to part (b)?

2.2 Separable Equations

INTRODUCTION We begin our study of solution methods with the simplest of all differential equations: first-order equations with separable variables. Because the method discussed in this section and many other methods for solving differential equations involve integration, you are urged to refresh your memory on important formulas (such as $\int du/u$) and techniques (such as integration by parts) by consulting a calculus text.

SOLUTION BY INTEGRATION Consider the first-order differential equation $dy/dx = f(x, y)$. When f does not depend on the variable y , that is, $f(x, y) = g(x)$, the differential equation

$$\frac{dy}{dx} = g(x) \quad (1)$$

can be solved by integration. If $g(x)$ is a continuous function, then integrating both sides of (1) gives $y = \int g(x) dx = G(x) + c$, where $G(x)$ is an antiderivative (indefinite integral) of $g(x)$. For example, if $dy/dx = 1 + e^{2x}$, then its solution is $y = \int (1 + e^{2x}) dx$ or $y = x + \frac{1}{2}e^{2x} + c$.

A DEFINITION Equation (1), as well as its method of solution, is just a special case when the function f in the normal form $dy/dx = f(x, y)$ can be factored into a function of x times a function of y .

DEFINITION 2.2.1 Separable Equation

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be **separable** or to have **separable variables**.

For example, the equations

$$\frac{dy}{dx} = y^2 x e^{3x+4y} \quad \text{and} \quad \frac{dy}{dx} = y + \sin x$$

are separable and nonseparable, respectively. In the first equation we can factor $f(x, y) = y^2 x e^{3x+4y}$ as

$$f(x, y) = y^2 x e^{3x+4y} = \overset{g(x)}{\underset{\downarrow}{(x e^{3x})}} \overset{h(y)}{\underset{\downarrow}{(y^2 e^{4y})}},$$

but in the second equation there is no way of expressing $y + \sin x$ as a product of a function of x times a function of y .

Observe that by dividing by the function $h(y)$, we can write a separable equation $dy/dx = g(x)h(y)$ as

$$p(y) \frac{dy}{dx} = g(x), \quad (2)$$

where, for convenience, we have denoted $1/h(y)$ by $p(y)$. From this last form we can see immediately that (2) reduces to (1) when $h(y) = 1$.

Now if $y = \phi(x)$ represents a solution of (2), we must have $p(\phi(x))\phi'(x) = g(x)$, and therefore

$$\int p(\phi(x))\phi'(x) dx = \int g(x) dx. \quad (3)$$

But $dy = \phi'(x) dx$, and so (3) is the same as

$$\int p(y) dy = \int g(x) dx \quad \text{or} \quad H(y) = G(x) + c, \quad (4)$$

where $H(y)$ and $G(x)$ are antiderivatives of $p(y) = 1/h(y)$ and $g(x)$, respectively.

METHOD OF SOLUTION Equation (4) indicates the procedure for solving separable equations. A one-parameter family of solutions, usually given implicitly, is obtained by integrating both sides of $p(y) dy = g(x) dx$.

NOTE There is no need to use two constants in the integration of a separable equation, because if we write $H(y) + c_1 = G(x) + c_2$, then the difference $c_2 - c_1$ can be replaced by a single constant c , as in (4). In many instances throughout the chapters that follow, we will relabel constants in a manner convenient to a given equation. For example, multiples of constants or combinations of constants can sometimes be replaced by a single constant.

EXAMPLE 1 Solving a Separable DE

Solve $(1 + x) dy - y dx = 0$.

SOLUTION Dividing by $(1 + x)y$, we can write $dy/y = dx/(1 + x)$, from which it follows that

$$\begin{aligned} \int \frac{dy}{y} &= \int \frac{dx}{1+x} \\ \ln|y| &= \ln|1+x| + c_1 \\ |y| &= e^{\ln|1+x|+c_1} = e^{\ln|1+x|} \cdot e^{c_1} && \leftarrow \text{laws of exponents} \\ &= |1+x| e^{c_1} \\ y &= \pm e^{c_1}(1+x). && \leftarrow \begin{cases} |1+x| = 1+x, & x \geq -1 \\ |1+x| = -(1+x), & x < -1 \end{cases} \end{aligned}$$

Relabeling $\pm e^{c_1}$ as c then gives $y = c(1+x)$. ■

In the solution of Example 1 because each integral results in a logarithm, a judicious choice for the constant of integration is $\ln|c|$ rather than c . Rewriting the second line of the solution as $\ln|y| = \ln|1 + x| + \ln|c|$ enables us to combine the terms on the right-hand side by the properties of logarithms. From $\ln|y| = \ln|c(1 + x)|$ we immediately get $y = c(1 + x)$. Even if the indefinite integrals are not *all* logarithms, it may still be advantageous to use $\ln|c|$. However, no firm rule can be given.

In Section 1.1 we saw that a solution curve may be only a segment or an arc of the graph of an implicit solution $G(x, y) = 0$.

EXAMPLE 2 Solution Curve

Solve the initial-value problem $\frac{dy}{dx} = -\frac{x}{y}$, $y(4) = -3$.

SOLUTION Rewriting the equation as $y \, dy = -x \, dx$, we get

$$\int y \, dy = -\int x \, dx \quad \text{and} \quad \frac{y^2}{2} = -\frac{x^2}{2} + c_1.$$

We can write the result of the integration as $x^2 + y^2 = c^2$ by replacing the constant $2c_1$ by c^2 . This solution of the differential equation represents a one-parameter family of concentric circles centered at the origin.

Now when $x = 4$, $y = -3$, so $16 + 9 = 25 = c^2$. Thus the initial-value problem determines the circle $x^2 + y^2 = 25$ with radius 5. Because of its simplicity we can solve this implicit solution for an explicit solution that satisfies the initial condition. We saw this solution as $y = \phi_2(x)$ or $y = -\sqrt{25 - x^2}$, $-5 < x < 5$ in Example 7 of Section 1.1. A solution curve is the graph of a differentiable function. In this case the solution curve is the lower semicircle, shown in dark blue in Figure 2.2.1 containing the point $(4, -3)$.

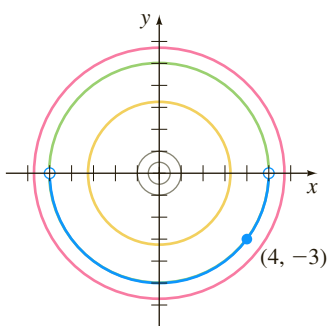


FIGURE 2.2.1 Solution curve for the IVP in Example 2

LOSING A SOLUTION Some care should be exercised in separating variables, since the variable divisors could be zero at a point. Specifically, if r is a zero of the function $h(y)$, then substituting $y = r$ into $dy/dx = g(x)h(y)$ makes both sides zero; in other words, $y = r$ is a constant solution of the differential equation.

But after variables are separated, the left-hand side of $\frac{dy}{h(y)} = g(x) \, dx$ is undefined at r .

As a consequence, $y = r$ might not show up in the family of solutions that are obtained after integration and simplification. Recall that such a solution is called a singular solution.

EXAMPLE 3 Losing a Solution

Solve $\frac{dy}{dx} = y^2 - 4$.

SOLUTION We put the equation in the form

$$\frac{dy}{y^2 - 4} = dx \quad \text{or} \quad \left[\frac{\frac{1}{4}}{y - 2} - \frac{\frac{1}{4}}{y + 2} \right] dy = dx. \quad (5)$$

The second equation in (5) is the result of using partial fractions on the left-hand side of the first equation. Integrating and using the laws of logarithms gives

$$\begin{aligned} \frac{1}{4} \ln|y - 2| - \frac{1}{4} \ln|y + 2| &= x + c_1 \\ \ln \left| \frac{y - 2}{y + 2} \right| &= 4x + c_2 \quad \text{or} \quad \frac{y - 2}{y + 2} = \pm e^{4x + c_2}. \end{aligned}$$

Here we have replaced $4c_1$ by c_2 . Finally, after replacing $\pm e^{c_2}$ by c and solving the last equation for y , we get the one-parameter family of solutions

$$y = 2 \frac{1 + ce^{4x}}{1 - ce^{4x}}. \quad (6)$$

Now if we factor the right-hand side of the differential equation as $dy/dx = (y - 2)(y + 2)$, we know from the discussion of critical points in Section 2.1 that $y = 2$ and $y = -2$ are two constant (equilibrium) solutions. The solution $y = 2$ is a member of the family of solutions defined by (6) corresponding to the value $c = 0$. However, $y = -2$ is a singular solution; it cannot be obtained from (6) for any choice of the parameter c . This latter solution was lost early on in the solution process. Inspection of (5) clearly indicates that we must preclude $y = \pm 2$ in these steps. ■

EXAMPLE 4 An Initial-Value Problem

Solve $(e^{2y} - y) \cos x \frac{dy}{dx} = e^y \sin 2x$, $y(0) = 0$.

SOLUTION Dividing the equation by $e^y \cos x$ gives

$$\frac{e^{2y} - y}{e^y} dy = \frac{\sin 2x}{\cos x} dx.$$

Before integrating, we use termwise division on the left-hand side and the trigonometric identity $\sin 2x = 2 \sin x \cos x$ on the right-hand side. Then

$$\text{integration by parts} \rightarrow \int (e^y - ye^{-y}) dy = 2 \int \sin x dx$$

$$\text{yields} \quad e^y + ye^{-y} + e^{-y} = -2 \cos x + c. \quad (7)$$

The initial condition $y = 0$ when $x = 0$ implies $c = 4$. Thus a solution of the initial-value problem is

$$e^y + ye^{-y} + e^{-y} = 4 - 2 \cos x. \quad (8) \quad \blacksquare$$

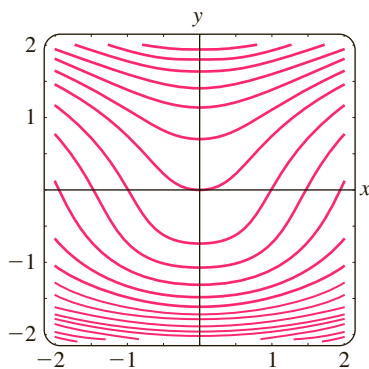


FIGURE 2.2.2 Level curves of $G(x, y) = e^y + ye^{-y} + e^{-y} + 2 \cos x$

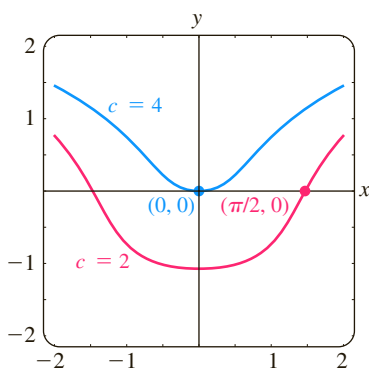


FIGURE 2.2.3 Level curves $c = 2$ and $c = 4$

USE OF COMPUTERS The *Remarks* at the end of Section 1.1 mentioned that it may be difficult to use an implicit solution $G(x, y) = 0$ to find an explicit solution $y = \phi(x)$. Equation (8) shows that the task of solving for y in terms of x may present more problems than just the drudgery of symbol pushing—sometimes it simply cannot be done! Implicit solutions such as (8) are somewhat frustrating; neither the graph of the equation nor an interval over which a solution satisfying $y(0) = 0$ is defined is apparent. The problem of “seeing” what an implicit solution looks like can be overcome in some cases by means of technology. One way* of proceeding is to use the contour plot application of a computer algebra system (CAS). Recall from multivariate calculus that for a function of two variables $z = G(x, y)$ the *two-dimensional* curves defined by $G(x, y) = c$, where c is constant, are called the *level curves* of the function. With the aid of a CAS, some of the level curves of the function $G(x, y) = e^y + ye^{-y} + e^{-y} + 2 \cos x$ have been reproduced in Figure 2.2.2. The family of solutions defined by (7) is the level curves $G(x, y) = c$. Figure 2.2.3 illustrates the level curve $G(x, y) = 4$, which is the particular solution (8), in blue color. The other curve in Figure 2.2.3 is the level curve $G(x, y) = 2$, which is the member of the family $G(x, y) = c$ that satisfies $y(\pi/2) = 0$.

*In Section 2.6 we will discuss several other ways of proceeding that are based on the concept of a numerical solver.

If an initial condition leads to a particular solution by yielding a specific value of the parameter c in a family of solutions for a first-order differential equation, there is a natural inclination for most students (and instructors) to relax and be content. However, a solution of an initial-value problem might not be unique. We saw in Example 4 of Section 1.2 that the initial-value problem

$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0 \quad (9)$$

has at least two solutions, $y = 0$ and $y = \frac{1}{16}x^4$. We are now in a position to solve the equation. Separating variables and integrating $y^{-1/2} dy = x dx$ gives $2y^{1/2} = \frac{1}{2}x^2 + c_1$. Solving for y and replacing $\frac{1}{2}c_1$ by c yields

$$y = \left(\frac{1}{4}x^2 + c \right)^2. \quad (10)$$

Each of the functions in the family given in (10) is a solution of the DE defined on the interval $(-\infty, \infty)$ provided we take $c \geq 0$. See Problem 52 in Exercises 2.2. Now when we substitute $x = 0, y = 0$ in (10) we see that $c = 0$. Therefore $y = \frac{1}{16}x^4$ is a solution of the IVP. The trivial solution $y = 0$ was lost by dividing by $y^{1/2}$. The initial-value problem (9) actually possesses many more solutions, since for any choice of the parameter $a \geq 0$ the piecewise-defined function

$$y = \begin{cases} 0, & x < a \\ \frac{1}{16}(x^2 - a^2)^2, & x \geq a \end{cases}$$

satisfies both the differential equation and the initial condition. See Figure 2.2.4.

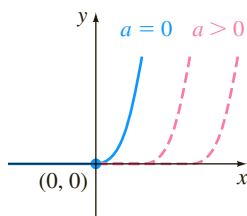


FIGURE 2.2.4 Piecewise-defined solutions of (9)

AN INTEGRAL-DEFINED FUNCTION In (ii) of the *Remarks* at the end of Section 1.1 it was pointed out that a solution method for a certain kind of differential equation may lead to an integral-defined function. This is especially true for separable differential equations because integration is the method of solution. For example, if g is continuous on some interval I containing x_0 and x , then a solution of the simple initial-value problem $dy/dx = g(x)$, $y(x_0) = y_0$, defined on I , is given by

$$y(x) = y_0 + \int_{x_0}^x g(t) dt.$$

To see this, we have immediately from (12) of Section 1.1 that $dy/dx = g(x)$ and $y(x_0) = y_0$ because $\int_{x_0}^{x_0} g(t) dt = 0$. When $\int g(t) dt$ is nonelementary—that is, cannot be expressed in terms of elementary functions—the form $y(x) = y_0 + \int_{x_0}^x g(t) dt$ may be the best we can do in obtaining an explicit solution of an IVP. The next example illustrates this idea.

EXAMPLE 5 An Initial-Value Problem

Solve $\frac{dy}{dx} = e^{-x^2}$, $y(3) = 5$.

SOLUTION The function $g(x) = e^{-x^2}$ is continuous on $(-\infty, \infty)$, but its antiderivative is not an elementary function. Using t as dummy variable of integration, we can write

$$\begin{aligned} \int_3^x \frac{dy}{dt} dt &= \int_3^x e^{-t^2} dt \\ y(t) \Big|_3^x &= \int_3^x e^{-t^2} dt \\ y(x) - y(3) &= \int_3^x e^{-t^2} dt \\ y(x) &= y(3) + \int_3^x e^{-t^2} dt. \end{aligned}$$

Using the initial condition $y(3) = 5$, we obtain the solution

$$y(x) = 5 + \int_3^x e^{-t^2} dt.$$

The procedure demonstrated in Example 5 works equally well on separable equations $dy/dx = g(x)f(y)$ where, say, $f(y)$ possesses an elementary antiderivative but $g(x)$ does not possess an elementary antiderivative. See Problems 29 and 30 in Exercises 2.2.

REMARKS

In some of the preceding examples we saw that the constant in the one-parameter family of solutions for a first-order differential equation can be relabeled when convenient. Also, it can easily happen that two individuals solving the same equation correctly arrive at dissimilar expressions for their answers. For example, by separation of variables we can show that one-parameter families of solutions for the DE $(1 + y^2) dx + (1 + x^2) dy = 0$ are

$$\arctan x + \arctan y = c \quad \text{or} \quad \frac{x + y}{1 - xy} = c.$$

As you work your way through the next several sections, bear in mind that families of solutions may be equivalent in the sense that one family may be obtained from another by either relabeling the constant or applying algebra and trigonometry. See Problems 27 and 28 in Exercises 2.2.

EXERCISES 2.2

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1–22 solve the given differential equation by separation of variables.

1. $\frac{dy}{dx} = \sin 5x$
2. $\frac{dy}{dx} = (x + 1)^2$
3. $dx + e^{3x} dy = 0$
4. $dy - (y - 1)^2 dx = 0$
5. $x \frac{dy}{dx} = 4y$
6. $\frac{dy}{dx} + 2xy^2 = 0$
7. $\frac{dy}{dx} = e^{3x+2y}$
8. $e^{xy} \frac{dy}{dx} = e^{-y} + e^{-2x-y}$
9. $y \ln x \frac{dx}{dy} = \left(\frac{y+1}{x}\right)^2$
10. $\frac{dy}{dx} = \left(\frac{2y+3}{4x+5}\right)^2$
11. $\csc y dx + \sec^2 x dy = 0$
12. $\sin 3x dx + 2y \cos^3 3x dy = 0$
13. $(e^y + 1)^2 e^{-y} dx + (e^x + 1)^3 e^{-x} dy = 0$
14. $x(1 + y^2)^{1/2} dx = y(1 + x^2)^{1/2} dy$
15. $\frac{dS}{dr} = kS$
16. $\frac{dQ}{dt} = k(Q - 70)$
17. $\frac{dP}{dt} = P - P^2$
18. $\frac{dN}{dt} + N = Nte^{t+2}$
19. $\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$
20. $\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3}$

$$21. \frac{dy}{dx} = x\sqrt{1 - y^2} \qquad 22. (e^x + e^{-x}) \frac{dy}{dx} = y^2$$

In Problems 23–28 find an explicit solution of the given initial-value problem.

23. $\frac{dx}{dt} = 4(x^2 + 1), \quad x(\pi/4) = 1$
24. $\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1}, \quad y(2) = 2$
25. $x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$
26. $\frac{dy}{dt} + 2y = 1, \quad y(0) = \frac{5}{2}$
27. $\sqrt{1 - y^2} dx - \sqrt{1 - x^2} dy = 0, \quad y(0) = \frac{\sqrt{3}}{2}$
28. $(1 + x^4) dy + x(1 + 4y^2) dx = 0, \quad y(1) = 0$

In Problems 29 and 30 proceed as in Example 5 and find an explicit solution of the given initial-value problem.

29. $\frac{dy}{dx} = ye^{-x^2}, \quad y(4) = 1$
30. $\frac{dy}{dx} = y^2 \sin x^2, \quad y(-2) = \frac{1}{3}$

In Problems 31–34 find an explicit solution of the given initial-value problem. Determine the exact interval I of definition by analytical methods. Use a graphing utility to plot the graph of the solution.

31. $\frac{dy}{dx} = \frac{2x+1}{2y}$, $y(-2) = -1$

32. $(2y-2)\frac{dy}{dx} = 3x^2 + 4x + 2$, $y(1) = -2$

33. $e^y dx - e^{-x} dy = 0$, $y(0) = 0$

34. $\sin x dx + y dy = 0$, $y(0) = 1$

35. (a) Find a solution of the initial-value problem consisting of the differential equation in Example 3 and each of the initial conditions: $y(0) = 2$, $y(0) = -2$, and $y(\frac{1}{4}) = 1$.

(b) Find the solution of the differential equation in Example 3 when $\ln c_1$ is used as the constant of integration on the left-hand side in the solution and $4 \ln c_1$ is replaced by $\ln c$. Then solve the same initial-value problems in part (a).

36. Find a solution of $x \frac{dy}{dx} = y^2 - y$ that passes through the indicated points.

(a) $(0, 1)$ (b) $(0, 0)$ (c) $(\frac{1}{2}, \frac{1}{2})$ (d) $(2, \frac{1}{4})$

37. Find a singular solution of Problem 21. Of Problem 22.

38. Show that an implicit solution of

$$2x \sin^2 y dx - (x^2 + 10) \cos y dy = 0$$

is given by $\ln(x^2 + 10) + \csc y = c$. Find the constant solutions, if any, that were lost in the solution of the differential equation.

Often a radical change in the form of the solution of a differential equation corresponds to a very small change in either the initial condition or the equation itself. In Problems 39–42 find an explicit solution of the given initial-value problem. Use a graphing utility to plot the graph of each solution. Compare each solution curve in a neighborhood of $(0, 1)$.

39. $\frac{dy}{dx} = (y-1)^2$, $y(0) = 1$

40. $\frac{dy}{dx} = (y-1)^2$, $y(0) = 1.01$

41. $\frac{dy}{dx} = (y-1)^2 + 0.01$, $y(0) = 1$

42. $\frac{dy}{dx} = (y-1)^2 - 0.01$, $y(0) = 1$

43. Every autonomous first-order equation $dy/dx = f(y)$ is separable. Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the differential equation $dy/dx = y - y^3$ that satisfy, in turn, the initial conditions $y_1(0) = 2$, $y_2(0) = \frac{1}{2}$, $y_3(0) = -\frac{1}{2}$, and $y_4(0) = -2$. Use a graphing utility to plot the graphs of each solution. Compare these graphs with those predicted in Problem 19 of Exercises 2.1. Give the exact interval of definition for each solution.

44. (a) The autonomous first-order differential equation $dy/dx = 1/(y-3)$ has no critical points. Nevertheless, place 3 on the phase line and obtain a phase portrait of the equation. Compute d^2y/dx^2 to determine where solution curves are concave up and where they are concave down

(see Problems 35 and 36 in Exercises 2.1). Use the phase portrait and concavity to sketch, by hand, some typical solution curves.

(b) Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the differential equation in part (a) that satisfy, in turn, the initial conditions $y_1(0) = 4$, $y_2(0) = 2$, $y_3(1) = 2$, and $y_4(-1) = 4$. Graph each solution and compare with your sketches in part (a). Give the exact interval of definition for each solution.

In Problems 45–50 use a technique of integration or a substitution to find an explicit solution of the given differential equation or initial-value problem.

45. $\frac{dy}{dx} = \frac{1}{1 + \sin x}$

46. $\frac{dy}{dx} = \frac{\sin \sqrt{x}}{\sqrt{y}}$

47. $(\sqrt{x} + x) \frac{dy}{dx} = \sqrt{y} + y$

48. $\frac{dy}{dx} = y^{2/3} - y$

49. $\frac{dy}{dx} = \frac{e^{\sqrt{x}}}{y}$, $y(1) = 4$

50. $\frac{dy}{dx} = \frac{x \tan^{-1} x}{y}$, $y(0) = 3$

Discussion Problems

51. (a) Explain why the interval of definition of the explicit solution $y = \phi_2(x)$ of the initial-value problem in Example 2 is the open interval $(-5, 5)$.

(b) Can any solution of the differential equation cross the x -axis? Do you think that $x^2 + y^2 = 1$ is an implicit solution of the initial-value problem $dy/dx = -x/y$, $y(1) = 0$?

52. On page 51 we showed that a one-parameter family of solutions of the first-order differential equation $dy/dx = xy^{1/2}$ is $y = (\frac{1}{4}x^4 + c)^2$ for $c \geq 0$. Each solution in this family is defined on $(-\infty, \infty)$. The last statement is not true if we choose c to be negative. For $c = -1$, explain why $y = (\frac{1}{4}x^4 - 1)^2$ is not a solution of the DE on the interval $(-\infty, \infty)$. Find an interval of definition I on which $y = (\frac{1}{4}x^4 - 1)^2$ is a solution of the DE.

53. In Problems 43 and 44 we saw that every autonomous first-order differential equation $dy/dx = f(y)$ is separable. Does this fact help in the solution of the initial-value problem

$\frac{dy}{dx} = \sqrt{1 + y^2} \sin^2 y$, $y(0) = \frac{1}{2}$? Discuss. Sketch, by hand, a plausible solution curve of the problem.

54. (a) Solve the two initial-value problems:

$$\frac{dy}{dx} = y, \quad y(0) = 1$$

and

$$\frac{dy}{dx} = y + \frac{y}{x \ln x}, \quad y(e) = 1.$$

(b) Show that there are more than 1.65 million digits in the y -coordinate of the point of intersection of the two solution curves in part (a).

55. Find a function whose square plus the square of its derivative is 1.
56. (a) The differential equation in Problem 27 is equivalent to the normal form

$$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

in the square region in the xy -plane defined by $|x| < 1$, $|y| < 1$. But the quantity under the radical is nonnegative also in the regions defined by $|x| > 1$, $|y| > 1$. Sketch all regions in the xy -plane for which this differential equation possesses real solutions.

- (b) Solve the DE in part (a) in the regions defined by $|x| > 1$, $|y| > 1$. Then find an implicit and an explicit solution of the differential equation subject to $y(2) = 2$.

Mathematical Model

57. **Suspension Bridge** In (16) of Section 1.3 we saw that a mathematical model for the shape of a flexible cable strung between two vertical supports is

$$\frac{dy}{dx} = \frac{W}{T_1}, \quad (11)$$

where W denotes the portion of the total vertical load between the points P_1 and P_2 shown in Figure 1.3.7. The DE (11) is separable under the following conditions that describe a suspension bridge.

Let us assume that the x - and y -axes are as shown in Figure 2.2.5—that is, the x -axis runs along the horizontal roadbed, and the y -axis passes through $(0, a)$, which is the lowest point on one cable over the span of the bridge, coinciding with the interval $[-L/2, L/2]$. In the case of a suspension bridge, the usual assumption is that the vertical load in (11) is only a uniform roadbed distributed along the horizontal axis. In other words, it is assumed that the weight of all cables is negligible in comparison to the weight of the roadbed and that the weight per unit length of the roadbed (say, pounds per horizontal foot) is a constant p . Use this information to set up and solve an appropriate initial-value problem from which the shape (a curve with equation $y = \phi(x)$) of each of the two cables in a suspension bridge is determined. Express your solution of the IVP in terms of the sag h and span L . See Figure 2.2.5.

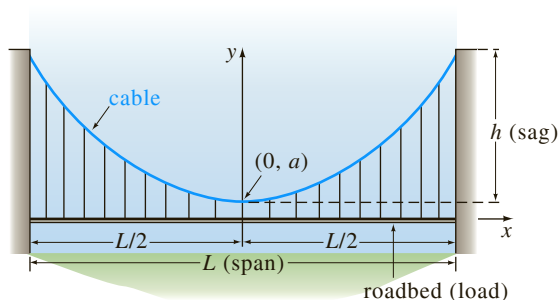


FIGURE 2.2.5 Shape of a cable in Problem 57

Computer Lab Assignments

58. (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation $\frac{dy}{dx} = -\frac{8x+5}{3y^2+1}$. Experiment with different numbers of level curves as well as various rectangular regions defined by $a \leq x \leq b$, $c \leq y \leq d$.
- (b) On separate coordinate axes plot the graphs of the particular solutions corresponding to the initial conditions: $y(0) = -1$; $y(0) = 2$; $y(-1) = 4$; $y(-1) = -3$.
59. (a) Find an implicit solution of the IVP
- $$(2y + 2) dy - (4x^3 + 6x) dx = 0, \quad y(0) = -3.$$
- (b) Use part (a) to find an explicit solution $y = \phi(x)$ of the IVP.
- (c) Consider your answer to part (b) as a *function* only. Use a graphing utility or a CAS to graph this function, and then use the graph to estimate its domain.
- (d) With the aid of a root-finding application of a CAS, determine the approximate largest interval I of definition of the solution $y = \phi(x)$ in part (b). Use a graphing utility or a CAS to graph the solution curve for the IVP on this interval.
60. (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation

$$\frac{dy}{dx} = \frac{x(1-x)}{y(-2+y)}.$$

Experiment with different numbers of level curves as well as various rectangular regions in the xy -plane until your result resembles Figure 2.2.6.

- (b) On separate coordinate axes, plot the graph of the implicit solution corresponding to the initial condition $y(0) = \frac{3}{2}$. Use a colored pencil to mark off that segment of the graph that corresponds to the solution curve of a solution ϕ that satisfies the initial condition. With the aid of a root-finding application of a CAS, determine the approximate largest interval I of definition of the solution ϕ . [Hint: First find the points on the curve in part (a) where the tangent is vertical.]
- (c) Repeat part (b) for the initial condition $y(0) = -2$.

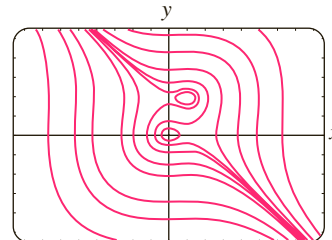


FIGURE 2.2.6 Level curves in Problem 60

2.3 Linear Equations

INTRODUCTION We continue our quest for solutions of first-order differential equations by next examining linear equations. Linear differential equations are an especially “friendly” family of differential equations, in that, given a linear equation, whether first order or a higher-order kin, there is always a good possibility that we can find some sort of solution of the equation that we can examine.

A DEFINITION The form of a linear first-order DE was first given in (7) of Section 1.1. This form, in the case when $n = 1$ in (6) of that section, is reproduced here for convenience.

DEFINITION 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \quad (1)$$

is said to be a **linear equation** in the variable y .

STANDARD FORM By dividing both sides of (1) by the lead coefficient $a_1(x)$, we obtain a more useful form, the **standard form**, of a linear equation:

$$\frac{dy}{dx} + P(x)y = f(x). \quad (2)$$

We seek a solution of (2) on an interval I for which both coefficient functions P and f are continuous.

Before we examine a general procedure for solving equations of form (2) we note that in some instances (2) can be solved by separation of variables. For example, you should verify that the equations

We match each equation with (2). 
In the first equation $P(x) = 2x$,
 $f(x) = 0$ and in the second
 $P(x) = -1$, $f(x) = 5$.

$$\frac{dy}{dx} + 2xy = 0 \quad \text{and} \quad \frac{dy}{dx} = y + 5$$

are both linear and separable, but that the linear equation

$$\frac{dy}{dx} + y = x$$

is not separable.

METHOD OF SOLUTION The method for solving (2) hinges on a remarkable fact that the *left-hand side* of the equation can be recast into the form of the exact derivative of a product by multiplying the both sides of (2) by a special function $\mu(x)$. It is relatively easy to find the function $\mu(x)$ because we want

$$\frac{d}{dx} [\underbrace{\mu(x)y}_{\text{product}}] = \underbrace{\mu \frac{dy}{dx} + \frac{d\mu}{dx} y}_{\text{product rule}} = \underbrace{\mu \frac{dy}{dx} + \mu P y}_{\substack{\text{left-hand side of} \\ \text{(2) multiplied by } \mu(x)}}.$$

↑
these must be equal

The equality is true provided that

$$\frac{d\mu}{dx} = \mu P.$$

The last equation can be solved by separation of variables. Integrating

$$\frac{d\mu}{\mu} = P dx \quad \text{and solving} \quad \ln|\mu(x)| = \int P(x) dx + c_1$$

See Problem 55
in Exercises 2.3



gives $\mu(x) = c_2 e^{\int P(x) dx}$. Even though there are infinite choices of $\mu(x)$ (all constant multiples of $e^{\int P(x) dx}$), all produce the same desired result. Hence we can simplify life and choose $c_2 = 1$. The function

$$\mu(x) = e^{\int P(x) dx} \quad (3)$$

is called an **integrating factor** for equation (2).

Here is what we have so far: We multiplied both sides of (2) by (3) and, by construction, the left-hand side is the derivative of a product of the integrating factor and y :

$$e^{\int P(x) dx} \frac{dy}{dx} + P(x) e^{\int P(x) dx} y = e^{\int P(x) dx} f(x)$$

$$\frac{d}{dx} [e^{\int P(x) dx} y] = e^{\int P(x) dx} f(x).$$

Finally, we discover why (3) is called an *integrating* factor. We can integrate both sides of the last equation,

$$e^{\int P(x) dx} y = \int e^{\int P(x) dx} f(x) dx + c$$

and solve for y . The result is a one-parameter family of solutions of (2):

$$y = e^{-\int P(x) dx} \int e^{\int P(x) dx} f(x) dx + c e^{-\int P(x) dx}. \quad (4)$$

We emphasize that you should **not memorize** formula (4). The following procedure should be worked through each time.

Solving a Linear First-Order Equation

- (i) Remember to put a linear equation into the standard form (2).
- (ii) From the standard form of the equation identify $P(x)$ and then find the integrating factor $e^{\int P(x) dx}$. No constant need be used in evaluating the indefinite integral $\int P(x) dx$.
- (iii) Multiply the both sides of the standard form equation by the integrating factor. The left-hand side of the resulting equation is automatically the derivative of the product of the integrating factor $e^{\int P(x) dx}$ and y :

$$\frac{d}{dx} [e^{\int P(x) dx} y] = e^{\int P(x) dx} f(x). \quad (5)$$

- (iv) Integrate both sides of the last equation and solve for y .

EXAMPLE 1 Solving a Linear Equation

Solve $\frac{dy}{dx} - 3y = 0$.

SOLUTION This linear equation can be solved by separation of variables. Alternatively, since the differential equation is already in standard form (2), we identify

$P(x) = -3$, and so the integrating factor is $e^{\int(-3)dx} = e^{-3x}$. We then multiply the given equation by this factor and recognize that

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = e^{-3x} \cdot 0 \quad \text{is the same as} \quad \frac{d}{dx} [e^{-3x} y] = 0.$$

Integration of the last equation,

$$\int \frac{d}{dx} [e^{-3x} y] dx = \int 0 dx$$

then yields $e^{-3x} y = c$ or $y = ce^{3x}$, $-\infty < x < \infty$. ■

EXAMPLE 2 Solving a Linear Equation

Solve $\frac{dy}{dx} - 3y = 6$.

SOLUTION This linear equation, like the one in Example 1, is already in standard form with $P(x) = -3$. Thus the integrating factor is again e^{-3x} . This time multiplying the given equation by this factor gives

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = 6e^{-3x} \quad \text{and so} \quad \frac{d}{dx} [e^{-3x} y] = 6e^{-3x}.$$

Integrating the last equation,

$$\int \frac{d}{dx} [e^{-3x} y] dx = 6 \int e^{-3x} dx \quad \text{gives} \quad e^{-3x} y = -6 \left(\frac{e^{-3x}}{3} \right) + c,$$

or $y = -2 + ce^{3x}$, $-\infty < x < \infty$. ■

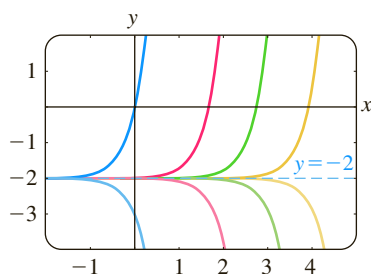


FIGURE 2.3.1 Solution curves of DE in Example 2

When a_1 , a_0 , and g in (1) are constants, the differential equation is autonomous. In Example 2 you can verify from the normal form $dy/dx = 3(y + 2)$ that -2 is a critical point and that it is unstable (a repeller). Thus a solution curve with an initial point either above or below the graph of the equilibrium solution $y = -2$ pushes away from this horizontal line as x increases. Figure 2.3.1, obtained with the aid of a graphing utility, shows the graph of $y = -2$ along with some additional solution curves.

GENERAL SOLUTION Suppose again that the functions P and f in (2) are continuous on a common interval I . In the steps leading to (4) we showed that if (2) has a solution on I , then it must be of the form given in (4). Conversely, it is a straightforward exercise in differentiation to verify that any function of the form given in (4) is a solution of the differential equation (2) on I . In other words, (4) is a one-parameter family of solutions of equation (2) and *every solution of (2) defined on I is a member of this family*. Therefore we call (4) the **general solution** of the differential equation on the interval I . (See the *Remarks* at the end of Section 1.1.) Now by writing (2) in the normal form $y' = F(x, y)$, we can identify $F(x, y) = -P(x)y + f(x)$ and $\partial F/\partial y = -P(x)$. From the continuity of P and f on the interval I we see that F and $\partial F/\partial y$ are also continuous on I . With Theorem 1.2.1 as our justification, we conclude that there exists one and only one solution of the first-order initial-value problem

$$\frac{dy}{dx} + P(x)y = f(x), \quad y(x_0) = y_0 \quad (6)$$

defined on *some* interval I_0 containing x_0 . But when x_0 is in I , finding a solution of (6) is just a matter of finding an appropriate value of c in (4)—that is, to each x_0 in I there

corresponds a distinct c . In other words, the interval I_0 of existence and uniqueness in Theorem 1.2.1 for the initial-value problem (6) is the entire interval I .

EXAMPLE 3 General Solution

Solve $x \frac{dy}{dx} - 4y = x^6 e^x$.

SOLUTION Dividing by x , the standard form of the given DE is

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x. \quad (7)$$

From this form we identify $P(x) = -4/x$ and $f(x) = x^5 e^x$ and further observe that P and f are continuous on $(0, \infty)$. Hence the integrating factor is

we can use $\ln x$ instead of $\ln |x|$ since $x > 0$

$$e^{-4 \int dx/x} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4}.$$

Here we have used the basic identity $b^{\log_b N} = N$, $N > 0$. Now we multiply (7) by x^{-4} and rewrite

$$x^{-4} \frac{dy}{dx} - 4x^{-5}y = xe^x \quad \text{as} \quad \frac{d}{dx}[x^{-4}y] = xe^x.$$

It follows from integration by parts that the general solution defined on the interval $(0, \infty)$ is $x^{-4}y = xe^x - e^x + c$ or $y = x^5 e^x - x^4 e^x + cx^4$. ■

In case you are wondering why the interval $(0, \infty)$ is important in Example 3, read this paragraph and the paragraph following Example 4.



Except in the case in which the lead coefficient is 1, the recasting of equation (1) into the standard form (2) requires division by $a_1(x)$. Values of x for which $a_1(x) = 0$ are called **singular points** of the equation. Singular points are potentially troublesome. Specifically, in (2), if $P(x)$ (formed by dividing $a_0(x)$ by $a_1(x)$) is discontinuous at a point, the discontinuity may carry over to solutions of the differential equation.

EXAMPLE 4 General Solution

Find the general solution of $(x^2 - 9) \frac{dy}{dx} + xy = 0$.

SOLUTION We write the differential equation in standard form

$$\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0 \quad (8)$$

and identify $P(x) = x/(x^2 - 9)$. Although P is continuous on $(-\infty, -3)$, $(-3, 3)$, and $(3, \infty)$, we shall solve the equation on the first and third intervals. On these intervals the integrating factor is

$$e^{\int x dx/(x^2-9)} = e^{1/2 \int 2x dx/(x^2-9)} = e^{1/2 \ln|x^2-9|} = \sqrt{x^2 - 9}.$$

After multiplying the standard form (8) by this factor, we get

$$\frac{d}{dx} \left[\sqrt{x^2 - 9} y \right] = 0.$$

Integrating both sides of the last equation gives $\sqrt{x^2 - 9} y = c$. Thus on either

$(-\infty, -3)$ or $(3, \infty)$ the general solution of the equation is $y = \frac{c}{\sqrt{x^2 - 9}}$. ■

Notice in Example 4 that $x = 3$ and $x = -3$ are singular points of the equation and that every function in the general solution $y = c/\sqrt{x^2 - 9}$ is discontinuous at these points. On the other hand, $x = 0$ is a singular point of the differential equation in Example 3, but the general solution $y = x^5e^x - x^4e^x + cx^4$ is noteworthy in that every function in this one-parameter family is continuous at $x = 0$ and is defined on the interval $(-\infty, \infty)$ and not just on $(0, \infty)$, as stated in the solution. However, the family $y = x^5e^x - x^4e^x + cx^4$ defined on $(-\infty, \infty)$ cannot be considered the general solution of the DE, since the singular point $x = 0$ still causes a problem. See Problems 50 and 51 in Exercises 2.3.

EXAMPLE 5 An Initial-Value Problem

Solve $\frac{dy}{dx} + y = x$, $y(0) = 4$.

SOLUTION The equation is in standard form, and $P(x) = 1$ and $f(x) = x$ are continuous on $(-\infty, \infty)$. The integrating factor is $e^{\int dx} = e^x$, so integrating

$$\frac{d}{dx}[e^xy] = xe^x$$

gives $e^xy = xe^x - e^x + c$. Solving this last equation for y yields the general solution $y = x - 1 + ce^{-x}$. But from the initial condition we know that $y = 4$ when $x = 0$. Substituting these values into the general solution implies that $c = 5$. Hence the solution of the problem on the interval $(-\infty, \infty)$ is

$$y = x - 1 + 5e^{-x}. \quad (9) \quad \blacksquare$$

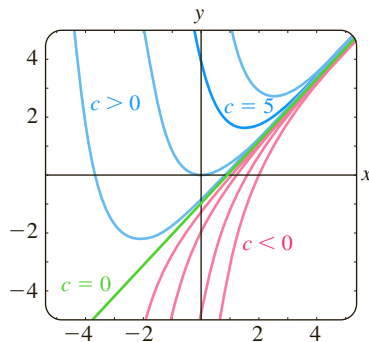


FIGURE 2.3.2 Solution curves of DE in Example 5

Figure 2.3.2, obtained with the aid of a graphing utility, shows the graph of the solution (9) in dark blue along with the graphs of other members of the one-parameter family of solutions $y = x - 1 + ce^{-x}$. It is interesting to observe that as x increases, the graphs of *all* members of this family are close to the graph of the solution $y = x - 1$. The last solution corresponds to $c = 0$ in the family and is shown in dark green in Figure 2.3.2. This asymptotic behavior of solutions is due to the fact that the contribution of ce^{-x} , $c \neq 0$, becomes negligible for increasing values of x . We say that ce^{-x} is a **transient term**, since $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. While this behavior is not characteristic of all general solutions of linear equations (see Example 2), the notion of a transient is often important in applied problems.

PIECEWISE-LINEAR DIFFERENTIAL EQUATION In the construction of mathematical models (especially in the biological sciences and engineering) it can happen that one or more coefficients in a differential equation is a piecewise-defined function. In particular, when either $P(x)$ or $f(x)$ in (2) is a piecewise-defined function the equation is then referred to as a **piecewise-linear differential equation**. In the next example, $f(x)$ is piecewise continuous on the interval $[0, \infty)$ with a single jump discontinuity at $x = 1$. The basic idea is to solve the initial-value problem in two parts corresponding to the two intervals over which $f(x)$ is defined; each part consists of a linear equation solvable by the method of this section. As we will see, it is then possible to piece the two solutions together at $x = 1$ so that $y(x)$ is continuous on $[0, \infty)$. See Problems 37–42 in Exercises 2.3.

EXAMPLE 6 An Initial-Value Problem

Solve $\frac{dy}{dx} + y = f(x)$, $y(0) = 0$ where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1. \end{cases}$

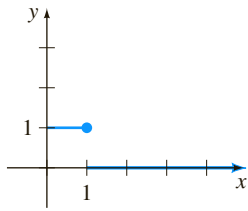


FIGURE 2.3.3 Discontinuous $f(x)$ in Example 6

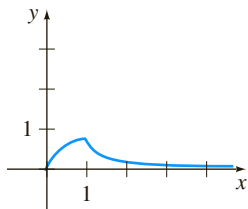


FIGURE 2.3.4 Graph of (10) in Example 6

SOLUTION The graph of the discontinuous function f is shown in Figure 2.3.3. We solve the DE for $y(x)$ first on the interval $[0, 1]$ and then on the interval $(1, \infty)$. For $0 \leq x \leq 1$ we have

$$\frac{dy}{dx} + y = 1 \quad \text{or, equivalently,} \quad \frac{d}{dx}[e^x y] = e^x.$$

Integrating this last equation and solving for y gives $y = 1 + c_1 e^{-x}$. Since $y(0) = 0$, we must have $c_1 = -1$, and therefore $y = 1 - e^{-x}$, $0 \leq x \leq 1$. Then for $x > 1$ the equation

$$\frac{dy}{dx} + y = 0$$

leads to $y = c_2 e^{-x}$. Hence we can write

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ c_2 e^{-x}, & x > 1. \end{cases}$$

By appealing to the definition of continuity at a point, it is possible to determine c_2 so that the foregoing function is continuous at $x = 1$. The requirement that $\lim_{x \rightarrow 1^+} y(x) = y(1)$ implies that $c_2 e^{-1} = 1 - e^{-1}$ or $c_2 = e - 1$. As seen in Figure 2.3.4, the function

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases} \quad (10)$$

is continuous on $(0, \infty)$. ■

It is worthwhile to think about (10) and Figure 2.3.4 a little bit; you are urged to read and answer Problem 53 in Exercises 2.3.

ERROR FUNCTION In mathematics, science, and engineering some important functions are defined in terms of nonelementary integrals. Two such special functions are the **error function** and **complementary error function**:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \text{and} \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \quad (11)$$

From the known result $\int_0^\infty e^{-t^2} dt = \sqrt{\pi}/2^*$ we can write $(2/\sqrt{\pi}) \int_0^\infty e^{-t^2} dt = 1$.

Using the additive interval property of definite integrals $\int_0^\infty = \int_0^x + \int_x^\infty$ we can rewrite the last result in the alternative form

$$\frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt = \overbrace{\frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt}^{\operatorname{erf}(x)} + \overbrace{\frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt}^{\operatorname{erfc}(x)} = 1. \quad (12)$$

It is seen from (12) that the error function $\operatorname{erf}(x)$ and complementary error function $\operatorname{erfc}(x)$ are related by the identity

$$\operatorname{erf}(x) + \operatorname{erfc}(x) = 1.$$

Because of its importance in probability, statistics, and applied partial differential equations, the error function has been extensively tabulated. Note that $\operatorname{erf}(0) = 0$ is one obvious function value. Numerical values of $\operatorname{erf}(x)$ can also be found using a CAS such as *Mathematica*.

*This result is proved in Appendix A.

If we are solving an initial-value problem (6) and recognize that indefinite integration of the right-hand side of (5) would lead to a nonelementary integral, then as we saw in Example 5 of Section 2.2 it is convenient to use instead *definite* integration over the interval $[x_0, x]$. The last example illustrates that this procedure automatically incorporates the initial condition at x_0 into the solution of the DE; in other words, we do not have to solve for the constant c in its general solution.

EXAMPLE 7 The Error Function

Solve the initial-value problem $\frac{dy}{dx} - 2xy = 2$, $y(0) = 1$.

SOLUTION The differential equation is already in standard form, and so we see that the integrating factor is $e^{\int(-2x dx)} = e^{-x^2}$. Multiplying both sides of the equation by this factor then gives $e^{-x^2} \frac{dy}{dx} - 2xe^{-x^2} y = 2e^{-x^2}$, which is the same as

$$\frac{d}{dx}[e^{-x^2}y] = 2e^{-x^2}. \quad (13)$$

Because indefinite integration of both sides of equation (13) leads to the nonelementary integral $\int e^{-x^2} dx$ we identify $x_0 = 0$ and use definite integration over the interval $[0, x]$:

$$\int_0^x \frac{d}{dt}(e^{-t^2}y(t)) dt = 2 \int_0^x e^{-t^2} dt \quad \text{or} \quad e^{-x^2}y(x) - y(0) = 2 \int_0^x e^{-t^2} dt.$$

Using the initial condition $y(0) = 1$ the last expression yields the solution

$$y = e^{x^2} + 2e^{x^2} \int_0^x e^{-t^2} dt. \quad (14)$$

Then by inserting the factor $\sqrt{\pi}/\sqrt{\pi}$ into this solution in the following manner—

$$y = e^{x^2} + 2e^{x^2} \int_0^x e^{-t^2} dt = e^{x^2} \left[1 + \sqrt{\pi} \left(\frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \right) \right]$$

$\text{erf}(x)$

we see from (11) that (14) can be rewritten in terms of the error function as

$$y = e^{x^2} [1 + \sqrt{\pi} \text{erf}(x)]. \quad (15)$$

The graph of solution (15), obtained with the aid of a CAS, is given in Figure 2.3.5.

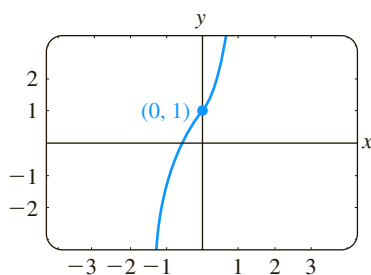


FIGURE 2.3.5 Graph of (15) in Example 7

See Problems 43–48 in Exercises 2.3.

USE OF COMPUTERS The computer algebra systems *Mathematica* and *Maple* are capable of producing implicit or explicit solutions for some kinds of differential equations using their *dsolve* commands.*

*Certain commands have the same spelling, but in *Mathematica* commands begin with a capital letter (**DSolve**), whereas in *Maple* the same command begins with a lower case letter (**dsolve**). When discussing such common syntax, we compromise and write, for example, *dsolve*. See the *Student Resource Manual* for the complete input commands used to solve a linear first-order DE.

REMARKS

(i) Occasionally, a first-order differential equation is not linear in one variable but is linear in the other variable. For example, the differential equation

$$\frac{dy}{dx} = \frac{1}{x + y^2}$$

is not linear in the variable y . But its reciprocal

$$\frac{dx}{dy} = x + y^2 \quad \text{or} \quad \frac{dx}{dy} - x = y^2$$

is recognized as linear in the variable x . You should verify that the integrating factor $e^{\int(-1)dy} = e^{-y}$ and integration by parts yield the explicit solution $x = -y^2 - 2y - 2 + ce^y$ for the second equation. This expression is then an implicit solution of the first equation.

(ii) Mathematicians have adopted as their own certain words from engineering, which they found appropriately descriptive. The word *transient*, used earlier, is one of these terms. In future discussions the words *input* and *output* will occasionally pop up. The function f in (2) is called the **input** or **driving function**; a solution $y(x)$ of the differential equation for a given input is called the **output** or **response**.

(iii) The term **special functions** mentioned in conjunction with the error function also applies to the **sine integral function** and the **Fresnel sine integral** introduced in Problems 47 and 48 in Exercises 2.3. “Special Functions” is actually a well-defined branch of mathematics. More special functions are studied in Section 6.4.

EXERCISES 2.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$

2. $\frac{dy}{dx} + 2y = 0$

3. $\frac{dy}{dx} + y = e^{3x}$

4. $3\frac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x\frac{dy}{dx} - y = x^2\sin x$

10. $x\frac{dy}{dx} + 2y = 3$

11. $x\frac{dy}{dx} + 4y = x^3 - x$

12. $(1+x)\frac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x+2)y = e^x$

14. $xy' + (1+x)y = e^{-x}\sin 2x$

15. $y\,dx - 4(x+y^6)\,dy = 0$

16. $y\,dx = (ye^y - 2x)\,dy$

17. $\cos x \frac{dy}{dx} + (\sin x)y = 1$

18. $\cos^2 x \sin x \frac{dy}{dx} + (\cos^3 x)y = 1$

19. $(x+1)\frac{dy}{dx} + (x+2)y = 2xe^{-x}$

20. $(x+2)^2\frac{dy}{dx} = 5 - 8y - 4xy$

21. $\frac{dr}{d\theta} + r \sec \theta = \cos \theta$

22. $\frac{dP}{dt} + 2tP = P + 4t - 2$

23. $x\frac{dy}{dx} + (3x+1)y = e^{-3x}$

24. $(x^2-1)\frac{dy}{dx} + 2y = (x+1)^2$

In Problems 25–36 solve the given initial-value problem. Give the largest interval I over which the solution is defined.

25. $\frac{dy}{dx} = x + 5y, \quad y(0) = 3$

26. $\frac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$

27. $xy' + y = e^x, \quad y(1) = 2$

28. $y \frac{dx}{dy} - x = 2y^2, \quad y(1) = 5$

29. $L \frac{di}{dt} + Ri = E, \quad i(0) = i_0, \quad L, R, E, i_0 \text{ constants}$

30. $\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0, \quad k, T_m, T_0 \text{ constants}$

31. $x \frac{dy}{dx} + y = 4x + 1, \quad y(1) = 8$

32. $y' + 4xy = x^3 e^{x^2}, \quad y(0) = -1$

33. $(x + 1) \frac{dy}{dx} + y = \ln x, \quad y(1) = 10$

34. $x(x + 1) \frac{dy}{dx} + xy = 1, \quad y(e) = 1$

35. $y' - (\sin x)y = 2 \sin x, \quad y(\pi/2) = 1$

36. $y' + (\tan x)y = \cos^2 x, \quad y(0) = -1$

In Problems 37–40 proceed as in Example 6 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

37. $\frac{dy}{dx} + 2y = f(x), \quad y(0) = 0, \text{ where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$$

38. $\frac{dy}{dx} + y = f(x), \quad y(0) = 1, \text{ where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ -1, & x > 1 \end{cases}$$

39. $\frac{dy}{dx} + 2xy = f(x), \quad y(0) = 2, \text{ where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

40. $(1 + x^2) \frac{dy}{dx} + 2xy = f(x), \quad y(0) = 0, \text{ where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & x \geq 1 \end{cases}$$

In Problems 41 and 42 proceed as in Example 6 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

41. $\frac{dy}{dx} + P(x)y = 4x, \quad y(0) = 3, \text{ where}$

$$P(x) = \begin{cases} 2, & 0 \leq x \leq 1 \\ -\frac{2}{x}, & x > 1 \end{cases}$$

42. $\frac{dy}{dx} + P(x)y = 0, \quad y(0) = 4, \text{ where}$

$$P(x) = \begin{cases} 1, & 0 \leq x \leq 2 \\ 5, & x > 2 \end{cases}$$

In Problems 43 and 44 proceed as in Example 7 and express the solution of the given initial-value problem in terms of $\operatorname{erf}(x)$ (Problem 43) and $\operatorname{erfc}(x)$ (Problem 44).

43. $\frac{dy}{dx} - 2xy = 1, \quad y(1) = 1$

44. $\frac{dy}{dx} - 2xy = -1, \quad y(0) = \sqrt{\pi}/2$

In Problems 45 and 46 proceed as in Example 7 and express the solution of the given initial-value problem in terms of an integral-defined function.

45. $\frac{dy}{dx} + e^x y = 1, \quad y(0) = 1$

46. $x^2 \frac{dy}{dx} - y = x^3, \quad y(1) = 0$

47. The **sine integral function** is defined as

$$\operatorname{Si}(x) = \int_0^x \frac{\sin t}{t} dt,$$

where the integrand is defined to be 1 at $x = 0$. See Appendix A. Express the solution of the initial-value problem

$$x^3 \frac{dy}{dx} + 2x^2 y = 10 \sin x, \quad y(1) = 0$$

in terms of $\operatorname{Si}(x)$.

48. The **Fresnel sine integral function** is defined as

$$S(x) = \int_0^x \sin\left(\frac{\pi}{2} t^2\right) dt.$$

See Appendix A. Express the solution of the initial-value problem

$$\frac{dy}{dx} - (\sin x^2)y = 0, \quad y(0) = 5$$

in terms of $S(x)$.

Discussion Problems

49. Reread the discussion following Example 2. Construct a linear first-order differential equation for which all nonconstant solutions approach the horizontal asymptote $y = 4$ as $x \rightarrow \infty$.

50. Reread Example 3 and then discuss, with reference to Theorem 1.2.1, the existence and uniqueness of a solution of the initial-value problem consisting of $xy' - 4y = x^6e^x$ and the given initial condition.
- (a) $y(0) = 0$
- (b) $y(0) = y_0, y_0 > 0$
- (c) $y(x_0) = y_0, x_0 > 0, y_0 > 0$
51. Reread Example 4 and then find the general solution of the differential equation on the interval $(-3, 3)$.
52. Reread the discussion following Example 5. Construct a linear first-order differential equation for which all solutions are asymptotic to the line $y = 3x - 5$ as $x \rightarrow \infty$.
53. Reread Example 6 and then discuss why it is technically incorrect to say that the function in (10) is a “solution” of the IVP on the interval $[0, \infty)$.
54. (a) Construct a linear first-order differential equation of the form $xy' + 3y = g(x)$ for which $y = x^3 + c/x^3$ is its general solution. Give an interval I of definition of this solution.
- (b) Give an initial condition $y(x_0) = y_0$ for the DE found in part (a) so that the solution of the IVP is $y = x^3 - 1/x^3$. Repeat if the solution is $y = x^3 + 2/x^3$. Give an interval I of definition of each of these solutions. Graph the solution curves. Is there an initial-value problem whose solution is defined on $(-\infty, \infty)$?
- (c) Is each IVP found in part (b) unique? That is, can there be more than one IVP for which, say, $y = x^3 - 1/x^3, x$ in some interval I , is the solution?
55. In determining the integrating factor (3), we did not use a constant of integration in the evaluation of $\int P(x) dx$. Explain why using $\int P(x) dx + c_1$ has no effect on the solution of (2).
56. Suppose $P(x)$ is continuous on some interval I and a is a number in I . What can be said about the solution of the initial-value problem $y' + P(x)y = 0, y(a) = 0$?

Mathematical Models

57. **Radioactive Decay Series** The following system of differential equations is encountered in the study of the decay of a special type of radioactive series of elements:

$$\begin{aligned}\frac{dx}{dt} &= -\lambda_1 x \\ \frac{dy}{dt} &= \lambda_1 x - \lambda_2 y,\end{aligned}$$

where λ_1 and λ_2 are constants. Discuss how to solve this system subject to $x(0) = x_0, y(0) = y_0$. Carry out your ideas.

58. **Heart Pacemaker** A heart pacemaker consists of a switch, a battery of constant voltage E_0 , a capacitor with constant capacitance C , and the heart as a resistor with constant resistance R . When the switch is closed, the capacitor charges; when the switch is open, the capacitor discharges, sending an electrical stimulus to the heart. During the time the heart is being stimulated, the voltage E across the heart satisfies the linear differential equation

$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

Solve the DE, subject to $E(4) = E_0$.

Computer Lab Assignments

59. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 44 on the interval $(-\infty, \infty)$.
- (b) Use tables or a CAS to value the value $y(2)$.
60. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 47 on the interval $[0, \infty)$.
- (b) Use a CAS to find the value of the absolute maximum of the solution $y(x)$ on the interval.
61. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 48 on the interval $(-\infty, \infty)$.
- (b) It is known that Fresnel sine integral $S(x) \rightarrow \frac{1}{2}$ as $x \rightarrow \infty$ and $S(x) \rightarrow -\frac{1}{2}$ as $x \rightarrow -\infty$. What does the solution $y(x)$ approach as $x \rightarrow \infty$? As $x \rightarrow -\infty$?
- (c) Use a CAS to find the values of the absolute maximum and the absolute minimum of the solution $y(x)$ on the interval.

2.4 Exact Equations

INTRODUCTION Although the simple first-order differential equation

$$ydx + xdy = 0$$

is both separable and linear, we can solve the equation by an alternative manner by recognizing that the expression on the left-hand side of the equality is the differential of the function $f(x, y) = xy$; that is,

$$d(xy) = ydx + xdy.$$

In this section we examine first-order equations written in differential form $M(x, y)dx + N(x, y)dy = 0$. By applying a simple test to the coefficients M and N , we can determine whether $M(x, y)dx + N(x, y)dy$ is the differential of a function $f(x, y)$. If the answer is yes, we can construct f by partial integration.

DIFFERENTIAL OF A FUNCTION OF TWO VARIABLES If $z = f(x, y)$ is a function of two variables with continuous first partial derivatives in a region R of the xy -plane, then recall from calculus that its **differential** is defined to be

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \quad (1)$$

In the special case when $f(x, y) = c$, where c is a constant, then (1) gives

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0. \quad (2)$$

In other words, given a one-parameter family of functions $f(x, y) = c$, we can generate a first-order differential equation by computing the differential of both sides of the equality. For example, if $x^2 - 5xy + y^3 = c$, then (2) gives the first-order DE

$$(2x - 5y) dx + (-5x + 3y^2) dy = 0. \quad (3)$$

A DEFINITION Of course, not every first-order DE written in differential form $M(x, y) dx + N(x, y) dy = 0$ corresponds to a differential of $f(x, y) = c$. So for our purposes it is more important to turn the foregoing example around; namely, if we are given a first-order DE such as (3), is there some way we can recognize that the differential expression $(2x - 5y) dx + (-5x + 3y^2) dy$ is the differential $d(x^2 - 5xy + y^3)$? If there is, then an implicit solution of (3) is $x^2 - 5xy + y^3 = c$. We answer this question after the next definition.

DEFINITION 2.4.1 Exact Equation

A differential expression $M(x, y) dx + N(x, y) dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$ defined in R . A first-order differential equation of the form

$$M(x, y) dx + N(x, y) dy = 0$$

is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

For example, $x^2y^3 dx + x^3y^2 dy = 0$ is an exact equation, because its left-hand side is an exact differential:

$$d\left(\frac{1}{3} x^3 y^3\right) = x^2 y^3 dx + x^3 y^2 dy.$$

Notice that if we make the identifications $M(x, y) = x^2y^3$ and $N(x, y) = x^3y^2$, then $\partial M/\partial y = 3x^2y^2 = \partial N/\partial x$. Theorem 2.4.1, given next, shows that the equality of the partial derivatives $\partial M/\partial y$ and $\partial N/\partial x$ is no coincidence.

THEOREM 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition that $M(x, y) dx + N(x, y) dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad (4)$$

PROOF OF THE NECESSITY For simplicity let us assume that $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives for all (x, y) . Now if the expression $M(x, y) dx + N(x, y) dy$ is exact, there exists some function f such that for all x in R ,

$$M(x, y) dx + N(x, y) dy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

Therefore
$$M(x, y) = \frac{\partial f}{\partial x}, \quad N(x, y) = \frac{\partial f}{\partial y},$$

and
$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial N}{\partial x}.$$

The equality of the mixed partials is a consequence of the continuity of the first partial derivatives of $M(x, y)$ and $N(x, y)$. ■

The sufficiency part of Theorem 2.4.1 consists of showing that there exists a function f for which $\partial f / \partial x = M(x, y)$ and $\partial f / \partial y = N(x, y)$ whenever (4) holds. The construction of the function f actually reflects a basic procedure for solving exact equations.

METHOD OF SOLUTION Given an equation in the differential form $M(x, y) dx + N(x, y) dy = 0$, determine whether the equality in (4) holds. If it does, then there exists a function f for which

$$\frac{\partial f}{\partial x} = M(x, y).$$

We can find f by integrating $M(x, y)$ with respect to x while holding y constant:

$$f(x, y) = \int M(x, y) dx + g(y), \quad (5)$$

where the arbitrary function $g(y)$ is the “constant” of integration. Now differentiate (5) with respect to y and assume that $\partial f / \partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M(x, y) dx + g'(y) = N(x, y).$$

This gives
$$g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx. \quad (6)$$

Finally, integrate (6) with respect to y and substitute the result in (5). The implicit solution of the equation is $f(x, y) = c$.

Some observations are in order. First, it is important to realize that the expression $N(x, y) - (\partial / \partial y) \int M(x, y) dx$ in (6) is independent of x , because

$$\frac{\partial}{\partial x} \left[N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx \right] = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \int M(x, y) dx \right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0.$$

Second, we could just as well start the foregoing procedure with the assumption that $\partial f / \partial y = N(x, y)$. After integrating N with respect to y and then differentiating that result, we would find the analogues of (5) and (6) to be, respectively,

$$f(x, y) = \int N(x, y) dy + h(x) \quad \text{and} \quad h'(x) = M(x, y) - \frac{\partial}{\partial x} \int N(x, y) dy.$$

In either case *none of these formulas should be memorized.*

EXAMPLE 1 Solving an Exact DE

Solve $2xy \, dx + (x^2 - 1) \, dy = 0$.

SOLUTION With $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ we have

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

Thus the equation is exact, and so by Theorem 2.4.1 there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 2xy \quad \text{and} \quad \frac{\partial f}{\partial y} = x^2 - 1.$$

From the first of these equations we obtain, after integrating,

$$f(x, y) = x^2y + g(y).$$

Taking the partial derivative of the last expression with respect to y and setting the result equal to $N(x, y)$ gives

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1. \quad \leftarrow N(x, y)$$

It follows that $g'(y) = -1$ and $g(y) = -y$. Hence $f(x, y) = x^2y - y$, so the solution of the equation in implicit form is $x^2y - y = c$. The explicit form of the solution is easily seen to be $y = c/(x^2 - 1)$ and is defined on any interval not containing either $x = 1$ or $x = -1$. ■

NOTE The solution of the DE in Example 1 is *not* $f(x, y) = x^2y - y$. Rather, it is $f(x, y) = c$; if a constant is used in the integration of $g'(y)$, we can then write the solution as $f(x, y) = 0$.

EXAMPLE 2 Solving an Exact DE

Solve $(e^{2y} - y \cos xy) \, dx + (2xe^{2y} - x \cos xy + 2y) \, dy = 0$.

SOLUTION The equation is exact because

$$\frac{\partial M}{\partial y} = 2e^{2y} + xy \sin xy - \cos xy = \frac{\partial N}{\partial x}.$$

Hence a function $f(x, y)$ exists for which

$$M(x, y) = \frac{\partial f}{\partial x} \quad \text{and} \quad N(x, y) = \frac{\partial f}{\partial y}.$$

Now, for variety, we shall start with the assumption that $\partial f / \partial y = N(x, y)$; that is,

$$\frac{\partial f}{\partial y} = 2xe^{2y} - x \cos xy + 2y$$

$$f(x, y) = 2x \int e^{2y} \, dy - x \int \cos xy \, dy + 2 \int y \, dy + h(x).$$

Remember, the reason x can come out in front of the symbol $\int$ is that in the integration with respect to y , x is treated as an ordinary constant. It follows that

$$f(x, y) = xe^{2y} - \sin xy + y^2 + h(x)$$

$$\frac{\partial f}{\partial x} = e^{2y} - y \cos xy + h'(x) = e^{2y} - y \cos xy, \quad \leftarrow M(x, y)$$

and so $h'(x) = 0$ or $h(x) = c$. Hence a family of solutions is

$$xe^{2y} - \sin xy + y^2 + c = 0.$$

EXAMPLE 3 An Initial-Value Problem

Solve $\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2.$

SOLUTION By writing the differential equation in the form

$$(\cos x \sin x - xy^2) dx + y(1 - x^2) dy = 0,$$

we recognize that the equation is exact because

$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}.$$

Now $\frac{\partial f}{\partial y} = y(1 - x^2)$

$$f(x, y) = \frac{y^2}{2}(1 - x^2) + h(x)$$

$$\frac{\partial f}{\partial x} = -xy^2 + h'(x) = \cos x \sin x - xy^2.$$

The last equation implies that $h'(x) = \cos x \sin x$. Integrating gives

$$h(x) = -\int (\cos x)(-\sin x dx) = -\frac{1}{2} \cos^2 x.$$

Thus $\frac{y^2}{2}(1 - x^2) - \frac{1}{2} \cos^2 x = c_1 \quad \text{or} \quad y^2(1 - x^2) - \cos^2 x = c, \quad (7)$

where $2c_1$ has been replaced by c . The initial condition $y = 2$ when $x = 0$ demands that $4(1) - \cos^2(0) = c$, and so $c = 3$. An implicit solution of the problem is then $y^2(1 - x^2) - \cos^2 x = 3$.

The solution curve of the IVP is the curve drawn in blue in Figure 2.4.1; it is part of an interesting family of curves. The graphs of the members of the one-parameter family of solutions given in (7) can be obtained in several ways, two of which are using software to graph level curves (as discussed in Section 2.2) and using a graphing utility to carefully graph the explicit functions obtained for various values of c by solving $y^2 = (c + \cos^2 x)/(1 - x^2)$ for y .

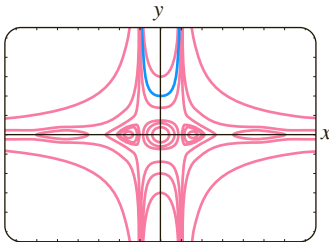


FIGURE 2.4.1 Solution curves of DE in Example 3

INTEGRATING FACTORS Recall from Section 2.3 that the left-hand side of the linear equation $y' + P(x)y = f(x)$ can be transformed into a derivative when we multiply the equation by an integrating factor. The same basic idea sometimes works for a nonexact differential equation $M(x, y) dx + N(x, y) dy = 0$. That is, it is sometimes possible to find an **integrating factor** $\mu(x, y)$ so that after multiplying, the left-hand side of

$$\mu(x, y)M(x, y) dx + \mu(x, y)N(x, y) dy = 0 \quad (8)$$

is an exact differential. In an attempt to find μ , we turn to the criterion (4) for exactness. Equation (8) is exact if and only if $(\mu M)_y = (\mu N)_x$, where the subscripts denote partial derivatives. By the Product Rule of differentiation the last equation is the same as $\mu M_y + \mu_y M = \mu N_x + \mu_x N$ or

$$\mu_x N - \mu_y M = (M_y - N_x)\mu. \quad (9)$$

Although M , N , M_y , and N_x are known functions of x and y , the difficulty here in determining the unknown $\mu(x, y)$ from (9) is that we must solve a partial differential equation. Since we are not prepared to do that, we make a simplifying assumption. Suppose μ is a function of one variable; for example, say that μ depends only on x . In this case, $\mu_x = d\mu/dx$ and $\mu_y = 0$, so (9) can be written as

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N} \mu. \quad (10)$$

We are still at an impasse if the quotient $(M_y - N_x)/N$ depends on both x and y . However, if after all obvious algebraic simplifications are made, the quotient $(M_y - N_x)/N$ turns out to depend solely on the variable x , then (10) is a first-order ordinary differential equation. We can finally determine μ because (10) is *separable* as well as *linear*. It follows from either Section 2.2 or Section 2.3 that $\mu(x) = e^{\int ((M_y - N_x)/N) dx}$. In like manner, it follows from (9) that if μ depends only on the variable y , then

$$\frac{d\mu}{dy} = \frac{N_x - M_y}{M} \mu. \quad (11)$$

In this case, if $(N_x - M_y)/M$ is a function of y only, then we can solve (11) for μ .

We summarize the results for the differential equation

$$M(x, y) dx + N(x, y) dy = 0. \quad (12)$$

- If $(M_y - N_x)/N$ is a function of x alone, then an integrating factor for (12) is

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}. \quad (13)$$

- If $(N_x - M_y)/M$ is a function of y alone, then an integrating factor for (12) is

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}. \quad (14)$$

EXAMPLE 4 A Nonexact DE Made Exact

The nonlinear first-order differential equation

$$xy dx + (2x^2 + 3y^2 - 20) dy = 0$$

is not exact. With the identifications $M = xy$, $N = 2x^2 + 3y^2 - 20$, we find the partial derivatives $M_y = x$ and $N_x = 4x$. The first quotient from (13) gets us nowhere, since

$$\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

depends on x and y . However, (14) yields a quotient that depends only on y :

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}.$$

The integrating factor is then $e^{\int 3 dy/y} = e^{3 \ln y} = e^{\ln y^3} = y^3$. After we multiply the given DE by $\mu(y) = y^3$, the resulting equation is

$$xy^4 dx + (2x^2y^3 + 3y^5 - 20y^3) dy = 0.$$

You should verify that the last equation is now exact as well as show, using the method of this section, that a family of solutions is $\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = c$. ■

REMARKS

(i) When testing an equation for exactness, make sure it is of the precise form $M(x, y)dx + N(x, y)dy = 0$. Sometimes a differential equation is written $G(x, y)dx = H(x, y)dy$. In this case, first rewrite it as $G(x, y)dx - H(x, y)dy = 0$ and then identify $M(x, y) = G(x, y)$ and $N(x, y) = -H(x, y)$ before using (4).

(ii) In some texts on differential equations the study of exact equations precedes that of linear DEs. Then the method for finding integrating factors just discussed can be used to derive an integrating factor for $y' + P(x)y = f(x)$. By rewriting the last equation in the differential form $(P(x)y - f(x))dx + dy = 0$, we see that

$$\frac{M_y - N_x}{N} = P(x).$$

From (13) we arrive at the already familiar integrating factor $e^{\int P(x)dx}$ used in Section 2.3.

EXERCISES 2.4

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–20 determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1)dx + (3y + 7)dy = 0$
2. $(2x + y)dx - (x + 6y)dy = 0$
3. $(5x + 4y)dx + (4x - 8y^3)dy = 0$
4. $(\sin y - y \sin x)dx + (\cos x + x \cos y - y)dy = 0$
5. $(2xy^2 - 3)dx + (2x^2y + 4)dy = 0$
6. $\left(2y - \frac{1}{x} + \cos 3x\right)\frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$
7. $(x^2 - y^2)dx + (x^2 - 2xy)dy = 0$
8. $\left(1 + \ln x + \frac{y}{x}\right)dx = (1 - \ln x)dy$
9. $(x - y^3 + y^2 \sin x)dx = (3xy^2 + 2y \cos x)dy$
10. $(x^3 + y^3)dx + 3xy^2dy = 0$
11. $(y \ln y - e^{-xy})dx + \left(\frac{1}{y} + x \ln y\right)dy = 0$
12. $(3x^2y + e^y)dx + (x^3 + xe^y - 2y)dy = 0$
13. $x \frac{dy}{dx} = 2xe^x - y + 6x^2$
14. $\left(1 - \frac{3}{y} + x\right)\frac{dy}{dx} + y = \frac{3}{x} - 1$
15. $\left(x^2y^3 - \frac{1}{1 + 9x^2}\right)\frac{dx}{dy} + x^3y^2 = 0$
16. $(5y - 2x)y' - 2y = 0$
17. $(\tan x - \sin x \sin y)dx + \cos x \cos y dy = 0$

18. $(2y \sin x \cos x - y + 2y^2e^{xy^2})dx = (x - \sin^2 x - 4xye^{xy^2})dy$
19. $(4t^3y - 15t^2 - y)dt + (t^4 + 3y^2 - t)dy = 0$
20. $\left(\frac{1}{t} + \frac{1}{t^2} - \frac{y}{t^2 + y^2}\right)dt + \left(ye^y + \frac{t}{t^2 + y^2}\right)dy = 0$

In Problems 21–26 solve the given initial-value problem.

21. $(x + y)^2 dx + (2xy + x^2 - 1)dy = 0, \quad y(1) = 1$
22. $(e^x + y)dx + (2 + x + ye^y)dy = 0, \quad y(0) = 1$
23. $(4y + 2t - 5)dt + (6y + 4t - 1)dy = 0, \quad y(-1) = 2$
24. $\left(\frac{3y^2 - t^2}{y^5}\right)\frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1$
25. $(y^2 \cos x - 3x^2y - 2x)dx + (2y \sin x - x^3 + \ln y)dy = 0, \quad y(0) = e$
26. $\left(\frac{1}{1 + y^2} + \cos x - 2xy\right)\frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

In Problems 27 and 28 find the value of k so that the given differential equation is exact.

27. $(y^3 + kxy^4 - 2x)dx + (3xy^2 + 20x^2y^3)dy = 0$
28. $(6xy^3 + \cos y)dx + (2kx^2y^2 - x \sin y)dy = 0$

In Problems 29 and 30 verify that the given differential equation is not exact. Multiply the given differential equation by the indicated integrating factor $\mu(x, y)$ and verify that the new equation is exact. Solve.

29. $(-xy \sin x + 2y \cos x)dx + 2x \cos x dy = 0;$
 $\mu(x, y) = xy$

30. $(x^2 + 2xy - y^2) dx + (y^2 + 2xy - x^2) dy = 0$;
 $\mu(x, y) = (x + y)^{-2}$

In Problems 31–36 solve the given differential equation by finding, as in Example 4, an appropriate integrating factor.

31. $(2y^2 + 3x) dx + 2xy dy = 0$

32. $y(x + y + 1) dx + (x + 2y) dy = 0$

33. $6xy dx + (4y + 9x^2) dy = 0$

34. $\cos x dx + \left(1 + \frac{2}{y}\right) \sin x dy = 0$

35. $(10 - 6y + e^{-3x}) dx - 2 dy = 0$

36. $(y^2 + xy^3) dx + (5y^2 - xy + y^3 \sin y) dy = 0$

In Problems 37 and 38 solve the given initial-value problem by finding, as in Example 4, an appropriate integrating factor.

37. $x dx + (x^2y + 4y) dy = 0$, $y(4) = 0$

38. $(x^2 + y^2 - 5) dx = (y + xy) dy$, $y(0) = 1$

39. (a) Show that a one-parameter family of solutions of the equation

$$(4xy + 3x^2) dx + (2y + 2x^2) dy = 0$$

$$\text{is } x^3 + 2x^2y + y^2 = c.$$

- (b) Show that the initial conditions $y(0) = -2$ and $y(1) = 1$ determine the same implicit solution.

- (c) Find explicit solutions $y_1(x)$ and $y_2(x)$ of the differential equation in part (a) such that $y_1(0) = -2$ and $y_2(1) = 1$. Use a graphing utility to graph $y_1(x)$ and $y_2(x)$.

Discussion Problems

40. Consider the concept of an integrating factor used in Problems 29–38. Are the two equations $M dx + N dy = 0$ and $\mu M dx + \mu N dy = 0$ necessarily equivalent in the sense that a solution of one is also a solution of the other? Discuss.
41. Reread Example 3 and then discuss why we can conclude that the interval of definition of the explicit solution of the IVP (the blue curve in Figure 2.4.1) is $(-1, 1)$.
42. Discuss how the functions $M(x, y)$ and $N(x, y)$ can be found so that each differential equation is exact. Carry out your ideas.

(a) $M(x, y) dx + \left(xe^{xy} + 2xy + \frac{1}{x}\right) dy = 0$

(b) $\left(x^{-1/2}y^{1/2} + \frac{x}{x^2 + y}\right) dx + N(x, y) dy = 0$

43. Differential equations are sometimes solved by having a clever idea. Here is a little exercise in cleverness: Although the differential equation $(x - \sqrt{x^2 + y^2}) dx + y dy = 0$ is not exact, show how the rearrangement $(x dx + y dy)/\sqrt{x^2 + y^2} = dx$ and the observation $\frac{1}{2}d(x^2 + y^2) = x dx + y dy$ can lead to a solution.

44. True or False: Every separable first-order equation $dy/dx = g(x)h(y)$ is exact.

Mathematical Model

45. **Falling Chain** A portion of a uniform chain of length 8 ft is loosely coiled around a peg at the edge of a high horizontal platform, and the remaining portion of the chain hangs at rest over the edge of the platform. See Figure 2.4.2. Suppose that the length of the overhanging chain is 3 ft, that the chain weighs 2 lb/ft, and that the positive direction is downward. Starting at $t = 0$ seconds, the weight of the overhanging portion causes the chain on the table to uncoil smoothly and to fall to the floor. If $x(t)$ denotes the length of the chain overhanging the table at time $t > 0$, then $v = dx/dt$ is its velocity. When all resistive forces are ignored, it can be shown that a mathematical model relating v to x is given by

$$xv \frac{dv}{dx} + v^2 = 32x.$$

- (a) Rewrite this model in differential form. Proceed as in Problems 31–36 and solve the DE for v in terms of x by finding an appropriate integrating factor. Find an explicit solution $v(x)$.
- (b) Determine the velocity with which the chain leaves the platform.

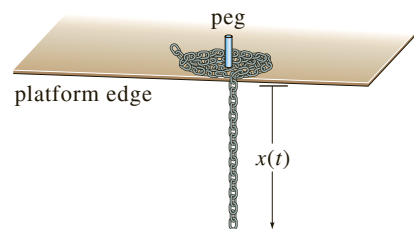


FIGURE 2.4.2 Uncoiling chain in Problem 45

Computer Lab Assignments

46. Streamlines

- (a) The solution of the differential equation

$$\frac{2xy}{(x^2 + y^2)^2} dx + \left[1 + \frac{y^2 - x^2}{(x^2 + y^2)^2}\right] dy = 0$$

is a family of curves that can be interpreted as streamlines of a fluid flow around a circular object whose boundary is described by the equation $x^2 + y^2 = 1$. Solve this DE and note the solution $f(x, y) = c$ for $c = 0$.

- (b) Use a CAS to plot the streamlines for $c = 0, \pm 0.2, \pm 0.4, \pm 0.6$, and ± 0.8 in three different ways. First, use the *contourplot* of a CAS. Second, solve for x in terms of the variable y . Plot the resulting two functions of y for the given values of c , and then combine the graphs. Third, use the CAS to solve a cubic equation for y in terms of x .

2.5 Solutions by Substitutions

INTRODUCTION We usually solve a differential equation by recognizing it as a certain kind of equation (say, separable, linear, exact, and so on) and then carrying out a procedure consisting of *equation-specific mathematical steps* (say, separating variables and integrating) that yields a solution of the equation. But it is not uncommon to be stumped by a differential equation because it does not fall into one of the classes of equations that we know how to solve. The substitution procedures that are discussed in this section may be helpful in this situation.

SUBSTITUTIONS Often the first step in solving a differential equation consists of transforming it into another differential equation by means of a **substitution**. For example, suppose we wish to transform the first-order differential equation $dy/dx = f(x, y)$ by the substitution $y = g(x, u)$, where u is regarded as a function of the variable x . If g possesses first-partial derivatives, then the Chain Rule

$$\frac{dy}{dx} = \frac{\partial g}{\partial x} \frac{dx}{dx} + \frac{\partial g}{\partial u} \frac{du}{dx} \quad \text{gives} \quad \frac{dy}{dx} = g_x(x, u) + g_u(x, u) \frac{du}{dx}.$$

If we replace dy/dx by the foregoing derivative and replace y in $f(x, y)$ by $g(x, u)$, then the DE $dy/dx = f(x, y)$ becomes $g_x(x, u) + g_u(x, u) \frac{du}{dx} = f(x, g(x, u))$, which, solved for du/dx , has the form $\frac{du}{dx} = F(x, u)$. If we can determine a solution $u = \phi(x)$ of this last equation, then a solution of the original differential equation is $y = g(x, \phi(x))$.

In the discussion that follows we examine three different kinds of first-order differential equations that are solvable by means of a substitution.

HOMOGENEOUS EQUATIONS If a function f possesses the property $f(tx, ty) = t^\alpha f(x, y)$ for some real number α , then f is said to be a **homogeneous function** of degree α . For example, $f(x, y) = x^3 + y^3$ is a homogeneous function of degree 3, since

$$f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y),$$

whereas $f(x, y) = x^3 + y^3 + 1$ is not homogeneous. A first-order DE in differential form

$$M(x, y) dx + N(x, y) dy = 0 \quad (1)$$

is said to be **homogeneous** if both coefficient functions M and N are homogeneous functions of the *same* degree. In other words, (1) is homogeneous if

$$M(tx, ty) = t^\alpha M(x, y) \quad \text{and} \quad N(tx, ty) = t^\alpha N(x, y).$$

In addition, if M and N are homogeneous functions of degree α , we can also write

$$M(x, y) = x^\alpha M(1, u) \quad \text{and} \quad N(x, y) = x^\alpha N(1, u), \quad \text{where } u = y/x, \quad (2)$$

and

$$M(x, y) = y^\alpha M(v, 1) \quad \text{and} \quad N(x, y) = y^\alpha N(v, 1), \quad \text{where } v = x/y. \quad (3)$$

See Problem 31 in Exercises 2.5. Properties (2) and (3) suggest the substitutions that can be used to solve a homogeneous differential equation. Specifically, *either* of the substitutions $y = ux$ or $x = vy$, where u and v are new dependent variables, will reduce a homogeneous equation to a *separable* first-order differential

equation. To show this, observe that as a consequence of (2) a homogeneous equation $M(x, y) dx + N(x, y) dy = 0$ can be rewritten as

$$x^\alpha M(1, u) dx + x^\alpha N(1, u) dy = 0 \quad \text{or} \quad M(1, u) dx + N(1, u) dy = 0,$$

where $u = y/x$ or $y = ux$. By substituting the differential $dy = u dx + x du$ into the last equation and gathering terms, we obtain a separable DE in the variables u and x :

$$M(1, u) dx + N(1, u)[u dx + x du] = 0$$

$$[M(1, u) + uN(1, u)] dx + xN(1, u) du = 0$$

or
$$\frac{dx}{x} + \frac{N(1, u) du}{M(1, u) + uN(1, u)} = 0.$$

At this point we offer the same advice as in the preceding sections: Do not memorize anything here (especially the last formula); rather, *work through the procedure each time*. The proof that the substitutions $x = vy$ and $dx = v dy + y dv$ also lead to a separable equation follows in an analogous manner from (3).

EXAMPLE 1 Solving a Homogeneous DE

Solve $(x^2 + y^2) dx + (x^2 - xy) dy = 0$.

SOLUTION Inspection of $M(x, y) = x^2 + y^2$ and $N(x, y) = x^2 - xy$ shows that these coefficients are homogeneous functions of degree 2. If we let $y = ux$, then $dy = u dx + x du$, so after substituting, the given equation becomes

$$(x^2 + u^2x^2) dx + (x^2 - ux^2)[u dx + x du] = 0$$

$$x^2(1 + u) dx + x^3(1 - u) du = 0$$

$$\frac{1 - u}{1 + u} du + \frac{dx}{x} = 0$$

$$\left[-1 + \frac{2}{1 + u} \right] du + \frac{dx}{x} = 0. \quad \leftarrow \text{long division}$$

After integration the last line gives

$$-u + 2 \ln|1 + u| + \ln|x| = \ln|c|$$

$$-\frac{y}{x} + 2 \ln\left|1 + \frac{y}{x}\right| + \ln|x| = \ln|c|. \quad \leftarrow \text{resubstituting } u = y/x$$

Using the properties of logarithms, we can write the preceding solution as

$$\ln\left|\frac{(x + y)^2}{cx}\right| = \frac{y}{x} \quad \text{or} \quad (x + y)^2 = cxe^{y/x}. \quad \blacksquare$$

Although either of the indicated substitutions can be used for every homogeneous differential equation, in practice we try $x = vy$ whenever the function $M(x, y)$ is simpler than $N(x, y)$. Also it could happen that after using one substitution, we may encounter integrals that are difficult or impossible to evaluate in closed form; switching substitutions may result in an easier problem.

BERNOULLI'S EQUATION The differential equation

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \quad (4)$$

where n is any real number, is called **Bernoulli's equation**. Note that for $n = 0$ and $n = 1$, equation (4) is linear. For $n \neq 0$ and $n \neq 1$ the substitution $u = y^{1-n}$ reduces any equation of form (4) to a linear equation.

EXAMPLE 2 Solving a Bernoulli DE

Solve $x \frac{dy}{dx} + y = x^2 y^2$.

SOLUTION We begin by rewriting the equation in the form given in (4) by dividing by x :

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2.$$

With $n = 2$ we have $u = y^{-1}$ or $y = u^{-1}$. We then substitute

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = -u^{-2} \frac{du}{dx} \quad \leftarrow \text{Chain Rule}$$

into the given equation and simplify. The result is

$$\frac{du}{dx} - \frac{1}{x}u = -x.$$

The integrating factor for this linear equation on, say, $(0, \infty)$ is

$$e^{-\int dx/x} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1}.$$

Integrating
$$\frac{d}{dx} [x^{-1}u] = -1$$

gives $x^{-1}u = -x + c$ or $u = -x^2 + cx$. Since $u = y^{-1}$, we have $y = 1/u$, so a solution of the given equation is $y = 1/(-x^2 + cx)$. ■

Note that we have not obtained the general solution of the original nonlinear differential equation in Example 2, since $y = 0$ is a singular solution of the equation.

REDUCTION TO SEPARATION OF VARIABLES A differential equation of the form

$$\frac{dy}{dx} = f(Ax + By + C) \quad (5)$$

can always be reduced to an equation with separable variables by means of the substitution $u = Ax + By + C$, $B \neq 0$. Example 3 illustrates the technique.

EXAMPLE 3 An Initial-Value Problem

Solve $\frac{dy}{dx} = (-2x + y)^2 - 7$, $y(0) = 0$.

SOLUTION If we let $u = -2x + y$, then $du/dx = -2 + dy/dx$, so the differential equation is transformed into

$$\frac{du}{dx} + 2 = u^2 - 7 \quad \text{or} \quad \frac{du}{dx} = u^2 - 9.$$

The last equation is separable. Using partial fractions

$$\frac{du}{(u-3)(u+3)} = dx \quad \text{or} \quad \frac{1}{6} \left[\frac{1}{u-3} - \frac{1}{u+3} \right] du = dx$$

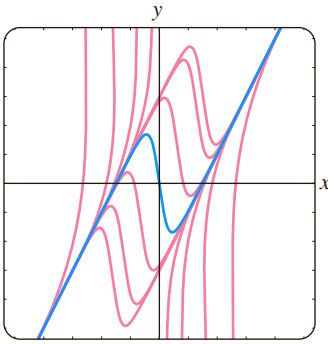


FIGURE 2.5.1 Solutions of DE in Example 3

and then integrating yields

$$\frac{1}{6} \ln \left| \frac{u-3}{u+3} \right| = x + c_1 \quad \text{or} \quad \frac{u-3}{u+3} = e^{6x+6c_1} = ce^{6x}. \quad \leftarrow \text{replace } e^{6c_1} \text{ by } c$$

Solving the last equation for u and then resubstituting gives the solution

$$u = \frac{3(1 + ce^{6x})}{1 - ce^{6x}} \quad \text{or} \quad y = 2x + \frac{3(1 + ce^{6x})}{1 - ce^{6x}}. \quad (6)$$

Finally, applying the initial condition $y(0) = 0$ to the last equation in (6) gives $c = -1$. Figure 2.5.1, obtained with the aid of a graphing utility, shows the graph of the particular solution $y = 2x + \frac{3(1 - e^{6x})}{1 + e^{6x}}$ in blue, along with the graphs of some other members of the family of solutions (6). ■

EXERCISES 2.5

Answers to selected odd-numbered problems begin on page ANS-2.

Each DE in Problems 1–14 is homogeneous.

In Problems 1–10 solve the given differential equation by using an appropriate substitution.

1. $(x - y) dx + x dy = 0$
2. $(x + y) dx + x dy = 0$
3. $x dx + (y - 2x) dy = 0$
4. $y dx = 2(x + y) dy$
5. $(y^2 + yx) dx - x^2 dy = 0$
6. $(y^2 + yx) dx + x^2 dy = 0$
7. $\frac{dy}{dx} = \frac{y - x}{y + x}$
8. $\frac{dy}{dx} = \frac{x + 3y}{3x + y}$
9. $-y dx + (x + \sqrt{xy}) dy = 0$
10. $x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$

In Problems 11–14 solve the given initial-value problem.

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$
12. $(x^2 + 2y^2) \frac{dx}{dy} = xy, \quad y(-1) = 1$
13. $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, \quad y(1) = 0$
14. $y dx + x(\ln x - \ln y - 1) dy = 0, \quad y(1) = e$

Each DE in Problems 15–22 is a Bernoulli equation.

In Problems 15–20 solve the given differential equation by using an appropriate substitution.

15. $x \frac{dy}{dx} + y = \frac{1}{y^2}$
16. $\frac{dy}{dx} - y = e^x y^2$
17. $\frac{dy}{dx} = y(xy^3 - 1)$
18. $x \frac{dy}{dx} - (1 + x)y = xy^2$

19. $t^2 \frac{dy}{dt} + y^2 = ty$
20. $3(1 + t^2) \frac{dy}{dt} = 2ty(y^3 - 1)$

In Problems 21 and 22 solve the given initial-value problem.

21. $x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$
22. $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$

Each DE in Problems 23–30 is of the form given in (5).

In Problems 23–28 solve the given differential equation by using an appropriate substitution.

23. $\frac{dy}{dx} = (x + y + 1)^2$
24. $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$
25. $\frac{dy}{dx} = \tan^2(x + y)$
26. $\frac{dy}{dx} = \sin(x + y)$
27. $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$
28. $\frac{dy}{dx} = 1 + e^{y-x+5}$

In Problems 29 and 30 solve the given initial-value problem.

29. $\frac{dy}{dx} = \cos(x + y), \quad y(0) = \pi/4$
30. $\frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$

Discussion Problems

31. Explain why it is always possible to express any homogeneous differential equation $M(x, y) dx + N(x, y) dy = 0$ in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

You might start by proving that

$$M(x, y) = x^\alpha M(1, y/x) \quad \text{and} \quad N(x, y) = x^\alpha N(1, y/x).$$

32. Put the homogeneous differential equation $(5x^2 - 2y^2) dx - xy dy = 0$ into the form given in Problem 31.

33. (a) Determine two singular solutions of the DE in Problem 10.

- (b) If the initial condition $y(5) = 0$ is as prescribed in Problem 10, then what is the largest interval I over which the solution is defined? Use a graphing utility to graph the solution curve for the IVP.

34. In Example 3 the solution $y(x)$ becomes unbounded as $x \rightarrow \pm\infty$. Nevertheless, $y(x)$ is asymptotic to a curve as $x \rightarrow -\infty$ and to a different curve as $x \rightarrow \infty$. What are the equations of these curves?

35. The differential equation $dy/dx = P(x) + Q(x)y + R(x)y^2$ is known as **Riccati's equation**.

- (a) A Riccati equation can be solved by a succession of two substitutions *provided* that we know a particular solution y_1 of the equation. Show that the substitution $y = y_1 + u$ reduces Riccati's equation to a Bernoulli equation (4) with $n = 2$. The Bernoulli equation can then be reduced to a linear equation by the substitution $w = u^{-1}$.

- (b) Find a one-parameter family of solutions for the differential equation

$$\frac{dy}{dx} = -\frac{4}{x^2} - \frac{1}{x}y + y^2$$

where $y_1 = 2/x$ is a known solution of the equation.

36. Determine an appropriate substitution to solve

$$xy' = y \ln(xy).$$

Mathematical Models

37. **Falling Chain** In Problem 45 in Exercises 2.4 we saw that a mathematical model for the velocity v of a chain slipping off the edge of a high horizontal platform is

$$xv \frac{dv}{dx} + v^2 = 32x.$$

In that problem you were asked to solve the DE by converting it into an exact equation using an integrating factor. This time solve the DE using the fact that it is a Bernoulli equation.

38. **Population Growth** In the study of population dynamics one of the most famous models for a growing but bounded population is the **logistic equation**

$$\frac{dP}{dt} = P(a - bP),$$

where a and b are positive constants. Although we will come back to this equation and solve it by an alternative method in Section 3.2, solve the DE this first time using the fact that it is a Bernoulli equation.

2.6 A Numerical Method

INTRODUCTION A differential equation can be a source of information. We started this chapter by observing that we could garner *qualitative* information from a first-order DE about its solutions even before we attempted to solve the equation. Then in Sections 2.2–2.5 we examined first-order DEs *analytically*—that is, we developed some procedures for obtaining explicit and implicit solutions. But a differential equation can possess a solution and yet it may not be possible to obtain it analytically. So to round out the picture of the different types of analyses of differential equations, we conclude this chapter with a method by which we can “solve” the differential equation *numerically*; this means that the DE is used as the foundation on which we construct an algorithm for approximating the unknown solution.

In this section we are going to develop only the simplest of numerical methods—a method that utilizes the idea that a tangent line can be used to approximate the values of a function in a small neighborhood of the point of tangency. A more extensive treatment of numerical methods for ordinary differential equations is given in Chapter 9.

USING THE TANGENT LINE Let us assume that the first-order initial-value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \quad (1)$$

possesses a solution. One way of approximating this solution is to use tangent lines. For example, let $y(x)$ denote the unknown solution of the first-order initial-value problem $y' = 0.1\sqrt{y} + 0.4x^2$, $y(2) = 4$. The nonlinear differential equation in this IVP cannot be solved directly by any of the methods considered in

Sections 2.2, 2.4, and 2.5; nevertheless, we can still find approximate numerical values of the unknown $y(x)$. Specifically, suppose we wish to know the value of $y(2.5)$. The IVP has a solution, and as the flow of the direction field of the DE in Figure 2.6.1(a) suggests, a solution curve must have a shape similar to the curve shown in blue.

The direction field in Figure 2.6.1(a) was generated with lineal elements passing through points in a grid with integer coordinates. As the solution curve passes through the initial point $(2, 4)$, the lineal element at this point is a tangent line with slope given by $f(2, 4) = 0.1\sqrt{4} + 0.4(2)^2 = 1.8$. As is apparent in Figure 2.6.1(a) and the “zoom in” in Figure 2.6.1(b), when x is close to 2, the points on the solution curve are close to the points on the tangent line (the lineal element). Using the point $(2, 4)$, the slope $f(2, 4) = 1.8$, and the point-slope form of a line, we find that an equation of the tangent line is $y = L(x)$, where $L(x) = 1.8x + 0.4$. This last equation, called a **linearization** of $y(x)$ at $x = 2$, can be used to approximate values of $y(x)$ within a small neighborhood of $x = 2$. If $y_1 = L(x_1)$ denotes the y -coordinate on the tangent line and $y(x_1)$ is the y -coordinate on the solution curve corresponding to an x -coordinate x_1 that is close to $x = 2$, then $y(x_1) \approx y_1$. If we choose, say, $x_1 = 2.1$, then $y_1 = L(2.1) = 1.8(2.1) + 0.4 = 4.18$, so $y(2.1) \approx 4.18$.

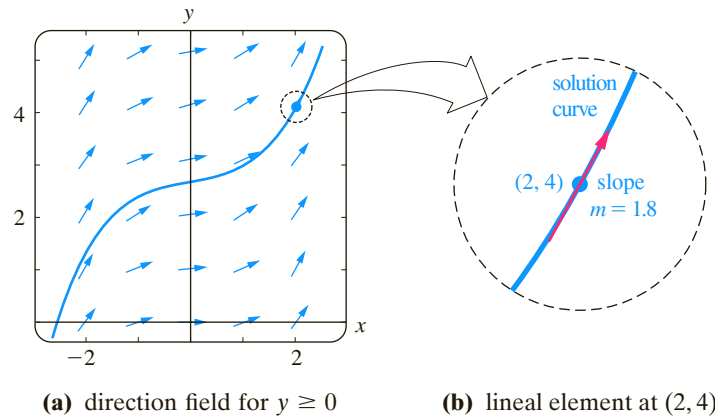


FIGURE 2.6.1 Magnification of a neighborhood about the point $(2, 4)$

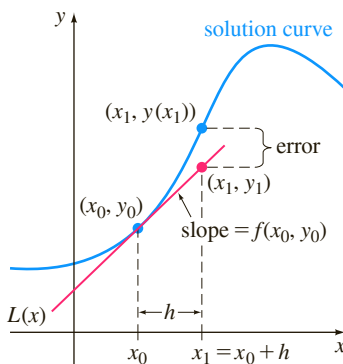


FIGURE 2.6.2 Approximating $y(x_1)$ using a tangent line

EULER'S METHOD To generalize the procedure just illustrated, we use the linearization of the unknown solution $y(x)$ of (1) at $x = x_0$:

$$L(x) = y_0 + f(x_0, y_0)(x - x_0). \quad (2)$$

The graph of this linearization is a straight line tangent to the graph of $y = y(x)$ at the point (x_0, y_0) . We now let h be a positive increment of the x -axis, as shown in Figure 2.6.2. Then by replacing x by $x_1 = x_0 + h$ in (2), we get

$$L(x_1) = y_0 + f(x_0, y_0)(x_0 + h - x_0) \quad \text{or} \quad y_1 = y_0 + hf(x_0, y_0),$$

where $y_1 = L(x_1)$. The point (x_1, y_1) on the tangent line is an approximation to the point $(x_1, y(x_1))$ on the solution curve. Of course, the accuracy of the approximation $L(x_1) \approx y(x_1)$ or $y_1 \approx y(x_1)$ depends heavily on the size of the increment h . Usually, we must choose this **step size** to be “reasonably small.” We now repeat the process using a second “tangent line” at (x_1, y_1) .* By identifying the new starting

*This is not an actual tangent line, since (x_1, y_1) lies on the first tangent and not on the solution curve.

point as (x_1, y_1) with (x_0, y_0) in the above discussion, we obtain an approximation $y_2 \approx y(x_2)$ corresponding to two steps of length h from x_0 , that is, $x_2 = x_1 + h = x_0 + 2h$, and

$$y(x_2) = y(x_0 + 2h) = y(x_1 + h) \approx y_2 = y_1 + hf(x_1, y_1).$$

Continuing in this manner, we see that $y_1, y_2, y_3, \dots$, can be defined recursively by the general formula

$$y_{n+1} = y_n + hf(x_n, y_n), \quad (3)$$

where $x_n = x_0 + nh$, $n = 0, 1, 2, \dots$. This procedure of using successive “tangent lines” is called **Euler’s method**.

EXAMPLE 1 Euler’s Method

Consider the initial-value problem $y' = 0.1\sqrt{y} + 0.4x^2$, $y(2) = 4$. Use Euler’s method to obtain an approximation of $y(2.5)$ using first $h = 0.1$ and then $h = 0.05$.

SOLUTION With the identification $f(x, y) = 0.1\sqrt{y} + 0.4x^2$, (3) becomes

$$y_{n+1} = y_n + h(0.1\sqrt{y_n} + 0.4x_n^2).$$

Then for $h = 0.1$, $x_0 = 2$, $y_0 = 4$, and $n = 0$ we find

$$y_1 = y_0 + h(0.1\sqrt{y_0} + 0.4x_0^2) = 4 + 0.1(0.1\sqrt{4} + 0.4(2)^2) = 4.18,$$

which, as we have already seen, is an estimate to the value of $y(2.1)$. However, if we use the smaller step size $h = 0.05$, it takes two steps to reach $x = 2.1$. From

$$y_1 = 4 + 0.05(0.1\sqrt{4} + 0.4(2)^2) = 4.09$$

$$y_2 = 4.09 + 0.05(0.1\sqrt{4.09} + 0.4(2.05)^2) = 4.18416187$$

we have $y_1 \approx y(2.05)$ and $y_2 \approx y(2.1)$. The remainder of the calculations were carried out by using software. The results are summarized in Tables 2.6.1 and 2.6.2, where each entry has been rounded to four decimal places. We see in Tables 2.6.1 and 2.6.2 that it takes five steps with $h = 0.1$ and 10 steps with $h = 0.05$, respectively, to get to $x = 2.5$. Intuitively, we would expect that $y_{10} = 5.0997$ corresponding to $h = 0.05$ is the better approximation of $y(2.5)$ than the value $y_5 = 5.0768$ corresponding to $h = 0.1$. ■

In Example 2 we apply Euler’s method to a differential equation for which we have already found a solution. We do this to compare the values of the approximations y_n at each step with the true or actual values of the solution $y(x_n)$ of the initial-value problem.

EXAMPLE 2 Comparison of Approximate and Actual Values

Consider the initial-value problem $y' = 0.2xy$, $y(1) = 1$. Use Euler’s method to obtain an approximation of $y(1.5)$ using first $h = 0.1$ and then $h = 0.05$.

SOLUTION With the identification $f(x, y) = 0.2xy$, (3) becomes

$$y_{n+1} = y_n + h(0.2x_n y_n)$$

where $x_0 = 1$ and $y_0 = 1$. Again with the aid of computer software we obtain the values in Tables 2.6.3 and 2.6.4 on page 79.

In Example 1 the true or actual values were calculated from the known solution $y = e^{0.1(x^2-1)}$. (Verify.) The **absolute error** is defined to be

$$|\text{actual value} - \text{approximation}|.$$

TABLE 2.6.1 $h = 0.1$

x_n	y_n
2.00	4.0000
2.10	4.1800
2.20	4.3768
2.30	4.5914
2.40	4.8244
2.50	5.0768

TABLE 2.6.2 $h = 0.05$

x_n	y_n
2.00	4.0000
2.05	4.0900
2.10	4.1842
2.15	4.2826
2.20	4.3854
2.25	4.4927
2.30	4.6045
2.35	4.7210
2.40	4.8423
2.45	4.9686
2.50	5.0997

TABLE 2.6.3 $h = 0.1$

x_n	y_n	Actual value	Abs. error	% Rel. error
1.00	1.0000	1.0000	0.0000	0.00
1.10	1.0200	1.0212	0.0012	0.12
1.20	1.0424	1.0450	0.0025	0.24
1.30	1.0675	1.0714	0.0040	0.37
1.40	1.0952	1.1008	0.0055	0.50
1.50	1.1259	1.1331	0.0073	0.64

TABLE 2.6.4 $h = 0.05$

x_n	y_n	Actual value	Abs. error	% Rel. error
1.00	1.0000	1.0000	0.0000	0.00
1.05	1.0100	1.0103	0.0003	0.03
1.10	1.0206	1.0212	0.0006	0.06
1.15	1.0318	1.0328	0.0009	0.09
1.20	1.0437	1.0450	0.0013	0.12
1.25	1.0562	1.0579	0.0016	0.16
1.30	1.0694	1.0714	0.0020	0.19
1.35	1.0833	1.0857	0.0024	0.22
1.40	1.0980	1.1008	0.0028	0.25
1.45	1.1133	1.1166	0.0032	0.29
1.50	1.1295	1.1331	0.0037	0.32

The **relative error** and **percentage relative error** are, in turn,

$$\frac{\text{absolute error}}{|\text{actual value}|} \quad \text{and} \quad \frac{\text{absolute error}}{|\text{actual value}|} \times 100.$$

It is apparent from Tables 2.6.3 and 2.6.4 that the accuracy of the approximations improves as the step size h decreases. Also, we see that even though the percentage relative error is growing with each step, it does not appear to be that bad. But you should not be deceived by one example. If we simply change the coefficient of the right side of the DE in Example 2 from 0.2 to 2, then at $x_n = 1.5$ the percentage relative errors increase dramatically. See Problem 4 in Exercises 2.6.

A CAVEAT Euler's method is just one of many different ways in which a solution of a differential equation can be approximated. Although attractive for its simplicity, *Euler's method is seldom used in serious calculations*. It was introduced here simply to give you a first taste of numerical methods. We will go into greater detail in discussing numerical methods that give significantly greater accuracy, notably the **fourth order Runge-Kutta method**, referred to as the **RK4 method**, in Chapter 9.

NUMERICAL SOLVERS Regardless of whether we can actually find an explicit or implicit solution, if a solution of a differential equation exists, it represents a smooth curve in the Cartesian plane. The basic idea behind *any* numerical method for first-order ordinary differential equations is to somehow approximate the y -values of a solution for preselected values of x . We start at a specified initial point (x_0, y_0) on a solution curve and proceed to calculate in a step-by-step fashion a sequence of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ whose y -coordinates y_i approximate the y -coordinates $y(x_i)$ of points $(x_1, y(x_1)), (x_2, y(x_2)), \dots, (x_n, y(x_n))$ that lie on the graph of the usually unknown solution $y(x)$. By taking the x -coordinates close together (that is, for small values of h) and by joining the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with short line segments, we obtain a polygonal curve whose qualitative characteristics we hope are close to those of an actual solution curve. Drawing curves is something that is well suited to a computer. A computer program written to either implement a numerical method or render a visual representation of an approximate solution curve fitting the numerical data produced by this method is referred to as a **numerical solver**. Many different numerical solvers are commercially available, either embedded in a larger software package, such as a computer algebra system, or provided as a stand-alone package. Some software packages simply plot the generated numerical approximations, whereas others generate hard numerical data as well as the corresponding approximate or

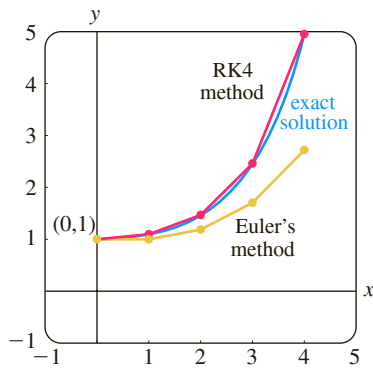


FIGURE 2.6.3 Comparison of the Runge-Kutta (RK4) and Euler methods

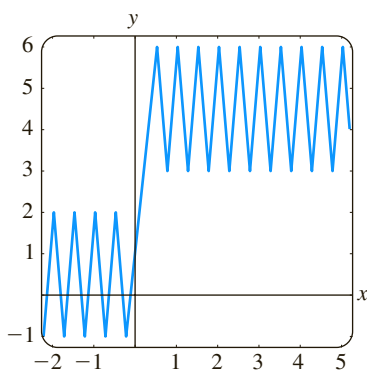


FIGURE 2.6.4 A not-very helpful numerical solution curve

numerical solution curves. By way of illustration of the connect-the-dots nature of the graphs produced by a numerical solver, the two colored polygonal graphs in Figure 2.6.3 are the numerical solution curves for the initial-value problem $y' = 0.2xy$, $y(0) = 1$ on the interval $[0, 4]$ obtained from Euler's method and the RK4 method using the step size $h = 1$. The blue smooth curve is the graph of the exact solution $y = e^{0.1x^2}$ of the IVP. Notice in Figure 2.6.3 that, even with the ridiculously large step size of $h = 1$, the RK4 method produces the more believable "solution curve." The numerical solution curve obtained from the RK4 method is indistinguishable from the actual solution curve on the interval $[0, 4]$ when a more typical step size of $h = 0.1$ is used.

USING A NUMERICAL SOLVER Knowledge of the various numerical methods is not necessary in order to use a numerical solver. A solver usually requires that the differential equation be expressed in normal form $dy/dx = f(x, y)$. Numerical solvers that generate only curves usually require that you supply $f(x, y)$ and the initial data x_0 and y_0 and specify the desired numerical method. If the idea is to approximate the numerical value of $y(a)$, then a solver may additionally require that you state a value for h or, equivalently, give the number of steps that you want to take to get from $x = x_0$ to $x = a$. For example, if we wanted to approximate $y(4)$ for the IVP illustrated in Figure 2.6.3, then, starting at $x = 0$ it would take four steps to reach $x = 4$ with a step size of $h = 1$; 40 steps is equivalent to a step size of $h = 0.1$. Although we will not delve here into the many problems that one can encounter when attempting to approximate mathematical quantities, you should at least be aware of the fact that a numerical solver may break down near certain points or give an incomplete or misleading picture when applied to some first-order differential equations in the normal form. Figure 2.6.4 illustrates the graph obtained by applying Euler's method to a certain first-order initial-value problem $dy/dx = f(x, y)$, $y(0) = 1$. Equivalent results were obtained using three different commercial numerical solvers, yet the graph is hardly a plausible solution curve. (Why?) There are several avenues of recourse when a numerical solver has difficulties; three of the more obvious are decrease the step size, use another numerical method, and try a different numerical solver.

EXERCISES 2.6

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1 and 2 use Euler's method to obtain a four-decimal approximation of the indicated value. Carry out the recursion of (3) by hand, first using $h = 0.1$ and then using $h = 0.05$.

1. $y' = 2x - 3y + 1$, $y(1) = 5$; $y(1.2)$
2. $y' = x + y^2$, $y(0) = 0$; $y(0.2)$

In Problems 3 and 4 use Euler's method to obtain a four-decimal approximation of the indicated value. First use $h = 0.1$ and then use $h = 0.05$. Find an explicit solution for each initial-value problem and then construct tables similar to Tables 2.6.3 and 2.6.4.

3. $y' = y$, $y(0) = 1$; $y(1.0)$
4. $y' = 2xy$, $y(1) = 1$; $y(1.5)$

In Problems 5–10 use a numerical solver and Euler's method to obtain a four-decimal approximation of the indicated value. First use $h = 0.1$ and then use $h = 0.05$.

5. $y' = e^{-y}$, $y(0) = 0$; $y(0.5)$

6. $y' = x^2 + y^2$, $y(0) = 1$; $y(0.5)$
7. $y' = (x - y)^2$, $y(0) = 0.5$; $y(0.5)$
8. $y' = xy + \sqrt{y}$, $y(0) = 1$; $y(0.5)$
9. $y' = xy^2 - \frac{y}{x}$, $y(1) = 1$; $y(1.5)$
10. $y' = y - y^2$, $y(0) = 0.5$; $y(0.5)$

In Problems 11 and 12 use a numerical solver to obtain a numerical solution curve for the given initial-value problem. First use Euler's method and then the RK4 method. Use $h = 0.25$ in each case. Superimpose both solution curves on the same coordinate axes. If possible, use a different color for each curve. Repeat, using $h = 0.1$ and $h = 0.05$.

11. $y' = 2(\cos x)y$, $y(0) = 1$
12. $y' = y(10 - 2y)$, $y(0) = 1$

Discussion Problems

13. Use a numerical solver and Euler's method to approximate $y(1.0)$, where $y(x)$ is the solution to $y' = 2xy^2$, $y(0) = 1$. First use $h = 0.1$ and then use $h = 0.05$. Repeat, using the RK4 method. Discuss what might cause the approximations to $y(1.0)$ to differ so greatly.

Computer Lab Assignments

14. (a) Use a numerical solver and the RK4 method to graph the solution of the initial-value problem $y' = -2xy + 1$, $y(0) = 0$.
- (b) Solve the initial-value problem by one of the analytic procedures developed earlier in this chapter.
- (c) Use the analytic solution $y(x)$ found in part (b) and a CAS to find the coordinates of all relative extrema.

Chapter 2 In Review

Answers to selected odd-numbered problems begin on page ANS-3.

Answer Problems 1–12 without referring back to the text. Fill in the blanks or answer true or false.

- The linear DE, $y' - ky = A$, where k and A are constants, is autonomous. The critical point _____ of the equation is a(n) _____ (attractor or repeller) for $k > 0$ and a(n) _____ (attractor or repeller) for $k < 0$.
- The initial-value problem $x \frac{dy}{dx} - 4y = 0$, $y(0) = k$, has an infinite number of solutions for $k =$ _____ and no solution for $k =$ _____.
- The linear DE, $y' + k_1y = k_2$, where k_1 and k_2 are nonzero constants, always possesses a constant solution. _____
- The linear DE, $a_1(x)y' + a_0(x)y = 0$ is also separable. _____
- An example of a nonlinear third-order differential equation in normal form is _____.
- The first-order DE, $\frac{dr}{d\theta} = r\theta + r + \theta + 1$ is not separable. _____
- Every autonomous DE $dy/dx = f(y)$ is separable. _____
- By inspection, two solutions of the differential equation $y' + |y| = 2$ are _____.
- If $y' = e^xy$, then $y =$ _____.
- If a differentiable function $y(x)$ satisfies $y' = |x|$, $y(-1) = 2$, then $y(x) =$ _____.
- $y = e^{\cos x} \int_0^x te^{-\cos t} dt$ is a solution of the linear first-order differential equation _____.
- An example of an autonomous linear first-order DE with a single critical point -3 is _____, whereas an autonomous nonlinear first-order DE with a single critical point -3 is _____.

In Problems 13 and 14 construct an autonomous first-order differential equation $dy/dx = f(y)$ whose phase portrait is consistent with the given figure.



FIGURE 2.R.1 Graph for Problem 13



FIGURE 2.R.2 Graph for Problem 14

15. The number 0 is a critical point of the autonomous differential equation $dx/dt = x^n$, where n is a positive integer. For what values of n is 0 asymptotically stable? Semi-stable? Unstable? Repeat for the differential equation $dx/dt = -x^n$.
16. Consider the differential equation $dP/dt = f(P)$, where
- $$f(P) = -0.5P^3 - 1.7P + 3.4.$$
- The function $f(P)$ has one real zero, as shown in Figure 2.R.3. Without attempting to solve the differential equation, estimate the value of $\lim_{t \rightarrow \infty} P(t)$.

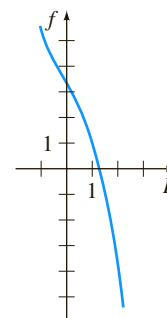


FIGURE 2.R.3 Graph for Problem 16

17. Figure 2.R.4 is a portion of a direction field of a differential equation $dy/dx = f(x, y)$. By hand, sketch two different solution curves—one that is tangent to the lineal element shown in black and one that is tangent to the lineal element shown in red.

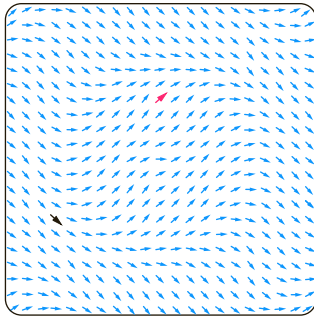


FIGURE 2.R.4 Portion of a direction field for Problem 17

18. Classify each differential equation as separable, exact, linear, homogeneous, or Bernoulli. Some equations may be more than one kind. Do not solve.

(a) $\frac{dy}{dx} = \frac{x-y}{x}$

(b) $\frac{dy}{dx} = \frac{1}{y-x}$

(c) $(x+1)\frac{dy}{dx} = -y+10$

(d) $\frac{dy}{dx} = \frac{1}{x(x-y)}$

(e) $\frac{dy}{dx} = \frac{y^2+y}{x^2+x}$

(f) $\frac{dy}{dx} = 5y+y^2$

(g) $y dx = (y - xy^2) dy$

(h) $x \frac{dy}{dx} = ye^{x/y} - x$

(i) $xy' + y^2 = 2x$

(j) $2xy' + y^2 = 2x^2$

(k) $y dx + x dy = 0$

(l) $\left(x^2 + \frac{2y}{x}\right) dx = (3 - \ln x^2) dy$

(m) $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x} + 1$

(n) $\frac{y}{x^2} \frac{dy}{dx} + e^{2x^3+y^2} = 0$

In Problems 19–26 solve the given differential equation.

19. $(y^2 + 1) dx = y \sec^2 x dy$

20. $y(\ln x - \ln y) dx = (x \ln x - x \ln y - y) dy$

21. $(6x+1)y^2 \frac{dy}{dx} + 3x^2 + 2y^3 = 0$

22. $\frac{dx}{dy} = -\frac{4y^2 + 6xy}{3y^2 + 2x}$

23. $t \frac{dQ}{dt} + Q = t^4 \ln t$

24. $(2x + y + 1)y' = 1$

25. $(x^2 + 4) dy = (2x - 8xy) dx$

26. $(2r^2 \cos \theta \sin \theta + r \cos \theta) d\theta + (4r + \sin \theta - 2r \cos^2 \theta) dr = 0$

In Problems 27–30 express the solution of the given initial-value problem in terms of an integral-defined function.

27. $2 \frac{dy}{dx} + (4 \cos x)y = x, \quad y(0) = 1$

28. $\frac{dy}{dx} - 4xy = \sin x^2, \quad y(0) = 7$

29. $x \frac{dy}{dx} + 2y = xe^{x^2}, \quad y(1) = 3$

30. $x \frac{dy}{dx} + (\sin x)y = 0, \quad y(0) = 10$

In Problems 31 and 32 solve the given initial-value problem.

31. $\frac{dy}{dx} + y = f(x), \quad y(0) = 5, \text{ where}$

$$f(x) = \begin{cases} e^{-x}, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

32. $\frac{dy}{dx} + P(x)y = e^x, \quad y(0) = -1, \text{ where}$

$$P(x) = \begin{cases} 1, & 0 \leq x < 1 \\ -1, & x \geq 1 \end{cases}$$

In Problems 33 and 34 solve the given initial-value problem and give the largest interval I on which the solution is defined.

33. $\sin x \frac{dy}{dx} + (\cos x)y = 0, \quad y(7\pi/6) = -2$

34. $\frac{dy}{dt} + 2(t+1)y^2 = 0, \quad y(0) = -\frac{1}{8}$

35. (a) Without solving, explain why the initial-value problem

$$\frac{dy}{dx} = \sqrt{y}, \quad y(x_0) = y_0$$

has no solution for $y_0 < 0$.

- (b) Solve the initial-value problem in part (a) for $y_0 > 0$ and find the largest interval I on which the solution is defined.

36. (a) Find an implicit solution of the initial-value problem

$$\frac{dy}{dx} = \frac{y^2 - x^2}{xy}, \quad y(1) = -\sqrt{2}.$$

- (b) Find an explicit solution of the problem in part (a) and give the largest interval I over which the solution is defined. A graphing utility may be helpful here.

37. Graphs of some members of a family of solutions for a first-order differential equation $dy/dx = f(x, y)$ are shown in Figure 2.R.5. The graphs of two implicit solutions, one that passes through the point $(1, -1)$ and one that passes through $(-1, 3)$, are shown in blue. Reproduce the figure on a piece of paper. With colored pencils trace out the solution curves for the solutions $y = y_1(x)$ and $y = y_2(x)$ defined by the implicit solutions such that

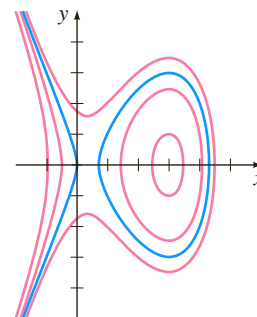


FIGURE 2.R.5 Graph for Problem 37

$y_1(1) = -1$ and $y_2(-1) = 3$, respectively. Estimate the intervals on which the solutions $y = y_1(x)$ and $y = y_2(x)$ are defined.

38. Use Euler's method with step size $h = 0.1$ to approximate $y(1.2)$, where $y(x)$ is a solution of the initial-value problem $y' = 1 + x\sqrt{y}$, $y(1) = 9$.

In Problems 39 and 40 each figure represents a portion of a direction field of an autonomous first-order differential equation $dy/dx = f(y)$. Reproduce the figure on a separate piece of paper and then complete the direction field over the grid. The points of the grid are (mh, nh) , where $h = \frac{1}{2}$, m and n integers, $-7 \leq m \leq 7$, $-7 \leq n \leq 7$. In each direction field, sketch by hand an approximate solution curve that passes through each of the solid points shown in red. Discuss: Does it appear that the DE possesses critical points in the interval $-3.5 \leq y \leq 3.5$? If so, classify the critical points as asymptotically stable, unstable, or semi-stable.

39.

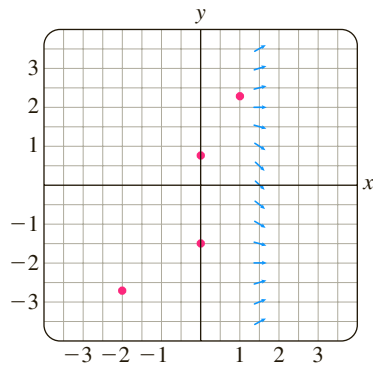


FIGURE 2.R.6 Portion of a direction field for Problem 39

40.

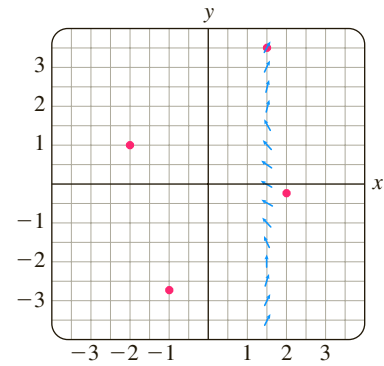


FIGURE 2.R.7 Portion of a direction field for Problem 40

Modeling with First-Order Differential Equations



Fotos593/Shutterstock.com

- 3.1 Linear Models
- 3.2 Nonlinear Models
- 3.3 Modeling with Systems of First-Order DEs

CHAPTER 3 IN REVIEW

In Section 1.3 we saw how a first-order differential equation could be used as a mathematical model in the study of population growth, radioactive decay, cooling of bodies, mixtures, chemical reactions, fluid draining from a tank, velocity of a falling body, and current in a series circuit. Using the methods discussed in Chapter 2, we are now able to solve some of the linear DEs in Section 3.1 and nonlinear DEs in Section 3.2 that frequently appear in applications.

3.1 Linear Models

INTRODUCTION In this section we solve some of the linear first-order models that were introduced in Section 1.3.

GROWTH AND DECAY The initial-value problem

$$\frac{dx}{dt} = kx, \quad x(t_0) = x_0, \quad (1)$$

where k is a constant of proportionality, serves as a model for diverse phenomena involving either growth or decay. We saw in Section 1.3 that in biological applications the rate of growth of certain populations (bacteria, small animals) over short periods of time is proportional to the population present at time t . Knowing the population at some arbitrary initial time t_0 , we can then use the solution of (1) to predict the population in the future—that is, at times $t > t_0$. The constant of proportionality k in (1) can be determined from the solution of the initial-value problem, using a subsequent measurement of x at a time $t_1 > t_0$. In physics and chemistry (1) is seen in the form of a *first-order reaction*—that is, a reaction whose rate, or velocity, dx/dt is directly proportional to the amount x of a substance that is unconverted or remaining at time t . The decomposition, or decay, of U-238 (uranium) by radioactivity into Th-234 (thorium) is a first-order reaction.

EXAMPLE 1 Bacterial Growth

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

SOLUTION We first solve the differential equation in (1), with the symbol x replaced by P . With $t_0 = 0$ the initial condition is $P(0) = P_0$. We then use the empirical observation that $P(1) = \frac{3}{2}P_0$ to determine the constant of proportionality k .

Notice that the differential equation $dP/dt = kP$ is both separable and linear. When it is put in the standard form of a linear first-order DE,

$$\frac{dP}{dt} - kP = 0,$$

we can see by inspection that the integrating factor is e^{-kt} . Multiplying both sides of the equation by this term and integrating gives, in turn,

$$\frac{d}{dt} [e^{-kt}P] = 0 \quad \text{and} \quad e^{-kt}P = c.$$

Therefore $P(t) = ce^{kt}$. At $t = 0$ it follows that $P_0 = ce^0 = c$, so $P(t) = P_0e^{kt}$. At $t = 1$ we have $\frac{3}{2}P_0 = P_0e^k$ or $e^k = \frac{3}{2}$. From the last equation we get $k = \ln \frac{3}{2} = 0.4055$, so $P(t) = P_0e^{0.4055t}$. To find the time at which the number of bacteria has tripled, we solve $3P_0 = P_0e^{0.4055t}$ for t . It follows that $0.4055t = \ln 3$, or

$$t = \frac{\ln 3}{0.4055} \approx 2.71 \text{ h}.$$

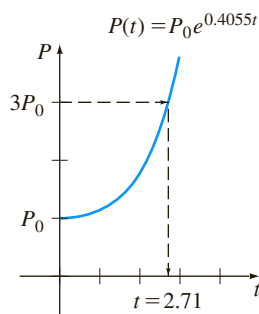


FIGURE 3.1.1 Time in which population triples in Example 1

See Figure 3.1.1.

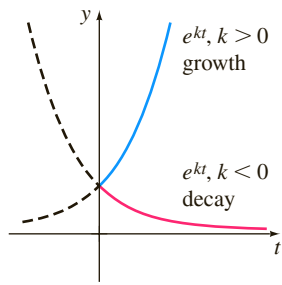


FIGURE 3.1.2 Growth ($k > 0$) and decay ($k < 0$)

Notice in Example 1 that the actual number P_0 of bacteria present at time $t = 0$ played no part in determining the time required for the number in the culture to triple. The time necessary for an initial population of, say, 100 or 1,000,000 bacteria to triple is still approximately 2.71 hours.

As shown in Figure 3.1.2, the exponential function e^{kt} increases as t increases for $k > 0$ and decreases as t increases for $k < 0$. Thus problems describing growth (whether of populations, bacteria, or even capital) are characterized by a positive value of k , whereas problems involving decay (as in radioactive disintegration) yield a negative k value. Accordingly, we say that k is either a **growth constant** ($k > 0$) or a **decay constant** ($k < 0$).

HALF-LIFE In physics the **half-life** is a measure of the stability of a radioactive substance. The half-life is simply the time it takes for one-half of the atoms in an initial amount A_0 to disintegrate, or transmute, into the atoms of another element. The longer the half-life of a substance, the more stable it is. For example, the half-life of highly radioactive radium, Ra-226, is about 1700 years. In 1700 years one-half of a given quantity of Ra-226 is transmuted into radon, Rn-222. The most commonly occurring uranium isotope, U-238, has a half-life of approximately 4,500,000,000 years. In about 4.5 billion years, one-half of a quantity of U-238 is transmuted into lead, Pb-206.

EXAMPLE 2 Half-Life of Plutonium

A breeder reactor converts relatively stable uranium-238 into the isotope plutonium-239. After 15 years it is determined that 0.043% of the initial amount A_0 of plutonium has disintegrated. Find the half-life of this isotope if the rate of disintegration is proportional to the amount remaining.

SOLUTION Let $A(t)$ denote the amount of plutonium remaining at time t . As in Example 1 the solution of the initial-value problem

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

is $A(t) = A_0 e^{kt}$. If 0.043% of the atoms of A_0 have disintegrated, then 99.957% of the substance remains. To find the decay constant k , we use $0.99957A_0 = A(15)$ —that is, $0.99957A_0 = A_0 e^{15k}$. Solving for k then gives $k = \frac{1}{15} \ln 0.99957 = -0.00002867$. Hence $A(t) = A_0 e^{-0.00002867t}$. Now the half-life is the corresponding value of time at which $A(t) = \frac{1}{2}A_0$. Solving for t gives $\frac{1}{2}A_0 = A_0 e^{-0.00002867t}$, or $\frac{1}{2} = e^{-0.00002867t}$. The last equation yields

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ yr.}$$



Jack Fields/Science Source

FIGURE 3.1.3 Willard Libby (1908–1980)

CARBON DATING About 1950, a team of scientists at the University of Chicago led by the chemist **Willard Libby** devised a method using a radioactive isotope of carbon as a means of determining the approximate ages of carbonaceous fossilized matter. See Figure 3.1.3. The theory of **carbon dating** is based on the fact that the radioisotope carbon-14 is produced in the atmosphere by the action of cosmic radiation on nitrogen-14. The ratio of the amount of C-14 to the stable C-12 in the atmosphere appears to be a constant, and as a consequence the proportionate amount of the isotope present in all living organisms is the same as that in the atmosphere. When a living organism dies, the absorption of C-14, by breathing, eating, or photosynthesis, ceases. By comparing the proportionate amount of C-14, say, in a fossil with the constant amount ratio found in the atmosphere, it is possible to obtain a reasonable estimation of its age. The method is based on the knowledge of the



Kenneth Garrett/National Geographic Creative

FIGURE 3.1.4 A page of the Gnostic Gospel of Judas

The size and location of the sample caused major difficulties when a team of scientists were invited to use carbon-14 dating on the Shroud of Turin in 1988.



The half-life of uranium-238 is about 4.47 billion years.



half-life of C-14. Libby's calculated value of the half-life of C-14 was approximately 5600 years, and is called the **Libby half-life**. Today the commonly accepted value for the half-life of C-14 is the **Cambridge half-life** that is close to 5730 years. For his work, Libby was awarded the Nobel Prize for chemistry in 1960. Libby's method has been used to date wooden furniture found in Egyptian tombs, the woven flax wrappings of the Dead Sea Scrolls, a recently discovered copy of the Gnostic Gospel of Judas written on papyrus, and the cloth of the enigmatic Shroud of Turin. See Figure 3.1.4 and Problem 12 in Exercises 3.1.

EXAMPLE 3 Age of a Fossil

A fossilized bone is found to contain 0.1% of its original amount of C-14. Determine the age of the fossil.

SOLUTION As in Example 2 the starting point is $A(t) = A_0 e^{kt}$. To determine the value of the decay constant k we use the fact that $\frac{1}{2} A_0 = A(5730)$ or $\frac{1}{2} A_0 = A_0 e^{5730k}$. The last equation implies $5730k = \ln \frac{1}{2} = -\ln 2$ and so we get $k = -(\ln 2)/5730 = -0.00012097$. Therefore $A(t) = A_0 e^{-0.00012097t}$. With $A(t) = 0.001A_0$ we have $0.001A_0 = A_0 e^{-0.00012097t}$ and $-0.00012097t = \ln(0.001) = -\ln 1000$. Thus

$$t = \frac{\ln 1000}{0.00012097} \approx 57,100 \text{ years.}$$

The age found in Example 3 is really at the border of accuracy of this method. The usual carbon-14 technique is limited to about 10 half-lives of the isotope, or roughly 60,000 years. One fundamental reason for this limitation is the relatively short half-life of C-14. There are other problems as well: The chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point $0.001A_0$. Moreover, this analysis requires the destruction of a rather large sample of the specimen. If this measurement is accomplished indirectly, based on the actual radioactivity of the specimen, then it is very difficult to distinguish between the radiation from the specimen and the normal background radiation. But recently the use of a particle accelerator has enabled scientists to separate the C-14 from the stable C-12 directly. When the precise value of the ratio of C-14 to C-12 is computed, the accuracy of the dating method can be extended to 70,000–100,000 years. For these reasons and the fact that the C-14 dating is restricted to organic materials, this method is used mainly by archeologists. On the other hand, geologists who are interested in questions about the age of rocks or the age of the Earth use **radiometric-dating** techniques. Radiometric dating, invented by the physicist chemist **Ernest Rutherford** (1871–1937) around 1905, is based on the radioactive decay of a naturally occurring radioactive isotope with a very long half-life and a comparison between a measured quantity of this decaying isotope and one of its decay products using known decay rates. Radiometric methods using potassium-argon, rubidium-strontium, or uranium-lead can give ages of certain kinds of rocks of several billion years. See Problems 5 and 6 in Exercises 3.3 for a brief discussion of the **potassium-argon method of dating**.

NEWTON'S LAW OF COOLING/WARMING In equation (3) of Section 1.3 we saw that the mathematical formulation of Newton's empirical law of cooling/warming of an object is given by the linear first-order differential equation

$$\frac{dT}{dt} = k(T - T_m), \quad (2)$$

where k is a constant of proportionality, $T(t)$ is the temperature of the object for $t > 0$, and T_m is the ambient temperature—that is, the temperature of the medium around the object. In Example 4 we assume that T_m is constant.

EXAMPLE 4 Cooling of a Cake

When a cake is removed from an oven, its temperature is measured at 300° F. Three minutes later its temperature is 200° F. How long will it take for the cake to cool off to a room temperature of 70° F?

SOLUTION In (2) we make the identification $T_m = 70$. We must then solve the initial-value problem

$$\frac{dT}{dt} = k(T - 70), \quad T(0) = 300 \quad (3)$$

and determine the value of k so that $T(3) = 200$.

Equation (3) is both linear and separable. If we separate variables,

$$\frac{dT}{T - 70} = k dt,$$

yields $\ln |T - 70| = kt + c_1$, and so $T = 70 + c_2 e^{kt}$. When $t = 0$, $T = 300$, so $300 = 70 + c_2$ gives $c_2 = 230$; therefore $T = 70 + 230e^{kt}$. Finally, the measurement $T(3) = 200$ leads to $e^{3k} = \frac{13}{23}$, or $k = \frac{1}{3} \ln \frac{13}{23} = -0.19018$. Thus

$$T(t) = 70 + 230e^{-0.19018t}. \quad (4)$$

We note that (4) furnishes no finite solution to $T(t) = 70$, since $\lim_{t \rightarrow \infty} T(t) = 70$. Yet we intuitively expect the cake to reach room temperature after a reasonably long period of time. How long is “long”? Of course, we should not be disturbed by the fact that the model (3) does not quite live up to our physical intuition. Parts (a) and (b) of Figure 3.1.5 clearly show that the cake will be approximately at room temperature in about one-half hour.

The ambient temperature in (2) need not be a constant but could be a function $T_m(t)$ of time t . See Problem 18 in Exercises 3.1.

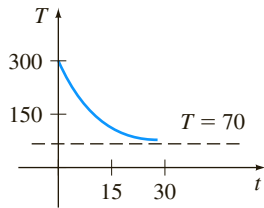
MIXTURES The mixing of two fluids sometimes gives rise to a linear first-order differential equation. When we discussed the mixing of two brine solutions in Section 1.3, we assumed that the rate $A'(t)$ at which the amount of salt in the mixing tank changes was a net rate:

$$\frac{dA}{dt} = (\text{input rate of salt}) - (\text{output rate of salt}) = R_{in} - R_{out}. \quad (5)$$

In Example 5 we solve equation (8) on page 25 of Section 1.3.

EXAMPLE 5 Mixture of Two Salt Solutions

Recall that the large tank considered in Section 1.3 held 300 gallons of a brine solution. Salt was entering and leaving the tank; a brine solution was being pumped into the tank at the rate of 3 gal/min; it mixed with the solution there, and then the mixture was pumped out at the rate of 3 gal/min. The concentration of the salt in the inflow, or solution entering, was 2 lb/gal, so salt was entering the tank at the rate $R_{in} = (2 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = 6 \text{ lb/min}$ and leaving the tank at the rate $R_{out} = (A/300 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = A/100 \text{ lb/min}$. From this data and (5) we get equation (8) of Section 1.3. Let us pose the question: If 50 pounds of salt were dissolved initially in the 300 gallons, how much salt is in the tank after a long time?

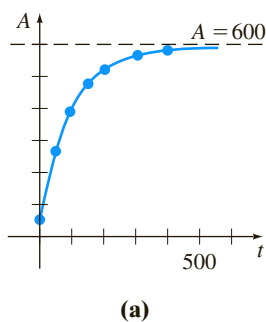


(a)

$T(t)$	t (min)
75°	20.1
74°	21.3
73°	22.8
72°	24.9
71°	28.6
70.5°	32.3

(b)

FIGURE 3.1.5 Temperature of cooling cake in Example 4



t (min)	A (lb)
50	266.41
100	397.67
150	477.27
200	525.57
300	572.62
400	589.93

(b)

FIGURE 3.1.6 Pounds of salt in the tank in Example 5

SOLUTION To find the amount of salt $A(t)$ in the tank at time t , we solve the initial-value problem

$$\frac{dA}{dt} + \frac{1}{100}A = 6, \quad A(0) = 50.$$

Note here that the side condition is the initial amount of salt $A(0) = 50$ in the tank and *not* the initial amount of liquid in the tank. Now since the integrating factor of the linear differential equation is $e^{t/100}$, we can write the equation as

$$\frac{d}{dt}[e^{t/100}A] = 6e^{t/100}.$$

Integrating the last equation and solving for A gives the general solution $A(t) = 600 + ce^{-t/100}$. When $t = 0$, $A = 50$, so we find that $c = -550$. Thus the amount of salt in the tank at time t is given by

$$A(t) = 600 - 550e^{-t/100}. \quad (6)$$

The solution (6) was used to construct the table in Figure 3.1.6(b). Also, it can be seen from (6) and Figure 3.1.6(a) that $A(t) \rightarrow 600$ as $t \rightarrow \infty$. Of course, this is what we would intuitively expect; over a long time the number of pounds of salt in the solution must be $(300 \text{ gal})(2 \text{ lb/gal}) = 600 \text{ lb}$. ■

In Example 5 we assumed that the rate at which the solution was pumped in was the same as the rate at which the solution was pumped out. However, this need not be the case; the mixed brine solution could be pumped out at a rate r_{out} that is faster or slower than the rate r_{in} at which the other brine solution is pumped in. The next example illustrates the case when the mixture is pumped out at rate that is *slower* than the rate at which the brine solution is being pumped into the tank.

EXAMPLE 6 Example 5 Revisited

If the well-stirred solution in Example 5 is pumped out at a slower rate of, say, $r_{out} = 2$ gal/min, then liquid will accumulate in the tank at the rate of

$$r_{in} - r_{out} = (3 - 2) \text{ gal/min} = 1 \text{ gal/min}.$$

After t minutes,

$$(1 \text{ gal/min}) \cdot (t \text{ min}) = t \text{ gal}$$

will accumulate, so the tank will contain $300 + t$ gallons of brine. The concentration of the outflow is then $c(t) = A/(300 + t)$ lb/gal, and the output rate of salt is $R_{out} = c(t) \cdot r_{out}$, or

$$R_{out} = \left(\frac{A}{300 + t} \text{ lb/gal} \right) \cdot (2 \text{ gal/min}) = \frac{2A}{300 + t} \text{ lb/min}.$$

Hence equation (5) becomes

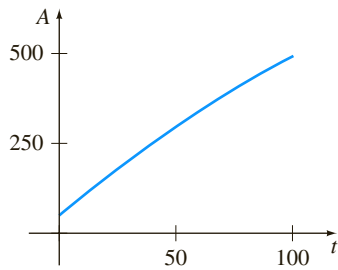
$$\frac{dA}{dt} = 6 - \frac{2A}{300 + t} \quad \text{or} \quad \frac{dA}{dt} + \frac{2}{300 + t}A = 6.$$

The integrating factor for the last equation is

$$e^{\int 2 dt/(300+t)} = e^{2 \ln(300+t)} = e^{\ln(300+t)^2} = (300 + t)^2$$

and so after multiplying by the factor the equation is cast into the form

$$\frac{d}{dt}[(300 + t)^2 A] = 6(300 + t)^2.$$

FIGURE 3.1.7 Graph of $A(t)$ in Example 6

Integrating the last equation gives $(300 + t)^2 A = 2(300 + t)^3 + c$. By applying the initial condition $A(0) = 50$ and solving for A yields the solution $A(t) = 600 + 2t - (4.95 \times 10^7)(300 + t)^{-2}$. As Figure 3.1.7 shows, not unexpectedly, salt builds up in the tank over time, that is, $A \rightarrow \infty$ as $t \rightarrow \infty$. ■

SERIES CIRCUITS For a series circuit containing only a resistor and an inductor, Kirchhoff's second law states that the sum of the voltage drop across the inductor ($L(di/dt)$) and the voltage drop across the resistor (iR) is the same as the impressed voltage ($E(t)$) on the circuit. See Figure 3.1.8.

Thus we obtain the linear differential equation for the current $i(t)$,

$$L \frac{di}{dt} + Ri = E(t), \quad (7)$$

where L and R are known as the inductance and the resistance, respectively. The current $i(t)$ is also called the **response** of the system.

The voltage drop across a capacitor with capacitance C is given by $q(t)/C$, where q is the charge on the capacitor. Hence, for the series circuit shown in Figure 3.1.9, Kirchhoff's second law gives

$$Ri + \frac{1}{C}q = E(t). \quad (8)$$

But current i and charge q are related by $i = dq/dt$, so (8) becomes the linear differential equation

$$R \frac{dq}{dt} + \frac{1}{C}q = E(t). \quad (9)$$

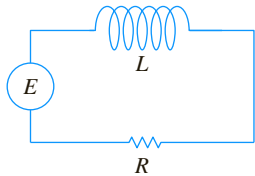


FIGURE 3.1.8 LR-series circuit

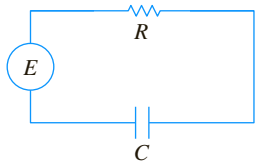


FIGURE 3.1.9 RC-series circuit

EXAMPLE 7 LR-Series Circuit

A 12-volt battery is connected to a series circuit in which the inductance is $\frac{1}{2}$ henry and the resistance is 10 ohms. Determine the current i if the initial current is zero.

SOLUTION From (7) we see that we must solve the equation

$$\frac{1}{2} \frac{di}{dt} + 10i = 12,$$

subject to $i(0) = 0$. First, we multiply the differential equation by 2 and read off the integrating factor e^{20t} . We then obtain

$$\frac{d}{dt} [e^{20t}i] = 24e^{20t}.$$

Integrating each side of the last equation and solving for i gives $i(t) = \frac{6}{5} + ce^{-20t}$. Now $i(0) = 0$ implies that $0 = \frac{6}{5} + c$ or $c = -\frac{6}{5}$. Therefore the response is $i(t) = \frac{6}{5} - \frac{6}{5}e^{-20t}$. ■

From (4) of Section 2.3 we can write a general solution of (7):

$$i(t) = \frac{e^{-(R/L)t}}{L} \int e^{(R/L)t} E(t) dt + ce^{-(R/L)t}. \quad (10)$$

In particular, when $E(t) = E_0$ is a constant, (10) becomes

$$i(t) = \frac{E_0}{R} + ce^{-(R/L)t}. \quad (11)$$

Note that as $t \rightarrow \infty$, the second term in equation (11) approaches zero. Such a term is usually called a **transient term**; any remaining terms are called the **steady-state** part of the solution. In this case E_0/R is also called the **steady-state current**; for large values of time it appears that the current in the circuit is simply governed by Ohm's law ($E = iR$).

REMARKS

The solution $P(t) = P_0 e^{0.4055t}$ of the initial-value problem in Example 1 described the population of a colony of bacteria at any time $t \geq 0$. Of course, $P(t)$ is a continuous function that takes on *all* real numbers in the interval $P_0 \leq P < \infty$. But since we are talking about a population, common sense dictates that P can take on only positive integer values. Moreover, we would not expect the population to grow continuously—that is, every second, every microsecond, and so on—as predicted by our solution; there may be intervals of time $[t_1, t_2]$ over which there is no growth at all. Perhaps, then, the graph shown in Figure 3.1.10(a) is a more realistic description of P than is the graph of an exponential function. Using a continuous function to describe a discrete phenomenon is often more a matter of convenience than of accuracy. However, for some purposes we may be satisfied if our model describes the system fairly closely when viewed macroscopically in time, as in Figures 3.1.10(b) and 3.1.10(c), rather than microscopically, as in Figure 3.1.10(a).

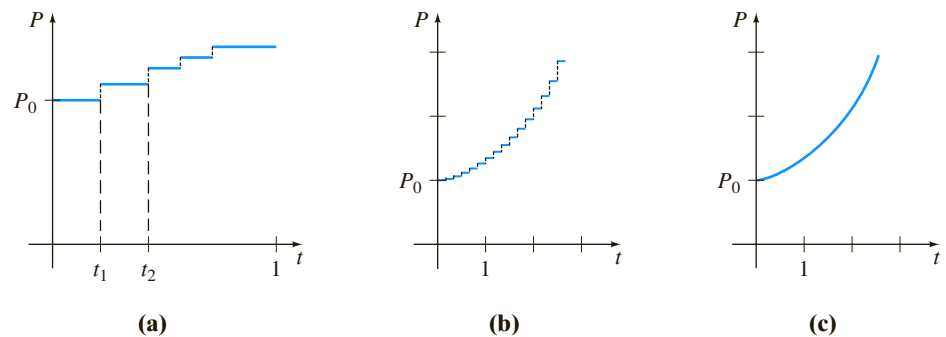


FIGURE 3.1.10 Population growth is a discrete process

EXERCISES 3.1

Answers to selected odd-numbered problems begin on page ANS-3.

Growth and Decay

- The population of a community is known to increase at a rate proportional to the number of people present at time t . If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?
- Suppose it is known that the population of the community in Problem 1 is 10,000 after 3 years. What was the initial population P_0 ? What will be the population in 10 years? How fast is the population growing at $t = 10$?
- The population of a town grows at a rate proportional to the population present at time t . The initial population of 500 increases by 15% in 10 years. What will be the population in 30 years? How fast is the population growing at $t = 30$?
- The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t . After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?
- The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?
- Initially 100 milligrams of a radioactive substance was present. After 6 hours the mass had decreased by 3%. If the rate of decay is proportional to the amount of the substance present at time t , find the amount remaining after 24 hours.

7. Determine the half-life of the radioactive substance described in Problem 6.
8. (a) Consider the initial-value problem $dA/dt = kA$, $A(0) = A_0$ as the model for the decay of a radioactive substance. Show that, in general, the half-life T of the substance is $T = -(\ln 2)/k$.
- (b) Show that the solution of the initial-value problem in part (a) can be written $A(t) = A_0 2^{-t/T}$.
- (c) If a radioactive substance has the half-life T given in part (a), how long will it take an initial amount A_0 of the substance to decay to $\frac{1}{8}A_0$?
9. When a vertical beam of light passes through a transparent medium, the rate at which its intensity I decreases is proportional to $I(t)$, where t represents the thickness of the medium (in feet). In clear seawater, the intensity 3 feet below the surface is 25% of the initial intensity I_0 of the incident beam. What is the intensity of the beam 15 feet below the surface?
10. When interest is compounded continuously, the amount of money increases at a rate proportional to the amount S present at time t , that is, $dS/dt = rS$, where r is the annual rate of interest.
- (a) Find the amount of money accrued at the end of 5 years when \$5000 is deposited in a savings account drawing $5\frac{3}{4}\%$ annual interest compounded continuously.
- (b) In how many years will the initial sum deposited have doubled?
- (c) Use a calculator to compare the amount obtained in part (a) with the amount $S = 5000(1 + \frac{1}{4}(0.0575))^{5(4)}$ that is accrued when interest is compounded quarterly.

Carbon Dating

11. Archaeologists used pieces of burned wood, or charcoal, found at the site to date prehistoric paintings and drawings on walls and ceilings of a cave in Lascaux, France. See Figure 3.1.11. Use the information on page 87 to determine the approximate age of a piece of burned wood, if it was found that 85.5% of the C-14 found in living trees of the same type had decayed.



Rock painting showing a horse and a cow, c. 17,000 BC (cave painting). Prehistoric/Caves of Lascaux, Dordogne, France/Bridgeman Images

FIGURE 3.1.11 Cave wall painting in Problem 11

12. The Shroud of Turin, which shows the negative image of the body of a man who appears to have been crucified, is believed by many to be the burial shroud of Jesus of Nazareth. See Figure 3.1.12. In 1988 the Vatican granted permission to have the shroud carbon-dated. Three independent scientific laboratories analyzed the cloth and concluded that the shroud was approximately 660 years old, an age consistent with its historical appearance. Using this age, determine what percentage of the original amount of C-14 remained in the cloth as of 1988.



Source: Wikipedia.org

FIGURE 3.1.12 Shroud image in Problem 12

Newton's Law of Cooling/Warming

13. A thermometer is removed from a room where the temperature is 70°F and is taken outside, where the air temperature is 10°F . After one-half minute the thermometer reads 50°F . What is the reading of the thermometer at $t = 1$ min? How long will it take for the thermometer to reach 15°F ?
14. A thermometer is taken from an inside room to the outside, where the air temperature is 5°F . After 1 minute the thermometer reads 55°F , and after 5 minutes it reads 30°F . What is the initial temperature of the inside room?
15. A small metal bar, whose initial temperature was 20°C , is dropped into a large container of boiling water. How long will it take the bar to reach 90°C if it is known that its temperature increases 2° in 1 second? How long will it take the bar to reach 98°C ?
16. Two large containers A and B of the same size are filled with different fluids. The fluids in containers A and B are maintained at 0°C and 100°C , respectively. A small metal bar, whose initial temperature is 100°C , is lowered into container A . After 1 minute the temperature of the bar is 90°C . After 2 minutes the bar is removed and instantly transferred to the other container. After 1 minute in container B the temperature of the bar rises 10° . How long, measured from the start of the entire process, will it take the bar to reach 99.9°C ?
17. A thermometer reading 70°F is placed in an oven preheated to a constant temperature. Through a glass window in the oven door, an observer records that the thermometer reads 110°F after $\frac{1}{2}$ minute and 145°F after 1 minute. How hot is the oven?
18. At $t = 0$ a sealed test tube containing a chemical is immersed in a liquid bath. The initial temperature of the chemical in the test tube is 80°F . The liquid bath has a controlled

temperature (measured in degrees Fahrenheit) given by $T_m(t) = 100 - 40e^{-0.1t}$, $t \geq 0$, where t is measured in minutes.

- (a) Assume that $k = -0.1$ in (2). Before solving the IVP, describe in words what you expect the temperature $T(t)$ of the chemical to be like in the short term. In the long term.
 - (b) Solve the initial-value problem. Use a graphing utility to plot the graph of $T(t)$ on time intervals of various lengths. Do the graphs agree with your predictions in part (a)?
19. A dead body was found within a closed room of a house where the temperature was a constant 70°F . At the time of discovery the core temperature of the body was determined to be 85°F . One hour later a second measurement showed that the core temperature of the body was 80°F . Assume that the time of death corresponds to $t = 0$ and that the core temperature at that time was 98.6°F . Determine how many hours elapsed before the body was found. [Hint: Let $t_1 > 0$ denote the time that the body was discovered.]
 20. The rate at which a body cools also depends on its exposed surface area S . If S is a constant, then a modification of (2) is

$$\frac{dT}{dt} = kS(T - T_m),$$

where $k < 0$ and T_m is a constant. Suppose that two cups A and B are filled with coffee at the same time. Initially, the temperature of the coffee is 150°F . The exposed surface area of the coffee in cup B is twice the surface area of the coffee in cup A . After 30 min the temperature of the coffee in cup A is 100°F . If $T_m = 70^\circ\text{F}$, then what is the temperature of the coffee in cup B after 30 min?

Mixtures

21. A tank contains 200 liters of fluid in which 30 grams of salt is dissolved. Brine containing 1 gram of salt per liter is then pumped into the tank at a rate of 4 L/min; the well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of grams of salt in the tank at time t .
22. Solve Problem 21 assuming that pure water is pumped into the tank.
23. A large tank is filled to capacity with 500 gallons of pure water. Brine containing 2 pounds of salt per gallon is pumped into the tank at a rate of 5 gal/min. The well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of pounds of salt in the tank at time t .
24. In Problem 23, what is the concentration $c(t)$ of the salt in the tank at time t ? At $t = 5$ min? What is the concentration of the salt in the tank after a long time, that is, as $t \rightarrow \infty$? At what time is the concentration of the salt in the tank equal to one-half this limiting value?
25. Solve Problem 23 under the assumption that the solution is pumped out at a faster rate of 10 gal/min. When is the tank empty?
26. Determine the amount of salt in the tank at time t in Example 5 if the concentration of salt in the inflow is variable and given by $c_{in}(t) = 2 + \sin(t/4)$ lb/gal. Without actually graphing, conjecture what the solution curve of the IVP should look like. Then use a graphing utility to plot the graph of the solution on the interval $[0, 300]$. Repeat for the interval $[0, 600]$ and compare your graph with that in Figure 3.1.6(a).
27. A large tank is partially filled with 100 gallons of fluid in which 10 pounds of salt is dissolved. Brine containing $\frac{1}{2}$ pound of salt per gallon is pumped into the tank at a rate of 6 gal/min. The well-mixed solution is then pumped out at a slower rate of 4 gal/min. Find the number of pounds of salt in the tank after 30 minutes.
28. In Example 5 the size of the tank containing the salt mixture was not given. Suppose, as in the discussion following Example 5, that the rate at which brine is pumped into the tank is 3 gal/min but that the well-stirred solution is pumped out at a rate of 2 gal/min. It stands to reason that since brine is accumulating in the tank at the rate of 1 gal/min, any finite tank must eventually overflow. Now suppose that the tank has an open top and has a total capacity of 400 gallons.
 - (a) When will the tank overflow?
 - (b) What will be the number of pounds of salt in the tank at the instant it overflows?
 - (c) Assume that although the tank is overflowing, brine solution continues to be pumped in at a rate of 3 gal/min and the well-stirred solution continues to be pumped out at a rate of 2 gal/min. Devise a method for determining the number of pounds of salt in the tank at $t = 150$ minutes.
 - (d) Determine the number of pounds of salt in the tank as $t \rightarrow \infty$. Does your answer agree with your intuition?
 - (e) Use a graphing utility to plot the graph of $A(t)$ on the interval $[0, 500]$.

Series Circuits

29. A 30-volt electromotive force is applied to an LR -series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current $i(t)$ if $i(0) = 0$. Determine the current as $t \rightarrow \infty$.
30. Solve equation (7) under the assumption that $E(t) = E_0 \sin \omega t$ and $i(0) = i_0$.
31. A 100-volt electromotive force is applied to an RC -series circuit in which the resistance is 200 ohms and the capacitance is 10^{-4} farad. Find the charge $q(t)$ on the capacitor if $q(0) = 0$. Find the current $i(t)$.
32. A 200-volt electromotive force is applied to an RC -series circuit in which the resistance is 1000 ohms and the capacitance is 5×10^{-6} farad. Find the charge $q(t)$ on the capacitor if $i(0) = 0.4$. Determine the charge and current at $t = 0.005$ s. Determine the charge as $t \rightarrow \infty$.
33. An electromotive force

$$E(t) = \begin{cases} 120, & 0 \leq t \leq 20 \\ 0, & t > 20 \end{cases}$$
 is applied to an LR -series circuit in which the inductance is 20 henries and the resistance is 2 ohms. Find the current $i(t)$ if $i(0) = 0$.
34. An LR -series circuit has a variable inductor with the inductance defined by

$$L(t) = \begin{cases} 1 - \frac{1}{10}t, & 0 \leq t < 10 \\ 0, & t > 10. \end{cases}$$

Find the current $i(t)$ if the resistance is 0.2 ohm, the impressed voltage is $E(t) = 4$, and $i(0) = 0$. Graph $i(t)$.

Additional Linear Models

35. Air Resistance In (14) of Section 1.3 we saw that a differential equation describing the velocity v of a falling mass subject to air resistance proportional to the instantaneous velocity is

$$m \frac{dv}{dt} = mg - kv,$$

where $k > 0$ is a constant of proportionality. The positive direction is downward.

- Solve the equation subject to the initial condition $v(0) = v_0$.
- Use the solution in part (a) to determine the limiting, or terminal, velocity of the mass. We saw how to determine the terminal velocity without solving the DE in Problem 40 in Exercises 2.1.
- If the distance s , measured from the point where the mass was released above ground, is related to velocity v by $ds/dt = v(t)$, find an explicit expression for $s(t)$ if $s(0) = 0$.

36. How High?—No Air Resistance Suppose a small cannonball weighing 16 pounds is shot vertically upward, as shown in Figure 3.1.13, with an initial velocity $v_0 = 300$ ft/s. The answer to the question “How high does the cannonball go?” depends on whether we take air resistance into account.

- Suppose air resistance is ignored. If the positive direction is upward, then a model for the state of the cannonball is given by $d^2s/dt^2 = -g$ (equation (12) of Section 1.3). Since $ds/dt = v(t)$ the last differential equation is the same as $dv/dt = -g$, where we take $g = 32$ ft/s². Find the velocity $v(t)$ of the cannonball at time t .
- Use the result obtained in part (a) to determine the height $s(t)$ of the cannonball measured from ground level. Find the maximum height attained by the cannonball.

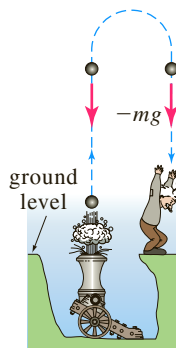


FIGURE 3.1.13 Find the maximum height of the cannonball in Problem 36

37. How High?—Linear Air Resistance Repeat Problem 36, but this time assume that air resistance is proportional to instantaneous velocity. It stands to reason that the maximum height attained by the cannonball must be *less* than that in part (b) of Problem 36. Show this by supposing that the constant of proportionality is $k = 0.0025$. [Hint: Slightly modify the differential equation in Problem 35.]

38. Skydiving A skydiver weighs 125 pounds, and her parachute and equipment combined weigh another 35 pounds. After

exiting from a plane at an altitude of 15,000 feet, she waits 15 seconds and opens her parachute. Assume that the constant of proportionality in the model in Problem 35 has the value $k = 0.5$ during free fall and $k = 10$ after the parachute is opened. Assume that her initial velocity on leaving the plane is zero. What is her velocity and how far has she traveled 20 seconds after leaving the plane? See Figure 3.1.14. How does her velocity at 20 seconds compare with her terminal velocity? How long does it take her to reach the ground? [Hint: Think in terms of two distinct IVPs.]

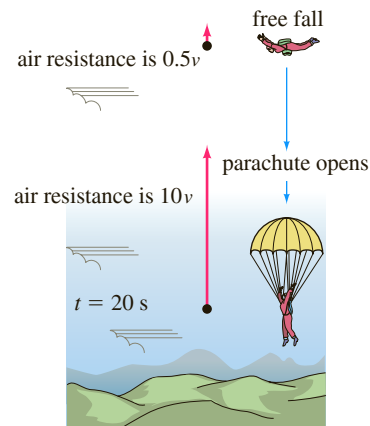


FIGURE 3.1.14 Find the time to reach the ground in Problem 38

39. Rocket Motion Suppose a small single-stage rocket of total mass $m(t)$ is launched vertically, the positive direction is upward, the air resistance is linear, and the rocket consumes its fuel at a constant rate. In Problem 22 of Exercises 1.3 you were asked to use Newton's second law of motion in the form given in (17) of that exercise set to show that a mathematical model for the velocity $v(t)$ of the rocket is given by

$$\frac{dv}{dt} + \frac{k - \lambda}{m_0 - \lambda t} v = -g + \frac{R}{m_0 - \lambda t},$$

where k is the air resistance constant of proportionality, λ is the constant rate at which fuel is consumed, R is the thrust of the rocket, $m(t) = m_0 - \lambda t$, m_0 is the total mass of the rocket at $t = 0$, and g is the acceleration due to gravity.

- Find the velocity $v(t)$ of the rocket if $m_0 = 200$ kg, $R = 2000$ N, $\lambda = 1$ kg/s, $g = 9.8$ m/s², $k = 3$ kg/s, and $v(0) = 0$.
- Use $ds/dt = v$ and the result in part (a) to find the height $s(t)$ of the rocket at time t .

40. Rocket Motion—Continued In Problem 39 suppose of the rocket's initial mass m_0 that 50 kg is the mass of the fuel.

- What is the burnout time t_b , or the time at which all the fuel is consumed?
- What is the velocity of the rocket at burnout?
- What is the height of the rocket at burnout?
- Why would you expect the rocket to attain an altitude higher than the number in part (b)?
- After burnout what is a mathematical model for the velocity of the rocket?

- 41. Evaporating Raindrop** As a raindrop falls, it evaporates while retaining its spherical shape. If we make the further assumptions that the rate at which the raindrop evaporates is proportional to its surface area and that air resistance is negligible, then a model for the velocity $v(t)$ of the raindrop is

$$\frac{dv}{dt} + \frac{3(k/\rho)}{(k/\rho)t + r_0}v = g.$$

Here ρ is the density of water, r_0 is the radius of the raindrop at $t = 0$, $k < 0$ is the constant of proportionality, and the downward direction is taken to be the positive direction.

- (a) Solve for $v(t)$ if the raindrop falls from rest.
- (b) Reread Problem 36 of Exercises 1.3 and then show that the radius of the raindrop at time t is $r(t) = (k/\rho)t + r_0$.
- (c) If $r_0 = 0.01$ ft and $r = 0.007$ ft 10 seconds after the raindrop falls from a cloud, determine the time at which the raindrop has evaporated completely.
- 42. Fluctuating Population** The differential equation $dP/dt = (k \cos t)P$, where k is a positive constant, is a mathematical model for a population $P(t)$ that undergoes yearly seasonal fluctuations. Solve the equation subject to $P(0) = P_0$. Use a graphing utility to graph the solution for different choices of P_0 .

- 43. Population Model** In one model of the changing population $P(t)$ of a community, it is assumed that

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt},$$

where dB/dt and dD/dt are the birth and death rates, respectively.

- (a) Solve for $P(t)$ if $dB/dt = k_1P$ and $dD/dt = k_2P$.
- (b) Analyze the cases $k_1 > k_2$, $k_1 = k_2$, and $k_1 < k_2$.
- 44. Constant-Harvest Model** A model that describes the population of a fishery in which harvesting takes place at a constant rate is given by

$$\frac{dP}{dt} = kP - h,$$

where k and h are positive constants.

- (a) Solve the DE subject to $P(0) = P_0$.
- (b) Describe the behavior of the population $P(t)$ for increasing time in the three cases $P_0 > h/k$, $P_0 = h/k$, and $0 < P_0 < h/k$.
- (c) Use the results from part (b) to determine whether the fish population will ever go extinct in finite time, that is, whether there exists a time $T > 0$ such that $P(T) = 0$. If the population goes extinct, then find T .
- 45. Drug Dissemination** A mathematical model for the rate at which a drug disseminates into the bloodstream is given by

$$\frac{dx}{dt} = r - kx,$$

where r and k are positive constants. The function $x(t)$ describes the concentration of the drug in the bloodstream at time t .

- (a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of $x(t)$ as $t \rightarrow \infty$.
- (b) Solve the DE subject to $x(0) = 0$. Sketch the graph of $x(t)$ and verify your prediction in part (a). At what time is the concentration one-half this limiting value?
- 46. Memorization** When forgetfulness is taken into account, the rate of memorization of a subject is given by

$$\frac{dA}{dt} = k_1(M - A) - k_2A,$$

where $k_1 > 0$, $k_2 > 0$, $A(t)$ is the amount memorized in time t , M is the total amount to be memorized, and $M - A$ is the amount remaining to be memorized.

- (a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of $A(t)$ as $t \rightarrow \infty$. Interpret the result.
- (b) Solve the DE subject to $A(0) = 0$. Sketch the graph of $A(t)$ and verify your prediction in part (a).
- 47. Heart Pacemaker** A heart pacemaker, shown in Figure 3.1.15, consists of a switch, a battery, a capacitor, and the heart as a resistor. When the switch S is at P , the capacitor charges; when S is at Q , the capacitor discharges, sending an electrical stimulus to the heart. In Problem 58 in Exercises 2.3 we saw that during this time the electrical stimulus is being applied to the heart, the voltage E across the heart satisfies the linear DE

$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

- (a) Let us assume that over the time interval of length t_1 , $0 < t < t_1$, the switch S is at position P shown in Figure 3.1.15 and the capacitor is being charged. When the switch is moved to position Q at time t_1 the capacitor discharges, sending an impulse to the heart over the time interval of length t_2 : $t_1 \leq t < t_1 + t_2$. Thus over the initial charging/discharging interval $0 < t < t_1 + t_2$ the voltage to the heart is actually modeled by the piecewise-linear differential equation

$$\frac{dE}{dt} = \begin{cases} 0, & 0 \leq t < t_1 \\ -\frac{1}{RC}E, & t_1 \leq t < t_1 + t_2. \end{cases}$$

By moving S between P and Q , the charging and discharging over time intervals of lengths t_1 and t_2 is

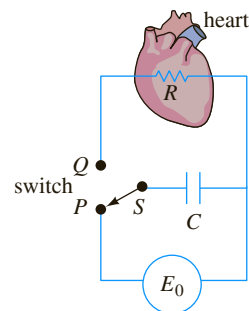


FIGURE 3.1.15 Model of a pacemaker in Problem 47

repeated indefinitely. Suppose $t_1 = 4$ s, $t_2 = 2$ s, $E_0 = 12$ V, and $E(0) = 0$, $E(4) = 12$, $E(6) = 0$, $E(10) = 12$, $E(12) = 0$, and so on. Solve for $E(t)$ for $0 \leq t \leq 24$.

- (b) Suppose for the sake of illustration that $R = C = 1$. Use a graphing utility to graph the solution for the IVP in part (a) for $0 \leq t \leq 24$.

48. Sliding Box (a) A box of mass m slides down an inclined plane that makes an angle θ with the horizontal as shown in Figure 3.1.16. Find a differential equation for the velocity $v(t)$ of the box at time t in each of the following three cases:

- (i) No sliding friction and no air resistance
- (ii) With sliding friction and no air resistance
- (iii) With sliding friction and air resistance

In cases (ii) and (iii), use the fact that the force of friction opposing the motion of the box is μN , where μ is the coefficient of sliding friction and N is the normal component of the weight of the box. In case (iii) assume that air resistance is proportional to the instantaneous velocity.

- (b) In part (a), suppose that the box weighs 96 pounds, that the angle of inclination of the plane is $\theta = 30^\circ$, that the coefficient of sliding friction is $\mu = \sqrt{3}/4$, and that the additional retarding force due to air resistance is numerically equal to $\frac{1}{4}v$. Solve the differential equation in each of the three cases, assuming that the box starts from rest from the highest point 50 ft above ground.

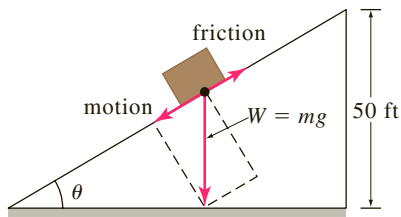


FIGURE 3.1.16 Box sliding down inclined plane in Problem 48

49. Sliding Box—Continued (a) In Problem 48 let $s(t)$ be the distance measured down the inclined plane from the highest point. Use $ds/dt = v(t)$ and the solution for each of the three cases in part (b) of Problem 48 to find the time that it takes the box to slide completely down the inclined plane. A root-finding application of a CAS may be useful here.

- (b) In the case in which there is friction ($\mu \neq 0$) but no air resistance, explain why the box will not slide down the plane starting from rest from the highest point above ground when the inclination angle θ satisfies $\tan \theta \leq \mu$.
- (c) The box will slide downward on the plane when $\tan \theta \leq \mu$ if it is given an initial velocity $v(0) = v_0 > 0$. Suppose that $\mu = \sqrt{3}/4$ and $\theta = 23^\circ$. Verify that $\tan \theta \leq \mu$. How far will the box slide down the plane if $v_0 = 1$ ft/s?
- (d) Using the values $\mu = \sqrt{3}/4$ and $\theta = 23^\circ$, approximate the smallest initial velocity v_0 that can be given to the box so that, starting at the highest point 50 ft above ground, it will slide completely down the inclined plane. Then find the corresponding time it takes to slide down the plane.

50. What Goes Up . . . (a) It is well known that the model in which air resistance is ignored, part (a) of Problem 36, predicts that the time t_a it takes the cannonball to attain its maximum height is the same as the time t_d it takes the cannonball to fall from the maximum height to the ground. Moreover, the magnitude of the impact velocity v_i will be the same as the initial velocity v_0 of the cannonball. Verify both of these results.

- (b) Then, using the model in Problem 37 that takes air resistance into account, compare the value of t_a with t_d and the value of the magnitude of v_i with v_0 . A root-finding application of a CAS (or graphic calculator) may be useful here.

3.2 Nonlinear Models

INTRODUCTION We finish our study of single first-order differential equations with an examination of some nonlinear models.

POPULATION DYNAMICS If $P(t)$ denotes the size of a population at time t , the model for exponential growth begins with the assumption that $dP/dt = kP$ for some $k > 0$. In this model, the **relative**, or **specific**, **growth rate** defined by

$$\frac{dP/dt}{P} \quad (1)$$

is a constant k . True cases of exponential growth over long periods of time are hard to find because the limited resources of the environment will at some time exert restrictions on the growth of a population. Thus for other models, (1) can be expected to decrease as the population P increases in size.

The assumption that the rate at which a population grows (or decreases) is dependent only on the number P present and not on any time-dependent mechanisms such as seasonal phenomena (see Problem 33 in Exercises 1.3) can be stated as

$$\frac{dP/dt}{P} = f(P) \quad \text{or} \quad \frac{dP}{dt} = Pf(P). \quad (2)$$

The differential equation in (2), which is widely assumed in models of animal populations, is called the **density-dependent hypothesis**.

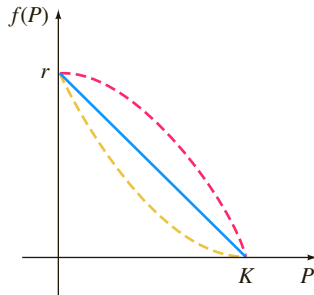


FIGURE 3.2.1 Simplest assumption for $f(P)$ is a straight line (blue color)

LOGISTIC EQUATION Suppose an environment is capable of sustaining no more than a fixed number K of individuals in its population. The quantity K is called the **carrying capacity** of the environment. Hence for the function f in (2) we have $f(K) = 0$, and we simply let $f(0) = r$. Figure 3.2.1 shows three functions f that satisfy these two conditions. The simplest assumption that we can make is that $f(P)$ is linear—that is, $f(P) = c_1P + c_2$. If we use the conditions $f(0) = r$ and $f(K) = 0$, we find, in turn, $c_2 = r$ and $c_1 = -r/K$, and so f takes on the form $f(P) = r - (r/K)P$. Equation (2) becomes

$$\frac{dP}{dt} = P \left(r - \frac{r}{K}P \right). \quad (3)$$

With constants relabeled, the nonlinear equation (3) is the same as

$$\frac{dP}{dt} = P(a - bP). \quad (4)$$

Around 1840 the Belgian mathematician-biologist **P. F. Verhulst** (1804–1849) was concerned with mathematical models for predicting the human populations of various countries. One of the equations he studied was (4), where $a > 0$ and $b > 0$. Equation (4) came to be known as the **logistic equation**, and its solution is called the **logistic function**. The graph of a logistic function is called a **logistic curve**.

The linear differential equation $dP/dt = kP$ does not provide a very accurate model for population when the population itself is very large. Overcrowded conditions, with the resulting detrimental effects on the environment such as pollution and excessive and competitive demands for food and fuel, can have an inhibiting effect on population growth. As we shall now see, the solution of (4) is bounded as $t \rightarrow \infty$. If we rewrite (4) as $dP/dt = aP - bP^2$, the nonlinear term $-bP^2$, $b > 0$, can be interpreted as an “inhibition” or “competition” term. Also, in most applications the positive constant a is much larger than the constant b .

Logistic curves have proved to be quite accurate in predicting the growth patterns, in a limited space, of certain types of bacteria, protozoa, water fleas (*Daphnia*), and fruit flies (*Drosophila*).

SOLUTION OF THE LOGISTIC EQUATION One method of solving (4) is separation of variables. Decomposing the left side of $dP/P(a - bP) = dt$ into partial fractions and integrating gives

$$\begin{aligned} \left(\frac{1/a}{P} + \frac{b/a}{a - bP} \right) dP &= dt \\ \frac{1}{a} \ln |P| - \frac{1}{a} \ln |a - bP| &= t + c \\ \ln \left| \frac{P}{a - bP} \right| &= at + ac \\ \frac{P}{a - bP} &= c_1 e^{at}. \end{aligned}$$

It follows from the last equation that

$$P(t) = \frac{ac_1 e^{at}}{1 + bc_1 e^{at}} = \frac{ac_1}{bc_1 + e^{-at}}.$$

If $P(0) = P_0$, $P_0 \neq a/b$, we find $c_1 = P_0/(a - bP_0)$, and so after substituting and simplifying, the solution becomes

$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}. \quad (5)$$

GRAPHS OF $P(t)$ The basic shape of the graph of the logistic function $P(t)$ can be obtained without too much effort. Although the variable t usually represents time and we are seldom concerned with applications in which $t < 0$, it is nonetheless of some interest to include this interval in displaying the various graphs of P . From (5) we see that

$$P(t) \rightarrow \frac{aP_0}{bP_0} = \frac{a}{b} \quad \text{as } t \rightarrow \infty \quad \text{and} \quad P(t) \rightarrow 0 \quad \text{as } t \rightarrow -\infty.$$

The dashed line $P = a/2b$ shown in Figure 3.2.2 corresponds to the ordinate of a point of inflection of the logistic curve. To show this, we differentiate (4) by the Product Rule:

$$\begin{aligned} \frac{d^2P}{dt^2} &= P \left(-b \frac{dP}{dt} \right) + (a - bP) \frac{dP}{dt} = \frac{dP}{dt} (a - 2bP) \\ &= P(a - bP)(a - 2bP) \\ &= 2b^2P \left(P - \frac{a}{b} \right) \left(P - \frac{a}{2b} \right). \end{aligned}$$

From calculus recall that the points where $d^2P/dt^2 = 0$ are possible points of inflection, but $P = 0$ and $P = a/b$ can obviously be ruled out. Hence $P = a/2b$ is the only possible ordinate value at which the concavity of the graph can change. For $0 < P < a/2b$ it follows that $P'' > 0$, and $a/2b < P < a/b$ implies that $P'' < 0$. Thus, as we read from left to right, the graph changes from concave up to concave down at the point corresponding to $P = a/2b$. When the initial value satisfies $0 < P_0 < a/2b$, the graph of $P(t)$ assumes the shape of an S, as we see in Figure 3.2.2(b). For $a/2b < P_0 < a/b$ the graph is still S-shaped, but the point of inflection occurs at a negative value of t , as shown in Figure 3.2.2(c).

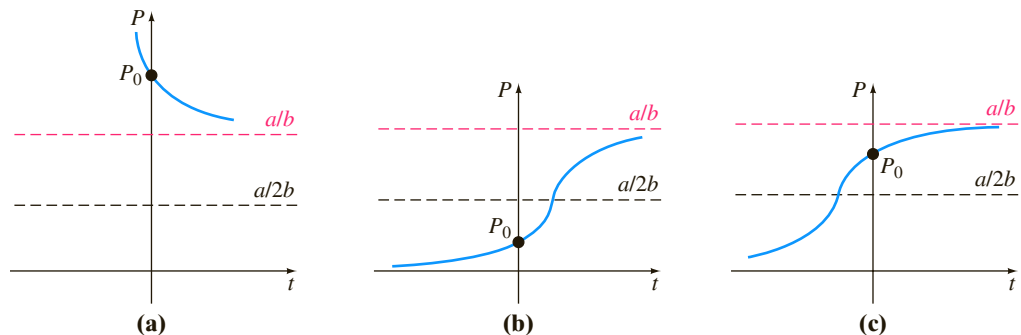


FIGURE 3.2.2 Logistic curves for different initial conditions

We have already seen equation (4) in (5) of Section 1.3 in the form $dx/dt = kx(n + 1 - x)$, $k > 0$. This differential equation provides a reasonable model for describing the spread of an epidemic brought about initially by introducing an infected individual into a static population. The solution $x(t)$ represents the number of individuals infected with the disease at time t .

EXAMPLE 1 Logistic Growth

Suppose a student carrying a flu virus returns to an isolated college campus of 1000 students. If it is assumed that the rate at which the virus spreads is proportional not only to the number x of infected students but also to the number of students not infected, determine the number of infected students after 6 days if it is further observed that after 4 days $x(4) = 50$.

SOLUTION Assuming that no one leaves the campus throughout the duration of the disease, we must solve the initial-value problem

$$\frac{dx}{dt} = kx(1000 - x), \quad x(0) = 1.$$

By making the identification $a = 1000k$ and $b = k$, we have immediately from (5) that

$$x(t) = \frac{1000k}{k + 999ke^{-1000kt}} = \frac{1000}{1 + 999e^{-1000kt}}.$$

Now, using the information $x(4) = 50$, we determine k from

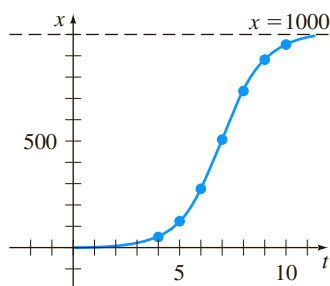
$$50 = \frac{1000}{1 + 999e^{-4000k}}.$$

We find $-1000k = \frac{1}{4} \ln \frac{19}{999} = -0.9906$. Thus

$$x(t) = \frac{1000}{1 + 999e^{-0.9906t}}.$$

Finally,
$$x(6) = \frac{1000}{1 + 999e^{-5.9436}} = 276 \text{ students.}$$

Additional calculated values of $x(t)$ are given in the table in Figure 3.2.3(b). Note that the number of infected students $x(t)$ approaches 1000 as t increases. ■



(a)

t (days)	x (number infected)
4	50 (observed)
5	124
6	276
7	507
8	735
9	882
10	953

(b)

FIGURE 3.2.3 Number of infected students in Example 1

MODIFICATIONS OF THE LOGISTIC EQUATION There are many variations of the logistic equation. For example, the differential equations

$$\frac{dP}{dt} = P(a - bP) - h \quad \text{and} \quad \frac{dP}{dt} = P(a - bP) + h \quad (6)$$

could serve, in turn, as models for the population in a fishery where fish are **harvested** or are **restocked** at rate h . When $h > 0$ is a constant, the DEs in (6) can be readily analyzed qualitatively or solved analytically by separation of variables. The equations in (6) could also serve as models of the human population decreased by **emigration** or increased by immigration, respectively. The rate h in (6) could be a function of time t or could be population dependent; for example, harvesting might be done periodically over time or might be done at a rate proportional to the population P at time t . In the latter instance, the model would look like $P' = P(a - bP) - cP$, $c > 0$. The human population of a community might change because of **immigration** in such a manner that the contribution due to immigration was large when the population P of the community was itself small but small when P was large; a reasonable model for the population of the community would then be $P' = P(a - bP) + ce^{-kP}$, $c > 0$, $k > 0$. See Problem 24 in Exercises 3.2. Another equation of the form given in (2),

$$\frac{dP}{dt} = P(a - b \ln P), \quad (7)$$

is a modification of the logistic equation known as the **Gompertz differential equation** named after the English mathematician **Benjamin Gompertz** (1779–1865). This DE is sometimes used as a model in the study of the growth or decline of populations, the growth of solid tumors, and certain kinds of actuarial predictions. See Problem 8 in Exercises 3.2.

CHEMICAL REACTIONS Suppose that a grams of chemical A are combined with b grams of chemical B . If there are M parts of A and N parts of B formed in the compound and $X(t)$ is the number of grams of chemical C formed, then the number of grams of chemical A and the number of grams of chemical B remaining at time t are, respectively,

$$a - \frac{M}{M+N}X \quad \text{and} \quad b - \frac{N}{M+N}X.$$

The law of mass action states that when no temperature change is involved, the rate at which the two substances react is proportional to the product of the amounts of A and B that are untransformed (remaining) at time t :

$$\frac{dX}{dt} \propto \left(a - \frac{M}{M+N}X\right)\left(b - \frac{N}{M+N}X\right). \quad (8)$$

If we factor out $M/(M+N)$ from the first factor and $N/(M+N)$ from the second and introduce a constant of proportionality $k > 0$, (8) has the form

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X), \quad (9)$$

where $\alpha = a(M+N)/M$ and $\beta = b(M+N)/N$. Recall from (6) of Section 1.3 that a chemical reaction governed by the nonlinear differential equation (9) is said to be a **second-order reaction**.

EXAMPLE 2 Second-Order Chemical Reaction

A compound C is formed when two chemicals A and B are combined. The resulting reaction between the two chemicals is such that for each gram of A , 4 grams of B is used. It is observed that 30 grams of the compound C is formed in 10 minutes. Determine the amount of C at time t if the rate of the reaction is proportional to the amounts of A and B remaining and if initially there are 50 grams of A and 32 grams of B . How much of the compound C is present at 15 minutes? Interpret the solution as $t \rightarrow \infty$.

SOLUTION Let $X(t)$ denote the number of grams of the compound C present at time t . Clearly, $X(0) = 0$ g and $X(10) = 30$ g.

If, for example, 2 grams of compound C is present, we must have used, say, a grams of A and b grams of B , so $a + b = 2$ and $b = 4a$. Thus we must use $a = \frac{2}{5} = 2(\frac{1}{5})$ g of chemical A and $b = \frac{8}{5} = 2(\frac{4}{5})$ g of B . In general, for X grams of C we must use

$$\frac{1}{5}X \text{ grams of } A \quad \text{and} \quad \frac{4}{5}X \text{ grams of } B.$$

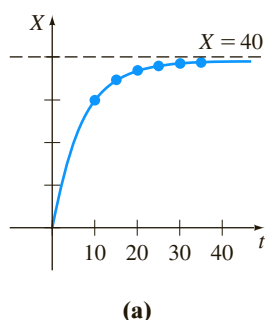
The amounts of A and B remaining at time t are then

$$50 - \frac{1}{5}X \quad \text{and} \quad 32 - \frac{4}{5}X,$$

respectively.

Now we know that the rate at which compound C is formed satisfies

$$\frac{dX}{dt} \propto \left(50 - \frac{1}{5}X\right)\left(32 - \frac{4}{5}X\right).$$



t (min)	X (g)
10	30 (measured)
15	34.78
20	37.25
25	38.54
30	39.22
35	39.59

(b)

FIGURE 3.2.4 Number of grams of compound C in Example 2

To simplify the subsequent algebra, we factor $\frac{1}{5}$ from the first term and $\frac{4}{5}$ from the second and then introduce the constant of proportionality:

$$\frac{dX}{dt} = k(250 - X)(40 - X).$$

By separation of variables and partial fractions we can write

$$-\frac{\frac{1}{210}}{250 - X} dX + \frac{\frac{1}{210}}{40 - X} dX = k dt.$$

Integrating gives

$$\ln \frac{250 - X}{40 - X} = 210kt + c_1 \quad \text{or} \quad \frac{250 - X}{40 - X} = c_2 e^{210kt}. \quad (10)$$

When $t = 0$, $X = 0$, so it follows at this point that $c_2 = \frac{25}{4}$. Using $X = 30$ g at $t = 10$, we find $210k = \frac{1}{10} \ln \frac{88}{25} = 0.1258$. With this information we solve the last equation in (10) for X :

$$X(t) = 1000 \frac{1 - e^{-0.1258t}}{25 - 4e^{-0.1258t}}. \quad (11)$$

From (11) we find $X(15) = 34.78$ grams. The behavior of X as a function of time is displayed in Figure 3.2.4. It is clear from the accompanying table and (11) that $X \rightarrow 40$ as $t \rightarrow \infty$. This means that 40 grams of compound C is formed, leaving

$$50 - \frac{1}{5}(40) = 42 \text{ g of A} \quad \text{and} \quad 32 - \frac{4}{5}(40) = 0 \text{ g of B.}$$

REMARKS

The indefinite integral $\int du/(a^2 - u^2)$ can be evaluated in terms of logarithms, the inverse hyperbolic tangent, or the inverse hyperbolic cotangent. For example, of the two results

$$\int \frac{du}{a^2 - u^2} = \frac{1}{a} \tanh^{-1} \frac{u}{a} + c, \quad |u| < a \quad (12)$$

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right| + c, \quad |u| \neq a, \quad (13)$$

(12) may be convenient in Problems 15 and 26 in Exercises 3.2, whereas (13) may be preferable in Problem 27.

EXERCISES 3.2

Answers to selected odd-numbered problems begin on page ANS-3.

Logistic Equation

1. The number $N(t)$ of supermarkets throughout the country that are using a computerized checkout system is described by the initial-value problem

$$\frac{dN}{dt} = N(1 - 0.0005N), \quad N(0) = 1.$$

- (a) Use the phase portrait concept of Section 2.1 to predict how many supermarkets are expected to adopt the new

procedure over a long period of time. By hand, sketch a solution curve of the given initial-value problem.

- (b) Solve the initial-value problem and then use a graphing utility to verify the solution curve in part (a). How many companies are expected to adopt the new technology when $t = 10$?

2. The number $N(t)$ of people in a community who are exposed to a particular advertisement is governed by the logistic equation. Initially, $N(0) = 500$, and it is observed that $N(1) = 1000$. Solve

for $N(t)$ if it is predicted that the limiting number of people in the community who will see the advertisement is 50,000.

3. A model for the population $P(t)$ in a suburb of a large city is given by the initial-value problem

$$\frac{dP}{dt} = P(10^{-1} - 10^{-7}P), \quad P(0) = 5000,$$

where t is measured in months. What is the limiting value of the population? At what time will the population be equal to one-half of this limiting value?

4. (a) Census data for the United States between 1790 and 1950 are given in Table 3.2.1. Construct a logistic population model using the data from 1790, 1850, and 1910.
- (b) Construct a table comparing actual census population with the population predicted by the model in part (a). Compute the error and the percentage error for each entry pair.

TABLE 3.2.1

Year	Population (in millions)
1790	3.929
1800	5.308
1810	7.240
1820	9.638
1830	12.866
1840	17.069
1850	23.192
1860	31.433
1870	38.558
1880	50.156
1890	62.948
1900	75.996
1910	91.972
1920	105.711
1930	122.775
1940	131.669
1950	150.697

Modifications of the Logistic Model

5. (a) If a constant number h of fish are harvested from a fishery per unit time, then a model for the population $P(t)$ of the fishery at time t is given by

$$\frac{dP}{dt} = P(a - bP) - h, \quad P(0) = P_0,$$

where a , b , h , and P_0 are positive constants. Suppose $a = 5$, $b = 1$, and $h = 4$. Since the DE is autonomous, use the phase portrait concept of Section 2.1 to sketch representative solution curves corresponding to the cases $P_0 > 4$, $1 < P_0 < 4$, and $0 < P_0 < 1$. Determine the long-term behavior of the population in each case.

- (b) Solve the IVP in part (a). Verify the results of your phase portrait in part (a) by using a graphing utility to plot the

graph of $P(t)$ with an initial condition taken from each of the three intervals given.

- (c) Use the information in parts (a) and (b) to determine whether the fishery population becomes extinct in finite time. If so, find that time.

6. Investigate the harvesting model in Problem 5 both qualitatively and analytically in the case $a = 5$, $b = 1$, $h = \frac{25}{4}$. Determine whether the population becomes extinct in finite time. If so, find that time.
7. Repeat Problem 6 in the case $a = 5$, $b = 1$, $h = 7$.
8. (a) Suppose $a = b = 1$ in the Gompertz differential equation (7). Since the DE is autonomous, use the phase portrait concept of Section 2.1 to sketch representative solution curves corresponding to the cases $P_0 > e$ and $0 < P_0 < e$.
- (b) Suppose $a = 1$, $b = -1$ in (7). Use a new phase portrait to sketch representative solution curves corresponding to the cases $P_0 > e^{-1}$ and $0 < P_0 < e^{-1}$.
- (c) Find an explicit solution of (7) subject to $P(0) = P_0$.

Chemical Reactions

9. Two chemicals A and B are combined to form a chemical C . The rate, or velocity, of the reaction is proportional to the product of the instantaneous amounts of A and B not converted to chemical C . Initially, there are 40 grams of A and 50 grams of B , and for each gram of B , 2 grams of A is used. It is observed that 10 grams of C is formed in 5 minutes. How much is formed in 20 minutes? What is the limiting amount of C after a long time? How much of chemicals A and B remains after a long time?
10. Solve Problem 9 if 100 grams of chemical A is present initially. At what time is chemical C half-formed?

Additional Nonlinear Models

11. **Leaking Cylindrical Tank** A tank in the form of a right-circular cylinder standing on end is leaking water through a circular hole in its bottom. As we saw in (10) of Section 1.3, when friction and contraction of water at the hole are ignored, the height h of water in the tank is described by

$$\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh},$$

where A_w and A_h are the cross-sectional areas of the water and the hole, respectively.

- (a) Solve the DE if the initial height of the water is H . By hand, sketch the graph of $h(t)$ and give its interval I of definition in terms of the symbols A_w , A_h , and H . Use $g = 32 \text{ ft/s}^2$.
- (b) Suppose the tank is 10 feet high and has radius 2 feet and the circular hole has radius $\frac{1}{2}$ inch. If the tank is initially full, how long will it take to empty?

- 12. Leaking Cylindrical Tank—Continued** When friction and contraction of the water at the hole are taken into account, the model in Problem 11 becomes

$$\frac{dh}{dt} = -c \frac{A_h}{A_w} \sqrt{2gh},$$

where $0 < c < 1$. How long will it take the tank in Problem 11(b) to empty if $c = 0.6$? See Problem 13 in Exercises 1.3.

- 13. Leaking Conical Tank** A tank in the form of a right-circular cone standing on end, vertex down, is leaking water through a circular hole in its bottom.

- (a) Suppose the tank is 20 feet high and has radius 8 feet and the circular hole has radius 2 inches. In Problem 14 in Exercises 1.3 you were asked to show that the differential equation governing the height h of water leaking from a tank is

$$\frac{dh}{dt} = -\frac{5}{6h^{3/2}}.$$

In this model, friction and contraction of the water at the hole were taken into account with $c = 0.6$, and g was taken to be 32 ft/s^2 . See Figure 1.3.12. If the tank is initially full, how long will it take the tank to empty?

- (b) Suppose the tank has a vertex angle of 60° and the circular hole has radius 2 inches. Determine the differential equation governing the height h of water. Use $c = 0.6$ and $g = 32 \text{ ft/s}^2$. If the height of the water is initially 9 feet, how long will it take the tank to empty?

- 14. Inverted Conical Tank** Suppose that the conical tank in Problem 13(a) is inverted, as shown in Figure 3.2.5, and that water leaks out a circular hole of radius 2 inches in the center of its circular base. Is the time it takes to empty a full tank the same as for the tank with vertex down in Problem 13? Take the friction/contraction coefficient to be $c = 0.6$ and $g = 32 \text{ ft/s}^2$.

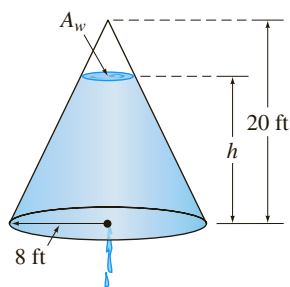


FIGURE 3.2.5 Inverted conical tank in Problem 14

- 15. Air Resistance** A differential equation for the velocity v of a falling mass m subjected to air resistance proportional to the square of the instantaneous velocity is

$$m \frac{dv}{dt} = mg - kv^2,$$

where $k > 0$ is a constant of proportionality. The positive direction is downward.

- (a) Solve the equation subject to the initial condition $v(0) = v_0$.
 (b) Use the solution in part (a) to determine the limiting, or terminal, velocity of the mass. We saw how to determine the terminal velocity without solving the DE in Problem 41 in Exercises 2.1.
 (c) If the distance s , measured from the point where the mass was released above ground, is related to velocity v by $ds/dt = v(t)$, find an explicit expression for $s(t)$ if $s(0) = 0$.

- 16. How High?—Nonlinear Air Resistance** Consider the 16-pound cannonball shot vertically upward in Problems 36 and 37 in Exercises 3.1 with an initial velocity $v_0 = 300 \text{ ft/s}$. Determine the maximum height attained by the cannonball if air resistance is assumed to be proportional to the square of the instantaneous velocity. Assume that the positive direction is upward and take $k = 0.0003$. [Hint: Slightly modify the DE in Problem 15.]

- 17. That Sinking Feeling** (a) Determine a differential equation for the velocity $v(t)$ of a mass m sinking in water that imparts a resistance proportional to the square of the instantaneous velocity and also exerts an upward buoyant force whose magnitude is given by Archimedes' principle. See Problem 18 in Exercises 1.3. Assume that the positive direction is downward.

- (b) Solve the differential equation in part (a).
 (c) Determine the limiting, or terminal, velocity of the sinking mass.

- 18. Solar Collector** The differential equation

$$\frac{dy}{dx} = \frac{-x + \sqrt{x^2 + y^2}}{y}$$

describes the shape of a plane curve C that will reflect all incoming light beams to the same point and could be a model for the mirror of a reflecting telescope, a satellite antenna, or a solar collector. See Problem 29 in Exercises 1.3. There are several ways of solving this DE.

- (a) Verify that the differential equation is homogeneous (see Section 2.5). Show that the substitution $y = ux$ yields

$$\frac{u du}{\sqrt{1+u^2}(1-\sqrt{1+u^2})} = \frac{dx}{x}.$$

Use a CAS (or another judicious substitution) to integrate the left-hand side of the equation. Show that the curve C must be a parabola with focus at the origin and is symmetric with respect to the x -axis.

- (b) Show that the first differential equation can also be solved by means of the substitution $u = x^2 + y^2$.

- 19. Tsunami** (a) A simple model for the shape of a tsunami is given by

$$\frac{dW}{dx} = W \sqrt{4 - 2W},$$

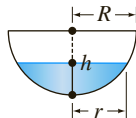
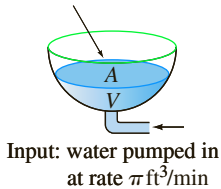
where $W(x) > 0$ is the height of the wave expressed as a function of its position relative to a point offshore. By inspection, find all constant solutions of the DE.

- (b) Solve the differential equation in part (a). A CAS may be useful for integration.
- (c) Use a graphing utility to obtain the graphs of all solutions that satisfy the initial condition $W(0) = 2$.

20. Evaporation An outdoor decorative pond in the shape of a hemispherical tank is to be filled with water pumped into the tank through an inlet in its bottom. Suppose that the radius of the tank is $R = 10$ ft, that water is pumped in at a rate of π ft³/min, and that the tank is initially empty. See Figure 3.2.6. As the tank fills, it loses water through evaporation. Assume that the rate of evaporation is proportional to the area A of the surface of the water and that the constant of proportionality is $k = 0.01$.

- (a) The rate of change dV/dt of the volume of the water at time t is a net rate. Use this net rate to determine a differential equation for the height h of the water at time t . The volume of the water shown in the figure is $V = \pi R h^2 - \frac{1}{3}\pi h^3$, where $R = 10$. Express the area of the surface of the water $A = \pi r^2$ in terms of h .
- (b) Solve the differential equation in part (a). Graph the solution.
- (c) If there were no evaporation, how long would it take the tank to fill?
- (d) With evaporation, what is the depth of the water at the time found in part (c)? Will the tank ever be filled? Prove your assertion.

Output: water evaporates
at rate proportional
to area A of surface



(a) hemispherical tank (b) cross-section of tank

FIGURE 3.2.6 Decorative pond in Problem 20

21. Doomsday Equation Consider the differential equation

$$\frac{dP}{dt} = kP^{1+c},$$

where $k > 0$ and $c \geq 0$. In Section 3.1 we saw that in the case $c = 0$ the linear differential equation $dP/dt = kP$ is a mathematical model of a population $P(t)$ that exhibits unbounded growth over the infinite time interval $[0, \infty)$, that is, $P(t) \rightarrow \infty$ as $t \rightarrow \infty$. See Example 1 on page 85.

- (a) Suppose for $c = 0.01$ that the nonlinear differential equation

$$\frac{dP}{dt} = kP^{1.01}, \quad k > 0,$$

is a mathematical model for a population of small animals, where time t is measured in months. Solve the differential

equation subject to the initial condition $P(0) = 10$ and the fact that the animal population has doubled in 5 months.

- (b) The differential equation in part (a) is called a **doomsday equation** because the population $P(t)$ exhibits unbounded growth over a finite time interval $(0, T)$, that is, there is some time T such that $P(t) \rightarrow \infty$ as $t \rightarrow T^-$. Find T .
- (c) From part (a), what is $P(50)$? $P(100)$?

22. Doomsday or Extinction Suppose the population model (4) is modified to be

$$\frac{dP}{dt} = P(bP - a).$$

- (a) If $a > 0$, $b > 0$ show by means of a phase portrait (see page 40) that, depending on the initial condition $P(0) = P_0$, the mathematical model could include a doomsday scenario ($P(t) \rightarrow \infty$) or an extinction scenario ($P(t) \rightarrow 0$).
- (b) Solve the initial-value problem

$$\frac{dP}{dt} = P(0.0005P - 0.1), \quad P(0) = 300.$$

Show that this model predicts a doomsday for the population in a finite time T .

- (c) Solve the differential equation in part (b) subject to the initial condition $P(0) = 100$. Show that this model predicts extinction for the population as $t \rightarrow \infty$.

Project Problems

23. Regression Line Read the documentation for your CAS on *scatter plots* (or *scatter diagrams*) and *least-squares linear fit*. The straight line that best fits a set of data points is called a **regression line** or a **least squares line**. Your task is to construct a logistic model for the population of the United States, defining $f(P)$ in (2) as an equation of a regression line based on the population data in the table in Problem 4. One way of doing this is to approximate the left-hand side $\frac{1}{P} \frac{dP}{dt}$ of the first equation in (2), using the forward difference quotient in place of dP/dt :

$$Q(t) = \frac{1}{P(t)} \frac{P(t+h) - P(t)}{h}.$$

- (a) Make a table of the values t , $P(t)$, and $Q(t)$ using $t = 0, 10, 20, \dots, 160$ and $h = 10$. For example, the first line of the table should contain $t = 0$, $P(0)$, and $Q(0)$. With $P(0) = 3.929$ and $P(10) = 5.308$,

$$Q(0) = \frac{1}{P(0)} \frac{P(10) - P(0)}{10} = 0.035.$$

Note that $Q(160)$ depends on the 1960 census population $P(170)$. Look up this value.

- (b) Use a CAS to obtain a scatter plot of the data $(P(t), Q(t))$ computed in part (a). Also use a CAS to find an equation of the regression line and to superimpose its graph on the scatter plot.

- (c) Construct a logistic model $dP/dt = Pf(P)$, where $f(P)$ is the equation of the regression line found in part (b).
- (d) Solve the model in part (c) using the initial condition $P(0) = 3.929$.
- (e) Use a CAS to obtain another scatter plot, this time of the ordered pairs $(t, P(t))$ from your table in part (a). Use your CAS to superimpose the graph of the solution in part (d) on the scatter plot.
- (f) Look up the U.S. census data for 1970, 1980, and 1990. What population does the logistic model in part (c) predict for these years? What does the model predict for the U.S. population $P(t)$ as $t \rightarrow \infty$?

24. Immigration Model (a) In Examples 3 and 4 of Section 2.1 we saw that any solution $P(t)$ of (4) possesses the asymptotic behavior $P(t) \rightarrow a/b$ as $t \rightarrow \infty$ for $P_0 > a/b$ and for $0 < P_0 < a/b$; as a consequence the equilibrium solution $P = a/b$ is called an attractor. Use a root-finding application of a CAS (or a graphic calculator) to approximate the equilibrium solution of the immigration model

$$\frac{dP}{dt} = P(1 - P) + 0.3e^{-P}.$$

- (b) Use a graphing utility to graph the function $F(P) = P(1 - P) + 0.3e^{-P}$. Explain how this graph can be used to determine whether the number found in part (a) is an attractor.
- (c) Use a numerical solver to compare the solution curves for the IVPs

$$\frac{dP}{dt} = P(1 - P), \quad P(0) = P_0$$

for $P_0 = 0.2$ and $P_0 = 1.2$ with the solution curves for the IVPs

$$\frac{dP}{dt} = P(1 - P) + 0.3e^{-P}, \quad P(0) = P_0$$

for $P_0 = 0.2$ and $P_0 = 1.2$. Superimpose all curves on the same coordinate axes but, if possible, use a different color for the curves of the second initial-value problem. Over a long period of time, what percentage increase does the immigration model predict in the population compared to the logistic model?

25. What Goes Up . . . In Problem 16 let t_a be the time it takes the cannonball to attain its maximum height and let t_d be the time it takes the cannonball to fall from the maximum height to the ground. Compare the value of t_a with the value of t_d and compare the magnitude of the impact velocity v_i with the initial velocity v_0 . See Problem 50 in Exercises 3.1. A root-finding application of a CAS might be useful here. [Hint: Use the model in Problem 15 when the cannonball is falling.]

26. Skydiving A skydiver is equipped with a stopwatch and an altimeter. As shown in Figure 3.2.7, he opens his parachute 25 seconds after exiting a plane flying at an altitude of 20,000 feet and observes that his altitude is 14,800 feet. Assume that air resistance is proportional to the square of the instantaneous

velocity, his initial velocity on leaving the plane is zero, and $g = 32 \text{ ft/s}^2$.

- (a) Find the distance $s(t)$, measured from the plane, the skydiver has traveled during freefall in time t . [Hint: The constant of proportionality k in the model given in Problem 15 is not specified. Use the expression for terminal velocity v_t obtained in part (b) of Problem 15 to eliminate k from the IVP. Then eventually solve for v_t .]
- (b) How far does the skydiver fall and what is his velocity at $t = 15$ s?

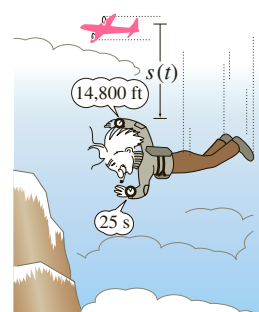


FIGURE 3.2.7 Skydiver in Problem 26

27. Hitting Bottom A helicopter hovers 500 feet above a large open tank full of liquid (not water). A dense compact object weighing 160 pounds is dropped (released from rest) from the helicopter into the liquid. Assume that air resistance is proportional to instantaneous velocity v while the object is in the air and that viscous damping is proportional to v^2 after the object has entered the liquid. For air take $k = \frac{1}{4}$, and for the liquid take $k = 0.1$. Assume that the positive direction is downward. If the tank is 75 feet high, determine the time and the impact velocity when the object hits the bottom of the tank. [Hint: Think in terms of two distinct IVPs. If you use (13), be careful in removing the absolute value sign. You might compare the velocity when the object hits the liquid—the initial velocity for the second problem—with the terminal velocity v_t of the object falling through the liquid.]

28. Old Man River . . . In Figure 3.2.8(a) suppose that the y -axis and the dashed vertical line $x = 1$ represent, respectively, the straight west and east beaches of a river that is 1 mile wide. The river flows northward with a velocity $\mathbf{v}_r$, where $|\mathbf{v}_r| = v_r$ mi/h is a constant. A man enters the current at the point $(1, 0)$ on the east shore and swims in a direction and rate relative to the river given by the vector $\mathbf{v}_s$, where the speed $|\mathbf{v}_s| = v_s$ mi/h is a constant. The man wants to reach the west beach exactly at $(0, 0)$ and so swims in such a manner that keeps his velocity vector $\mathbf{v}_s$ always directed toward the point $(0, 0)$. Use Figure 3.2.8(b) as an aid in showing that a mathematical model for the path of the swimmer in the river is

$$\frac{dy}{dx} = \frac{v_s y - v_r \sqrt{x^2 + y^2}}{v_s x}.$$

[Hint: The velocity $\mathbf{v}$ of the swimmer along the path or curve shown in Figure 3.2.8 is the resultant $\mathbf{v} = \mathbf{v}_s + \mathbf{v}_r$. Resolve $\mathbf{v}_s$ and $\mathbf{v}_r$ into components in the x - and y -directions. If $x = x(t)$, $y = y(t)$ are parametric equations of the swimmer's path, then $\mathbf{v} = (dx/dt, dy/dt)$.]

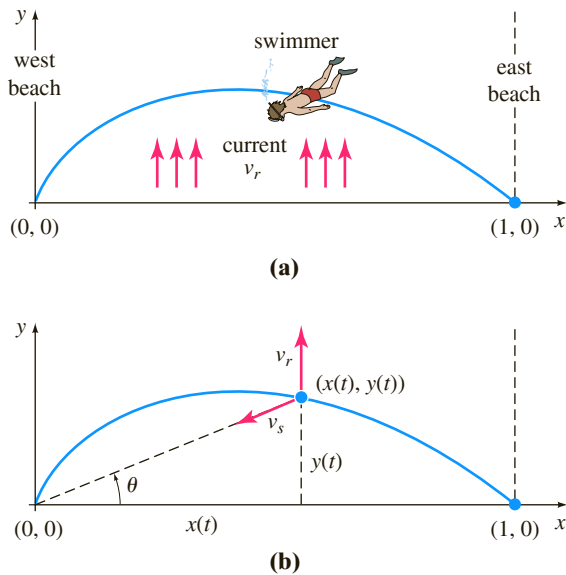


FIGURE 3.2.8 Path of swimmer in Problem 28

29. (a) Solve the DE in Problem 28 subject to $y(1) = 0$. For convenience let $k = v_r/v_s$.
- (b) Determine the values of v_s for which the swimmer will reach the point $(0, 0)$ by examining $\lim_{x \rightarrow 0^+} y(x)$ in the cases $k = 1$, $k > 1$, and $0 < k < 1$.
30. **Old Man River Keeps Moving . . .** Suppose the man in Problem 28 again enters the current at $(1, 0)$ but this time decides to swim so that his velocity vector $\mathbf{v}_s$ is always directed toward the west beach. Assume that the speed $|\mathbf{v}_s| = v_s$ mi/h is a constant. Show that a mathematical model for the path of the swimmer in the river is now
- $$\frac{dy}{dx} = -\frac{v_r}{v_s}.$$
31. The current speed v_r of a straight river such as that in Problem 28 is usually not a constant. Rather, an approximation to the current speed (measured in miles per hour) could be a function such as $v_r(x) = 30x(1 - x)$, $0 \leq x \leq 1$, whose values are small at the shores (in this case, $v_r(0) = 0$ and $v_r(1) = 0$) and largest in the middle of the river. Solve the DE in Problem 30 subject to $y(1) = 0$, where $v_s = 2$ mi/h and $v_r(x)$ is as given. When the swimmer makes it across the river, how far will he have to walk along the beach to reach the point $(0, 0)$?
32. **Raindrops Keep Falling . . .** When a bottle of liquid refreshment was opened recently, the following factoid was found inside the bottle cap:

The average velocity of a falling raindrop is 7 miles/hour.

A quick search of the Internet found that meteorologist Jeff Haby offers the additional information that an “average” spherical raindrop has a radius of 0.04 in. and an approximate volume of 0.00000155 ft^3 . Use this data and, if need be, dig up other data and make other reasonable assumptions to determine whether “average velocity of . . . 7 mi/h” is consistent with the models in Problems 35 and 36 in Exercises 3.1 and Problem 15 in this exercise set. Also see Problem 36 in Exercises 1.3.

33. Time Drips By The **clepsydra**, or water clock, was a device that the ancient Egyptians, Greeks, Romans, and Chinese used to measure the passage of time by observing the change in the height of water that was permitted to flow out of a small hole in the bottom of a container or tank.

- (a) Suppose a tank is made of glass and has the shape of a right-circular cylinder of radius 1 ft. Assume that $h(0) = 2$ ft corresponds to water filled to the top of the tank, a hole in the bottom is circular with radius $\frac{1}{32}$ in., $g = 32 \text{ ft/s}^2$, and $c = 0.6$. Use the differential equation in Problem 12 to find the height $h(t)$ of the water.
- (b) For the tank in part (a), how far up from its bottom should a mark be made on its side, as shown in Figure 3.2.9, that corresponds to the passage of one hour? Next determine where to place the marks corresponding to the passage of 2 hr, 3 hr, . . . , 12 hr. Explain why these marks are not evenly spaced.

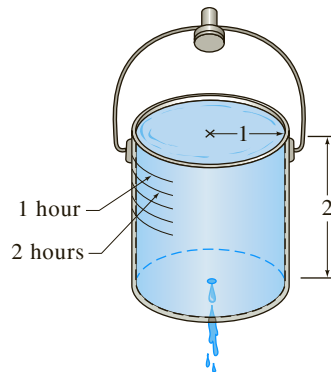


FIGURE 3.2.9 Clepsydra in Problem 33

34. (a) Suppose that a glass tank has the shape of a cone with circular cross section as shown in Figure 3.2.10. As in part (a) of Problem 33, assume that $h(0) = 2$ ft corresponds to water filled to the top of the tank, a hole in the bottom is circular with radius $\frac{1}{32}$ in., $g = 32 \text{ ft/s}^2$, and $c = 0.6$. Use the differential equation in Problem 12 to find the height $h(t)$ of the water.
- (b) Can this water clock measure 12 time intervals of length equal to 1 hour? Explain using sound mathematics.

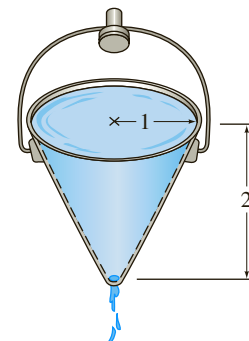


FIGURE 3.2.10 Clepsydra in Problem 34

35. Suppose that $r = f(h)$ defines the shape of a water clock for which the time marks are equally spaced. Use the differential equation in Problem 12 to find $f(h)$ and sketch a typical graph of h as a function of r . Assume that the cross-sectional area A_h of the hole is constant. [Hint: In this situation $dh/dt = -a$, where $a > 0$ is a constant.]

3.3

Modeling with Systems of First-Order DEs

INTRODUCTION This section is similar to Section 1.3 in that we are just going to discuss certain mathematical models, but instead of a single differential equation the models will be systems of first-order differential equations. Although some of the models will be based on topics that we explored in the preceding two sections, we are not going to develop any general methods for solving these systems. There are reasons for this: First, we do not possess the necessary mathematical tools for solving systems at this point. Second, some of the systems that we discuss—notably the systems of *nonlinear* first-order DEs—simply cannot be solved analytically. We shall examine solution methods for systems of *linear* DEs in Chapters 4, 7, and 8.

LINEAR/NONLINEAR SYSTEMS We have seen that a single differential equation can serve as a mathematical model for a single population in an environment. But if there are, say, two interacting and perhaps competing species living in the same environment (for example, rabbits and foxes), then a model for their populations $x(t)$ and $y(t)$ might be a system of two first-order differential equations such as

$$\begin{aligned}\frac{dx}{dt} &= g_1(t, x, y) \\ \frac{dy}{dt} &= g_2(t, x, y).\end{aligned}\tag{1}$$

When g_1 and g_2 are linear in the variables x and y —that is, g_1 and g_2 have the forms

$$g_1(t, x, y) = c_1x + c_2y + f_1(t) \quad \text{and} \quad g_2(t, x, y) = c_3x + c_4y + f_2(t),$$

where the coefficients c_i could depend on t —then (1) is said to be a **linear system**. A system of differential equations that is not linear is said to be **nonlinear**.

RADIOACTIVE SERIES In the discussion of radioactive decay in Sections 1.3 and 3.1 we assumed that the rate of decay was proportional to the number $A(t)$ of nuclei of the substance present at time t . When a substance decays by radioactivity, it usually doesn't just transmute in one step into a stable substance; rather, the first substance decays into another radioactive substance, which in turn decays into a third substance, and so on. This process, called a **radioactive decay series**, continues until a stable element is reached. For example, the uranium decay series is $\text{U-238} \rightarrow \text{Th-234} \rightarrow \cdots \rightarrow \text{Pb-206}$, where Pb-206 is a stable isotope of lead. The half-lives of the various elements in a radioactive series can range from billions of years (4.5×10^9 years for U-238) to a fraction of a second. Suppose a radioactive series is described schematically by $X \xrightarrow{-\lambda_1} Y \xrightarrow{-\lambda_2} Z$, where $k_1 = -\lambda_1 < 0$ and $k_2 = -\lambda_2 < 0$ are the decay constants for substances X and Y , respectively, and Z is a stable element. Suppose, too, that $x(t)$, $y(t)$, and $z(t)$ denote amounts of substances X , Y , and Z , respectively, remaining at time t . The decay of element X is described by

$$\frac{dx}{dt} = -\lambda_1x,$$

whereas the rate at which the second element Y decays is the net rate

$$\frac{dy}{dt} = \lambda_1x - \lambda_2y,$$

since Y is *gaining* atoms from the decay of X and at the same time *losing* atoms because of its own decay. Since Z is a stable element, it is simply gaining atoms from the decay of element Y :

$$\frac{dz}{dt} = \lambda_2 y.$$

In other words, a model of the radioactive decay series for three elements is the linear system of three first-order differential equations

$$\begin{aligned}\frac{dx}{dt} &= -\lambda_1 x \\ \frac{dy}{dt} &= \lambda_1 x - \lambda_2 y \\ \frac{dz}{dt} &= \lambda_2 y.\end{aligned}\tag{2}$$

MIXTURES Consider the two tanks shown in Figure 3.3.1. Let us suppose for the sake of discussion that tank A contains 50 gallons of water in which 25 pounds of salt is dissolved. Suppose tank B contains 50 gallons of pure water. Liquid is pumped into and out of the tanks as indicated in the figure; the mixture exchanged between the two tanks and the liquid pumped out of tank B are assumed to be well stirred. We wish to construct a mathematical model that describes the number of pounds $x_1(t)$ and $x_2(t)$ of salt in tanks A and B , respectively, at time t .

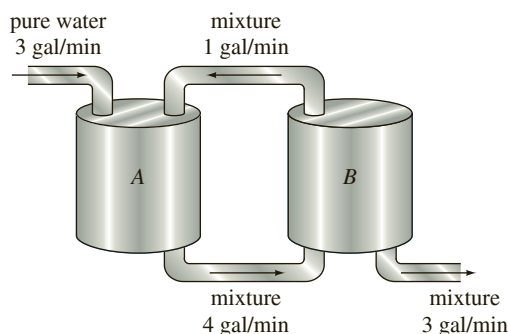


FIGURE 3.3.1 Connected mixing tanks

By an analysis similar to that on page 25 in Section 1.3 and Example 5 of Section 3.1 we see that the net rate of change of $x_1(t)$ for tank A is

$$\begin{aligned}\frac{dx_1}{dt} &= \overbrace{(3 \text{ gal/min}) \cdot (0 \text{ lb/gal}) + (1 \text{ gal/min}) \cdot \left(\frac{x_2}{50} \text{ lb/gal}\right)}^{\text{input rate of salt}} - \overbrace{(4 \text{ gal/min}) \cdot \left(\frac{x_1}{50} \text{ lb/gal}\right)}^{\text{output rate of salt}} \\ &= -\frac{2}{25}x_1 + \frac{1}{50}x_2.\end{aligned}$$

Similarly, for tank B the net rate of change of $x_2(t)$ is

$$\begin{aligned}\frac{dx_2}{dt} &= 4 \cdot \frac{x_1}{50} - 3 \cdot \frac{x_2}{50} - 1 \cdot \frac{x_2}{50} \\ &= \frac{2}{25}x_1 - \frac{2}{25}x_2.\end{aligned}$$

Thus we obtain the linear system

$$\begin{aligned}\frac{dx_1}{dt} &= -\frac{2}{25}x_1 + \frac{1}{50}x_2 \\ \frac{dx_2}{dt} &= \frac{2}{25}x_1 - \frac{2}{25}x_2.\end{aligned}\tag{3}$$

Observe that the foregoing system is accompanied by the initial conditions $x_1(0) = 25$, $x_2(0) = 0$.

A PREDATOR-PREY MODEL Suppose that two different species of animals interact within the same environment or ecosystem, and suppose further that the first species eats only vegetation and the second eats only the first species. In other words, one species is a predator, and the other is a prey. For example, wolves hunt grass-eating caribou, sharks devour little fish, and the snowy owl pursues an arctic rodent called the lemming. For the sake of discussion, let us imagine that the predators are foxes and the prey are rabbits.

Let $x(t)$ and $y(t)$ denote the fox and rabbit populations, respectively, at time t . If there were no rabbits, then one might expect that the foxes, lacking an adequate food supply, would decline in number according to

$$\frac{dx}{dt} = -ax, \quad a > 0. \quad (4)$$

When rabbits are present in the environment, however, it seems reasonable that the number of encounters or interactions between these two species per unit time is jointly proportional to their populations x and y —that is, proportional to the product xy . Thus when rabbits are present, there is a supply of food, so foxes are added to the system at a rate bxy , $b > 0$. Adding this last rate to (4) gives a model for the fox population:

$$\frac{dx}{dt} = -ax + bxy. \quad (5)$$

On the other hand, if there were no foxes, then the rabbits would, with an added assumption of unlimited food supply, grow at a rate that is proportional to the number of rabbits present at time t :

$$\frac{dy}{dt} = dy, \quad d > 0. \quad (6)$$

But when foxes are present, a model for the rabbit population is (6) decreased by cxy , $c > 0$ —that is, decreased by the rate at which the rabbits are eaten during their encounters with the foxes:

$$\frac{dy}{dt} = dy - cxy. \quad (7)$$

Equations (5) and (7) constitute a system of nonlinear differential equations

$$\begin{aligned} \frac{dx}{dt} &= -ax + bxy = x(-a + by) \\ \frac{dy}{dt} &= dy - cxy = y(d - cx), \end{aligned} \quad (8)$$

where a , b , c , and d are positive constants. This famous system of equations is known as the **Lotka-Volterra predator-prey model**.

Except for two constant solutions, $x(t) = 0$, $y(t) = 0$ and $x(t) = d/c$, $y(t) = a/b$, the nonlinear system (8) cannot be solved in terms of elementary functions. However, we can analyze such systems quantitatively and qualitatively. See Chapter 9, “Numerical Solutions of Ordinary Differential Equations,” and Chapter 10, “Systems of Nonlinear First-Order Differential Equations.”*

EXAMPLE 1 Predator-Prey Model

Suppose

$$\begin{aligned} \frac{dx}{dt} &= -0.16x + 0.08xy \\ \frac{dy}{dt} &= 4.5y - 0.9xy \end{aligned}$$

*Chapters 10–15 are in the expanded version of this text, *Differential Equations with Boundary-Value Problems*.

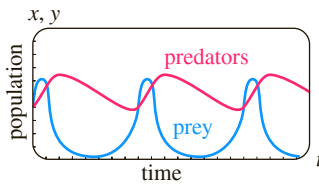


FIGURE 3.3.2 Populations of predators (red) and prey (blue) in Example 1

represents a predator-prey model. Because we are dealing with populations, we have $x(t) \geq 0$, $y(t) \geq 0$. Figure 3.3.2, obtained with the aid of a numerical solver, shows typical population curves of the predators and prey for this model superimposed on the same coordinate axes. The initial conditions used were $x(0) = 4$, $y(0) = 4$. The curve in red represents the population $x(t)$ of the predators (foxes), and the blue curve is the population $y(t)$ of the prey (rabbits). Observe that the model seems to predict that both populations $x(t)$ and $y(t)$ are periodic in time. This makes intuitive sense because as the number of prey decreases, the predator population eventually decreases because of a diminished food supply; but attendant to a decrease in the number of predators is an increase in the number of prey; this in turn gives rise to an increased number of predators, which ultimately brings about another decrease in the number of prey. ■

COMPETITION MODELS Now suppose two different species of animals occupy the same ecosystem, not as predator and prey but rather as competitors for the same resources (such as food and living space) in the system. In the absence of the other, let us assume that the rate at which each population grows is given by

$$\frac{dx}{dt} = ax \quad \text{and} \quad \frac{dy}{dt} = cy, \quad (9)$$

respectively.

Since the two species compete, another assumption might be that each of these rates is diminished simply by the influence, or existence, of the other population. Thus a model for the two populations is given by the linear system

$$\begin{aligned} \frac{dx}{dt} &= ax - by \\ \frac{dy}{dt} &= cy - dx, \end{aligned} \quad (10)$$

where a , b , c , and d are positive constants.

On the other hand, we might assume, as we did in (5), that each growth rate in (9) should be reduced by a rate proportional to the number of interactions between the two species:

$$\begin{aligned} \frac{dx}{dt} &= ax - bxy \\ \frac{dy}{dt} &= cy - dxy. \end{aligned} \quad (11)$$

Inspection shows that this nonlinear system is similar to the Lotka-Volterra predator-prey model. Finally, it might be more realistic to replace the rates in (9), which indicate that the population of each species in isolation grows exponentially, with rates indicating that each population grows logistically (that is, over a long time the population is bounded):

$$\frac{dx}{dt} = a_1x - b_1x^2 \quad \text{and} \quad \frac{dy}{dt} = a_2y - b_2y^2. \quad (12)$$

When these new rates are decreased by rates proportional to the number of interactions, we obtain another nonlinear model:

$$\begin{aligned} \frac{dx}{dt} &= a_1x - b_1x^2 - c_1xy = x(a_1 - b_1x - c_1y) \\ \frac{dy}{dt} &= a_2y - b_2y^2 - c_2xy = y(a_2 - b_2y - c_2x), \end{aligned} \quad (13)$$

where all coefficients are positive. The linear system (10) and the nonlinear systems (11) and (13) are, of course, called **competition models**.

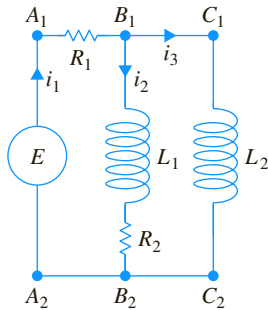


FIGURE 3.3.3 Network whose model is given in (17)

NETWORKS An electrical network having more than one loop also gives rise to simultaneous differential equations. As shown in Figure 3.3.3, the current $i_1(t)$ splits in the directions shown at point B_1 , called a *branch point* of the network. By **Kirchhoff's first law** we can write

$$i_1(t) = i_2(t) + i_3(t). \quad (14)$$

We can also apply **Kirchhoff's second law** to each loop. For loop $A_1B_1B_2A_2A_1$, summing the voltage drops across each part of the loop gives

$$E(t) = i_1R_1 + L_1 \frac{di_2}{dt} + i_2R_2. \quad (15)$$

Similarly, for loop $A_1B_1C_1C_2B_2A_2A_1$ we find

$$E(t) = i_1R_1 + L_2 \frac{di_3}{dt}. \quad (16)$$

Using (14) to eliminate i_1 in (15) and (16) yields two linear first-order equations for the currents $i_2(t)$ and $i_3(t)$:

$$\begin{aligned} L_1 \frac{di_2}{dt} + (R_1 + R_2)i_2 + R_1i_3 &= E(t) \\ L_2 \frac{di_3}{dt} + R_1i_2 + R_1i_3 &= E(t). \end{aligned} \quad (17)$$

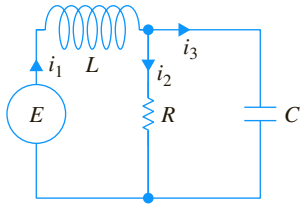


FIGURE 3.3.4 Network whose model is given in (18)

We leave it as an exercise (see Problem 16 in Exercises 3.3) to show that the system of differential equations describing the currents $i_1(t)$ and $i_2(t)$ in the network containing a resistor, an inductor, and a capacitor shown in Figure 3.3.4 is

$$\begin{aligned} L \frac{di_1}{dt} + Ri_2 &= E(t) \\ RC \frac{di_2}{dt} + i_2 - i_1 &= 0. \end{aligned} \quad (18)$$

EXERCISES 3.3

Answers to selected odd-numbered problems begin on page ANS-4.

Radioactive Series

1. We have not discussed methods by which systems of first-order differential equations can be solved. Nevertheless, systems such as (2) can be solved with no knowledge other than how to solve a single linear first-order equation. Find a solution of (2) subject to the initial conditions $x(0) = x_0$, $y(0) = 0$, $z(0) = 0$.
2. In Problem 1 suppose that time is measured in days, that the decay constants are $k_1 = -0.138629$ and $k_2 = -0.004951$, and that $x_0 = 20$. Use a graphing utility to obtain the graphs of the solutions $x(t)$, $y(t)$, and $z(t)$ on the same set of coordinate axes. Use the graphs to approximate the half-lives of substances X and Y.
3. Use the graphs in Problem 2 to approximate the times when the amounts $x(t)$ and $y(t)$ are the same, the times when the amounts $x(t)$ and $z(t)$ are the same, and the times when the amounts $y(t)$ and $z(t)$ are the same. Why does the time that is determined when the amounts $y(t)$ and $z(t)$ are the same make intuitive sense?
4. Construct a mathematical model for a radioactive series of four elements W, X, Y, and Z, where Z is a stable element.

5. **Potassium-40 Decay** The chemical element potassium is a soft metal that can be found extensively throughout the Earth's crust and oceans. Although potassium occurs naturally in the form of three isotopes, only the isotope potassium-40 (K-40) is radioactive. This isotope is also unusual in that it decays by two different nuclear reactions. Over time, by emitting beta particles a great percentage of an initial amount K_0 of K-40 decays into the stable isotope calcium-40 (Ca-40), whereas by electron capture a smaller percentage of K_0 decays into the stable isotope argon-40 (Ar-40). Because the rates at which the amounts $C(t)$ of Ca-40 and $A(t)$ of Ar-40 increase are proportional to the amount $K(t)$ of potassium present, and the rate at which $K(t)$ decays is also proportional to $K(t)$, we obtain the system of linear first-order equations

$$\begin{aligned} \frac{dC}{dt} &= \lambda_1 K \\ \frac{dA}{dt} &= \lambda_2 K \\ \frac{dK}{dt} &= -(\lambda_1 + \lambda_2)K, \end{aligned}$$

where λ_1 and λ_2 are positive constants of proportionality. By proceeding as in Problem 1 we can solve the foregoing mathematical model.

- (a) From the last equation in the given system of differential equations find $K(t)$ if $K(0) = K_0$. Then use $K(t)$ to find $C(t)$ and $A(t)$ from the first and second equations. Assume that $C(0) = 0$ and $A(0) = 0$.
 - (b) It is known that $\lambda_1 = 4.7526 \times 10^{-10}$ and $\lambda_2 = 0.5874 \times 10^{-10}$. Find the half-life of K-40.
 - (c) Use $C(t)$ and $A(t)$ found in part (a) to determine the percentage of an initial amount K_0 of K-40 that decays into Ca-40 and the percentage that decays into Ar-40 over a very long period of time.
- 6. Potassium-Argon Dating** The knowledge of how K-40 decays can be used to determine the age of very old igneous rocks. See Figure 3.3.5.

- (a) Use the solutions obtained in part (a) of Problem 5 to find the ratio $A(t)/K(t)$.
- (b) Use $A(t)/K(t)$ found in part (a) to show that

$$t = \frac{1}{\lambda_1 + \lambda_2} \ln \left[1 + \frac{\lambda_1 + \lambda_2}{\lambda_2} \frac{A(t)}{K(t)} \right].$$

- (c) Suppose it is determined that each gram of an igneous rock sample contains 8.5×10^{-7} grams of Ar-40 and 5.4×10^{-6} grams of K-40. Use the result in part (b) to find the approximate age of the rock.

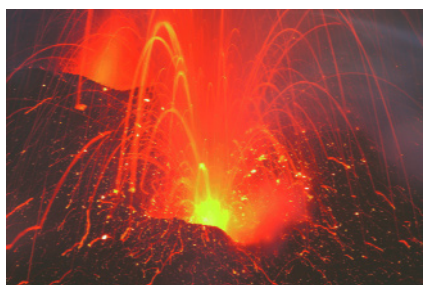


FIGURE 3.3.5 Igneous rocks are formed through solidification of volcanic lava

Mixtures

7. Consider two tanks A and B, with liquid being pumped in and out at the same rates, as described by the system of equations (3). What is the system of differential equations if, instead of pure water, a brine solution containing 2 pounds of salt per gallon is pumped into tank A?
8. Use the information given in Figure 3.3.6 to construct a mathematical model for the number of pounds of salt $x_1(t)$, $x_2(t)$, and $x_3(t)$ at time t in tanks A, B, and C, respectively.

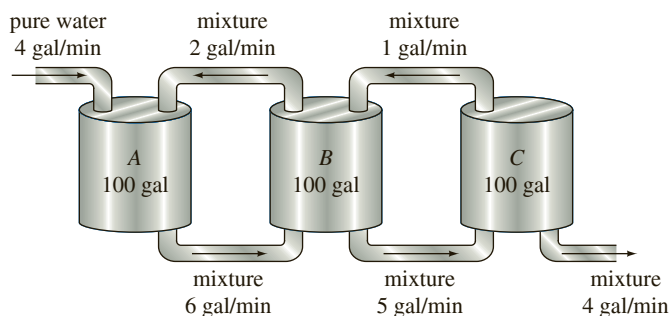


FIGURE 3.3.6 Mixing tanks in Problem 8

9. Two very large tanks A and B are each partially filled with 100 gallons of brine. Initially, 100 pounds of salt is dissolved in the solution in tank A and 50 pounds of salt is dissolved in the solution in tank B. The system is closed in that the well-stirred liquid is pumped only between the tanks, as shown in Figure 3.3.7.
- (a) Use the information given in the figure to construct a mathematical model for the number of pounds of salt $x_1(t)$ and $x_2(t)$ at time t in tanks A and B, respectively.
- (b) Find a relationship between the variables $x_1(t)$ and $x_2(t)$ that holds at time t . Explain why this relationship makes intuitive sense. Use this relationship to help find the amount of salt in tank B at $t = 30$ min.

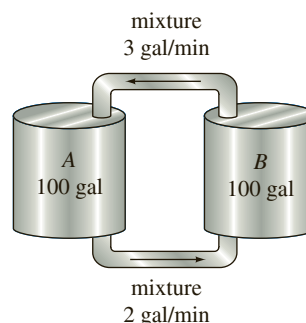


FIGURE 3.3.7 Mixing tanks in Problem 9

10. Three large tanks contain brine, as shown in Figure 3.3.8. Use the information in the figure to construct a mathematical model for the number of pounds of salt $x_1(t)$, $x_2(t)$, and $x_3(t)$ at time t in tanks A, B, and C, respectively. Without solving the system, predict limiting values of $x_1(t)$, $x_2(t)$, and $x_3(t)$ as $t \rightarrow \infty$.

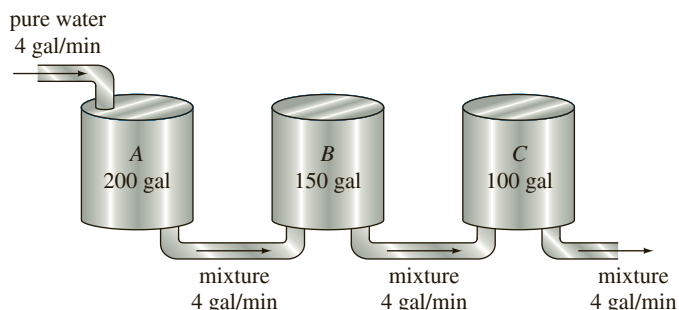


FIGURE 3.3.8 Mixing tanks in Problem 10

Predator-Prey Models

11. Consider the Lotka-Volterra predator-prey model defined by

$$\begin{aligned}\frac{dx}{dt} &= -0.1x + 0.02xy \\ \frac{dy}{dt} &= 0.2y - 0.025xy,\end{aligned}$$

where the populations $x(t)$ (predators) and $y(t)$ (prey) are measured in thousands. Suppose $x(0) = 6$ and $y(0) = 6$. Use a numerical solver to graph $x(t)$ and $y(t)$. Use the graphs to approximate the time $t > 0$ when the two populations are first equal. Use the graphs to approximate the period of each population.

Competition Models

12. Consider the competition model defined by

$$\begin{aligned}\frac{dx}{dt} &= x(2 - 0.4x - 0.3y) \\ \frac{dy}{dt} &= y(1 - 0.1y - 0.3x),\end{aligned}$$

where the populations $x(t)$ and $y(t)$ are measured in thousands and t in years. Use a numerical solver to analyze the populations over a long period of time for each of the following cases:

- (a) $x(0) = 1.5, \quad y(0) = 3.5$
- (b) $x(0) = 1, \quad y(0) = 1$
- (c) $x(0) = 2, \quad y(0) = 7$
- (d) $x(0) = 4.5, \quad y(0) = 0.5$

13. Consider the competition model defined by

$$\begin{aligned}\frac{dx}{dt} &= x(1 - 0.1x - 0.05y) \\ \frac{dy}{dt} &= y(1.7 - 0.1y - 0.15x),\end{aligned}$$

where the populations $x(t)$ and $y(t)$ are measured in thousands and t in years. Use a numerical solver to analyze the populations over a long period of time for each of the following cases:

- (a) $x(0) = 1, \quad y(0) = 1$
- (b) $x(0) = 4, \quad y(0) = 10$
- (c) $x(0) = 9, \quad y(0) = 4$
- (d) $x(0) = 5.5, \quad y(0) = 3.5$

Networks

14. Show that a system of differential equations that describes the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in Figure 3.3.9 is

$$\begin{aligned}L \frac{di_2}{dt} + L \frac{di_3}{dt} + R_1 i_2 &= E(t) \\ -R_1 \frac{di_2}{dt} + R_2 \frac{di_3}{dt} + \frac{1}{C} i_3 &= 0.\end{aligned}$$

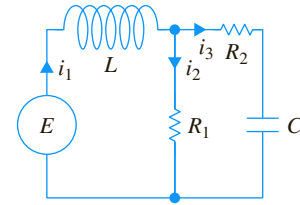


FIGURE 3.3.9 Network in Problem 14

15. Determine a system of first-order differential equations that describes the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in Figure 3.3.10.

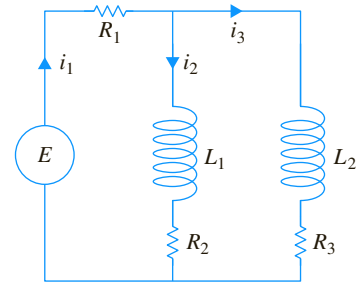


FIGURE 3.3.10 Network in Problem 15

16. Show that the linear system given in (18) describes the currents $i_1(t)$ and $i_2(t)$ in the network shown in Figure 3.3.4. [Hint: $dq/dt = i_3$.]

Additional Nonlinear Models

17. **SIR Model** A communicable disease is spread throughout a small community, with a fixed population of n people, by contact between infected individuals and people who are susceptible to the disease. Suppose that everyone is initially susceptible to the disease and that no one leaves the community while the epidemic is spreading. At time t , let $s(t)$, $i(t)$, and $r(t)$ denote, in turn, the number of people in the community (measured in hundreds) who are *susceptible* to the disease but not yet infected with it, the number of people who are *infected* with the disease, and the number of people who have *recovered* from the disease. Explain why the system of differential equations

$$\begin{aligned}\frac{ds}{dt} &= -k_1 si \\ \frac{di}{dt} &= -k_2 i + k_1 si \\ \frac{dr}{dt} &= k_2 i,\end{aligned}$$

where k_1 (called the *infection rate*) and k_2 (called the *removal rate*) are positive constants, is a reasonable mathematical model, commonly called a **SIR model**, for the spread of the epidemic throughout the community. Give plausible initial conditions associated with this system of equations.

18. (a) In Problem 17, explain why it is sufficient to analyze only

$$\begin{aligned}\frac{ds}{dt} &= -k_1 si \\ \frac{di}{dt} &= -k_2 i + k_1 si.\end{aligned}$$

- (b) Suppose $k_1 = 0.2$, $k_2 = 0.7$, and $n = 10$. Choose various values of $i(0) = i_0$, $0 < i_0 < 10$. Use a numerical solver to determine what the model predicts about the epidemic in the two cases $s_0 > k_2/k_1$ and $s_0 \leq k_2/k_1$. In the case of an epidemic, estimate the number of people who are eventually infected.

- 19. Concentration of a Nutrient** Suppose compartments A and B shown in Figure 3.3.11 are filled with fluids and are separated by a permeable membrane. The figure is a compartmental representation of the exterior and interior of a cell. Suppose, too, that a nutrient necessary for cell growth passes through the membrane. A model for the concentrations $x(t)$ and $y(t)$ of the nutrient in compartments A and B , respectively, at time t is given by the linear system of differential equations

$$\begin{aligned}\frac{dx}{dt} &= \frac{\kappa}{V_A}(y - x) \\ \frac{dy}{dt} &= \frac{\kappa}{V_B}(x - y),\end{aligned}$$

where V_A and V_B are the volumes of the compartments, and $\kappa > 0$ is a permeability factor. Let $x(0) = x_0$ and $y(0) = y_0$ denote the initial concentrations of the nutrient. Solely on the basis of the equations in the system and the assumption $x_0 > y_0 > 0$, sketch, on the same set of coordinate axes, possible solution curves of the system. Explain your reasoning. Discuss the behavior of the solutions over a long period of time.

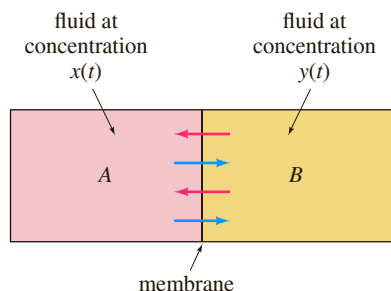


FIGURE 3.3.11 Nutrient flow through a membrane in Problem 19

- 20.** The system in Problem 19, like the system in (2), can be solved with no advanced knowledge. Solve for $x(t)$ and $y(t)$ and compare their graphs with your sketches in Problem 19. Determine the limiting values of $x(t)$ and $y(t)$ as $t \rightarrow \infty$. Explain why the answer to the last question makes intuitive sense.

- 21. Mixtures** Solely on the basis of the physical description of the mixture problem on page 108 and in Figure 3.3.1, discuss the nature of the functions $x_1(t)$ and $x_2(t)$. What is the behavior of each function over a long period of time? Sketch possible graphs of $x_1(t)$ and $x_2(t)$. Check your conjectures by using a numerical solver to obtain numerical solution curves of (3) subject to the initial conditions $x_1(0) = 25$, $x_2(0) = 0$.

- 22. Newton's Law of Cooling/Warming** As shown in Figure 3.3.12, a small metal bar is placed inside container A , and container A then is placed within a much larger container B . As the metal bar cools, the ambient temperature $T_A(t)$ of the medium within container A changes according to Newton's law of cooling. As container A cools, the temperature of the medium inside container B does not change significantly and can be considered to be a constant T_B . Construct a mathematical model for the temperatures $T(t)$ and $T_A(t)$, where $T(t)$ is the temperature of the metal bar inside container A . As in Problems 1, 5, and 20, this model can be solved by using prior knowledge. Find a solution of the system subject to the initial conditions $T(0) = T_0$, $T_A(0) = T_1$.

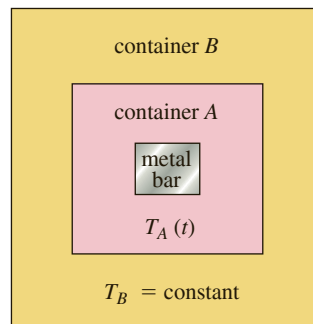


FIGURE 3.3.12 Container within a container in Problem 22

Chapter 3 In Review

Answers to selected odd-numbered problems begin on page ANS-4.

Answer Problems 1 and 2 without referring back to the text. Fill in the blank or answer true or false.

- If $P(t) = P_0 e^{0.15t}$ gives the population in an environment at time t , then a differential equation satisfied by $P(t)$ is _____.
- If the rate of decay of a radioactive substance is proportional to the amount $A(t)$ remaining at time t , then the half-life of the substance is necessarily $T = -(\ln 2)/k$. The rate of decay of the substance at time $t = T$ is one-half the rate of decay at $t = 0$. _____
- In March 1976 the world population reached 4 billion. At that time, a popular news magazine predicted that with an average

yearly growth rate of 1.8%, the world population would be 8 billion in 45 years. How does this value compare with the value predicted by the model that assumes that the rate of increase in population is proportional to the population present at time t ?

- Air containing 0.06% carbon dioxide is pumped into a room whose volume is 8000 ft³. The air is pumped in at a rate of 2000 ft³/min, and the circulated air is then pumped out at the same rate. If there is an initial concentration of 0.2% carbon dioxide in the room, determine the subsequent amount in the room at time t . What is the concentration of carbon dioxide at 10 minutes? What is the steady-state, or equilibrium, concentration of carbon dioxide?

- 5. Ötzi the Iceman** In September of 1991 two German tourists found the well-preserved body of a man partially frozen in a glacier in the Ötztal Alps on the border between Austria and Italy. See Figure 3.R.1. Through the carbon-dating technique it was found that the body of Ötzi the iceman—as he came to be called—contained 53% as much C-14 as found in a living person. Assume that the iceman was carbon dated in 1991. Use the method illustrated in Example 3 of Section 3.1 to find the approximate date of his death.



FIGURE 3.R.1 Ötzi the iceman in Problem 5

- 6.** In the treatment of cancer of the thyroid, the radioactive liquid Iodine-131 is often used. Suppose that after one day in storage, analysis shows that an initial amount A_0 of iodine-131 in a sample has decreased by 8.3%.

- Find the amount of iodine-131 remaining in the sample after 8 days.
- Explain the significance of the result in part (a).

- 7.** Solve the differential equation

$$\frac{dy}{dx} = -\frac{y}{\sqrt{a^2 - y^2}}$$

of the tractrix. See Problem 28 in Exercises 1.3. Assume that the initial point on the y -axis in $(0, 10)$ and that the length of the rope is $x = 10$ ft.

- 8.** Suppose a cell is suspended in a solution containing a solute of constant concentration C_s . Suppose further that the cell has constant volume V and that the area of its permeable membrane is the constant A . By **Fick's law** the rate of change of its mass m is directly proportional to the area A and the difference $C_s - C(t)$, where $C(t)$ is the concentration of the solute inside the cell at time t . Find $C(t)$ if $m = V \cdot C(t)$ and $C(0) = C_0$. See Figure 3.R.2.

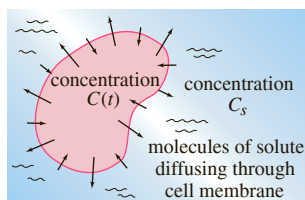


FIGURE 3.R.2 Cell in Problem 8

- 9.** Suppose that as a body cools, the temperature of the surrounding medium increases because it completely absorbs the heat being lost by the body. Let $T(t)$ and $T_m(t)$ be the temperatures of the body and the medium at time t , respectively.

If the initial temperature of the body is T_1 and the initial temperature of the medium is T_2 , then it can be shown in this case that Newton's law of cooling is $dT/dt = k(T - T_m)$, $k < 0$, where $T_m = T_2 + B(T_1 - T)$, $B > 0$ is a constant.

- The foregoing DE is autonomous. Use the phase portrait concept of Section 2.1 to determine the limiting value of the temperature $T(t)$ as $t \rightarrow \infty$. What is the limiting value of $T_m(t)$ as $t \rightarrow \infty$?
 - Verify your answers in part (a) by actually solving the differential equation.
 - Discuss a physical interpretation of your answers in part (a).
- 10.** According to **Stefan's law of radiation** the absolute temperature T of a body cooling in a medium at constant absolute temperature T_m is given by

$$\frac{dT}{dt} = k(T^4 - T_m^4),$$

where k is a constant. Stefan's law can be used over a greater temperature range than Newton's law of cooling.

- Solve the differential equation.
 - Show that when $T - T_m$ is small in comparison to T_m then Newton's law of cooling approximates Stefan's law. [Hint: Think binomial series of the right-hand side of the DE.]
- 11.** Suppose an RC -series circuit has a variable resistor. If the resistance at time t is defined by $R(t) = k_1 + k_2t$, where k_1 and k_2 are known positive constants, then the differential equation in (9) of Section 3.1 becomes

$$(k_1 + k_2t) \frac{dq}{dt} + \frac{1}{C} q = E(t),$$

where C is a constant. If $E(t) = E_0$ and $q(0) = q_0$, where E_0 and q_0 are constants, then show that

$$q(t) = E_0C + (q_0 - E_0C) \left(\frac{k_1}{k_1 + k_2t} \right)^{1/Ck_2}.$$

- 12.** A classical problem in the calculus of variations is to find the shape of a curve $\mathcal{C}$ such that a bead, under the influence of gravity, will slide from point $A(0, 0)$ to point $B(x_1, y_1)$ in the least time. See Figure 3.R.3. It can be shown that a nonlinear differential for the shape $y(x)$ of the path is $y[1 + (y')^2] = k$, where k is a constant. First solve for dx in terms of y and dy , and then use the substitution $y = k \sin^2\theta$ to obtain a parametric form of the solution. The curve $\mathcal{C}$ turns out to be a **cycloid**.

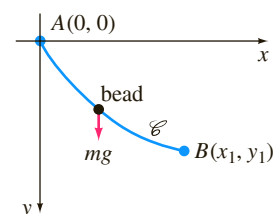


FIGURE 3.R.3 Sliding bead in Problem 12

13. A model for the populations of two interacting species of animals is

$$\frac{dx}{dt} = k_1 x (\alpha - x)$$

$$\frac{dy}{dt} = k_2xy.$$

Solve for x and y in terms of t .

14. Initially, two large tanks A and B each hold 100 gallons of brine. The well-stirred liquid is pumped between the tanks as shown in Figure 3.R.4. Use the information given in the figure to construct a mathematical model for the number of pounds of salt $x_1(t)$ and $x_2(t)$ at time t in tanks A and B , respectively.

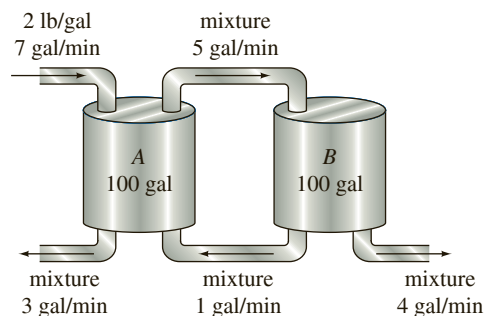


FIGURE 3.R.4 Mixing tanks in Problem 14

When all the curves in a family $G(x, y, c_1) = 0$ intersect orthogonally all the curves in another family $H(x, y, c_2) = 0$, the families are said to be **orthogonal trajectories** of each other. See Figure 3.R.5. If $dy/dx = f(x, y)$ is the differential equation of one family, then the differential equation for the orthogonal trajectories of this family is $dy/dx = -1/f(x, y)$. In Problems 15–18 find the differential equation of the given family by computing dy/dx and eliminating c_1 from this equation. Then find the orthogonal trajectories of the family. Use a graphing utility to graph both families on the same set of coordinate axes.

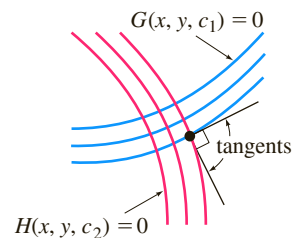


FIGURE 3.R.5 Orthogonal families

15. $y = c_1x$

16. $x^2 - 2y^2 = c_1$

17. $y = c_1 e^x$

18. $y = \frac{1}{x + c_1}$

19. With the identifications $a = r$, $b = r/K$, and $a/b = K$, Figures 2.1.7 and 3.2.2 show that the logistic population model, (3) of Section 3.2, predicts that for an initial population P_0 , $0 < P_0 < K$, regardless of how small P_0 is, the population increases over time but does not surpass the carrying capacity K . Also, for $P_0 > K$ the same model predicts that a population cannot sustain itself over time, so it decreases but yet never falls below the carrying capacity K of the ecosystem. The American ecologist **Warder Clyde Allee** (1885–1955) showed that by depleting certain fisheries beyond a certain level, the fish population never recovers. How would you modify the differential equation (3) to describe a population P that has these same two characteristics of (3) but additionally has a **threshold level** A , $0 < A < K$, below which the population cannot sustain itself and approaches extinction over time. [Hint: Construct a phase portrait of what you want and then form a differential equation.]

20. Sawing Wood A long *uniform* piece of wood (cross sections are the same) is cut through perpendicular to its length by a vertical saw blade. See Figure 3.R.6. If the friction between the sides of the saw blade and the wood through which the blade passes is ignored, then it can be assumed that the rate at which the saw blade moves through the piece of wood is

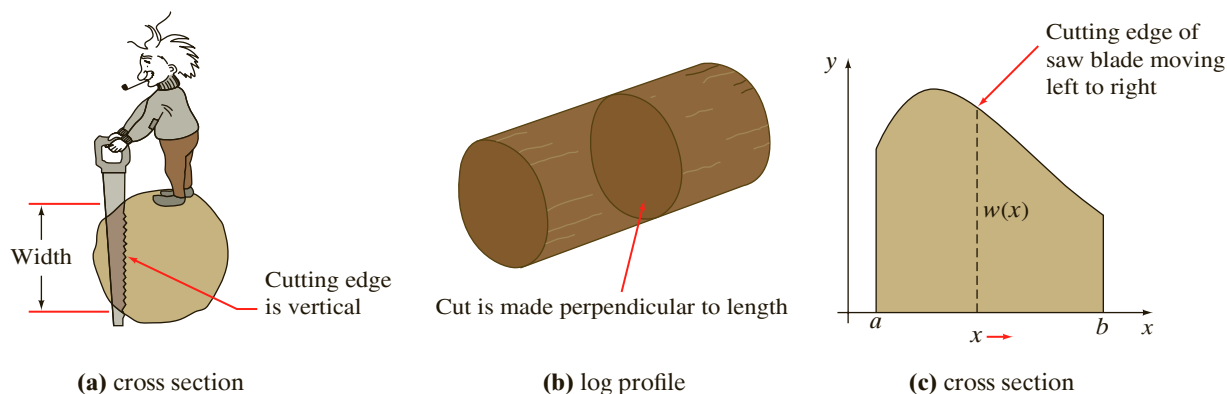


FIGURE 3.R.6 Sawing a log in Problem 20

inversely proportional to the width of the wood in contact with its cutting edge. As the blade advances through the wood (moving, say, left to right) the width of a cross section changes as a nonnegative continuous function w . If a cross section of the wood is described as a region in the xy -plane defined over an interval $[a, b]$ then, as shown in Figure 3.R.6(c), the position x of the saw blade is a function of time t and the vertical cut made by the blade can be represented by a vertical line segment. The length of this vertical line is the width $w(x)$ of the wood at that point. Thus the position $x(t)$ of the saw blade and the rate dx/dt at which it moves to the right are related to $w(x)$ by

$$w(x) \frac{dx}{dt} = k, \quad x(0) = a.$$

Here k represents the number of square units of the material removed by the saw blade per unit time.

- (a) Suppose the saw is computerized and can be programmed so that $k = 1$. Find an *implicit* solution of the foregoing initial-value problem when the piece of wood is a circular log. Assume a cross section is a circle of radius 2 centered at $(0, 0)$. [Hint: To save time see formula 41 in the table of integrals given on the right inside page of the front cover.]
- (b) Solve the implicit solution obtained in part (b) for time t as a function of x . Graph the function $t(x)$. With the aid of the graph, approximate the time that it takes the saw to cut through this piece of wood. Then find the exact value of this time.

21. Solve the initial-value problem in Problem 20 when a cross section of a uniform piece of wood is the triangular region given in Figure 3.R.7. Assume again that $k = 1$. How long does it take to cut through this piece of wood?

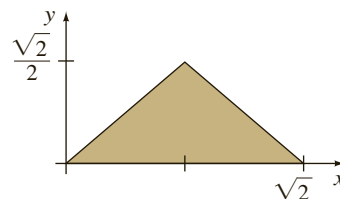


FIGURE 3.R.7 Triangular cross section in Problem 21

22. **Chemical Kinetics** Suppose a gas consists of molecules of type A. When the gas is heated a second substance B is formed by molecular collision. Let $A(t)$ and $B(t)$ denote, in turn, the number of molecules of types A and B present at time $t \geq 0$. A mathematical model for the rate at which the number of molecules of type A decreases is

$$\frac{dA}{dt} = -kA^2, \quad k > 0.$$

- (a) Determine $A(t)$ if $A(0) = A_0$.
- (b) Determine the number of molecules of substance B present at time t if it is assumed that $A(t) + B(t) = A_0$.
- (c) By hand, sketch rough graphs of $A(t)$ and $B(t)$ for $t \geq 0$.